CHIRAL KOSZUL DUALITY
NONCOMMUTATIVE FACTORISATION D-MODULES, ORDERED QUANTUM FIELDS, AND THE GEOMETRY OF DEFECTS

E_1-algebras in factorisable D-modules on ordered chiral configurations

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Abstract

Let X be a smooth complex algebraic curve. A protected quantum field theory on $X^{\mathrm{an}} \times \mathbb{R}$ has two local directions. Motion in X produces the singular operator product, while motion along the oriented real line orders the composition of observables. The algebraic object that carries both operations is an augmented algebra

$$A \in \mathrm{Alg}_{E_1}\big(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}\big),$$

where $\mathrm{FactD}(X)$ is the symmetric monoidal category of factorisable right D-modules. We call such an object an *ordered chiral algebra*. Its factorisation D-module records the complete diagonal-supported chiral operation, including every principal part and its coordinate covariance. Its internal E_1 multiplication records associative fusion in the transverse line.

The two operations admit a common geometric carrier. The block-ordered Ran space parametrises ordered sequences $(S_1, \dots, S_r)$ of finite chiral configurations. Its local operation category is the tensor product $E_1 \otimes \mathsf{Ch}_X$. A Boardman–Vogt resolution is indexed by planar levelled forests with a transverse edge colour and a chiral edge colour. Transverse contractions give the Stasheff identities, chiral contractions give the chiral Jacobi–Borcherds identities, and mixed contractions give the interchange law. The determinant orientation of the two edge sets makes the two cellular differentials anticommute.

The transverse bar construction is the two-sided simplicial object

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) = \big|[r] \longmapsto \mathbf{1} \otimes A^{\otimes r} \otimes \mathbf{1}\big| \simeq \int_{([0,1], \partial[0,1])} A.$$

Normalization identifies its chain object with the tensor coalgebra on $s\bar{A}$, and its differential multiplies adjacent transverse factors. The chiral operation remains a morphism of factorisation D-modules throughout this resolution. The pole filtration along a collision divisor identifies the coefficient of $(z-w)^{-m-1}$ with the $(m+1)$st normal symbol. The complete operator product therefore enters geometrically through the full principal-part filtration.

The Ran category carries pointwise tensor product and disjoint-union chiral convolution. Their Beck–Chevalley interchange makes the selected factorisation subcategory duoidal. An ordered chiral algebra is a bimonoid for these two products, and its bar carries commuting deconcatenation and chiral factorisation coproducts. The pointwise product governs the transverse associative bar. Chiral convolution governs the Francis–Gaitsgory chiral Chevalley and bar–cobar constructions. A

strong monoidal comparison functor relates the two duality theories while preserving their distinct geometric meanings.

The augmentation boundary condition has endomorphism algebra

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

For positive complete objects, the canonical map $A \to A^{!!}$ identifies the double dual with derived augmentation completion. If $C = \mathrm{Bar}_X^{\mathrm{ord}}(A)$, the universal twisting morphism identifies augmentation-complete A-modules with conilpotent C-comodules. Its two-sided form sends the diagonal bimodule to the diagonal bicomodule, carries derived tensor product to derived cotensor product, and identifies Hochschild theory with coHochschild theory. The interval, annulus, augmentation boundary, and derived centre are consequently four manifestations of one Koszul–Morita square.

Exchange arises from two-dimensional motion. An E_2 enhancement, a flat meromorphic connection on configuration space, or an equivalent braided line category supplies braid transport. The internal Hochschild cochains

$$Z_{\mathrm{ord,ch}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A)$$

form the universal E_2 algebra acting centrally on A. A rigid meromorphic tensor category with a faithful fibre functor reconstructs a coquasitriangular bialgebra by its Tannaka coend. KZ transport yields the Drinfeld–Jimbo quantum group through the Drinfeld–Kohno theorem. Rational difference transport together with PBW and evaluation-generation hypotheses yields Yangian recognition. Four-dimensional Chern–Simons theory supplies the corresponding physical line-operator construction.

A square-zero BRST derivation acts factorwise on the ordered bar and forms a bicomplex with the bar differential. Its spectral sequence compares the bar of the derived BRST reduction with BRST cohomology of the bar. Quantum Drinfeld–Sokolov reduction enters through this bicomplex, and a concentration theorem identifies its degree-zero cohomology with the corresponding W-algebra.

The theory carries two sewing geometries. A Calabi–Yau trace on the transverse E_1 algebra closes intervals into circles and produces Hochschild chains, annuli, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues pointed algebraic curves and produces conformal blocks over Deligne–Mumford moduli. Their simultaneous formal algebra is the two-edge-coloured Feynman transform, whose master equation is

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_X + \frac{1}{2}[\Theta, \Theta]_\perp + \hbar_X \Delta_X \Theta + \hbar_\perp \Delta_\perp \Theta = 0.$$

The two loop operators arise from distinct edge contractions and supercommute by the product orientation. Determinant-line curvature, BRST anomaly, BV anomaly, framing anomaly, and curved Koszul data appear as classes in their native deformation complexes; comparison morphisms transport them between mathematical and physical realizations.

A perturbatively renormalised holomorphic–topological BV theory on $X^{\mathrm{an}} \times \mathbb{R}$ produces an ordered chiral algebra whenever its observables are holomorphic in X, locally constant along $\mathbb{R}$, and satisfy the quantum master equation. Compactified configuration spaces resolve ultraviolet collisions, renormalised graph weights extend to their boundary strata, and the BV equation becomes the compatibility of those boundary restrictions. A second transverse direction produces line exchange and the Yang–Baxter equation. The bulk action factors canonically through the derived centre; an open–closed quasi-isomorphism identifies the selected physical bulk with that universal algebraic centre.

The resulting language unifies diagonal D-module operations, principal parts, partition lattices,

Fulton–MacPherson screens, Ran-space Koszul duality, ordered interval bars, Hochschild and Chevalley constructions, braid monodromy, Tannaka reconstruction, BRST reduction, ribbon sewing, stable curves, determinant lines, and perturbative quantum field theory. The detailed examples include matrix and quiver chiral algebras, Heisenberg and affine currents, $\beta\gamma$ and bc systems, Virasoro and principal W-algebras, KZ and KZB transport, Yangians, conformal blocks, Chern–Simons boundary algebras, Calabi–Yau pushforwards, Hall products, and Borcherds automorphic products.

Contents

Overture: five geometries of one quantum operation

Place two observables on a smooth complex curve X and on an oriented real line. Their approach in X probes the singular operator product. Their approach along the line composes them in a definite order. A second transverse direction permits exchange. Closing the line produces a trace. Varying the complex curve and plumbing nodes produces sewing. These five motions have five geometric carriers:

diagonal, interval, plane, circle, stable curve.

Each carrier contributes one algebraic operation.

On the curve, the singular product is the diagonal-supported morphism

$$\mu : j_*j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_!\mathfrak{g}. \tag{1}$$

On the oriented line, ordered fusion is the associative operation

$$m : A \otimes A \longrightarrow A. \tag{2}$$

On the plane, exchange is parallel transport in configuration space and gives a braid-group representation. On the circle, closure is factorisation homology,

$$\int_{S^1} A \simeq \mathrm{HH}_*(A). \tag{3}$$

On the boundary of moduli, sewing contracts the paired states carried by the two branches of a node.

The first two operations define the primitive object of the theory:

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}). \tag{4}$$

The factorisable D-module carries the complete chiral singularity. The internal E_1 structure carries transverse order. Exchange is supplied by two-dimensional transport, trace by cyclicity, and sewing by a modular action.

Theorem 0.1 (Five-carrier theorem)**.** *Let A be as in (4). The following constructions are functorial.*

The chiral coefficient is the system of D-modules on powers of X equipped with factorisation maps and diagonal singularities. The transverse bar is the derived tensor product

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) \simeq \mathbf{1} \otimes^L_A \mathbf{1}.$$

A locally constant analytic realization identifies this object with factorisation homology of an interval carrying the augmentation module at each endpoint. An E_2 enhancement or a flat meromorphic connection supplies braid transport. A cyclic trace supplies the circle object and ribbon contractions. A clutching-compatible modular action supplies stable-curve sewing. Each extension is classified by the corresponding geometric datum.

Proof. The coefficient statement is the definition of a factorisable D-module. The interval statement is the two-sided bar formula together with factorisation excision. The little two-disks operad supplies the exchange paths that generate braid transport. Evaluation against a cyclic trace closes an interval to a circle. A perfect pairing and the clutching correspondences contract the two flags of each stable-graph edge. Functoriality follows from functoriality of these operadic and correspondence constructions. □

The dependence of the constructions is

$$\begin{aligned} \text{chiral coefficient} &\xrightarrow{E_1\text{-fusion}} \text{ordered chiral algebra}, \\ \text{ordered chiral algebra} &\xrightarrow{E_2\text{-lift or transport}} \text{exchange category}, \\ \text{ordered chiral algebra} &\xrightarrow{\text{cyclic trace}} \text{ribbon amplitudes}, \\ \text{chiral coefficient} &\xrightarrow{\text{modular sewing}} \text{stable-curve amplitudes}. \end{aligned} \tag{5}$$

A theory carrying both trace and modular sewing has two compatible genus expansions, one transverse and ribbon, the other chiral and Deligne–Mumford.

The common law

All five carriers obey the same boundary law. Codimension-one strata carry elementary compositions. Codimension-two corners carry relations among pairs of compositions. The oriented boundary of an oriented boundary vanishes. Coefficient maps turn this topological identity into an algebraic master equation.

For the interval, the two endpoints of the one-dimensional associahedral boundary give

$$(ab)c - a(bc) = 0.$$

For three chiral points, the three partial diagonals give the chiral Jacobi identity. For a compactified configuration space of Feynman vertices, the collision faces give the classical or quantum BV equation after graph weights and counterterms are inserted. For a stable curve, separating and nonseparating nodes give the bracket and loop terms of the modular equation. Topology supplies the signed incidence relation; the theory supplies the coefficient maps.

Theorem 0.2 (Stokes-to-master theorem)**.** *Let $\overline{C}$ be an oriented stratified space whose codimension-one faces are labelled by elementary contractions, and let E be a coefficient complex. Suppose every oriented face F carries a degree-one contraction ρ_F on E, compatible with restriction to corners and satisfying*

$$d_E \rho_F + \rho_F d_E = 0.$$

Set

$$D = d_E + \sum_F \rho_F.$$

For every codimension-two stratum S, the component of D^2 supported on S is the signed sum of the composites associated with the incident faces. Consequently, $D^2 = 0$ is equivalent to the family of corner identities.

Proof. Expanding D^2 gives the internal square, the mixed chain-map terms, and ordered pairs of face contractions. The first two groups vanish by the hypotheses. The remaining ordered pairs are grouped by their common codimension-two stratum. The induced boundary orientations provide the signs in the corner sum. This is the cellular identity $\partial^2 = 0$ with coefficients. □

The theorem gives one algebraic shape to the bar differential, the Jacobi identity, the Yang–Baxter equation, the BV equation, and the modular sewing equation. The maps ρ_F carry the dynamics; the incidence relation carries the universal geometry.

Six realizations

A singular quantum field is a distribution. Its support and wave-front set belong to the observable. The D-module in (1) is the algebraic expression of this distributional structure.

A quantum amplitude composes by summing over intermediate states. The bar construction is the derived form of this composition. A bar word is an ordered sequence of intermediate boundary operations, and the bar differential fuses adjacent operations.

A conformal anomaly is the curvature of an invertible geometric object over moduli. Its representatives include determinant lines, Atiyah algebras, and invertible field theories. Tensoring with the inverse anomaly theory produces a flat family when the corresponding state line exists.

A topological field theory converts isotopy into algebra. Motion of line defects in a plane gives braid transport, and closure gives ribbon traces. The topology of the transverse space creates both operations.

A theory with boundaries and defects is naturally categorical. Boundary conditions are objects, boundary-changing fields are morphisms, interfaces are bimodules, and junctions are higher morphisms. Formula (4) is the endomorphism algebra of a chosen boundary object.

A perturbative quantum field theory is controlled by a BV complex and by extensions of graph weights across compactified collision strata. The quantum master equation gives a square-zero observable differential and compatible factorisation maps. Holomorphic dependence along X and topological dependence along the line produce the analytic realization of (4).

These six realizations express one geometric syntax in distributional, homological, anomalous, topological, categorical, and perturbative language.

The invariant sentence

The complete local statement is

> A noncommutative chiral field theory on a curve with one ordered topological direction is an E_1-algebra in factorisable D-modules.

Every construction that follows is obtained from this object by a specified functor, geometric extension, or trace. The geometry determines the admissible compositions. The coefficient determines their values. The boundary orientation determines their equations.

Part I
The noncommutative chiral object

1 The phenomenon and its invariant language

1.1 Two limits of the same experiment

Place local operators on a complex curve X and line defects on an oriented real line. There are two independent short-distance limits. In the first, two points of X approach one another while their transverse positions remain fixed. The product develops a finite principal part along the diagonal. In the second, two transverse intervals become adjacent while their positions on X remain separated. The product is associative and ordered. A theory in which both limits exist therefore carries two operations of different geometric origin:

$$\text{chiral collision on } X, \qquad \text{ordered fusion on } \mathbb{R}. \tag{7}$$

The first is encoded by a factorisation D-module. The second is encoded by an E_1-multiplication internal to the category of such modules.

This distinction is forced by topology. A punctured complex disc has a nontrivial fundamental group, so analytic continuation of insertions can produce monodromy. An ordered chamber

$$\mathrm{Conf}_n^{<}(\mathbb{R}) = \{t_1 < \cdots < t_n\}$$

is contractible. It supplies a canonical ordering and trivial parallel transport inside each chamber. Thickening the transverse line to a plane introduces exchange paths and braid monodromy. Chiral singularity, associative order, and braid exchange are therefore three independently specified structures.

Definition 1.1 (Ordered chiral algebra)**.** Let X be a smooth complex algebraic curve and let $\mathrm{FactD}(X)$ be the symmetric monoidal stable category of factorisable right D-modules defined in Section 2. An *ordered chiral algebra* on X is an augmented algebra

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X)).$$

Its multiplication is denoted $m : A \otimes A \to A$, and its augmentation is denoted $\varepsilon : A \to \mathbf{1}$.

The word *chiral* names the coefficient category. The word *ordered* names the internal E_1 operation. Each adjective records one geometric direction.

1.2 The diagonal and the interval

The diagonal is the carrier of the operator product expansion. Let

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2, \qquad \Delta : X \hookrightarrow X^2.$$

A chiral singular product is a morphism from an extension off the diagonal to a distribution supported on the diagonal. In the Lie form of Beilinson and Drinfeld it is

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}. \tag{8}$$

A coordinate expansion of (8) produces all OPE modes. The morphism itself is coordinate-free.

The interval is the carrier of the ordered product. If $I_1 < I_2$ are disjoint subintervals of an oriented interval I, factorisation gives

$$A(I_1) \otimes A(I_2) \longrightarrow A(I).$$

Shrinking I_1 and I_2 inside the chamber $I_1 < I_2$ produces the binary multiplication. Three ordered subintervals produce the two composites $(ab)c$ and $a(bc)$; the contractibility of the space of such shrinkings gives their coherent homotopy. After fixing the endpoints and quotienting by affine reparametrisation, the standard real compactification of ordered points is an associahedron.

The two geometries meet on $X \times \mathbb{R}$. A small rectangle $U \times I$ carries a complex of observables. Chiral factorisation varies U in X; ordered factorisation varies I in $\mathbb{R}$. Their interchange is the geometric origin of an E_1-algebra in $\mathrm{FactD}(X)$.

1.3 The unique language

A useful language must satisfy four requirements.

First, it must retain the entire singular product. A logarithmic residue detects a simple normal coefficient. A fourth-order Virasoro pole and a double affine pole require higher normal jets. A factorisation D-module contains all of them at once.

Second, it must retain noncommutative fusion before any symmetrisation. Tensor order, line order, and braid exchange are three different structures. The E_1-multiplication retains the first two; an E_2 enhancement supplies the third.

Third, it must separate algebraic and analytic completion. The bar construction is defined in the stable category. Trace-class sewing and convergence of conformal blocks are additional analytic theorems.

Fourth, it must type every anomaly. Determinant-line curvature, the BRST anomaly class, the BV anomaly class, and curved Koszul curvature live in distinct deformation complexes. A comparison morphism transports one class to another and states the precise sense in which they agree.

These requirements lead to the following square:

$$\begin{tikzcd}
\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \arrow[r, "\mathrm{Bar}_X^{\mathrm{ord}}"] \arrow[d, "\mathrm{an}"'] & \mathrm{Coalg}_{\mathrm{conil}}(\mathrm{FactD}(X)) \arrow[d, "\mathrm{an}"] \\
\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R}) \arrow[r, "\int_{([0,1],\partial)}"'] & \mathrm{Fact}_{\mathrm{hol}}(X^{\mathrm{an}}).
\end{tikzcd} \tag{9}$$

The upper row is the internal associative bar construction. For positive complete objects, cobar reconstructs it. The lower row is interval factorisation of a rectangular holomorphic–topological theory. On regular holonomic objects, Riemann–Hilbert identifies the vertical arrows. The square expresses one phenomenon in algebraic and analytic language.

1.4 What is genuinely noncommutative

A conventional factorisation algebra on a manifold has symmetric multiplication for disjoint opens because disjoint union is unordered. Noncommutativity enters when a codimension-one direction orders the opens. A boundary condition, an interface, or a family of parallel line defects supplies such a direction. The resulting multiplication is composition in the category of defects. The chiral OPE is the diagonal singularity of its factorisation coefficient.

Let $\mathcal{C}$ be a factorisation category on X and let b be an object representing a boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b) \tag{10}$$

is an E_1-algebra in the factorisation coefficient category. Composition of endomorphisms gives the E_1 product. Collision of their insertion points gives the chiral singularity. Formula (10) is the native source of noncommutative chiral algebras.

A simple algebraic model is obtained from a factorisation object V and an associative algebra B. If V is commutative as a factorisation object, then

$$A = B \otimes V$$

is an ordered chiral algebra with multiplication $m_B \otimes m_V$. Matrix-valued free fields and Chan–Paton factors have this form locally. The bar construction sees the noncommutative resolution of B while every term remains a chiral coefficient system on X.

1.5 The bar is transverse

The reduced bar construction of an augmented algebra is generated by ordered words

$$[a_1|\cdots|a_r],$$

and its differential multiplies adjacent letters. The word order is the order of intervals on $\mathbb{R}$. Each coefficient a_i and each factorisation map of $A^{\otimes r}$ retains the chiral singularities. The bar differential acts in the transverse word direction by multiplying adjacent letters.

The source and target of each operator now determine its role. The de Rham differential of a configuration space, the residue differential along a collision divisor, and the simplicial bar differential are three distinct operators. They can coexist in a total complex, but their source and target degrees must be written separately.

Proposition 1.2 (Low-arity bar identity)**.** *Let A be a dg E_1-algebra in a stable symmetric monoidal category. On homogeneous elements of the augmentation ideal,*

$$b_2[a|b|c] = (-1)^{\alpha}[(ab)|c] + (-1)^{\beta}[a|(bc)],$$

with the standard suspension signs, and

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

Hence the multiplication part of the bar differential squares to zero exactly by associativity.

Proof. There are two length-two contractions of a word of length three. The suspension orientation gives opposite signs. The two coefficient operations agree by associativity. □

The complete chiral identities are tested elsewhere: by the chiral operad on the formal disc, by factorisation on the Ran space, or by the partition collision complex. Associativity governs transverse contractions, the Borcherds identity governs chiral contractions, and the mixed forest faces impose their interchange.

1.6 From order to exchange

The distinction between order and exchange is the first physical test of the language. Parallel line defects on an oriented topological line have a fixed order and an associative fusion product. A second topological direction supplies an exchange path. The homotopy class of that path is a braid. Thus

$$E_1 \text{ fusion} \quad \xrightarrow{\text{one more topological direction}} \quad E_2 \text{ exchange.}$$

Dunn additivity identifies the second structure with an E_1-algebra object in E_1-algebras. The universal E_2 object acting on an E_1 algebra is its Hochschild centre.

This yields a precise physical dictionary:

ordered intervals	fusion of parallel defects,
chiral diagonals	singular OPE on the spectral curve,
second transverse direction	braid exchange,
Hochschild centre	universal bulk or exchange algebra,
Tannaka coend	reconstructed quantum symmetry.

Each row of the table denotes a functor determined by its geometric carrier.

1.7 The two notions of modularity

A trace on an E_1 algebra closes an interval into a circle. Iterated gluing produces ribbon surfaces. This is the modularity of open topological field theory and Hochschild homology.

A chiral algebra on a varying algebraic curve has a different modularity. Plumbing a node changes the complex curve. The relevant spaces are $\overline{\mathcal{M}}_{g,n}$, and the amplitudes are conformal blocks. The first genus is transverse and ribbon; the second is complex and Deligne–Mumford.

When an ordered chiral algebra carries both sewing structures and their mixed interchange maps, the two contractions supercommute with the product orientation. The resulting master equation has two loop operators. The separation of the two loop parameters is essential in gauge theory: one counts topological colour loops, the other counts handles of the chiral curve.

1.8 Mathematics and physics

The mathematical object is an E_1 algebra in factorisable D-modules. The physical object is a holomorphic–topological factorisation algebra with a boundary or line defect. The comparison is the restriction morphism from quantum factorisation observables to the defect factorisation algebra.

Let $\mathrm{Obs}_{\mathcal{T}}$ be the quantum observables of a renormalised BV theory $\mathcal{T}$ on $X \times \mathbb{R}_{\geq 0}$. A boundary condition gives a factorisation algebra Obs_∂ on $X \times \mathbb{R}$. When holomorphic dependence admits a factorisable D-module realization and the boundary theory is locally constant in the ordered direction, Theorem 10.5 gives

$$\mathrm{Obs}_\partial \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)). \tag{11}$$

The renormalized quantum master equation makes the observable differential square to zero and makes the factorisation maps compatible with ultraviolet contractions. The vanishing of the local BV anomaly class is the existence criterion for this quantum factorisation object.

The perturbative protected sectors of Chern–Simons theory, four-dimensional Chern–Simons theory, suitable holomorphic twists of supersymmetric gauge theories, and open string field theory furnish instances of this pattern after their boundary or defect observable comparisons are constructed. Their exchange geometries and spectral curves select the corresponding quantum symmetries, while the interval bar construction remains the same functor.

1.9 The theorem spine

The paper proves the following sequence of statements.

(i) Ordered block-Ran factorisation objects are equivalent to E_1-algebras in $\mathrm{FactD}(X)$.

(ii) Their ordered bar is interval factorisation homology and has an associahedral cofibrant model.

(iii) On connected positive-weight objects and their complete Mittag–Leffler limits, the internal associative bar and cobar reconstruct one another. Separately, Ran cardinality is pro-nilpotent for chiral convolution.

(iv) The augmentation boundary has endomorphism algebra $A^!$, and the double dual is the augmentation completion.

(v) An E_2 lift or flat two-dimensional transport supplies braid exchange; the Hochschild centre is the universal E_2 algebra acting centrally on the underlying E_1 algebra.

(vi) Rigid meromorphic line categories reconstruct quantum symmetries by Tannaka coends.

(vii) BRST derivations extend to the bar and yield a convergent comparison spectral sequence under boundedness and completeness.

(viii) Cyclic transverse sewing and chiral curve sewing form a bi-modular master equation with two independent loop operators.

(ix) Under the algebraisation, locality, gauge-fixing, and anomaly-cancellation hypotheses stated in Part V, factorisation observables realize the preceding constructions in perturbative holomorphic–topological field theory.

The subsequent constructions and examples exhibit these statements as functorial realizations of a single ordered chiral object.

2 Factorisable D-modules and the complete chiral coefficient

2.1 The finite-set presentation

Let X be a smooth complex algebraic curve. Write $\mathrm{Fin}^{\mathrm{surj}}$ for the category of nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \qquad (x_j)_{j\in J} \longmapsto (x_{\alpha(i)})_{i\in I}.$$

The Ran prestack is the colimit

$$\mathrm{Ran}(X) = \operatorname*{colim}_{I \in (\mathrm{Fin}^{\mathrm{surj}})^{\mathrm{op}}} X^I.$$

A right D-module on $\mathrm{Ran}(X)$ is represented by a compatible family $\mathcal{F}_I \in D\text{-mod}(X^I)$ with transition morphisms along the diagonals. All functors below are derived.

For a decomposition $I = I_1 \sqcup \cdots \sqcup I_r$, let

$$U(I_1, \ldots, I_r) \subset X^I$$

be the open locus on which points belonging to different blocks are disjoint; points in one block may collide. Denote the inclusion by $j_{I_\bullet}$.

Definition 2.1 (Factorisable D-module)**.** A factorisable right D-module $\mathcal{F}$ on X is a compatible Ran family $\{\mathcal{F}_I\}$ together with equivalences

$$j^!_{I_\bullet} \mathcal{F}_I \simeq j^!_{I_\bullet} (\mathcal{F}_{I_1} \boxtimes \cdots \boxtimes \mathcal{F}_{I_r}) \tag{12}$$

for every decomposition of I, compatible with refinement, permutation, and diagonal transition maps.

The equivalence (12) expresses multiplicativity on separated supports. The extension of $\mathcal{F}_I$ across partial diagonals carries the singular terms of the operator product. Coordinates and propagators provide local presentations of this intrinsic Ran-space datum.

Let $\mathrm{FactD}(X)$ denote a stable presentable enhancement of this category. Its *coefficient tensor product* is the componentwise !-tensor product

$$(\mathcal{F} \otimes^{\mathrm{pt}} \mathcal{G})_I := \mathcal{F}_I \otimes^! \mathcal{G}_I. \tag{13}$$

The unit is the vacuum factorisation module $\mathbf{1}$.

Proposition 2.2. *Assume that the chosen Ran enhancement is closed under componentwise !-tensor product and componentwise colimits. Then* (13) *makes* $\mathrm{FactD}(X)$ *a presentable stable symmetric monoidal category, and tensor product preserves colimits separately.*

Proof. The assertion holds on each X^I. The diagonal compatibilities and the restrictions in (12) are preserved by !-tensor product. The stated closure hypotheses therefore descend the componentwise structure to the factorisation subcategory. □

This is the symmetric monoidal category in which the transverse E_1 multiplication is formed. It must be distinguished from the chiral tensor product on $D(\mathrm{Ran}\, X)$, introduced below.

2.2 The complete singular coefficient

Consider the binary component $\mathcal{F}_{\{1,2\}}$ on X^2. Off the diagonal, factorisation identifies it with $\mathcal{F}_{\{1\}} \boxtimes \mathcal{F}_{\{2\}}$. The extension across the diagonal has local cohomology supported on $\Delta(X)$. In a formal coordinate it is expressed by principal parts

$$\sum_{m \geq 0} \frac{c_m(w)}{(z-w)^{m+1}}.$$

A coordinate change mixes the coefficients by an upper-triangular transformation and preserves the filtered distribution.

For $D = \Delta(X) \subset X^2$, the pole filtration is

$$F_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}(mD),$$

and

$$\mathrm{gr}^F_m \, \mathcal{O}_{X^2}(*D) \simeq N^{\otimes m}_{D/X^2}. \tag{14}$$

Thus the associated graded of a pole is a tensorial normal symbol; the filtered object retains the coordinate-dependent mixing of modes.

Principle 2.3 (Complete coefficient principle)**.** *The chiral coefficient of an ordered chiral algebra is the factorisable D-module itself. Residues, normal jets, formal modes, and distributional kernels are functorial realizations of its diagonal filtration.*

For an ordered chiral algebra, the transverse multiplication

$$m : A \otimes^{\mathrm{pt}} A \longrightarrow A$$

is a morphism of factorisable D-modules. It therefore respects the whole diagonal filtration and every disjoint-factorisation map. In coordinates it acts on the complete singular expansion, including every pole order and the regular term.

2.3 Transverse and chiral Lie operations

Restriction along the operad map $\mathsf{Lie} \to E_1$ gives the graded commutator

$$[a, b]_\perp = m(a, b) - (-1)^{|a||b|} m(b, a). \tag{15}$$

Proposition 2.4 (Internal commutator)**.** *Restriction of operads defines*

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}) \longrightarrow \mathrm{Alg}_{\mathsf{Lie}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}).$$

The bracket (15) *satisfies the graded Jacobi identity.*

Proof. It is the commutator bracket in an associative algebra object of a symmetric monoidal category. The usual cancellation proof of Jacobi is operadic and therefore applies internally. □

The bracket $[-,-]_\perp$ is a Lie operation on the *transverse multiplication.* A Beilinson–Drinfeld chiral Lie bracket is instead a morphism

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

in the chiral pseudo-tensor category. The factorisation coefficient of A may carry such a chiral operation. A comparison with (15) is a morphism between the corresponding operadic deformation problems. This distinction is essential in boundary field theory: the chiral bracket measures collision on X, while $[-,-]_\perp$ measures the discontinuity between two transverse orders.

2.4 Augmentations and boundary conditions

An augmentation $\varepsilon : A \to \mathbf{1}$ is a morphism of E_1 algebras in $\mathrm{FactD}(X)$. It makes $\mathbf{1}$ a left and right A-module, and the augmentation ideal is

$$\bar{A} = \mathrm{Fib}(A \xrightarrow{\varepsilon} \mathbf{1}).$$

For a right module R and a left module L, the two-sided bar $\mathrm{Bar}(R, A, L)$ computes interval factorisation homology with endpoint labels R and L. The reduced bar is the case $R = L = \mathbf{1}$.

The module formulation is physically primary. A boundary condition is an object whose local endomorphisms form an E_1 algebra. The augmentation selects the boundary state represented by the unit.

2.5 Two monoidal products on the Ran space

There are two different tensor products in the subject.

The coefficient tensor $\otimes^{\mathrm{pt}}$ of (13) is componentwise. Its support satisfies

$$\mathrm{supp}(\mathcal{F} \otimes^{\mathrm{pt}} \mathcal{G}) \subset \mathrm{supp}(\mathcal{F}) \cap \mathrm{supp}(\mathcal{G}).$$

It preserves the cardinality profile of the two factors. Consequently the cardinality filtration has degree zero for this tensor product.

The *chiral tensor product* on $D(\mathrm{Ran}\, X)$ is defined by the disjoint-union correspondence

$$\mathrm{Ran}(X) \times \mathrm{Ran}(X) \supset (\mathrm{Ran} \times \mathrm{Ran})_{\mathrm{disj}} \longrightarrow \mathrm{Ran}(X).$$

For reduced objects supported on subsets of cardinality at least r and s, respectively, chiral convolution is supported in cardinality at least $r + s$:

$$D(\mathrm{Ran}\, X)^{\geq r} \otimes^{\mathrm{ch}} D(\mathrm{Ran}\, X)^{\geq s} \subset D(\mathrm{Ran}\, X)^{\geq r+s}. \tag{16}$$

Theorem 2.5 (Ran chiral pro-nilpotence)**.** *For the chiral tensor product $\otimes^{\mathrm{ch}}$, the quotient by objects supported in cardinality $> N$ is nilpotent on reduced objects, and the cardinality completion is pro-nilpotent. The support growth in* (16) *is the mechanism of this chiral-convolution theorem.*

Proof. Equation (16) implies that an $(N + 1)$-fold chiral product of reduced objects is supported in cardinality at least $N + 1$ and hence vanishes in the Nth quotient. Taking the inverse limit gives the pro-nilpotent category. This is the support argument used in chiral Koszul duality on the Ran space [38]. □

The two products answer different questions. The pointwise tensor permits an E_1 algebra of defects with a fixed chiral coefficient. The chiral tensor assembles disjoint finite subsets and is the ambient product for the Francis–Gaitsgory Lie–commutative-coalgebra equivalence. Any comparison between their bar constructions must name the monoidal functor that relates them.

2.6 Regular holonomic realization

Assume that every $\mathcal{F}_I$ is regular holonomic and that the chosen subcategory is closed under the tensor operations used above. After the standard de Rham shifts, Riemann–Hilbert identifies $\mathcal{F}_I$ with a constructible complex on $X^{I,\mathrm{an}}$. Factorisation equivalences become constructible factorisation equivalences on the disjoint loci.

Proposition 2.6 (Analytic realization)**.** *Under these regular-holonomic closure hypotheses, Riemann–Hilbert gives a symmetric monoidal realization from* $(\mathrm{FactD}(X), \otimes^{\mathrm{pt}})$ *to constructible factorisation sheaves on* X^{an}*. It carries internal* E_1 *algebras, modules, bar constructions, and Hochschild cochains to their constructible realizations whenever it commutes with the required geometric realizations and internal Homs.*

The analytic realization reveals the monodromy of local systems on $\mathrm{Conf}_n(X)$. A transverse E_2 enhancement supplies the additional exchange paths of line defects.

2.7 Factorisation categories and Morita origin

Let $\mathcal{C}$ be a D-linear factorisation category on X, and let b be a compatible object. Its internal endomorphisms are

$$A_b(U) = \mathrm{RHom}_{\mathcal{C}(U)}(b|_U, b|_U).$$

Composition is ordered, while factorisation in U supplies the chiral coefficient.

Theorem 2.7 (Morita source)**.** *Let* $\mathcal{C}$ *be a presentable* D*-linear factorisation category and let* b *be a dualisable factorisation object. Then* $A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$ *is an* E_1 *algebra in* $\mathrm{FactD}(X)$*. If* b *is compact and generates each local category, and these local Morita equivalences commute with factorisation descent, then*

$$\mathrm{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \longrightarrow \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

is a factorisation Morita equivalence.

Proof. Composition gives the E_1 multiplication. Factorisation of b on disjoint opens gives the coefficient factorisation maps. The compact-generator Morita theorem applies locally; the final hypothesis identifies the local equivalences on overlaps and descends them. □

This is the source encountered in gauge theory: a boundary condition or line defect is b; local endomorphisms form A_b; transverse composition is noncommutative; singular dependence on the holomorphic insertion point is carried by the factorisable D-module.

2.8 Historical position

Dirac's distributional formalism made the singular support of products of quantum fields unavoidable. Wilson organised short-distance singularities by the operator product expansion. Borcherds and Kac encoded their algebraic identities. Beilinson and Drinfeld replaced coordinate series by D-module morphisms on diagonals. Factorisation algebras expressed the same local-to-global law for quantum observables. The present object adds an oriented topological direction whose associative order coexists with the diagonal chiral operation; exchange and symmetric descent are subsequent geometric functors.

3 The duoidal law: locality as a bialgebra identity

The primitive object admits a second, equivalent description which makes its geometry visible at once. The Ran category carries a pointwise tensor product and a convolution product along disjoint union. One product combines coefficients supported at the same finite subset; the other combines observables supported at disjoint finite subsets. Their compatibility is duoidal. An ordered chiral algebra is a monoid for the first product and a factorisation coalgebra for the second. The assertion that multiplication is a morphism of factorisation objects is therefore a bialgebra identity, but the multiplication and comultiplication belong to different monoidal structures.

3.1 The two products and their interchange

Let

$$\mathcal{D}_X = D(\operatorname{Ran} X).$$

Write $\otimes^{\mathrm{pt}}$ for the pointwise !-tensor product. Write $\otimes^{\mathrm{ch}}$ for chiral convolution along the correspondence

$$\operatorname{Ran} X \times \operatorname{Ran} X \xleftarrow{j} (\operatorname{Ran} X \times \operatorname{Ran} X)_{\mathrm{disj}} \xrightarrow{u} \operatorname{Ran} X, \qquad F \otimes^{\mathrm{ch}} G := u_! j^! (F \boxtimes G).$$

The diagonal of the Ran space is compatible with disjoint union. The projection formula and Beck–Chevalley base change therefore give a natural interchange morphism

$$(F_1 \otimes^{\mathrm{ch}} F_2) \otimes^{\mathrm{pt}} (G_1 \otimes^{\mathrm{ch}} G_2) \longrightarrow (F_1 \otimes^{\mathrm{pt}} G_1) \otimes^{\mathrm{ch}} (F_2 \otimes^{\mathrm{pt}} G_2), \tag{17}$$

with the middle factors permuted by the symmetry of $\otimes^{\mathrm{pt}}$. The six-functor formalism always supplies the lax interchange map (17). On the subcategory where the relevant base-change and projection-formula maps are equivalences, this interchange is strong on the disjoint locus.

Theorem 3.1 (Duoidal Ran theorem)**.** *Fix a sheaf theory on* $\operatorname{Ran} X$ *for which pointwise !-tensor, chiral convolution, the projection formula, and disjoint-union base change are defined. Then*

$$(\mathcal{D}_X, \otimes^{\mathrm{pt}}, \otimes^{\mathrm{ch}})$$

is a lax duoidal stable category. On the full subcategory on which the base-change morphisms in (17) *are equivalences, it is a strong duoidal category.*

Proof. Associativity and symmetry of $\otimes^{\mathrm{pt}}$ are inherited componentwise from $D(X^I)$. Associativity of $\otimes^{\mathrm{ch}}$ is the associativity of union of pairwise disjoint finite subsets. Apply the two chiral coproduct correspondences, tensor their outputs pointwise, and then use the diagonal base-change square to regroup the first and third, and the second and fourth, factors. The resulting natural transformation is (17). The coherence diagrams reduce to the associativity, symmetry, and base-change diagrams of the six-functor formalism. This is precisely the definition of a lax duoidal category; invertibility of the interchange maps gives the strong form. □

The theorem is an instance of the general theory of duoidal ∞-categories [79]. Its particular content here is geometric: the two products are the diagonal product and disjoint-union convolution on the Ran space.

3.2 Factorisation coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\mathrm{ch}}$. Its coproduct restricts, for disjoint finite subsets S and T, to a map

$$\Delta_{S,T} : F_{S \cup T} \longrightarrow F_S \boxtimes F_T.$$

It is a factorisation coalgebra when this map is an equivalence on every disjointness locus. Francis and Gaitsgory identify factorisation algebras with the corresponding full subcategory of chiral commutative coalgebras [38].

Let

$$\iota : \mathrm{FactD}(X) \hookrightarrow \mathrm{CoCAlg}^{\mathrm{ch}}(\mathcal{D}_X)$$

denote this realization. The pointwise tensor of two factorisation coalgebras is again a factorisation coalgebra: its coproduct is obtained by applying (17) to their two coproducts.

Theorem 3.2 (Ordered chirality as a duoidal bimonoid)**.** *Assume the strong duoidal hypotheses of Theorem 3.1 on the chosen factorisation subcategory. For an object $A \in \mathrm{FactD}(X)$, the following data are equivalent:*

(i) *an E_1 multiplication on A internal to* $\mathrm{FactD}(X)$*;*

(ii) *an E_1 monoid structure for $\otimes^{\mathrm{pt}}$ on ιA such that its multiplication and unit are morphisms of chiral commutative coalgebras;*

(iii) *a bimonoid in the duoidal category $(\mathcal{D}_X, \otimes^{\mathrm{pt}}, \otimes^{\mathrm{ch}})$ whose chiral coproduct is factorising on disjoint loci.*

The compatibility is the duoidal bialgebra identity

$$\Delta_{\mathrm{ch}} \circ m = (m \otimes^{\mathrm{ch}} m) \circ \chi \circ (\Delta_{\mathrm{ch}} \otimes^{\mathrm{pt}} \Delta_{\mathrm{ch}}), \tag{18}$$

where χ is the interchange morphism.

Proof. An E_1 algebra in the full subcategory of factorisation coalgebras is, by definition, an E_1 multiplication whose structure maps remain in that subcategory. Saying that multiplication is a morphism of coalgebras is exactly (18); saying that the unit is a coalgebra map is the corresponding unit identity. These are the axioms of a bimonoid in a duoidal category. The factorisation condition is the requirement that the chiral coproduct become an equivalence on disjoint loci. □

Equation (18) is the algebraic form of locality for the noncommutative theory. Separating two collections of insertions and then fusing them gives the same map as fusing inside each separated region and then applying factorisation. Cocommutativity belongs to a single coproduct, while an R-matrix belongs to an exchange theory; the duoidal equation expresses the interchange of support separation with ordered fusion.

3.3 The bar retains factorisation

The simplicial ordered bar is levelwise a chiral commutative coalgebra:

$$B_r(A) = \mathbf{1} \otimes^{\mathrm{pt}} A^{\otimes^{\mathrm{pt}} r} \otimes^{\mathrm{pt}} \mathbf{1}.$$

Every face is built from m or the augmentation. By Theorem 3.2, these are coalgebra maps. Every degeneracy is built from the unit and is likewise a coalgebra map.

Theorem 3.3 (Double-coalgebra bar theorem)**.** *Suppose geometric realization in $\mathcal{D}_X$ preserves chiral commutative coalgebras and commutes with the pointwise tensor products appearing in the simplicial bar. Then*

$$C = \mathrm{Bar}_X^{\mathrm{ord}}(A)$$

has two compatible coalgebra structures:

(i) *the coassociative deconcatenation coproduct of the E_1 bar;*

(ii) *the commutative chiral factorisation coproduct inherited levelwise from A.*

Deconcatenation is a morphism of chiral factorisation coalgebras. Equivalently, C is a coassociative coalgebra object in chiral commutative coalgebras.

Proof. The simplicial bar is a simplicial object in chiral commutative coalgebras because every simplicial operator is a coalgebra morphism. Geometric realization gives the second coproduct. Deconcatenation is defined by cutting an ordered word. The chiral coproduct acts independently on each letter, so cutting and factorising commute by the interchange law. The resulting square is (18) applied to the iterated pointwise tensor powers of A. □

This theorem explains why the ordered bar is simultaneously an interval object and a Ran-space coefficient. The coassociative coproduct remembers where the interval is cut. The chiral coproduct remembers where the curve is separated. Their compatibility is inherited functorially from the locality law of the original ordered chiral algebra.

3.4 A Fubini theorem

Let $p : X^{\mathrm{an}} \times \mathbb{R} \to X^{\mathrm{an}}$ and $q : X^{\mathrm{an}} \times \mathbb{R} \to \mathbb{R}$ be the projections. Factorisation pushforward along p integrates the locally constant real direction; pushforward along q integrates the chiral direction when the corresponding chiral homology is defined. Product excision gives a Fubini comparison.

Theorem 3.4 (Ordered chiral Fubini)**.** *Let A be an ordered chiral algebra whose analytic realization is rectangularly generated. Assume the relevant factorisation homologies exist and that the pushforwards preserve the geometric realizations of the bar. Then there are natural equivalences*

$$\int_X \mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \mathrm{Bar}^{\mathrm{ord}}\left(\int_X A\right), \tag{19}$$

$$\int_X \mathrm{HH}_*(A) \simeq \mathrm{HH}_*\left(\int_X A\right). \tag{20}$$

More generally, factorisation homology over a product rectangle may be computed in either order.

Proof. The bar is the geometric realization of the two-sided simplicial object $[r] \mapsto \mathbf{1} \otimes A^{\otimes^{\mathrm{pt}} r} \otimes \mathbf{1}$; its normalized chains are tensor powers of $\bar{A}$. Chiral factorisation homology is a colimit over chiral disks. Under the stated preservation hypotheses, the two colimits commute, giving (19). Hochschild chains are factorisation homology over the transverse circle, or equivalently the cyclic realization of the two-sided bar. Commuting that realization with chiral factorisation homology gives (20). On the analytic product, both assertions are the Fubini theorem for factorisation homology and follow from excision on product-disk covers. □

The first equivalence says that one may integrate the chiral curve before or after resolving ordered fusion. The second says the same after the interval has been closed into a circle. In conformal field theory this is the equality between first forming the open-string trace and then taking chiral coinvariants, or first taking chiral coinvariants and then closing the ordered state space. Trace-class or nuclearity estimates supply the analytic convergence of either trace.

3.5 Primitives and quantum symmetry

The chiral factorisation coproduct in Theorem 3.3 separates support on the curve. A quantum-group coproduct acts on tensor products of line representations and is reconstructed from a tensor-compatible exchange theory. A gauge-theoretic comparison functor can relate these coproducts in a specified model.

The primitive part of the chiral coalgebra records connected support. The primitive part of the coassociative bar coalgebra records indecomposable interval states. Their intersection consists of states that are connected in the curve and indecomposable along the interval. This bidegree is the natural home of a binary collision kernel or a first line-defect interaction. An exchange functor transports these primitives around a two-dimensional collision and thereby produces an R-matrix.

3.6 Physical meaning

The two products of the duoidal category are two experimental operations. Pointwise tensor places two coefficient systems on the same chiral support and then fuses them in transverse order. Chiral convolution places them on separated supports and then forgets the separation by union. Equation (18) says that a local experiment performed in two separated laboratories is independent of whether fusion is described before or after the laboratories are separated.

This is the geometric content of locality in the ordered theory. It refines associativity by controlling support dependence. Braiding is the further structure supplied by exchange paths. The duoidal law places Beilinson's factorisation language and the noncommutative language of defects in one equation.

4 The block-ordered chiral configuration category

4.1 What is ordered

Let Ch_X denote a fixed operadic enhancement of the Beilinson–Drinfeld chiral pseudo-tensor structure whose algebras in the chosen derived system of right D-modules are the factorisable D-modules of Theorem 2.1. The noncommutative chiral operad is

$$\mathsf{OCh}_X := E_1 \otimes \mathsf{Ch}_X, \tag{21}$$

where $\otimes$ is the Boardman–Vogt tensor product of ∞-operads. An algebra over (21) has a chiral factorisation structure and an associative transverse multiplication, with the interchange relation built into the operad.

The word *ordered* refers to the sequence of E_1 input blocks. The points inside each block retain the ordinary chiral finite-set symmetry. A typical operation is represented by an ordered sequence

$$(I_1, \ldots, I_r), \tag{22}$$

where the finite set I_a labels the chiral insertions carried by the ath transverse interval. Points belonging to different blocks may have the same coordinate on X because their transverse coordinates are distinct. Collisions inside the chiral direction are carried by the D-module extension across diagonals; composition in the transverse direction is carried by E_1.

4.2 The operadic Grothendieck construction

The block-ordered chiral configuration category $\mathbf{BRan}_X$ is the operadic Grothendieck construction of the associative operad acting on the chiral configuration category. Its objects are pairs

$$([r]; I_1, \ldots, I_r),$$

with $[r]$ a finite ordinal and each I_a a finite chiral label set. Its generating morphisms are of three kinds:

(i) chiral refinements, collisions, and permutations inside one I_a;

(ii) adjacent composition of two E_1 slots;

(iii) insertion or removal of the unit slot.

The relations are the chiral relations, the simplicial identities of the associative nerve, and the interchange squares between the two kinds of operation.

The object space of a fixed block profile (22) is

$$X^{I_1} \times \cdots \times X^{I_r}. \tag{23}$$

The product (23) is the geometric carrier of a block profile. The monoidal structure of $\mathrm{FactD}(X)$ is supplied by the chosen operadic correspondences. Adjacent block composition first places the chiral inputs in a common output profile and then applies the internal E_1 multiplication. In a six-functor model, this operation is the associated pull–tensor–push correspondence.

Definition 4.1 (Block-ordered factorisation D-module)**.** A block-ordered factorisation D-module is an algebra over $\mathsf{OCh}_X = E_1 \otimes \mathsf{Ch}_X$ in the chosen derived D-module system. Equivalently, it is a coCartesian operadic section over $\mathbf{BRan}_X$ whose restriction to each block is chiral factorisable, whose restriction to the ordinal direction is Segal, and whose mixed squares satisfy interchange.

Write $\mathrm{OrdFactD}(X)$ for this category. The definition includes the coefficient correspondences needed to compare the different spaces (23); it is therefore stronger than the datum of a simplicial family of sheaves on their underlying products.

Theorem 4.2 (Block-ordered recognition)**.** *There are natural equivalences*

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \simeq \mathrm{Alg}_{E_1 \otimes \mathsf{Ch}_X}(\mathcal{V}_X) \simeq \mathrm{OrdFactD}(X), \tag{24}$$

where $\mathcal{V}_X$ is the chosen finite-set D-module system presenting $\mathrm{FactD}(X)$*.*

Proof. By definition, $\mathrm{FactD}(X) \simeq \mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V}_X)$. The exponential law for ∞-operads gives

$$\mathrm{Alg}_{E_1}\Big(\mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V}_X)\Big) \simeq \mathrm{Alg}_{E_1\otimes\mathsf{Ch}_X}(\mathcal{V}_X).$$

Unstraightening an algebra over the tensor-product operad gives a coCartesian section over its operadic Grothendieck construction. The restrictions to the two operadic factors give the chiral and Segal conditions; the tensor-product relation gives interchange. Straightening reverses the construction. ☐

The theorem gives the invariant meaning of factorisation over an ordered chiral configuration space. The block picture is the geometric presentation of the tensor-product operad.

4.3 The strict and homotopy-coherent forms

In a strict dg model, an ordered chiral algebra is a factorisable D-module A with maps

$$m : A \otimes^{\mathrm{pt}} A \longrightarrow A, \qquad \eta : \mathbf{1} \longrightarrow A,$$

satisfying strict associativity and unitality. In the invariant ∞-category it is an E_1 algebra. A cofibrant A_∞ model replaces the single multiplication by operations indexed by planar trees. The chiral coefficient is unchanged.

Proposition 4.3 (Interchange)**.** *Let γ be a chiral operation and let m be the transverse multiplication. Whenever the two composites are defined by one product-disk embedding, their images satisfy*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \tag{25}$$

with the symmetry isomorphism that places equal-coloured inputs together.

Proof. Equation (25) is the defining interchange relation of $E_1 \otimes \mathsf{Ch}_X$. In the analytic product-disk model, both sides are the factorisation map associated with the same embedding, evaluated by composing first in opposite coordinate directions. ☐

The proposition upgrades the two operations to a compatible pair: holomorphic collision commutes with transverse composition. Exchange paths enter through an E_2 enhancement of the transverse geometry.

4.4 Analytic chamber model

For $r \geq 1$, let

$$C_r^< = \{(t_1, \ldots, t_r) \in \mathbb{R}^r \mid t_1 < \cdots < t_r\}.$$

This chamber is contractible. A profile (22) has analytic carrier

$$X^{I_1,\mathrm{an}} \times \cdots \times X^{I_r,\mathrm{an}} \times C_r^<. \tag{26}$$

Local constancy in $C_r^<$ erases the distances while retaining the order. The X-coordinates remain holomorphic and may approach any partial diagonal allowed by the coefficient D-module.

Compactifying the configurations of subintervals in one interval modulo translation and positive dilation gives associahedra. Their strata are indexed by planar rooted trees. This compactification resolves the homotopy coherence of the E_1 multiplication.

Proposition 4.4 (Associahedral resolution). *For a locally constant factorisation algebra on an oriented line, cellular chains on compactified ordered interval configurations act by the A_∞ bar operations. Contracting every associahedral cell to its corolla gives the ordinary simplicial bar construction through a quasi-isomorphism of operadic resolutions.*

Proof. Factorisation assigns an operation to every ordered embedding of disjoint intervals. The corresponding embedding spaces are contractible, and their compactified boundary strata are planar nested collisions. Stokes' formula is the sum of edge contractions, hence the Stasheff identity. The cellular chains form a cofibrant resolution of the associative operad, while the simplicial bar is the standard resolution of an algebra over that operad. The comparison is the augmentation of the cofibrant resolution. □

The simplex and the associahedron therefore play different roles. The simplex is the indexing object of the simplicial bar. The associahedron is a geometric cofibrant resolution of associative coherence.

4.5 Mixed collision geometry

A chain model for OCh_X may be built from two-coloured forests. Chiral edges describe nested partial diagonals in X; transverse edges describe nested intervals in the ordered line. A face contracts one edge of one colour. On a total complex, the two boundary operators anticommute by the product orientation.

Lemma 4.5 (Mixed square-zero identity). *Let d_X be the differential supplied by a chain model of the chiral coefficient and let $b_\perp$ be the E_1 bar differential. If both are morphisms for the interchange structure, then*

$$d_X b_\perp + b_\perp d_X = 0$$

after the standard totalization sign. Hence

$$(d_X + b_\perp)^2 = d_X^2 + b_\perp^2.$$

Proof. The two operators are boundaries in independent factors of the mixed operadic resolution. Exchanging their conormal orientation lines contributes the totalization sign. Compatibility with interchange identifies the two coefficient composites. □

Associativity proves $b_\perp^2 = 0$, the chiral identities prove $d_X^2 = 0$, and interchange proves the mixed anticommutator identity.

4.6 Modules, interfaces, and marked blocks

A right module is an algebra over the two-coloured operad with ordered algebra inputs followed by one module input. A left module uses the opposite order, and a bimodule has both boundary colours. For a right module R and a left module L, the simplicial two-sided bar is

$$B_r(R, A, L) = R \otimes A^{\otimes r} \otimes L. \tag{27}$$

Its realization is interval factorisation homology with endpoint labels R and L.

Proposition 4.6 (Marked-block recognition)**.** *The category of right, left, and bi-modules over A is equivalent to the corresponding category of marked coCartesian sections over the coloured block-ordered configuration category. Relative tensor product of bimodules is gluing of marked intervals.*

Proof. Apply the straightening argument of Theorem 4.2 to the standard coloured associative operads for modules and bimodules. Excision for the glued interval computes the relative tensor product. □

This is the native geometry of boundary-changing fields and defect junctions.

4.7 Opposite order

Reflection of the transverse line reverses the ordinal factor while leaving X unchanged. It induces

$$\mathrm{rev} : \mathsf{OCh}_X \longrightarrow \mathsf{OCh}_X$$

and sends multiplication to its opposite.

Proposition 4.7 (Geometric opposite)**.** *Under Theorem 4.2, transverse reflection sends A to A^{op}. Its chiral coefficient is unchanged.*

This is the canonical order reversal at the E_1 level. An E_2 structure supplies braiding, and reflection then acts on that additional transport datum.

4.8 Three quotient and descent operations

Three functors organize the passage from ordered to less structured data.

First, the forgetful functor retains the underlying factorisable D-module and discards the transverse multiplication from the target structure.

Second, derived commutativisation is the left adjoint

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_\infty}(\mathrm{FactD}(X))$$

when it exists in the selected presentable category. It imposes coherent commutativity on the transverse product through the universal $E_1 o E_\infty$ change-of-operad adjunction.

Third, descent from labelled to unlabelled configurations concerns monodromy of a coefficient local system on $\mathrm{Conf}_n(X)$. It asks for extension from the pure braid group to the full braid group; symmetric exchange further asks that the action factor through Σ_n. This descent depends on the configuration-space local system and may be combined with any chosen transverse E_1 multiplication.

The separation of these three operations is the first safeguard of noncommutative chiral Koszul duality.

5 The mixed operad of chiral collision and ordered fusion

5.1 The invariant operad

Let Ch_X be an ∞-operad whose algebras in the chosen coefficient category are factorisable D-modules on X. The ordered chiral operad is defined by the Boardman–Vogt tensor product

$$\mathsf{OrdCh}_X := E_1 \otimes \mathsf{Ch}_X. \tag{28}$$

The tensor product is characterized by interchange: an operation in the E_1 direction commutes with an operation in the chiral direction after the canonical permutation and base-change maps.

Theorem 5.1 (Operadic additivity)**.** *Let $\mathcal{V}$ be a presentable symmetric monoidal ∞-category for which the indicated algebra categories exist. Then*

$$\mathrm{Alg}_{\mathsf{OrdCh}_X}(\mathcal{V}) \simeq \mathrm{Alg}_{E_1}(\mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V})). \tag{29}$$

For the D-module realization of Ch_X, the right-hand side is $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. Equation (29) is the universal property of the tensor product of ∞-operads. It identifies an algebra carrying two compatible operadic actions with an algebra over their tensor product. □

For locally constant operations, Dunn additivity identifies $E_1 \otimes E_n$ with E_{n+1} [28, 54]. The chiral operation on X varies through prescribed diagonal singularities, so (28) retains one holomorphic singular direction and one ordered topological direction.

5.2 Three geometric models

Three spaces occur for three different reasons.

The simplex indexes the monadic simplicial bar. The associahedron resolves coherent E_1 multiplication. The Fulton–MacPherson compactification resolves relative chiral configurations near collision diagonals. Their distinct universal properties determine their distinct roles.

Fix an ordered partition $I = I_1 \sqcup \cdots \sqcup I_r$. On that stratum, a local product chart has the form

$$K_r \times \prod_{a=1}^{r} \mathrm{FM}_{I_a}(X), \tag{30}$$

up to the usual base and normal-bundle factors. When a transverse face merges two blocks, the partition changes and the corresponding Fulton–MacPherson factors are replaced by the factor for their union. Thus the global resolution is the Grothendieck assembly of the charts (30) over the category of ordered partitions, with transition maps that replace merged block factors by the factor of their union.

5.3 Two-coloured levelled forests

Choose a cofibrant dg-operadic presentation

$$\mathsf{Q}_X \longrightarrow \mathsf{Ch}_X$$

whose chiral cells are represented locally by Fulton–MacPherson incidence cells with their screen-chain coefficients. Apply a functorial Boardman–Vogt construction to the tensor-product operad:

$$W(E_1 \otimes \mathsf{Q}_X) \longrightarrow E_1 \otimes \mathsf{Ch}_X. \tag{31}$$

A cellular basis element of this resolution is represented by a rooted levelled forest with two kinds of internal edge:

- a transverse edge, recording composition of consecutive E_1 blocks;

- a chiral edge, recording a nested collision inside a chiral coefficient.

A vertex carries the screen-chain coefficient prescribed by Q_X. The determinant line is

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)). \tag{32}$$

Definition 5.2 (Mixed forest complex)**.** For a finite label set I, let

$$W\mathsf{OrdCh}_X(I) = \bigoplus_F C_*(\mathrm{Screen}(F)) \otimes \mathrm{or}(F), \tag{33}$$

where F ranges over the two-coloured levelled forests in the chosen cellular presentation. The differential is the sum of the internal screen differential and the signed contractions of one transverse or one chiral edge.

Definition 5.2 is the canonical chain model of the operadic resolution. A smooth compactification equipped with the same face category and screen coefficients supplies a geometric realization of this model.

5.4 The boundary identity

Write

$$d = d_{\mathrm{scr}} + d_\perp + d_{\mathrm{ch}}.$$

Theorem 5.3 (Mixed boundary theorem)**.** *With the orientation convention* (32)*,*

$$d_\perp^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0,$$

and the screen differential supercommutes with both edge-contraction differentials. Hence $d^2 = 0$.

Proof. A codimension-two forest is obtained either by contracting two transverse edges, two chiral edges, or one edge of each colour. In the first two cases, the two orders of contraction induce opposite boundary orientations in the corresponding one-colour Boardman–Vogt complex. In the mixed case, the final forest is the same in either order, while exchanging the two one-dimensional determinant factors in (32) changes the sign. The screen differential is compatible with every operadic face map and contributes the standard total-complex signs. □

When endomorphism operations are attached to vertices, the transverse part of this identity becomes the Stasheff relations, the chiral part becomes the identities of the chosen chiral coefficient model, and the mixed part says that transverse multiplication is a morphism of chiral objects.

5.5 Cofibrant comparison

Theorem 5.4 (Mixed Boardman–Vogt resolution)**.** *Assume that chain operads are taken in an admissible monoidal model category, that Q_X is cofibrant, and that the tensor-product operad and functorial W-construction preserve weak equivalences between the relevant cofibrant objects. Then the augmentation*

$$W(E_1 \otimes \mathsf{Q}_X) \longrightarrow E_1 \otimes \mathsf{Ch}_X \tag{34}$$

is a cofibrant resolution. Its algebras are homotopy-coherent ordered chiral algebras. When rectification holds in the coefficient model category, their localized category is equivalent to $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. The W-construction replaces each internal edge by a length parameter and has contractible length fibres. Under the stated model-categorical hypotheses it is cofibrant and its edge-contraction augmentation is an aritywise weak equivalence. The tensor-product universal property identifies its strict algebras with compatible E_1 and chiral structures. Rectification is the standard change-of-operad equivalence along a weak equivalence between admissible cofibrant operads. □

The theorem derives cofibrancy from the admissible tensor-product and functorial W-construction hypotheses stated above. These hypotheses are the model-categorical input that controls the tensor product of the two resolutions.

5.6 Incidence and screen geometry

A collision forest has an incidence type and a screen space. The incidence category records which clusters collide and in what order. Its fibre over a fixed root partition has a terminal corolla and hence a contractible nerve. The screen space records relative positions inside each cluster and generally has nontrivial cohomology.

For three points on an affine complex line modulo translation and dilation, the open screen is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore a proper pushforward of the geometric stratum retains more information than the pushforward of its incidence nerve. The incidence differential governs cluster contractions. The screen coefficient in (33) carries Arnold classes, configuration-space periods, and propagator forms.

5.7 Homotopy transfer and Feynman trees

Let a deformation retract of ordered chiral complexes be given. Homological perturbation transfers the structure by summing over the two-coloured forests: vertices carry original operations, internal edges carry the contracting homotopy, and screen chains carry the chiral coefficient. The resulting operations satisfy the complete forest boundary identity because Theorem 5.3 is the differential of the cofibrant resolution.

This is the precise coincidence between Feynman trees and homotopy transfer. Once propagator and vertex maps are supplied, both constructions evaluate the same two-coloured forest sum. Configuration-space forms remain the geometric edge coefficients, while OPE operations remain the vertex coefficients.

5.8 Renormalised Feynman weights

In perturbative field theory, an edge of a graph carries a propagator and a vertex carries an interaction. Renormalisation extends the resulting distribution to a compactified configuration space. Boundary restriction factors through contracted subgraphs.

Proposition 5.5 (Feynman weight morphism)**.** *Let a holomorphic–topological BV theory on $X \times \mathbb{R}$ have renormalised graph weights extending to a geometric realization of the mixed forest model. Assume that restriction to every boundary stratum factorizes by graph contraction and that the renormalised quantum master equation holds. Then the graph weights define a morphism from the mixed cofibrant operad to the endomorphism operad of the boundary observables.*

Proof. Factorisation on a boundary stratum identifies the weight with the composite of the weights of the contracted graph and the quotient graph. Stokes' formula, including the BV boundary and counterterm contributions, is exactly compatibility with the differential of the mixed forest complex. These two identities are the operadic composition and chain-map axioms. □

5.9 Three reversals

Reflection of the transverse line sends an E_1 algebra to its opposite. Reversal of an exchange path, when an E_2 enhancement exists, inverts the braid. Verdier duality reverses !- and *-extensions of the chiral coefficient. The two-colour grading records the separate domains of transverse reflection, exchange-path reversal, and Verdier duality, together with every comparison map among them.

6 The ordered chiral bar construction

6.1 The simplicial object

Let A be an augmented ordered chiral algebra with augmentation $\varepsilon : A \to \mathbf{1}$, and put $\bar{A} = \mathrm{Fib}(\varepsilon)$. The two-sided simplicial bar object in $\mathrm{FactD}(X)$ is

$$B_r^{\mathrm{ord}}(A) = \mathbf{1} \otimes A^{\otimes r} \otimes \mathbf{1}, \qquad r \geq 0. \tag{35}$$

The middle faces multiply adjacent copies of A, the first and last faces use the two augmentation actions on $\mathbf{1}$, and the degeneracies insert the unit of A. Its geometric realization defines the ordered chiral bar:

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) := |B_\bullet^{\mathrm{ord}}(A)|. \tag{36}$$

Every simplicial term is a factorisable D-module and every simplicial operator is a morphism in $\mathrm{FactD}(X)$. Normalization removes the degenerate unit factors and identifies the reduced chain object with tensor powers of $\bar{A}$.

In cochain notation, suspension by [1] places the multiplication coderivation in degree +1, and normalization gives

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \left(\bigoplus_{r \geq 0} (s\bar{A})^{\otimes r},\ d_A + b_m \right), \tag{37}$$

where b_m is the coderivation induced by the E_1 multiplication. Completion replaces the direct sum by a product.

6.2 The chiral coefficient throughout the bar

Formula (37) uses the internal multiplication morphism in $\mathrm{FactD}(X)$. If a and b have a fourth-order OPE pole, the complete principal part belongs to the factorisation D-module $A \otimes A$ and is transported by m. The bar face lowers transverse word length while preserving the full chiral coefficient.

A residue complex can be paired with the bar as a separate geometric realization. Such a pairing has the form

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow \Gamma(\mathrm{FM}_I(X), \mathcal{K}_I \otimes A_I),$$

where $\mathcal{K}_I$ is a chosen complex of forms, currents, or local cohomology classes. The map must specify how each principal part is paired with a normal jet. Its square-zero identity uses both the bar differential and the differential of $\mathcal{K}_I$. The intrinsic bar exists before this choice.

Proposition 6.1 (Bidegrees of bar fusion and collision residue)**.** *Let a geometric coefficient model carry a logarithmic or principal-part complex $\mathcal{K}_I$ along the chiral collision divisors. In the bicomplex*

$$(s\bar{A})^{\otimes r} \otimes \mathcal{K}_I,$$

the bar coderivation has bidegree $(-1, 0)$ in transverse word length and coefficient degree, while Poincaré residue has bidegree $(0, -1)$ in those gradings and takes values on the corresponding boundary divisor. A geometric comparison consists of a pairing between principal parts and normal jets together with the mixed chain-map identity.

Proof. The bidegree of the bar coderivation follows from adjacent multiplication. The bidegree of residue follows from the residue exact sequence for logarithmic forms. The mixed identity is precisely the condition that the chosen principal-part pairing define a morphism from the internal bar to the geometric coefficient complex. □

6.3 Coalgebra structure

Deconcatenation gives

$$\Delta[a_1|\cdots|a_r] = \sum_{p=0}^{r} [a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r]. \tag{38}$$

It is coassociative and conilpotent. The bar differential is a coderivation.

Proposition 6.2 (Internal bar coalgebra)**.** *The ordered chiral bar is a conilpotent coassociative coalgebra object in* $\mathrm{FactD}(X)$*. The coaugmentation is the empty word, and the coradical filtration is the transverse word-length filtration.*

Proof. Deconcatenation is defined componentwise in every symmetric monoidal category. Its coassociativity is the equality of the two ways of cutting a word twice. The multiplication coderivation is local with respect to cuts, so it satisfies the coderivation identity. Iterating the reduced coproduct of a word of length r more than r times gives zero. □

The coalgebra structure records interval splitting. The duoidal factorisation structure of Theorem 3.3 supplies a second, chiral coproduct. A bialgebra structure consists of a multiplication compatible with the relevant coproduct through the stated interchange law.

6.4 Interval factorisation homology

Let $([0, 1], \partial)$ denote an oriented interval whose endpoints are labelled by the augmentation module $\mathbf{1}$. Excision for factorisation homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes^{L}_{A} \mathbf{1}. \tag{39}$$

The standard two-sided simplicial resolution of the derived tensor product is precisely (35).

Theorem 6.3 (Bar as interval factorisation homology)**.** *There is a canonical equivalence in* $\mathrm{FactD}(X)$

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) \simeq \int_{([0,1],\partial)} A. \tag{40}$$

For a right module R and a left module L,

$$\mathrm{Bar}(R, A, L) \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Choose a Weiss cover of the interval by an ordered chain of small intervals together with collars at the endpoints. Factorisation excision identifies the colimit of the Čech diagram with the derived relative tensor product. Refining the cover gives the simplicial two-sided bar. The argument takes place in the coefficient category $\mathrm{FactD}(X)$ and is therefore compatible with chiral factorisation. □

The theorem is the physical meaning of the bar construction. It is the amplitude of an interval whose interior carries the boundary operator algebra and whose endpoints carry the vacuum boundary state.

6.5 The associahedral model

The simplicial bar is the derived tensor product associated with a strict E_1 multiplication. A homotopy-coherent multiplication is an algebra over a cofibrant resolution $W\mathsf{Ass} \to \mathsf{Ass}$. Its bar coderivation contains the transferred operations m_k,

$$b = \sum_{k\geq 1} b_{m_k}, \tag{41}$$

and $b^2 = 0$ is the family of Stasheff identities.

Let K_r denote the associahedron in the convention in which its cells are planar trees with r leaves. The cellular chains $C_*(K_r)$ form one presentation of $W\mathsf{Ass}(r)$. They resolve the homotopy-coherent E_1 operation; the simplicial degree continues to index the two-sided bar.

Theorem 6.4 (Associahedral comparison)**.** *Assume an admissible monoidal dg model of* $\mathrm{FactD}(X)$*, let A be cofibrant as a $W\mathsf{Ass}$-algebra, and let A^{str} be a strictification along $W\mathsf{Ass} \to \mathsf{Ass}$. The operadic bar construction of A has a cellular filtration with terms*

$$\bigoplus_{r\geq 0} C_*(K_r) \otimes (s\bar{A})^{\otimes r}, \tag{42}$$

whose differential contains the cellular boundary and the operations m_k. Change of operad and the simplicial two-sided bar give a natural zigzag of weak equivalences

$$\mathrm{Bar}_{W\mathsf{Ass}}(A) \simeq \mathrm{Bar}_{\mathsf{Ass}}(A^{\mathrm{str}}) \simeq \mathrm{Bar}^{\mathrm{ord}}_X(A^{\mathrm{str}}).$$

Proof. Cellular chains of associahedra present the Boardman–Vogt resolution of the associative operad. The change-of-operad adjunction is a Quillen equivalence under the stated admissibility and cofibrancy hypotheses. Bar constructions are homotopy invariant on cofibrant algebras and compute the same derived relative tensor product $\mathbf{1} \otimes^L_A \mathbf{1}$. This yields the zigzag. □

The theorem identifies the simplicial and associahedral constructions as two homotopy-invariant models of the same interval amplitude, with their respective indexing roles retained.

6.6 Low arity

For a strict dg algebra, the normalized reduced bar begins

$$s\bar{A} \xleftarrow{b} (s\bar{A})^{\otimes 2} \xleftarrow{b} (s\bar{A})^{\otimes 3} \xleftarrow{b} \cdots .$$

The binary differential is

$$b[a|b] = (-1)^{|a|}[ab].$$

The length-three component of b^2 is the signed associator. In higher length, every term is indexed by two adjacent multiplications; the two orders of contracting the corresponding planar edges occur with opposite bar signs.

For an A_∞ algebra, m_3 fills the homotopy between the two binary composites. At the first two-dimensional associahedral cell, m_4 and composites of m_2, m_3 fill the pentagon. The geometry is exactly the cellular boundary of associahedra.

The chiral coefficient adds an independent filtration. A word $[a|b]$ can carry an arbitrary OPE singularity in the X variables. The bar differential applies the transverse product ab as a morphism of those singular coefficient systems.

6.7 Functoriality

An augmented morphism $f : A \to B$ of ordered chiral algebras induces

$$\mathrm{Bar}(f) : \mathrm{Bar}^{\mathrm{ord}}_X(A) \longrightarrow \mathrm{Bar}^{\mathrm{ord}}_X(B)$$

by applying f to every bar factor. A homotopy-coherent morphism induces the corresponding morphism of cellular bars.

A square-zero derivation Q of A induces a coderivation Q_{Bar} acting on each factor. Since Q is a derivation of multiplication,

$$[b_m, Q_{\mathrm{Bar}}] = 0.$$

This identity is the basis of BRST descent.

A monoidal functor $F : \mathrm{FactD}(X) \to \mathcal{V}$ preserving geometric realizations satisfies

$$F(\mathrm{Bar}^{\mathrm{ord}}_X(A)) \simeq \mathrm{Bar}_{\mathcal{V}}(F(A)). \tag{43}$$

Thus restriction to a formal disc, analytic realization, taking a fibre, and factorisation homology commute with the ordered bar when their continuity hypotheses hold.

6.8 Fubini in the two directions

Let $\mathcal{A}$ be the analytic factorisation algebra on $X^{\mathrm{an}} \times \mathbb{R}$ corresponding to A. For a compact one-manifold M with boundary labels, Fubini for factorisation homology gives

$$\int_{U\times M} \mathcal{A} \simeq \int_U \left(\int_M \mathcal{A} \right). \tag{44}$$

Taking $M = [0, 1]$ with vacuum endpoints yields the analytic realization of (40). Taking $M = S^1$ yields internal Hochschild chains whenever the circle factorisation homology exists. A cyclic trace or trace-class analytic completion is additional data needed only to turn those chains into scalar amplitudes.

Formula (44) relates chiral homology and topological factorisation by integrating one direction and retaining the other as its coefficient system.

6.9 Completion

The bar length filtration has finite quotients

$$\mathrm{Bar}_{\leq N}(A) = \bigoplus_{r=0}^{N} (s\bar{A})^{\otimes r}.$$

The completed bar is

$$\widehat{\mathrm{Bar}}_X^{\mathrm{ord}}(A) = \prod_{r\geq 0} (s\bar{A})^{\otimes r}. \tag{45}$$

The coderivation is continuous when every component has finite output in a fixed total filtration degree. This holds automatically for strict E_1 multiplication and for filtered A_∞ operations that raise a complete positive filtration.

If conformal weight is unbounded, one can first restrict to a finite weight window and then complete. The two inverse limits commute under the Mittag–Leffler condition described in Theorem 7.3.

Proposition 6.5 (Cohomology of completion)**.** *Let C_N be the tower of finite bar-length and finite-weight truncations with surjective transition maps on cohomology. Then*

$$H^*(\varprojlim_N C_N) \simeq \varprojlim_N H^*(C_N).$$

Proof. The Milnor exact sequence has a $\varprojlim^1$ term. Surjectivity gives the Mittag–Leffler condition and annihilates that term. □

Completion is the derived inverse limit of the finite bar-length and weight truncations, and Proposition 6.5 controls its cohomology.

6.10 The bar of a boundary endomorphism algebra

Let $A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$ be the endomorphism algebra of a compact boundary condition. The two-sided bar

$$\mathrm{Bar}(b, A_b, b)$$

resolves the identity kernel of the full subcategory generated by b. Under the Morita equivalence of Theorem 2.7, it is the factorisation analogue of the diagonal bimodule resolution.

Applying internal Hom to this resolution computes Hochschild cochains. Applying a trace computes Hochschild chains. Applying a second boundary label computes defect-changing operators. Thus the ordered bar is the common resolution behind the centre, the trace, and the module category.

6.11 The perturbative reading

In a perturbative boundary field theory, a bar word is a sequence of boundary insertions ordered along a collar. The bar differential fuses adjacent insertions. A Feynman propagator on X controls the chiral singularity of that fusion. Counterterms extend the coefficient distribution to the collision diagonal. The two operations are separate but compatible.

For a free Gaussian theory, Wick contraction gives an explicit morphism from the bar to the configuration-space de Rham complex. For an interacting renormalisable theory, Costello's renormalisation constructs the corresponding factorisation map order by order in $\hbar$. The quantum master equation is precisely the condition that the resulting differential squares to zero.

This is the exact sense in which a bar word is an interval amplitude. The bar differential is composition of boundary operations; the propagator supplies the coefficient of that composition.

7 Ordered chiral bar–cobar reconstruction

7.1 The internal adjunction

Put

$$\mathcal{C}_X := (\mathrm{FactD}(X), \otimes^{\mathrm{pt}}, \mathbf{1}).$$

Assume throughout this section that $\mathcal{C}_X$ admits the geometric realizations and totalizations used below, and that tensor product preserves geometric realizations separately. These hypotheses hold in the standard presentable derived enhancements after restricting to the complete subcategories on which the indicated limits exist.

For a coaugmented conilpotent coassociative coalgebra C with coaugmentation coideal $\bar{C}$, define

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = \left(\bigoplus_{r \geq 0} (s^{-1}\bar{C})^{\otimes r},\ d_C + d_\Delta \right), \tag{46}$$

where d_Δ is the derivation induced by the reduced coproduct. A complete cobar uses the product over tensor lengths and the continuous extension of the same derivation.

A degree-one morphism $\tau : C \to A$ is a twisting morphism when

$$d\tau + m(\tau \otimes \tau)\bar{\Delta} = 0. \tag{47}$$

The universal twisting morphisms give natural equivalences of mapping spaces

$$\mathrm{Map}_{\mathrm{Alg}_{E_1}}(\mathrm{Cobar}_X^{\mathrm{ord}} C, A) \simeq \mathrm{Tw}(C, A) \simeq \mathrm{Map}_{\mathrm{Coalg}_{\mathrm{coAss}}}(C, \mathrm{Bar}_X^{\mathrm{ord}} A).$$

Hence

$$\mathrm{Cobar}_X^{\mathrm{ord}} : \mathrm{Coalg}_{\mathrm{coAss}}^{\mathrm{conil}}(\mathcal{C}_X) \rightleftarrows \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathcal{C}_X) : \mathrm{Bar}_X^{\mathrm{ord}} \tag{48}$$

is an adjunction. Its unit and counit become equivalences on the positive complete subcategories defined next.

7.2 The connected positive-weight theorem

Definition 7.1 (Admissible positive object)**.** An augmented algebra A in $\mathcal{C}_X$ is admissibly positive if it has a complete weight decomposition

$$A \simeq \mathbf{1} \oplus \prod_{w \geq 1} A_w, \qquad A_p \otimes A_q \longrightarrow A_{p+q},$$

with each finite weight truncation dualisable and bounded below after every component evaluation $\mathrm{ev}_I : \mathrm{FactD}(X) \to D\text{-mod}(X^I)$. The evaluations are required to be conservative and to commute with the bar and cobar constructions. A coalgebra is admissibly positive when its coaugmentation coideal has the analogous positive, complete, conilpotent weight grading.

The positivity condition makes every fixed weight of a bar or cobar contain only finitely many tensor lengths. It is the internal analogue of a connected weight-graded augmented dg algebra.

Theorem 7.2 (Positive ordered bar–cobar reconstruction)**.** *For an admissibly positive augmented ordered chiral algebra A, the counit*

$$\mathrm{Cobar}_X^{\mathrm{ord}} \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow A \tag{49}$$

is an equivalence. For an admissibly positive conilpotent coalgebra C, the unit

$$C \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}} \mathrm{Cobar}_X^{\mathrm{ord}}(C) \tag{50}$$

is an equivalence.

Proof. Filter the two composites by tensor length and then take a fixed total weight. Positivity makes the resulting filtration finite. After applying ev_I, the associated graded is the ordinary connected associative bar–cobar resolution in complexes; its augmentation has the standard extra-degeneracy contraction. Thus each evaluated unit and counit is a quasi-isomorphism in every finite weight. Conservativity of the evaluations gives the equivalence in $\mathcal{C}_X$, and completeness reconstructs the full object from its finite weight truncations. □

Corollary 7.3 (Completed reconstruction)**.** *Let $A \simeq \varprojlim_N A_{\leq N}$ be an inverse limit of admissibly positive finite truncations, and suppose the transition maps are Mittag–Leffler on cohomology and tensor product commutes with the inverse limit. Then the continuous counit*

$$\widehat{\mathrm{Cobar}}_X^{\mathrm{ord}} \widehat{\mathrm{Bar}}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is an equivalence. The analogous assertion holds for a complete conilpotent coalgebra.

Proof. The theorem applies at every finite stage. The Milnor exact sequence identifies cohomology of the inverse limit with the inverse limit of cohomology because the $\varprojlim^1$ term vanishes under the Mittag–Leffler hypothesis. □

This is the associative bar–cobar theorem internal to the pointwise coefficient category. Weight and augmentation filtrations control its convergence. The Ran-cardinality filtration controls the separate chiral-convolution theorem.

7.3 The separate Ran theorem

The Ran space carries another monoidal structure, the chiral tensor product $\otimes^{\mathrm{ch}}$. By Theorem 2.5, its reduced cardinality quotients are nilpotent. Francis and Gaitsgory prove the enhanced bar–cobar equivalence there for operads satisfying their reduced or pro-nilpotent hypotheses. The reduced associative operad gives the outer E_1 case, while their chiral Lie construction gives the Lie–commutative-coalgebra equivalence [38].

Theorem 7.4 (Enhanced Ran bar–cobar theorem)**.** *In the pro-nilpotent monoidal category* $(D(\operatorname{Ran} X), \otimes^{\mathrm{ch}})$*, enhanced bar and cobar are inverse for the reduced associative operad, and more generally for the operads covered by the Francis–Gaitsgory enhanced theorem. The corresponding Lie case identifies chiral Lie algebras with the appropriate conilpotent commutative coalgebras. A restriction to a factorisation subcategory is valid when that subcategory is closed under the enhanced constructions and the unit and counit remain factorisation equivalences.*

Theorem 7.4 uses chiral convolution on the Ran space. Theorem 7.2 uses pointwise tensor product and resolves the transverse E_1 multiplication. A strong monoidal comparison functor transports bar and cobar objects between these two settings when its essential image is closed under the constructions.

7.4 The augmentation boundary

Regard the tensor unit $\mathbf{1}$ as a right A-module through the augmentation. Its derived endomorphisms define

$$A^! := \operatorname{RHom}_A(\mathbf{1}, \mathbf{1}). \tag{51}$$

Composition gives $A^!$ an E_1 multiplication. Evaluation makes $\mathbf{1}$ an $(A^!, A)$-bimodule: $A^!$ acts on the left by endomorphisms and A acts on the right through the augmentation. When the bar terms are dualisable, or when continuous internal Hom is used, the bar resolution gives

$$A^! \simeq \operatorname{RHom}_{\mathcal{C}_X}(\operatorname{Bar}_X^{\mathrm{ord}}(A), \mathbf{1}). \tag{52}$$

Theorem 7.5 (Boundary Koszul duality)**.** *Let A be an admissibly positive augmented ordered chiral algebra, regard $\mathbf{1}$ as a right A-module, and put $A^! = \operatorname{RHom}_A(\mathbf{1}, \mathbf{1})$. With the $(A^!, A)$-bimodule structure above, the functors*

$$\Phi = \operatorname{RHom}_A(\mathbf{1}, -) : \operatorname{Mod}_A \longrightarrow \operatorname{Mod}_{A^!}, \qquad \Psi = (-) \otimes^L_{A^!} \mathbf{1} : \operatorname{Mod}_{A^!} \longrightarrow \operatorname{Mod}_A$$

form an adjunction $\Psi \dashv \Phi$. The adjunction restricts to an equivalence between $\operatorname{thick}_{\operatorname{Mod}_A}(\mathbf{1})$ *and* $\operatorname{Perf}(A^!) = \operatorname{thick}_{\operatorname{Mod}_{A^!}}(A^!)$*. If $\mathbf{1}$ is a compact generator of a presentable stable subcategory $\mathcal{M} \subset \operatorname{Mod}_A$, then Φ and Ψ extend to inverse equivalences $\mathcal{M} \simeq \operatorname{Mod}_{A^!}$.*

Proof. The adjunction is the tensor–Hom adjunction for the $(A^!, A)$-bimodule $\mathbf{1}$. Its unit is an equivalence on $A^!$, because $\Phi(\mathbf{1}) = A^!$, and its counit is an equivalence on $\mathbf{1}$. Exactness and preservation of finite colimits and retracts extend the two equivalences to the thick closures. When $\mathbf{1}$ compactly generates $\mathcal{M}$, the Barr–Beck–Morita theorem identifies $\mathcal{M}$ with modules over its endomorphism algebra $A^!$. □

Thus $A^!$ is the algebra of local operators on the augmentation boundary condition, computed from the ordered multiplication by derived endomorphisms.

7.5 The double dual and completion

There is a canonical map

$$A \longrightarrow A^{!!} := \operatorname{RHom}_{A^!}(\mathbf{1}, \mathbf{1}). \tag{53}$$

Theorem 7.6 (Double dual as augmentation completion)**.** *Let A be admissibly positive, with dualisable finite weight pieces. Then (53) identifies $A^{!!}$ with the derived completion of A for the augmentation ideal. In particular, if A is complete for its positive weight filtration, then $A \simeq A^{!!}$.*

Proof. At every finite weight, the bar resolution is finite and dualisable. Dualising it once computes $A^!$; dualising the dual bar computes the finite augmentation quotient of A. The maps are compatible with the weight tower. Taking the inverse limit gives the derived augmentation completion, and completeness gives the final equivalence. □

The theorem identifies $A^{!!}$ with the inverse limit of the finite augmentation quotients. Completeness of A is exactly the condition that the canonical map from A to this limit be an equivalence.

7.6 Quadratic comparison

Suppose that A has a connected quadratic presentation

$$A = T_{\mathcal{C}_X}(V)/(R)$$

with dualisable positive weight pieces. The quadratic coalgebra

$$A^{¡} = C(sV, s^2R) \subset T^c(sV)$$

has a canonical twisting morphism and a comparison

$$q_A : A^{¡} \longrightarrow \mathrm{Bar}^{\mathrm{ord}}_X(A). \tag{54}$$

Theorem 7.7 (Quadratic recognition)**.** *Assume that component evaluation is conservative and preserves the quadratic twisting constructions. The following conditions are equivalent:*

(i) q_A *is an equivalence;*

(ii) $\mathrm{Cobar}(A^{i}) \to A$ *is an equivalence;*

(iii) *the left quadratic twisted tensor product is acyclic;*

(iv) *the right quadratic twisted tensor product is acyclic;*

(v) *the evaluated bar homology is concentrated on the quadratic diagonal in every finite weight.*

Proof. After component evaluation this is the fundamental theorem for Koszul twisting morphisms. Conservativity lifts the equivalences to $\mathcal{C}_X$. Finite positive weights justify the passage between the quadratic coalgebra and the full bar. □

Quadratic Koszulness concerns the comparison (54). It is stronger than the universal reconstruction of Theorem 7.2.

7.7 PBW and nonhomogeneous relations

Let $F_\bullet A$ be a complete exhaustive multiplicative filtration and suppose $\mathrm{gr}_F A \cong A_0$. Filter the bar by total PBW degree.

Theorem 7.8 (Filtered PBW comparison)**.** *Assume that the filtered comparison is complete and separated, its spectral sequence converges strongly in every finite chiral-cardinality and conformal-weight window, and A_0 is quadratic Koszul. Then the nonhomogeneous Koszul coalgebra determined by the quadratic, linear, and constant parts of the relations maps equivalently to* $\mathrm{Bar}_X^{\mathrm{ord}}(A)$*.*

Proof. The associated graded of the comparison is the quadratic equivalence for A_0. Strong convergence lifts it through the filtration. Linear terms become part of the coalgebra differential, and constant terms become its curvature functional. □

For an inhomogeneous algebra, the quadratic, linear, and constant relation terms respectively determine the coproduct, differential, and curvature of the nonhomogeneous Koszul coalgebra.

7.8 Curved coalgebras

A curved coalgebra has a coderivation d_C and a curvature functional $h : C \to \mathbf{1}[2]$ satisfying

$$d_C^2 = (h \otimes \mathrm{id} - \mathrm{id} \otimes h)\bar{\Delta}, \qquad hd_C = 0.$$

Its curved cobar has a square-zero total differential because the curvature functional supplies the constant part of the cobar derivation. This curvature is the algebraic datum attached to an inhomogeneous presentation. Moduli-space curvature is represented separately by a horizontal two-form in a family superconnection.

Curved module theory uses the coderived or contraderived category when the chosen null system is the coacyclic or contraacyclic class [60]. The coalgebra, module variance, and completion determine this choice.

7.9 Verdier duality

Let $\mathbb{D}_{\mathrm{Ran}}$ denote Verdier duality. If the finite bar stages are dualisable and duality commutes with the completion, then

$$A^{\vee}_{\mathrm{Verdier}} := \mathbb{D}_{\mathrm{Ran}}\,\mathrm{Bar}_X^{\mathrm{ord}}(A) \tag{55}$$

acquires an algebra structure by dualising deconcatenation.

A continuous perfect pairing provides the comparison between the Verdier algebra (55) and the boundary endomorphism algebra (51):

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \otimes A^! \longrightarrow \mathbf{1}.$$

Proposition 7.9 (Three dual constructions)**.** *For an augmented ordered chiral algebra, the following three constructions have the displayed sources and targets:*

(i) *the covariant reconstruction* $\mathrm{Cobar}\,\mathrm{Bar}(A) \to A$*;*

(ii) *the boundary algebra* $A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$*;*

(iii) *the Verdier dual algebra* $\mathbb{D}_{\mathrm{Ran}}\,\mathrm{Bar}(A)$*, when defined.*

The source and target of each construction determine the additional comparison needed to relate opposite order, inverse braiding, level reflection, and Verdier duality.

7.10 The complete deformation complex

The deformation complex of the transverse E_1 multiplication is the shifted internal Hochschild complex

$$\mathrm{Def}_{E_1}(A) = \mathrm{HH}^{\bullet}_{\mathrm{FactD}(X)}(A, A)[1]. \tag{56}$$

The chiral coefficient has its own deformation complex. On a formal disc it is the chiral operadic complex governing vertex or nonlocal-vertex operations. The deformation theory of the full ordered chiral object is controlled by a cofibrant resolution of the tensor-product operad $E_1 \otimes \mathsf{Fact}_X$.

Proposition 7.10 (Bigraded deformation complex)**.** *Choose an admissible cofibrant resolution of $E_1 \otimes \mathsf{Fact}_X$. The convolution deformation complex of an ordered chiral algebra has a complete filtration by the number of transverse and chiral vertices. Its associated graded is spanned by two-coloured trees; the horizontal axis restricts to the Hochschild deformation complex of the E_1 multiplication, the vertical axis restricts to the deformation complex of the chiral coefficient, and the mixed components impose the interchange law.*

Proof. The cellular Boardman–Vogt resolution is filtered by the two edge colours. Taking convolution with the endomorphism operad sends its cellular boundary to the deformation differential. Forests containing only transverse edges give Hochschild cochains; forests containing only chiral edges give coefficient deformations; forests of both colours give the mixed compatibility terms. □

A Morita equivalence identifies the Hochschild axis. An equivalence of the full ordered chiral deformation problem consists of that Morita equivalence together with an identification of the chiral coefficient and the mixed interchange maps.

8 The Koszul–Morita square and the universal boundary state

The bar coalgebra is simultaneously the resolution of the augmentation and the coalgebraic form of the universal boundary state. Once this point is made explicit, the interval, the diagonal bimodule, the annulus, Hochschild theory, and the derived centre become different shadows of one two-sided construction.

Throughout this section let

$$\mathcal{C}_X = (\mathrm{FactD}(X), \otimes^{\mathrm{pt}}, \mathbf{1}),$$

let A be an admissibly positive augmented E_1 algebra in $\mathcal{C}_X$, and put

$$C = \mathrm{Bar}^{\mathrm{ord}}_X(A).$$

Let $\tau : C \to A$ be the universal twisting morphism. All tensor and cotensor products below are derived in the complete subcategories selected by the positive weight filtration.

8.1 One-sided Koszul transforms

For a left A-module M, define the twisted left C-comodule

$$\mathbb{K}_A(M) = (C \otimes M, d_C + d_M + d_\tau), \tag{57}$$

where d_τ applies the reduced coproduct to C, evaluates one factor by τ, and lets the result act on M. For a conilpotent left C-comodule N, define

$$\mathbb{L}_A(N) = \big(A \otimes N, d_A + d_N + d_\tau\big). \tag{58}$$

The Maurer–Cartan equation for τ is exactly the statement that both displayed differentials square to zero.

Theorem 8.1 (Module–comodule Koszul equivalence)**.** *The functors $\mathbb{K}_A$ and $\mathbb{L}_A$ are inverse equivalences between the derived category of augmentation-complete left A-modules and the coderived category of conilpotent left C-comodules. The same statement holds for right modules and right comodules.*

Proof. Filter both composites by the coradical degree of C. In each fixed positive weight, the filtration is finite. The associated graded twisting differential vanishes, and the unit and counit reduce to the ordinary bar–cobar contractions. The extra degeneracy of the simplicial bar gives the contraction at each finite weight. Conservativity of component evaluation and completeness lift the equivalence to $\mathcal{C}_X$. When the coalgebra differential is curved or infinite products are essential, the same filtered contraction has coacyclic cone and therefore gives the asserted coderived equivalence. □

The theorem is the module form of ordered chiral Koszul duality. It identifies the complete module sector generated by the augmentation boundary with the conilpotent comodule sector of the bar coalgebra. Ext-group comparisons are its numerical shadows, while a Morita equivalence of entire module categories requires the augmentation object to generate the chosen category.

8.2 The two-sided transform

For an A-bimodule M, set

$$\mathbb{K}_A^{\mathrm{bi}}(M) = C \otimes_\tau M \otimes_\tau C. \tag{59}$$

The two outer factors carry the left and right C-coactions. The differential has four terms: the differentials of C and M, the left twisting action, and the right twisting action. The two twisting terms anticommute because the left and right A-actions commute.

Conversely, for a conilpotent C-bicomodule N, set

$$\mathbb{L}_A^{\mathrm{bi}}(N) = A \otimes_\tau N \otimes_\tau A. \tag{60}$$

Theorem 8.2 (Bimodule–bicomodule equivalence)**.** *The functors $\mathbb{K}_A^{\mathrm{bi}}$ and $\mathbb{L}_A^{\mathrm{bi}}$ are inverse equivalences between augmentation-complete A-bimodules and conilpotent C-bicomodules in the corresponding derived or coderived categories.*

Proof. Apply Theorem 8.1 first to the left action and then to the right action. The two transforms commute because their twisting coderivations act on different outer coalgebra factors and the bimodule identity makes their mixed commutator zero. Equivalently, filter (59) by the sum of the two coradical degrees and use the two-sided extra-degeneracy contraction on the associated graded. □

8.3 The diagonal bicomodule

The diagonal A-bimodule is A itself. The diagonal C-bicomodule is the coalgebra C equipped with the two coactions

$$C \xrightarrow{\Delta} C \otimes C, \qquad C \xrightarrow{\Delta} C \otimes C,$$

one read as a left coaction and the other as a right coaction. Denote it by C_Δ.

Theorem 8.3 (Diagonal correspondence)**.** *There is a canonical equivalence of C-bicomodules*

$$\mathbb{K}^{\mathrm{bi}}_A(A) \simeq C_\Delta. \tag{61}$$

Proof. The two-sided transform of A is

$$C \otimes_\tau A \otimes_\tau C.$$

The bar–cobar counit contracts the middle A successively against the left and right twisting cochains. Filter by the total coradical degree of the two outer factors. On the associated graded, the twisting terms vanish and the contraction identifies the complex with one copy of C. Under this identification the two surviving coactions are the two legs of Δ_C. Finite positive weights and completeness lift the associated-graded equivalence. □

The theorem gives a precise mathematical form to the phrase "universal boundary state." The diagonal bimodule is the identity kernel for tensoring over A; its Koszul transform is the identity kernel for cotensoring over C.

Corollary 8.4 (Composition unit)**.** *For every conilpotent C-bicomodule N,*

$$C_\Delta \,\square^L_C\, N \simeq N, \qquad N \,\square^L_C\, C_\Delta \simeq N.$$

8.4 Tensor becomes cotensor

Let M be an A–B bimodule and N a B–D bimodule, with all algebras admissibly positive and all modules complete. Relative tensor product composes defects. The coalgebraic operation dual to it is derived cotensor product.

Theorem 8.5 (Tensor–cotensor gluing)**.** *Let C_A, C_B, C_D be the ordered bar coalgebras. Then there is a canonical equivalence*

$$\mathbb{K}^{\mathrm{bi}}(M \otimes^L_B N) \simeq \mathbb{K}^{\mathrm{bi}}(M) \,\square^L_{C_B}\, \mathbb{K}^{\mathrm{bi}}(N). \tag{62}$$

Proof. Choose two-sided bar resolutions for the relative tensor product and cobar resolutions for the cotensor product. After inserting the universal twisting morphisms, both total complexes are indexed by the same strings of B-operations between M and N. The Eilenberg–Zilber comparison identifies the two bisimplicial totalizations. In fixed positive weight the comparison is finite and is a quasi-isomorphism by the standard two-sided Koszul contraction. Completion gives (62). □

This theorem is the algebraic form of cutting and regluing an interval. Tensoring sums over intermediate open states. Cotensoring imposes equality of the two coalgebraic boundary states.

8.5 Hochschild and coHochschild theories

Let
$$C^e = C \otimes C^{\mathrm{cop}}.$$
For a C-bicomodule N, define
$$\mathrm{coHH}_*(C, N) := C_\Delta \,\square^L_{C^e}\, N,$$
and
$$\mathrm{coHH}^*(C, N) := \mathrm{RHom}_{C^e}(C_\Delta, N)$$
in the selected coderived category.

Theorem 8.6 (Chiral Hochschild–coHochschild duality)**.** *For every complete A-bimodule M there are natural equivalences*

$$\mathrm{HH}_*(A, M) \simeq \mathrm{coHH}_*(C, \mathbb{K}^{\mathrm{bi}}_A(M)), \tag{63}$$
$$\mathrm{HH}^*(A, M) \simeq \mathrm{coHH}^*(C, \mathbb{K}^{\mathrm{bi}}_A(M)). \tag{64}$$

In particular,

$$\mathrm{HH}_*(A) \simeq C_\Delta \,\square^L_{C^e}\, C_\Delta,$$
$$\mathrm{HH}^*(A) \simeq \mathrm{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Proof. Hochschild homology is the derived tensor product of the diagonal bimodule with M over A^e. Transport this expression through Theorem 8.2. By Theorem 8.3, the diagonal bimodule maps to C_Δ, and by Theorem 8.5, derived tensor becomes derived cotensor. This proves (63). Derived Hom is preserved by an equivalence of derived categories, which proves (64). □

The annulus is the self-cotensor of the universal boundary state, and the derived centre is its coalgebraic self-extension. The bimodule–bicomodule equivalence identifies these descriptions term by term in the derived category.

8.6 The Koszul–Morita square

The preceding theorems assemble into a commutative square of functors:

$$\begin{tikzcd}
D(A^e\text{-mod})_{\mathrm{comp}} \arrow[r, "\mathbb{K}^{\mathrm{bi}}_A", "\sim"'] \arrow[d, "A\otimes^L_{A^e}(-)"'] & D^{\mathrm{co}}(C^e\text{-comod})_{\mathrm{conil}} \arrow[d, "C_\Delta\square^L_{C^e}(-)"] \\
D(\mathcal{C}_X) \arrow[r, equal] & D(\mathcal{C}_X).
\end{tikzcd} \tag{65}$$

The corresponding square for derived Hom computes the centre. This is the Koszul–Morita square. Its horizontal maps change algebraic presentation. Its vertical maps close an interval into a circle. The square says that duality commutes with closure.

Corollary 8.7 (Centre as the closed endomorphism of the boundary state)**.** *The ordered chiral centre* *has the equivalent descriptions*

$$Z_{\mathrm{ord,ch}}(A) \simeq \mathrm{RHom}_{A^e}(A, A) \simeq \mathrm{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Its E_2 operations are transported by the bimodule–bicomodule equivalence to the brace operations of the diagonal bicomodule.

8.7 Calabi–Yau transport

Suppose the perfect A-module category carries a d-Calabi–Yau trace. Equivalently, the diagonal bimodule has a nondegenerate cyclic pairing of degree d compatible with composition. Transporting the pairing through Theorem 8.2 gives a cyclic copairing on C_Δ.

Theorem 8.8 (Transport of the open trace)**.** *Assume that the bimodule–bicomodule equivalence lifts to cyclic objects and commutes with geometric realization of the cyclic bar constructions. Then a perfect d-Calabi–Yau structure on the augmentation-generated A-module category induces a nondegenerate degree-d cyclic structure on the diagonal bicomodule C_Δ. Under the resulting S^1-equivariant Hochschild–coHochschild equivalence, the Connes operator and the pair-of-pants product correspond; when the framed circle action is defined, so do the induced Batalin–Vilkovisky operators.*

Proof. A Calabi–Yau trace is a cyclic natural transformation on the Morita category of perfect modules. Theorem 8.5 transports composition. By the assumed lift to cyclic objects, the cyclic bar construction, its rotation, and its gluing maps are carried to their coHochschild counterparts. The Connes and pair-of-pants operations therefore agree under the equivalence. The same argument applies to the BV operator when the framed circle action has been supplied. ☐

This is the exact bridge from ordered Koszul duality to ribbon geometry. The transverse trace generates open topological surfaces. Modular geometry of varying complex curves enters through the separate chiral sewing structure of Part IV.

8.8 Physical interpretation

Let a factorization-homology realization identify a boundary condition b with the endomorphism algebra $A = \mathrm{RHom}(b, b)$. Under interval excision, the bar coalgebra C is the state object of an interval ending on the augmentation boundary. An A–B bimodule is an interface. Relative tensor product composes interfaces. The Koszul transform turns the same interface into a bicomodule of boundary states, and cotensor product performs the same gluing in coalgebraic variables.

Closing an interface into an annulus gives Hochschild homology. Inserting a local bulk operation gives Hochschild cohomology. Theorem 8.6 says that both amplitudes can be computed before or after ordered chiral Koszul duality. The operation is the same; only the variables have changed.

The square (65) is the categorical form of open–closed complementarity on the augmentation-generated sector. Applying a specified trace produces its numerical invariants.

9 Exchange begins beyond the ordered line

9.1 Order and the geometric origin of braiding

For every n, the chamber

$$\mathrm{Conf}_n^<(\mathbb{R}) = \{t_1 < \cdots < t_n\}$$

is convex. It remembers the order of factors but has trivial fundamental groupoid.

Theorem 9.1 (Exchange-data theorem)**.** *The ordered chamber* $\mathrm{Conf}_n^<(\mathbb{R})$ *is contractible and supplies canonical composition with trivial chamber transport. An* E_2 *enhancement, a flat connection on a two-dimensional configuration space, or an equivalent braided tensor datum supplies the exchange paths and the resulting braid-group action.*

Proof. Convexity contracts every path inside $\mathrm{Conf}_n^<(\mathbb{R})$. In a transverse plane, a path interchanging two points represents a braid generator. Functorial parallel transport along these paths gives the braid action, and isotopy of three-point motions gives the braid relation. □

Associative multiplication governs ordered fusion. Two-dimensional transport governs braiding. An R-matrix is the algebraic image of the latter structure on a chosen tensor representation.

9.2 The E_2 enhancement

Replacing the transverse line by an oriented plane gives a forgetful functor

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

An E_2 lift supplies exchange paths. Three-disk isotopy gives the braid relation. Passing to module categories produces the pentagon and hexagon coherences. Analytically, the lift is a factorization algebra holomorphic on X and locally constant in two real transverse directions.

9.3 Internal Hochschild cochains

For an E_1 algebra A in $\mathrm{FactD}(X)$, define

$$Z_{\mathrm{ord,ch}}(A) := \mathrm{RHom}_{A\otimes A^{\mathrm{op}}}(A, A). \tag{66}$$

The diagonal bar resolution gives the cup product, braces, and Gerstenhaber operations internally in the coefficient category.

Theorem 9.2 (Universal E_2 centre)**.** *Assume that the diagonal bimodule admits a homotopically correct bar resolution in* $\mathrm{FactD}(X)$*. Then* $Z_{\mathrm{ord,ch}}(A)$ *is an* E_2 *algebra. For every* E_2 *algebra* B *acting on* A *through the Swiss-cheese operad, there is a contractibly unique* E_2 *morphism*

$$B \longrightarrow Z_{\mathrm{ord,ch}}(A)$$

that induces the given action.

Proof. The higher Deligne theorem equips endomorphisms of the unit object A in the monoidal category of A–A bimodules with an E_2 structure. The Swiss-cheese universal property identifies a central action with a morphism to these endomorphisms. The mapping space of such factorizations is contractible by the universal property of the endomorphism object. □

The theorem identifies the centre as the universal recipient of every central bulk action. A physical bulk is identified with this centre by an open–closed equivalence.

9.4 Bimodules and categorical centres

The category $\mathrm{Bimod}_A(\mathrm{FactD}(X))$ is monoidal under relative tensor product, with unit the diagonal bimodule A.

Proposition 9.3 (Hochschild cochains as unit endomorphisms)**.** *There is a natural equivalence*

$$\mathrm{End}_{\mathrm{Bimod}_A}(A) \simeq Z_{\mathrm{ord,ch}}(A),$$

and the E_2 structure on the right is the endomorphism E_2 structure of the unit in the monoidal bimodule category. The Drinfeld centre $Z(\mathrm{Bimod}_A)$ is braided and acts on this unit endomorphism algebra. A monoidal-affineness theorem, when available, identifies

$$Z(\mathrm{Bimod}_A) \simeq \mathrm{Mod}_{Z_{\mathrm{ord,ch}}(A)}.$$

The relative tensor product makes A-bimodules into a monoidal category. Its Drinfeld centre, or the centre of another explicitly monoidal line-defect category, is the corresponding categorical centre.

9.5 Meromorphic exchange

Suppose line defects are parametrized by points of a spectral curve. A meromorphic exchange is a family

$$R_{L,M}(z,w) : L_z \otimes M_w \longrightarrow M_w \otimes L_z \tag{67}$$

defined away from the collision divisor. Isotopy of three lines gives

$$R_{12}(z_1,z_2)R_{13}(z_1,z_3)R_{23}(z_2,z_3) = R_{23}(z_2,z_3)R_{13}(z_1,z_3)R_{12}(z_1,z_2). \tag{68}$$

The poles are chiral singularities on the spectral curve; the exchange path is topological in the transverse plane.

9.6 Pure-braid discrepancy

Let M_n be a complex with a B_n action and let $P_n = \ker(B_n \to \Sigma_n)$. Its derived pure-braid invariants and coinvariants are

$$M_n^{hP_n} = \mathrm{RHom}_{\mathbb{C}[P_n]}(\mathbb{C}, M_n), \qquad (M_n)_{hP_n} = \mathbb{C} \otimes^L_{\mathbb{C}[P_n]} M_n,$$

with residual Σ_n actions.

Definition 9.4 (Pure-braid discrepancy)**.** Writing $q : B_n \to \Sigma_n$ and q^* for inflation, define

$$\mathfrak{C}_n^{\mathrm{inv}}(M) := \mathrm{Cofib}\big(q^* M_n^{hP_n} \longrightarrow M_n\big), \tag{69}$$

$$\mathfrak{C}_n^{\mathrm{coinv}}(M) := \mathrm{Fib}\big(M_n \longrightarrow q^*(M_n)_{hP_n}\big). \tag{70}$$

The two objects are different derived measures of what is lost when pure-braid transport is removed. They are defined only after a braid action has been supplied.

Proposition 9.5 (Relative group model)**.** *Let $C_*(P_n; M_n) = \mathbb{C} \otimes^L_{\mathbb{C}[P_n]} M_n$ and $C^*(P_n; M_n) = \mathrm{RHom}_{\mathbb{C}[P_n]}(\mathbb{C}, M_n)$. There are canonical equivalences*

$$\mathfrak{C}^{\mathrm{coinv}}_n(M) \simeq \mathrm{Cone}\big(M_n \longrightarrow C_*(P_n; M_n)\big)[-1], \qquad \mathfrak{C}^{\mathrm{inv}}_n(M) \simeq \mathrm{Cone}\big(C^*(P_n; M_n) \longrightarrow M_n\big).$$

The two cones are the reduced group-bar and reduced group-cochain models with their displayed shifts. Conjugation in B_n induces the residual Σ_n action.

A map from collision strata to braid cells induces a partition filtration on these relative group complexes. The cone models retain the full homotopy type of pure-braid descent.

9.7 Meromorphic Tannaka reconstruction

Let K be the function field of the spectral curve. Let $\mathcal{L}$ be a small rigid K-linear tensor category with finite-dimensional Hom spaces. A fibre functor may carry either strong monoidal or only quasi-monoidal comparison maps.

Assume first that

$$F : \mathcal{L} \longrightarrow \mathrm{Vect}_K$$

is faithful, exact, and strong monoidal, and that the coend

$$\mathcal{H} = \int^{L \in \mathcal{L}} F(L)^\vee \otimes F(L) \tag{71}$$

exists. A meromorphic braiding on $\mathcal{L}$ is understood over the complement of the collision divisor.

Theorem 9.6 (Meromorphic Tannaka reconstruction)**.** *The strong monoidal pair $(\mathcal{L}, F)$ reconstructs a bialgebra $\mathcal{H}$ over K; rigidity supplies an antipode whenever the coend lies in the rigid finite or continuous-dual class. A braiding reconstructs a coquasitriangular structure. If the fibre functor is only quasi-monoidal, the same coend carries a coquasi-bialgebra structure whose coassociator is the transported tensor constraint. When the relevant dual exists, these statements dualize to quasitriangular Hopf or quasi-Hopf algebras, respectively.*

Proof. The coend represents natural endomorphisms of the fibre functor. The strong tensor constraints define a coassociative coproduct and counit; rigidity defines the antipode. The braiding gives the universal R-form, and the hexagon identities give coquasitriangularity. For a quasi-monoidal fibre functor, the defect of strict coassociativity is exactly the transported associator, which satisfies the pentagon and hence defines a coquasi-bialgebra. □

A presentation as a Yangian, quantum loop algebra, elliptic quantum group, or Hall double is an additional recognition theorem.

9.8 Morita invariance

Derived Morita equivalent E_1 algebras have equivalent bimodule categories and equivalent Hochschild cochains as E_2 algebras.

Theorem 9.7 (Centre transport through a Morita context)**.** *Let A and B be ordered chiral algebras joined by an invertible A–B bimodule in the Morita $(\infty, 2)$-category of complete factorisation algebras. Then*

$$Z_{\mathrm{ord,ch}}(A) \simeq Z_{\mathrm{ord,ch}}(B)$$

as E_2 algebras, and the induced monoidal equivalence of bimodule categories identifies their Drinfeld centres.

For $B = A^!$, the conclusion applies when the boundary Koszul equivalence of Theorem 7.5 extends to an invertible bimodule Morita context. The invertible bimodule is the additional datum that transports the full centre.

The theorem is the precise algebraic content of two boundary presentations of the same universal central bulk. Numerical complementarity is obtained by applying a Morita-invariant trace. A Koszul-dual pair acquires this conclusion exactly on the locus where its Koszul transform extends to the stated Morita context.

9.9 Topological enhancement

The centre is E_2 in the two transverse directions and remains holomorphic on the complex curve. Suppose both real translations underlying the holomorphic coordinate become null-homotopic, coherently with all E_2 operations.

Proposition 9.8 (Dimension of the topological centre)**.** *Under this coherent topologization, the analytic centre is locally constant on the four-real-dimensional manifold $X^{\mathrm{an}} \times \mathbb{R}^2$. It therefore determines an E_4 algebra.*

Proof. The null-homotopies make the factorization algebra locally constant in the two real directions of X. It was already locally constant in two transverse directions. The recognition theorem for locally constant factorization algebras gives an E_4 algebra; equivalently, Dunn additivity combines the two E_2 structures. □

Topologizing the two real directions of the complex curve together with the two transverse directions gives an E_4 algebra. Topologizing one additional real direction gives the intermediate E_3 structure.

9.10 The bulk-to-centre map

Let a bulk factorization algebra $\mathrm{Obs}_{\mathrm{bulk}}$ act on the boundary algebra A. Bringing a bulk observable to the boundary gives a central action and hence a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord,ch}}(A). \tag{72}$$

Criterion 9.9 (Algebraically complete boundary)**.** *The boundary condition reconstructs the chosen local bulk observables exactly when (72) is an equivalence in the relevant factorization category. For every bulk action, the same map exhibits the centre as its universal central recipient.*

A decoupled bulk factor lies in the fibre of (72). A generation or nondegeneracy theorem identifies the vanishing of this fibre and thereby upgrades the boundary action to bulk reconstruction.

10 Holomorphic–topological recognition

10.1 Product-generated local observables

Let $M = X^{\mathrm{an}} \times \mathbb{R}$. A product disk is an open set $U \times I$, with $U \subset X^{\mathrm{an}}$ a complex disc and $I \subset \mathbb{R}$ an oriented interval. Let $\mathsf{Disk}^{\mathrm{rect}}_{X\times\mathbb{R}}$ be the symmetric monoidal category generated by finite disjoint unions of product disks and their embeddings. The recognition theorem concerns the local class generated from product disks by factorization left Kan extension.

Definition 10.1 (Rectangular holomorphic–topological factorization algebra)**.** A rectangular holomorphic–topological factorization algebra is a factorization algebra $\mathcal{F}$ on $X^{\mathrm{an}} \times \mathbb{R}$ such that:

(i) $\mathcal{F}$ is the factorization left Kan extension of its restriction to $\mathsf{Disk}^{\mathrm{rect}}_{X\times\mathbb{R}}$;

(ii) for fixed I, the dependence on U is the analytic realization of a factorisable right D-module;

(iii) for fixed U, the dependence on I is locally constant and orientation preserving.

Write $\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R})$ for this category.

The first condition extends product-disk data to all opens. The second supplies chiral regularity. The third supplies the little-interval action. All three conditions are intrinsic to the complex and oriented structures.

10.2 The exponential law

The local operation category of product disks is the tensor product of the chiral disk category with the little-interval operad:

$$\mathsf{Disk}^{\mathrm{ch}}_X \otimes E_1.$$

The tensor-product universal property gives an exponential law for algebras.

Theorem 10.2 (Holomorphic–topological recognition)**.** *There is a natural equivalence*

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \simeq \mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R}). \tag{73}$$

On the analytic side, the statement is restricted to coefficient systems in the image of the chosen Riemann–Hilbert realization. On the algebraic side, it is an equivalence of local factorization structures generated by product disks.

Proof. An object on the left is an algebra over E_1 in algebras over the chiral disk category. By the tensor-product universal property,

$$\mathrm{Alg}_{E_1}(\mathrm{Alg}_{\mathsf{Disk}^{\mathrm{ch}}_X}(\mathcal{V})) \simeq \mathrm{Alg}_{E_1\otimes\mathsf{Disk}^{\mathrm{ch}}_X}(\mathcal{V}).$$

The latter is exactly a factorization algebra on the rectangular disk category that is chiral in U and locally constant in I. Factorization left Kan extension gives the right-hand side of (73). Restriction back to product disks is inverse because rectangular generation is part of its definition. Riemann–Hilbert identifies the regular-holonomic algebraic coefficient with its analytic constructible realization. □

The theorem identifies the product-generated local factorization structure with its ordered chiral algebra. A Lagrangian presentation, global state space, and physical integration cycle are additional realizations of that local object.

10.3 The two short-distance limits

Let A correspond to $\mathcal{F}$. For disjoint intervals $I_1 < I_2 \subset I$ and discs $U_1, U_2 \subset U$, factorization gives

$$\mathcal{F}(U_1 \times I_1) \otimes \mathcal{F}(U_2 \times I_2) \longrightarrow \mathcal{F}(U \times I). \tag{74}$$

Approaching the diagonal in $U_1 \times U_2$ probes the singular chiral coefficient. Bringing I_1 and I_2 together inside the chamber $I_1 < I_2$ probes the E_1 multiplication. The two operations satisfy interchange because they are the two factors of one product-disk operation.

Proposition 10.3 (Interchange of short-distance limits)**.** *Let m be the transverse multiplication and let γ be a chiral factorization map. On every product-disjoint locus where both composites are defined,*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \tag{75}$$

with the canonical permutation of coefficient factors.

Proof. Both sides are the image of one embedding of four product disks into a larger product disk. They are equal by functoriality of the factorization algebra. □

This interchange is stronger than the coexistence of two products and weaker than braiding. Braiding requires motion in a second transverse direction.

10.4 Boundary and line-defect origins

A boundary condition b in a holomorphic–topological theory has a factorisable endomorphism object

$$A_b = \mathrm{RHom}(b, b).$$

Composition in the normal or topological direction is E_1 multiplication; collision of insertion points on X is the chiral coefficient. Boundary-changing operators are bimodules.

Parallel line defects in a theory on $X \times \mathbb{R}^2$ give the same structure after their transverse positions are restricted to an oriented line. Allowing the lines to move in the full plane may promote fusion to an E_2 structure and supply exchange. The E_1 fusion exists before that promotion.

Principle 10.4 (Defect interpretation)**.** *The E_1 product is composition of boundary operators or line defects. The factorisable D-module records holomorphic collision. An exchange operator is additional codimension-two data.*

10.5 Classical and quantum observables

Let $(\mathcal{E}, Q, \{-,-\})$ be the local fields of a classical BV theory. A local interaction I satisfies

$$QI + \frac{1}{2}\{I, I\} = 0. \tag{76}$$

The classical observables have differential $Q + \{I, -\}$. A perturbative quantization is represented by $I_\hbar = I + \hbar I_1 + \cdots$ satisfying the renormalized quantum master equation

$$QI_\hbar + \frac{1}{2}\{I_\hbar, I_\hbar\} + \hbar\Delta I_\hbar = 0. \tag{77}$$

The equation makes the renormalized observable differential square to zero and makes graphwise factorization compatible with contraction of ultraviolet subgraphs.

Theorem 10.5 (BV realization of an ordered chiral algebra)**.** *Let a perturbatively renormalized BV theory on $X \times \mathbb{R}$ satisfy the quantum master equation. Assume that its quantum observables are rectangularly generated, locally constant in the real direction, and admit a factorisable D-module realization in the holomorphic direction. Then they determine*

$$\mathrm{Obs}^{\mathrm{q}}_{X\times\mathbb{R}} \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X)).$$

A weak equivalence of renormalized theories that induces an objectwise quasi-isomorphism of these factorization observables induces an equivalence of ordered chiral algebras.

Proof. The quantum master equation supplies the square-zero differential and compatibility of renormalized factorization maps. The three additional hypotheses place the observable factorization algebra in the right-hand side of Theorem 10.2. Applying that equivalence gives the ordered chiral algebra. Objectwise quasi-isomorphisms become equivalences after localization. □

A BV realization of a given ordered chiral algebra consists of fields, gauge symmetry, a BV pairing, a real form, counterterm data, state spaces, integration cycles, and an equivalence between the resulting local observables and the prescribed algebra.

10.6 Interval evolution

With augmentation boundary conditions at both ends of an interval, excision gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes^{L}_{A} \mathbf{1} \simeq \mathrm{Bar}^{\mathrm{ord}}_{X}(A). \tag{78}$$

The ordered bar is therefore the complex of interval amplitudes between the two chosen boundaries. Cutting the interval gives deconcatenation.

Corollary 10.6 (Orientation reversal)**.** *Reflection of the interval identifies the underlying bar objects of A and A^{op} by reversing every word and inserting the Koszul sign of the reversal permutation.*

The corollary produces opposite multiplication. When an E_2 enhancement is also present, reflection acts separately on its braid transport.

Part II

The chiral coefficient and its geometric resolutions

11 The diagonal as the primitive chiral coefficient

11.1 From separated observables to the collision operation

Place two quantum observables at distinct points of a smooth complex curve. Their factorization product is defined on the configuration space of two points. Its singular extension to coincidence is a distribution supported on the diagonal, while the unital factorization structure supplies the regular product. The pair consisting of the diagonal distribution and the factorization product is coordinate-free.

Let

$$j_2 : X^2 \setminus \Delta(X) \hookrightarrow X^2, \qquad \Delta : X \hookrightarrow X^2$$

be the open inclusion and the diagonal embedding. A binary chiral operation on a right D_X-module $\mathfrak{g}$ is a morphism

$$\mu : j_{2,*}j_2^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}. \tag{79}$$

The source contains finite principal parts along the diagonal. The target places their local coefficient on the collision locus. Antisymmetry and the chiral Jacobi identity make $\mathfrak{g}$ a chiral Lie algebra in the sense of Beilinson and Drinfeld [5].

The morphism (79) admits several faithful realizations. A formal coordinate expands it into vertex-algebra modes. Distribution theory expands it into normal derivatives of a delta distribution. Perturbative field theory reads it as a local contact operation. Factorization theory uses it as the infinitesimal collision map. Horizontal sections turn it into differential equations for conformal blocks. Each realization preserves the same singular support and composition law.

Principle 11.1 (Diagonal principle)**.** *The invariant local datum of a chiral product is the diagonal-supported morphism* (79)*. Laurent coefficients, residues, modes, propagators, and contact terms arise from functorial presentations of this morphism.*

The normal filtration resolves the singular operator product by pole order. Its first graded piece is represented locally by $d\log(z-w)$, and its mth graded piece carries the coefficient of a pole of order m. The unital factorization product supplies the regular coefficient, which becomes the normally ordered product in a formal-coordinate realization. The filtered singular distribution together with this regular factorization map organizes the complete OPE.

11.2 Historical synthesis

Dirac's delta distribution turned singular support into an operator calculus. Wilson's operator product expansion organized short-distance behavior by local fields. Borcherds expressed the consistency of all modes in one vertex identity. Beilinson and Drinfeld encoded the same local law by D-modules and diagonal correspondences. Fulton and MacPherson resolved simultaneous collisions by a normal-crossing geometry. Segal and Tsuchiya–Ueno–Yamada organized sewing and conformal blocks. Getzler and Kapranov encoded stable sewing by modular operads. Drinfeld and Kohno related braid transport to quantum-group representation theory. Costello and Gwilliam constructed perturbative observables as factorization algebras. The present construction joins these lines at the diagonal operation and combines it with an independent transverse E_1 multiplication.

The local operation enters a sequence of exact functors:

$$\mathrm{LieAlg}^{\mathrm{ch}}(X) \xrightarrow{\mathrm{coChev}^{\mathrm{ch}}} \mathrm{CoFact}_{\mathrm{conil}}(\mathrm{Ran}\, X) \xrightarrow{\mathrm{Prim}} \mathrm{LieAlg}^{\mathrm{ch}}(X). \tag{80}$$

It also admits three geometrically independent extensions:

$$\begin{tikzcd}
 & \text{labelled configurations} \arrow[d, "\text{parallel transport}"] & \\
\text{formal disc} \arrow[r, "\text{coordinate expansion}"] & \text{chiral operation} \arrow[l, bend left, "\text{localization}"] \arrow[r, "\text{cyclic sewing}"] \arrow[d, "\text{Ran globalization}"'] & \text{stable curves} \arrow[d, "\text{graph master equation}"] \\
 & \text{factorization coalgebra} & \text{modular graph operations.}
\end{tikzcd} \tag{81}$$

Every arrow in (80)–(81) is a specified functor, comparison morphism, or transport construction.

11.3 The geometric collision theorem

The incidence geometry between the diagonal operation and the Ran coalgebra has a finite-set model.

Theorem 11.2 (Geometric collision theorem). *Let X be a smooth algebraic curve over a field of characteristic zero and let $\mathfrak{g}$ be a chiral Lie algebra on X. For every nonempty finite set I define the following complexes.*

(i) *The partition collision complex*

$$C_I^{\mathrm{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \mathrm{Part}(I)} \Delta_{\pi,!}\left(\bigotimes_{B\in\pi}^{\mathrm{unord}} \mathfrak{g}[1]\right),$$

whose differential merges two blocks and applies the suspended chiral bracket.

(ii) *The corolla-normalized forest complex $W_I^{\mathrm{FM}}(\mathfrak{g})$, obtained by resolving every block by normalized chains on its rooted-tree category and coupling the tree differential to the collision differential.*

(iii) *The labelled component $\mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g})$ of the chiral Chevalley coalgebra.*

The differential on $C_I^{\mathrm{coll}}(\mathfrak{g})$ satisfies

$$(d_I^{\mathrm{coll}})^2 = \frac{1}{2} d_{J_\mu},$$

where J_μ is the suspended chiral Jacobiator. The chiral Jacobi identity therefore makes $C_I^{\mathrm{coll}}(\mathfrak{g})$ a complex. The terminal-corolla augmentations and the cofree-coalgebra decomposition give canonical maps

$$W_I^{\mathrm{FM}}(\mathfrak{g}) \xrightarrow{\sim} C_I^{\mathrm{coll}}(\mathfrak{g}) \xrightarrow{\cong} \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \tag{82}$$

The first map is a quasi-isomorphism, and the second is an isomorphism of labelled chiral coalgebra components.

The partition lattice carries the collision differential. The rooted-tree face category resolves its incidence relations. The Ran space globalizes the labelled complexes into a factorization coalgebra. Actual Fulton–MacPherson screen spaces contribute a further geometric coefficient system through their own cohomology and configuration-space forms.

11.4 The formal disc and the complete operator product

Choose a formal coordinate z. A weakly translation-equivariant chiral algebra yields a vertex algebra, and (79) acquires the expansion

$$Y(a,z)Y(b,w) \sim \sum_{n\geq 0} \frac{Y(a_{(n)}b,w)}{(z-w)^{n+1}}. \tag{83}$$

A change of coordinate acts triangularly on the modes $a_{(n)}b$. The complete pole filtration is coordinate invariant, and its order-$(n+1)$ symbol is a tensor power of the normal line to the diagonal. The triangular mode transformation is the coordinate expression of this filtered tensoriality.

The algebraic chiral operad $\mathcal{P}^{\mathrm{ch}}$ makes the vertex identity precise. Let V be a vector superspace with translation operator. An odd element $X \in W_1^{\mathrm{ch}}(\Pi V)$ defines a nonunital vertex algebra precisely when

$$[X,X] = 0. \tag{84}$$

The associated coderivation satisfies

$$d_X^2 = \frac{1}{2} d_{[X,X]}. \tag{85}$$

The Borcherds identity is the mode expansion of (84). Arnold's relation supplies the quadratic identity among configuration-space one-forms. The pole filtration supplies every higher normal symbol. Together these structures give the geometric and algebraic factors of the collision complex.

11.5 Five typed constructions

The common language organizes five distinct constructions.

First, the chiral Chevalley functor sends a chiral Lie algebra to a conilpotent factorization coalgebra and is defined in the pro-nilpotent chiral monoidal category. A quadratic dual algebra arises after choosing a quadratic presentation and a continuous duality functor.

Second, parallel transport on labelled configurations gives braid-group representations and braided categories. A strong or quasi-monoidal fibre functor reconstructs a bialgebra, Hopf algebra, coquasi-bialgebra, or quasi-Hopf algebra. A representation-theoretic recognition theorem identifies a named quantum group.

Third, projective curvature on moduli is the curvature of a connection, determinant line, or Atiyah algebra. An inverse anomaly line with opposite curvature produces a flat tensor product connection.

Fourth, the derived center is the universal recipient of homotopy-coherent central actions on an associative factorization algebra. A physical bulk supplies a bulk-to-boundary morphism into this center. A generation or nondegeneracy theorem promotes that morphism to an equivalence.

Fifth, numerical quantities arise by applying traces, characteristic classes, or asymptotic functionals to structured objects. Central charge, affine level, rank, Hodge classes, theta coefficients, and entropy coefficients retain the functor that produces each of them.

11.6 Mathematical conventions

All derived categories are stable and $\mathbb{C}$-linear. Right D-modules are used on algebraic curves. For a finite set I, X^I is the labelled Cartesian power, $\mathrm{Part}(I)$ is its partition lattice, and $\Delta_\pi : X^\pi \hookrightarrow X^I$ is the partial diagonal associated with π. Tensor products indexed by unordered finite sets are taken in the symmetric monoidal derived category; the suspension [1] carries the Koszul signs. The symbol $\mathrm{FM}_I(X)$ denotes the algebraic Fulton–MacPherson compactification. Real oriented compactifications occur in the analytic and field-theoretic realizations, where Stokes' theorem governs boundary integrals.

For a vertex algebra, modes follow the convention

$$Y(a,z)b = \sum_{n\in\mathbb{Z}} (a_{(n)}b)z^{-n-1}.$$

For a simple Lie algebra $\mathfrak{g}$, the invariant form and the dual Coxeter number are normalized so that the KZ parameter is $\hbar = (k+h^\vee)^{-1}$; a different normalization rescales the displayed exponential defining q.

12 Chiral operations on a curve

12.1 The chiral tensor product

Let X be a smooth algebraic curve. The pointwise tensor product of right D_X-modules describes coefficients located at one point. A chiral product begins with separated points and then extends their product toward a diagonal. For two right D_X-modules M and N, write

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2.$$

The source $j_*j^*(M \boxtimes N)$ carries meromorphic products on the complement of the diagonal. Evaluation of a chiral operation lands in a D-module supported on the diagonal.

For a finite set I, let

$$U_I = X^I \setminus \bigcup_{i\neq j} \Delta_{ij}, \qquad j_I : U_I \hookrightarrow X^I.$$

An I-ary chiral operation with values in a right D_X-module M is a morphism

$$\mu_I : j_{I,*}j_I^*(\boxtimes_{i\in I} M_i) \longrightarrow \Delta_{I,!}M, \tag{86}$$

where $\Delta_I : X \hookrightarrow X^I$ is the small diagonal. Partial composition first collapses a chosen block of labels, applies the corresponding operation, and then collapses the resulting block with the remaining labels. Base change for open embeddings and diagonals supplies associativity of these partial compositions.

Definition 12.1 (Chiral Lie algebra)**.** A chiral Lie algebra on X is a right D_X-module $\mathfrak{g}$ equipped with an antisymmetric operation

$$\mu : j_*j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_!\mathfrak{g}$$

whose three binary partial composites satisfy the chiral Jacobi identity.

The adjective "Lie" records the antisymmetry and Jacobi law of the diagonal operation. Its coefficients may form an infinite-dimensional mode algebra and may contain derivatives of every finite order. On a formal disc, the same identity becomes the integral vertex identity.

12.2 Jacobi from the geometry of partial diagonals

Let $I = \{1,2,3\}$. The three pair diagonals meet at the small diagonal. Colliding labels 1 and 2 first and then colliding the resulting point with label 3 gives a morphism supported on Δ_{123}. The other two intermediate pair collisions have the same target. After the symmetry signs are inserted, their sum is the Jacobiator

$$J_\mu = \mu \circ_{12} \mu + \mu \circ_{23} \mu + \mu \circ_{31} \mu. \tag{87}$$

The chiral Jacobi identity is $J_\mu = 0$.

The support geometry determines the square of a collision differential. Two disjoint collisions give two contraction orders with opposite Koszul signs. A three-point collision gives three intermediate pair collisions and hence the three terms of (87). The partition complex of Section 13 turns this observation into a chain identity.

12.3 Vacuum and augmentation

A unital chiral algebra contains a morphism from the dualizing module ω_X that represents the vacuum. The unit axiom identifies insertion next to the vacuum with regular extension across the diagonal. On the formal disc,

$$Y(\mathbf{1}, z) = \mathrm{id}, \qquad Y(a, z)\mathbf{1} \in a + zV[[z]].$$

The nonunital chiral operation contains the binary singular and regular products. The vacuum is nullary data. Passing to an augmentation ideal isolates the reduced object on which bar and Chevalley constructions are naturally formed.

12.4 Chiral and factorization structures

A factorization algebra assigns an object to every finite configuration and identifies the object on a disjoint union with the tensor product of the objects on its components. A chiral Lie algebra supplies the infinitesimal collision law that generates the corresponding factorization coalgebra through chiral Chevalley chains. Beilinson and Drinfeld construct chiral envelopes and factorization objects; Francis and Gaitsgory formulate the derived Koszul equivalence in the pro-nilpotent Ran category [5, 38].

The two directions of the equivalence are

$$\mathfrak{g} \longmapsto \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}), \qquad C \longmapsto \mathrm{Prim}(C).$$

They are mutually inverse on the conilpotent/pro-nilpotent subcategories specified by the chiral Koszul theorem. The construction is internal to the chiral monoidal category and uses its cardinality filtration; finite-dimensional linear duality plays a separate role only when a dual algebra is formed.

Historical note 12.2. Beilinson and Drinfeld place the coordinate-change law inside the right D_X-module. Coordinates then serve as calculational frames for an invariant diagonal operation. This passage turns a family of Laurent series into a geometric object that glues on the curve.

12.5 Distributions, principal parts, and contact terms

The analytic realization of (79) is a finite-order distribution supported on the diagonal. Locally it has the form

$$T(z, w) = \sum_{m=0}^{N} A_m(w)\, \partial_z^m \delta(z - w). \tag{88}$$

Cauchy's formula identifies the same coefficients with the principal parts of a meromorphic kernel. The formal-distribution identity

$$\partial_w^m \delta(z-w) = \iota_{z,w} \frac{(-1)^m m!}{(z-w)^{m+1}} - \iota_{w,z} \frac{(-1)^m m!}{(z-w)^{m+1}}$$

relates the two expansions. A chiral operation is therefore represented either by normal derivatives of the diagonal delta distribution or by finite principal parts, with identical coefficient fields.

Perturbative renormalization extends distributions from configuration space across collision strata. The space of extensions is affine over local distributions supported on those strata. The quantum master equation and locality constrain these local terms. A chiral algebra records the resulting compatible collision operations in the holomorphic sector.

12.6 The principal-part hierarchy

Let $D \subset Y$ be a smooth divisor with normal line $N_{D/Y}$. Poincaré residue is the quotient map

$$\mathrm{Res}_D : \Omega_Y^p(\log D) \longrightarrow \Omega_D^{p-1}$$

that extracts the coefficient of dt/t in a local defining equation t for D. Higher pole coefficients belong to the successive principal-part quotients

$$\mathcal{O}_Y(mD)/\mathcal{O}_Y((m-1)D) \cong N_{D/Y}^{\otimes m}, \qquad m \geq 1.$$

Proposition 12.3 (Principal-part hierarchy)**.** *The logarithmic residue and the higher normal symbols form distinct functorial operations:*

(i) Res_D *extracts the logarithmic normal coefficient in* $\Omega_Y^\bullet(\log D)$*;*

(ii) *the order-m symbol map extracts a section of* $N_{D/Y}^{\otimes m}$ *from* $\mathcal{O}_Y(mD)$*;*

(iii) *the direct system* $\{\mathcal{O}_Y(mD)\}_{m\geq 0}$*, together with its regular quotient, carries every finite principal part and the regular term.*

A coordinate t identifies these tensorial symbols with the coefficients of t^{-m}.

Proof. The exact sequence

$$0 \longrightarrow \mathcal{O}_Y((m-1)D) \longrightarrow \mathcal{O}_Y(mD) \longrightarrow N_{D/Y}^{\otimes m} \longrightarrow 0$$

constructs the order-m symbol. Under $t' = ut + O(t^2)$ its coefficient transforms by the mth tensor power of the normal transition function. The residue sequence for logarithmic forms constructs the first logarithmic coefficient. Taking the filtered union over m and adjoining the regular quotient assembles the complete local meromorphic expansion. □

The pole filtration of Section 15 applies this hierarchy simultaneously to every collision divisor.

12.7 The affine-line presentation

On $X = \mathbb{A}^1$ with coordinate z, translation-invariant sections of $j_*\mathcal{O}_{X^2\setminus\Delta}$ are Laurent polynomials in $z_1 - z_2$. A chiral operation is then represented by the lambda bracket

$$\{a_\lambda b\} = \sum_{n\geq 0} \frac{\lambda^n}{n!} a_{(n)}b, \tag{89}$$

together with the regular product. The identities

$$\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}, \qquad \{a_\lambda \partial b\} = (\partial + \lambda)\{a_\lambda b\}$$

express D-module covariance. The Jacobi identity becomes

$$\{a_\lambda\{b_\mu c\}\} - (-1)^{p(a)p(b)}\{b_\mu\{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}. \tag{90}$$

Expanding the translated spectral variable $\lambda + \mu$ gives the binomial coefficients in the Borcherds identity. Arnold's relation simultaneously governs the wedge products of logarithmic one-forms on configuration space. These two identities supply the coefficient and geometric factors of the collision differential.

12.8 Factorization as locality

Let $U_1, \ldots, U_n \subset X$ be pairwise disjoint discs contained in a larger disc V. A factorization algebra supplies a product

$$\mathcal{F}(U_1) \otimes \cdots \otimes \mathcal{F}(U_n) \longrightarrow \mathcal{F}(V).$$

Motion inside the configuration space transports this product as a local system. Approach to a collision divisor is governed by the chiral operation. The OPE is therefore the singular boundary expansion of the factorization product.

The factorization axiom compares every decomposition by disjoint subdiscs and is compatible with isotopy. Holomorphic dependence on a complex curve equips these products with the corresponding D-module covariance. The result is a geometric local-to-global object that admits transport, integration, and sewing.

12.9 Field-theoretic realization

In Euclidean quantum field theory, observables supported in a small disc form a cochain complex, and observables with disjoint supports multiply. A local perturbative quantization therefore defines a factorization algebra [14, 15]. For a holomorphic–topological theory, restriction to a holomorphic boundary produces a chiral factorization algebra. The operation (79) is the renormalized collision map of boundary observables.

A comparison morphism from the analytification of the algebraic D-module model to the analytic factorization algebra identifies the mathematical and physical realizations. Compatibility with differentials, factorization products, and diagonal extensions makes this morphism the precise bridge. Metric dependence, reality structures, positivity, and global state spaces remain additional structures on the physical realization.

13 The partition collision complex

13.1 Partitions and partial diagonals

Let I be a finite nonempty set. A partition π of I is a collection of disjoint nonempty blocks whose union is I. The refinement order is written $\pi \leq \rho$ when every block of π is contained in a block of ρ. The minimal partition has singleton blocks; the maximal partition has one block.

To π associate the partial diagonal

$$\Delta_\pi : X^\pi \hookrightarrow X^I,$$

where the coordinates indexed by a block are required to coincide. If $\pi \lessdot \rho$ is a cover relation, then ρ is obtained from π by merging exactly two blocks. Every saturated chain in $\mathrm{Part}(I)$ is an order of pairwise collisions.

Let $s\mathfrak{g} = \mathfrak{g}[1]$. For a partition π, set

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!}\left(\bigotimes_{B\in\pi}^{\mathrm{unord}} s\mathfrak{g}\right). \tag{91}$$

The unordered tensor product is the coinvariant form of the tensor product in the symmetric monoidal derived category. The suspension converts the antisymmetric chiral bracket into a symmetric degree-one contraction, exactly as in the Chevalley complex of an ordinary Lie algebra.

Definition 13.1 (Partition collision complex)**.** The I-labelled collision object is

$$C_I^{\mathrm{coll}}(\mathfrak{g}) = \bigoplus_{\pi\in\mathrm{Part}(I)} C_\pi(\mathfrak{g}). \tag{92}$$

For a cover $\pi \lessdot \rho$ that merges blocks B_1 and B_2, the component

$$d_{\pi,\rho} : C_\pi(\mathfrak{g}) \longrightarrow C_\rho(\mathfrak{g})$$

applies the suspended chiral bracket to the factors indexed by B_1, B_2 and acts identically on the remaining factors. The total differential is the internal differential plus the signed sum over covers.

The sign is determined by moving the two suspended factors next to one another, applying the degree-one bracket, and returning the merged factor to the position determined by the orientation line of the cover. Because the tensor product is unordered, the definition is independent of a chosen linear order on the blocks.

13.2 The square of the collision differential

Every length-two interval $[\pi, \sigma]$ in the partition lattice has one of two forms.

If σ is obtained by merging two disjoint pairs of blocks, there are exactly two intermediate partitions. The coefficient operations act on disjoint tensor factors and commute before suspension. Interchanging the two degree-one contractions changes the sign, so the two paths cancel.

If σ is obtained by merging three blocks into one, there are exactly three intermediate partitions. The three composites are the three terms of the suspended Jacobiator.

Theorem 13.2 (Square of the collision differential)**.** *Let μ be an antisymmetric chiral binary operation on a right D_X-module $\mathfrak{g}$. The collision operator defined from μ satisfies*

$$d_{\mathrm{coll}}^2 = \frac{1}{2} d_{[\mu,\mu]}. \tag{93}$$

Consequently $C_I^{\mathrm{coll}}(\mathfrak{g})$ is a complex for every finite I if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal differential squares to zero and its mixed commutator with d_{coll} vanishes precisely when μ is a chain map. The remaining square is a sum over length-two intervals in $\mathrm{Part}(I)$. Disjoint mergers cancel in pairs by the Koszul sign. Triple mergers sum to the three suspended partial composites of μ. This sum is $\frac{1}{2}[\mu, \mu]$ in the convolution Lie algebra of chiral operations. These two combinatorial types exhaust the length-two intervals of the partition lattice. $\square$

Corollary 13.3 (Selection by collisions)**.** *An antisymmetric collection of diagonal-supported binary operations defines a chiral Lie algebra exactly when the corresponding partition collision operators form differentials in all arities.*

The selection statement uses the complete chiral operation. The Jacobiator supplies its coefficient identity, while configuration-space forms may supply an additional geometric coefficient system. An ordinary Lie bracket already satisfies its Chevalley square-zero relation by Jacobi.

13.3 Arity two and three

For $I = \{1, 2\}$ there are two partitions. The complex is

$$\Delta_{1|2,!}(s\mathfrak{g} \boxtimes s\mathfrak{g}) \xrightarrow{s\mu} \Delta_{12,!}(s\mathfrak{g}).$$

The map is the collision operation.

For $I = \{1, 2, 3\}$, the Hasse diagram is

$$\begin{tikzcd} & 1|2|3 \arrow[dl] \arrow[d] \arrow[dr] & \\ 12|3 \arrow[dr] & 13|2 \arrow[d] & 23|1 \arrow[dl] \\ & 123 & \end{tikzcd} \tag{94}$$

The square from the top to the bottom is the sum of three composites. In a coordinate presentation these are the three expansions of the vertex Jacobi identity. In the D-module presentation they are three maps with the same source and target, supported on the small diagonal.

13.4 The cofree commutative coalgebra

Let $\mathcal{C}$ be a symmetric monoidal stable category. The cofree conilpotent commutative coalgebra on an object V has underlying object

$$\mathrm{coSym}(V) = \bigoplus_{n \geq 1} (V^{\otimes n})_{\Sigma_n}.$$

In the chiral monoidal category on the Ran space, the labelled component of $\mathrm{coSym}(s\mathfrak{g})$ decomposes by the ways in which labels are identified. Those ways are the partitions of I. The summand associated with π is exactly (91).

Proposition 13.4 (Chevalley identification)**.** *For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is canonically the I-component of its chiral Chevalley coalgebra:*

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \cong \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \tag{95}$$

Under this identification the collision differential is the Chevalley coderivation induced by the chiral bracket.

Proof. The cofree commutative coalgebra is the sum over finite collections of primitive factors. Pulling its Ran component back to X^I records a surjection from I to the set of primitive factors. A surjection modulo relabeling of its target is a partition of I. The Chevalley coderivation merges two primitive factors by the bracket, hence agrees with the cover differential of the partition lattice. □

13.5 The factorization coproduct

Suppose $I = I_1 \sqcup I_2$. Restriction to the open locus where the points of I_1 remain disjoint from the points of I_2 gives a coproduct

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \longrightarrow C_{I_1}^{\mathrm{coll}}(\mathfrak{g}) \boxtimes C_{I_2}^{\mathrm{coll}}(\mathfrak{g}). \tag{96}$$

On the disjointness locus, contributing partitions split uniquely as a partition of I_1 together with a partition of I_2. The coproduct is therefore deconcatenation at the level of primitive blocks. Coassociativity follows from the associativity of disjoint union.

Proposition 13.5. *The family $\{C_I^{\mathrm{coll}}(\mathfrak{g})\}_I$, with the coproducts (96), is a conilpotent commutative factorization coalgebra. Its primitive part is $\mathfrak{g}[1]$.*

Conilpotence is measured by the number of blocks. Every reduced coproduct separates a partition into strictly smaller collections of blocks, so an iterated reduced coproduct terminates after at most $|I|-1$ steps. The resulting coradical filtration is the block-number filtration in the ambient symmetric monoidal category.

13.6 Incidence geometry and Möbius inversion

The Euler characteristic of the collision complex is governed by the Möbius function of the partition lattice. For the interval from the discrete partition to the one-block partition,

$$\mu_{\mathrm{Part}(I)}(\hat{0}, \hat{1}) = (-1)^{|I|-1}(|I|-1)!.$$

This integer is also the top reduced Euler characteristic of the proper partition complex. It explains the factorial multiplicities that appear in Lie operad components and in the top cohomology of configuration spaces. The equality is combinatorial; the representations carried by the two sides require their respective coefficient systems.

The proper partition complex is homotopy equivalent to a wedge of $(|I|-1)!$ spheres of dimension $|I|-3$. Its top homology is the multilinear part of the Lie operad, up to the sign representation. This is the classical topological shadow of the Lie–commutative Koszul duality that becomes chiral on the Ran space.

13.7 Modules and marked points

Let M be a chiral module over $\mathfrak{g}$. Add a distinguished label $*$ and require every admissible partition to contain exactly one distinguished block, namely the block containing $*$. A partition of $I \sqcup \{*\}$ with a distinguished block containing $*$ indexes a summand in the module collision complex. Merging an ordinary block with the distinguished block applies the module action; merging two ordinary blocks applies the bracket. The same length-two interval analysis proves that the square is the sum of the Lie Jacobiator and the module identity.

This pointed complex is the geometric origin of derived coinvariants and conformal blocks with insertions. It is also the correct place to compare BRST reduction, because the BRST differential acts on both the algebra and the marked module.

13.8 The screen coefficient beyond incidence

The partition complex records the groups of colliding labels and their incidence pattern. Fulton–MacPherson geometry enriches this incidence datum by the relative shape of every collision. For three points on a complex line, translation and dilation leave one cross-ratio coordinate on the screen. Differential forms on such screens produce Arnold and Orlik–Solomon classes. The forest resolution in Section 14 carries the incidence complex, while the screen complexes supply the geometric coefficient system.

14 The Fulton–MacPherson forest resolution

14.1 Compactifying collisions

The configuration space $\mathrm{Conf}_I(X)$ is the complement of the diagonals in X^I. Its Fulton–MacPherson compactification $\mathrm{FM}_I(X)$ is obtained by successively blowing up partial diagonals in an order compatible with inclusion [40]. The result is smooth, and the boundary is a normal-crossing divisor. A boundary point records the colliding labels together with their limiting relative configuration at every scale.

Boundary strata are indexed by nested collections of subsets, equivalently by rooted forests whose leaves are the labels in I. An internal vertex represents a cluster. An edge represents one scale of collapse. Roots represent the final clusters that remain distinct in X.

Let F be such a forest. The root components determine a partition $q(F)$ of I. The corresponding stratum maps to the partial diagonal $\Delta_{q(F)}(X^{q(F)})$. Its fiber is a product of screen configuration spaces. On a curve, a screen for a cluster S is the compactification of configurations of $|S|$ points on a one-dimensional tangent space modulo translation and dilation.

14.2 Tree categories over collision blocks

For a nonempty finite set B, let $\mathrm{Tree}(B)$ be the category of rooted trees with leaves labelled by B and internal vertices of valence at least two. Morphisms contract internal edges. The corolla with leaf set B is terminal. Write

$$K(B) = N_*(N\,\mathrm{Tree}(B)), \tag{97}$$

for the normalized chain complex of its nerve, placed in nonpositive cohomological degrees. The augmentation and the terminal-corolla inclusion are chain maps

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C}, \qquad \mathbb{C} \xrightarrow{\iota_B} K(B), \qquad \epsilon_B \iota_B = \mathrm{id}. \tag{98}$$

Because $\mathrm{Tree}(B)$ has a terminal object, ϵ_B is a quasi-isomorphism.

For a partition π of I, define

$$K(\pi) = \bigotimes_{B \in \pi}^{\mathrm{unord}} K(B). \tag{99}$$

An element of $K(\pi)$ is a chain of contractions in a forest whose root components are the blocks of π. This complex is the incidence information carried by the face category of the Fulton–MacPherson boundary. Screen cohomology enters through the separate coefficient system described below.

14.3 The corolla-normalized forest resolution

Set

$$W_I^{\mathrm{FM}}(\mathfrak{g}) = \bigoplus_{\pi \in \mathrm{Part}(I)} C_\pi(\mathfrak{g}) \otimes K(\pi). \tag{100}$$

Its differential has three terms. The internal term is induced by the differential of $\mathfrak{g}$. The vertical term is the normalized nerve differential on the factors $K(B)$. For a cover $\pi \lessdot \rho$ merging blocks B, C, the horizontal term applies the suspended chiral bracket to the B, C coefficient factors and applies the chain map

$$K(B) \otimes K(C) \xrightarrow{\epsilon_B \otimes \epsilon_C} \mathbb{C} \xrightarrow{\iota_{B\cup C}} K(B \cup C) \tag{101}$$

while leaving the other tree factors unchanged. Thus a collision of two root components is represented by the terminal corolla of the merged block; all nonterminal tree data remains in the vertical resolution.

The maps (101) are symmetric and strictly associative because they factor through the augmentations. They commute with the vertical differential. Consequently the square of the horizontal differential is exactly the square of the partition collision differential, with a common terminal-corolla factor.

There is an augmentation

$$\epsilon_I : W_I^{\mathrm{FM}}(\mathfrak{g}) \longrightarrow C_I^{\mathrm{coll}}(\mathfrak{g}) \tag{102}$$

obtained by applying ϵ_B to every tree factor.

Theorem 14.1 (Forest resolution)**.** *For every chiral Lie algebra $\mathfrak{g}$ and finite set I, the differential on* (100) *squares to zero and the augmentation* (102) *is a quasi-isomorphism. The construction is natural under relabelling and compatible with disjoint union.*

Proof. The vertical differential squares to zero. Its mixed commutator with the horizontal differential vanishes because the augmentations and corolla inclusions in (101) are chain maps. Strict associativity of the corolla-normalized merge identifies the horizontal square with the Jacobiator computed in Theorem 13.2; it therefore vanishes for a chiral Lie algebra.

Filter by the total nerve degree of the tree factors. The filtration is finite for fixed I. Its first spectral sequence computes the homology of every $K(B)$. Since each ϵ_B is a quasi-isomorphism, the first page is precisely $C_I^{\mathrm{coll}}(\mathfrak{g})$, and the induced differential is its collision differential. The spectral-sequence comparison proves that ϵ_I is a quasi-isomorphism. Relabelling preserves terminal corollas, and disjoint union tensors the tree complexes, giving naturality and factorization compatibility. □

Corollary 14.2. *The corolla-normalized forest complex resolves the chiral Chevalley coalgebra:*

$$W_I^{\mathrm{FM}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}).$$

The adjective "geometric" refers here to the tree categories labelling the faces of the compactification. The actual screen fibers retain their de Rham and constructible cohomology, which is incorporated by the stratification spectral sequence.

14.4 Screen spaces as geometric coefficients

Let $p_I : \mathrm{FM}_I(X) \to X^I$ be the proper blowdown. The fiber over a point of a partial diagonal contains compactified screen configurations. For $|I| = 3$, translation and dilation reduce the screen to the moduli of one cross-ratio; its compactification is $\overline{M}_{0,4} \cong \mathbb{P}^1$. The derived pushforward $Rp_{I,*}\mathbb{C}$ therefore contains the cohomology of these screen fibers in addition to the partition incidence complex.

Proposition 14.3 (Screen-cohomology decomposition)**.** *The proper map p_I carries two functorial layers: the incidence complex of collision strata and the derived cohomology of their compactified screens. A comparison from a de Rham or constructible complex on $\mathrm{FM}_I(X)$ to the collision complex is specified by a morphism that projects or integrates the screen factor and is compatible with all incidence maps.*

The distinction separates two independent structures:

incidence of collision strata	geometry inside a screen
partition lattice	configuration space
edge contraction	de Rham differential
chiral Jacobi	Arnold–Orlik–Solomon relations
Chevalley coderivation	propagator and period classes

14.5 The stratification spectral sequence

Filter $\mathrm{FM}_I(X)$ by the codimension of its boundary strata. For a constructible complex $\mathcal{F}$ that is locally constant along strata, the standard spectral sequence of a filtered space has first page

$$E_1^{p,q} = \bigoplus_{\operatorname{codim} S = p} H_c^{p+q}(S; \mathcal{F}|_S \otimes \mathrm{or}_S) \Longrightarrow H_c^{p+q}(\mathrm{FM}_I(X); \mathcal{F}). \tag{103}$$

The differential is the signed sum of incidence maps between adjacent strata.

For coefficients pulled back from the root diagonal, a stratum splits locally into a base and a product of screens. The Kunneth theorem expresses its cohomology as

$$H_c^*(S; \mathcal{F}|_S) \cong H_c^*(\Delta_{q(F)}; \mathcal{F}_{q(F)}) \otimes \bigotimes_{v \in V_{\mathrm{int}}(F)} H_c^*(\mathrm{Screen}(v)). \tag{104}$$

After barycentric replacement of the face incidence category and corolla normalization, the degree-zero screen classes map to the forest incidence resolution. Positive-degree screen classes record the topology of relative configurations.

Theorem 14.4 (Incidence–screen separation)**.** *Let $\mathcal{F}$ be pulled back from every root diagonal and locally constant along each Fulton–MacPherson stratum. Assume that every compactified screen factor is connected and that the Kunneth morphism in* (104) *is an isomorphism for $\mathcal{F}$. Then the boundary-codimension spectral sequence carries a second filtration by total screen-cohomological degree. Its degree-zero screen row is canonically the barycentric face-incidence complex with coefficients on the root diagonals. Corolla normalization gives a quasi-isomorphism from that row to the forest resolution $W_I^{\mathrm{FM}}(\mathfrak{g})$.*

Proof. Connectedness identifies H^0 of every compactified screen with the constant line, and functoriality of restriction sends the constant generator to the constant generator on each boundary screen. The Kunneth isomorphism therefore decomposes every stratum term into a root-diagonal coefficient and tensor products of screen cohomology. Filtering by total screen degree makes the degree-zero row the signed incidence complex of the face category. Barycentric subdivision identifies that incidence complex with the normalized nerve of the rooted-forest category. Applying ϵ_B and ι_B to every root block gives the corolla-normalized horizontal maps of (101). The forest-resolution theorem then proves that the resulting map is a quasi-isomorphism. □

The theorem places the chiral Chevalley complex and configuration-space forms in one filtered complex. Its degree-zero screen row carries collision incidence, while its positive screen degrees carry Arnold–Orlik–Solomon and period classes.

14.6 Arnold forms on screens

On an affine coordinate chart, set

$$\eta_{ij} = d\log(z_i - z_j).$$

On $\mathrm{Conf}_3(\mathbb{C})$ the Arnold relation is

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0. \tag{105}$$

It follows by the elementary substitution $x = z_1 - z_2$, $y = z_2 - z_3$. The Orlik–Solomon algebra of the braid arrangement is generated by the classes of the η_{ij} with these three-term relations [4].

The Arnold relation expresses flatness of the universal logarithmic connection when its coefficients satisfy the infinitesimal braid relations. In the stratification spectral sequence, (105) belongs to the screen coefficient system, while the chiral Jacobi identity belongs to the coefficient operation on the root diagonal. The total differential combines these two identities according to their bidegrees.

14.7 Real oriented compactification and Stokes' theorem

For perturbative integrals one uses the real oriented blowup of the diagonals. The boundary hypersurfaces carry outward normal orientations. If α is a compactly supported form of top degree minus one, Stokes' theorem gives

$$\int_{\mathrm{FM}_I^{\mathrm{or}}(M)} d\alpha = \sum_{H \subset \partial\, \mathrm{FM}_I^{\mathrm{or}}(M)} \int_H \alpha.$$

Applying Stokes twice yields cancellation of codimension-two corners because each corner occurs with opposite induced orientations from its two parent hypersurfaces. This is the analytic form of the incidence differential squaring to zero.

When forms carry algebraic coefficients, the total square contains the de Rham square, the incidence square, the coefficient Jacobiator, and their mixed commutators. A complete proof assigns each term its bidegree and verifies the corresponding identity. Boundary-of-boundary supplies the incidence cancellation, while the algebraic coefficient maps supply Jacobi and module compatibility.

14.8 Renormalization forests

The same forest combinatorics appears in the BPHZ forest formula. A divergent subgraph determines a collision subconfiguration; nested divergent subgraphs determine a forest; subtraction is performed recursively from smaller clusters to larger clusters. Fulton–MacPherson compactification turns this recursion into geometry. Counterterms are supported on boundary strata, and locality is the compatibility of their restrictions under further collisions.

This parallel is exact at the level of incidence categories. A particular quantum field theory supplies a propagator, a density, a gauge fixing, and a renormalization prescription. Those analytic data form the screen coefficient system. The universal forest resolution supplies the combinatorics on which they live.

15 Pole filtrations and normal geometry

15.1 Principal parts along a divisor

Let Y be a smooth variety and $D \subset Y$ a smooth divisor. For $m \geq 0$ there is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \longrightarrow \mathcal{O}_D((m+1)D) \longrightarrow 0. \tag{106}$$

Since $\mathcal{O}_D(D) \cong N_{D/Y}$, the associated graded of the pole filtration is

$$\operatorname{gr}^F_{m+1} \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes(m+1)}. \tag{107}$$

If t is a local defining equation for D, the class of $t^{-(m+1)}$ is a local generator of this line. Under $t' = ut + O(t^2)$ it transforms by $u^{-(m+1)}$ modulo lower pole order. This is precisely the tensorial transformation law of the normal symbol.

For a normal-crossing divisor $D = \bigcup_{a \in A} D_a$, use the multi-filtration

$$F_{\boldsymbol{m}} \mathcal{O}_Y(*D) = \mathcal{O}_Y\left(\sum_{a \in A} m_a D_a\right).$$

On a stratum $D_J = \bigcap_{a \in J} D_a$, the multi-associated graded is

$$\operatorname{gr}^F_{\boldsymbol{m}} \mathcal{O}_Y(*D)|_{D_J} \cong \bigotimes_{a \in J} N_{D_a/Y}^{\otimes m_a} \otimes \mathcal{O}_{D_J}(*D_J^{\mathrm{res}}). \tag{108}$$

The residual divisor records further boundary components meeting the stratum.

15.2 The full operator product

Let D_{12} be the pair-collision divisor in $\mathrm{FM}_2(X) = X^2$. In a coordinate $t = z_1 - z_2$, the singular operator product is

$$\sum_{m\geq 1} A_m(z_2)t^{-m}.$$

The coefficient A_m belongs intrinsically to the mth normal symbol. For a vertex algebra, $A_m = a_{(m-1)}b$. Therefore

$$a_{(r)}b \quad \longleftrightarrow \quad \mathrm{gr}^F_{r+1}\,\mathcal{O}(*D_{12}) \cong N^{\otimes(r+1)}_{D_{12}/X^2}. \tag{109}$$

The correspondence is coordinate-free at the associated-graded level. A coordinate chooses a trivialization of the normal line and thereby names the coefficient as a mode.

The meromorphic filtration contains the singular modes in positive pole degree. A unital factorization or vertex-algebra realization supplies the regular coefficient in degree zero; in a formal coordinate this coefficient is the normally ordered product, and translation generates its regular Taylor descendants. Extension data between successive pole degrees records the triangular mixing of modes under coordinate changes.

15.3 Nested collisions

Let T be a rooted collision tree indexing a boundary stratum $D_T \subset \mathrm{FM}_I(X)$. Each internal edge e corresponds to one normal direction, hence to a line N_e. Locally the normal bundle splits

$$N_{D_T/\,\mathrm{FM}_I(X)} \cong \bigoplus_{e\in E_{\mathrm{int}}(T)} N_e. \tag{110}$$

The multi-associated graded of a meromorphic coefficient is therefore

$$\bigotimes_{e\in E_{\mathrm{int}}(T)} N_e^{\otimes m_e}. \tag{111}$$

Assigning a positive integer m_e to every internal edge is the geometric version of assigning a pole order to every intermediate OPE channel.

Tree substitution contracts a subtree and replaces it by one incoming branch. On an arbitrary curve the colour of a rooted collision consists of its root point together with the tangent line at that point. With these colours, nested diagonal embeddings define the intrinsic collision coefficient system. A tangent framing identifies the colours locally and produces the familiar uncoloured formal-disc presentation.

Theorem 15.1 (Normal-cone collision operad)**.** *The multi-normal symbols of nested collision strata form a coloured operad over X. For a collision tree T, its coefficient line is*

$$\bigotimes_{e\in E_{\mathrm{int}}(T)} N_e^{\otimes m_e},$$

with one positive pole order m_e on every internal edge. Substitution is induced by the canonical morphisms of Rees algebras for nested regular embeddings. It is associative and independent of local coordinates.

Proof. A nested collision stratum is an iterated regular embedding. The associated graded of its formal neighborhood is the symmetric algebra of the conormal bundle, and (110) identifies the independent conormal directions with internal edges. For a chain of nested regular embeddings, deformation to the normal cone is functorial: composing the corresponding Rees morphisms either before or after grouping a subtree gives the same morphism. Tensoring the resulting normal lines gives the stated substitution law and its associativity. The construction uses only ideals of diagonals and their normal bundles, hence is coordinate-free. □

Compatible local coordinates or tubular neighborhoods give strict filtered identifications of formal neighborhoods. Two choices differ by a filtration-positive automorphism. The associated-graded coloured operad is canonical, and the full filtered object is the coordinate-free chiral distribution represented by those local series.

15.4 The collision Rees space

The pole filtration can be assembled into a Rees algebra

$$\mathcal{R}_D(\mathcal{O}_Y(*D)) = \bigoplus_{m\geq 0} F_m\mathcal{O}_Y(*D)\, u^m. \tag{112}$$

For $u \neq 0$ its localization recovers the meromorphic coefficient algebra. At $u = 0$ it becomes the normal-symbol algebra. Locally, the deformation is the familiar equation

$$tv = u,$$

where v represents the inverse normal coordinate. The generic fiber contains the full pole; the special fiber records its normal cone.

For the full boundary of $\mathrm{FM}_I(X)$ one obtains a multi-Rees space. Collision scales become independent deformation parameters. The special fiber is a union of normal-cone pieces indexed by forests. This geometry explains why classical limits are controlled by trees and why quantum corrections mix different pole orders.

15.5 The chiral and classical operads

The algebraic chiral operad is filtered by the order of poles along diagonals. Bakalov–De Sole–Heluani–Kac construct the corresponding classical operad and a canonical comparison from chiral symbols to classical operations [6, 7]. Geometrically, a chiral product passes to its normal symbols; when a filtration on the state space is compatible with the products, the regular multiplication becomes commutative and the leading singular symbols define the Poisson vertex bracket.

Theorem 15.2 (Chiral-to-classical symbol map)**.** *On a translation-invariant formal disc, the pole filtration of the chiral operad has a canonical morphism of operads*

$$\mathrm{gr}^F \mathcal{P}^{\mathrm{ch}} \longrightarrow \mathcal{P}^{\mathrm{cl}}.$$

Under the normal-symbol identification (109)*, this is the symbol morphism of the geometric pole filtration. If a filtered vertex algebra is compatible with its filtration, the image of its chiral Maurer–Cartan operation is the Poisson vertex algebra operation on the associated graded.*

For the Li filtration and every other compatible exhaustive filtration, the theorem produces the associated Poisson vertex algebra. Flatness of a chosen quantization and bijectivity of the symbol morphism are additional properties expressed by the corresponding Rees module.

15.6 Residues, jets, and differential operators

Higher coefficients may be extracted by pairing principal parts with jets. If t is a coordinate normal to D, then

$$A_m = \frac{1}{2\pi i} \oint t^{m-1} \left(\sum_{r \geq 1} A_r t^{-r} \right) dt. \tag{113}$$

The factor t^{m-1} is a choice of jet. Intrinsically, the pairing is between $N_D^{\otimes m}$ and $N_D^{\vee \otimes m}$. The contour integral is the analytic realization of this pairing. Ordinary residue is $m = 1$.

In a D-module, normal derivatives of delta distributions supply the dual description. The identity between principal parts and delta derivatives is a local Fourier transform in the normal direction. This is why the lambda bracket encodes all singular modes polynomially in λ.

15.7 Coordinate changes and the stress tensor

A projective connection is required to make the stress tensor transform tensorially. Under $z \mapsto w(z)$,

$$T(z) = \left(\frac{dw}{dz} \right)^2 T(w) + \frac{c}{12} \{w, z\}, \tag{114}$$

where $\{w, z\}$ is the Schwarzian derivative. The inhomogeneous Schwarzian term is the coordinate expression of the central extension of the vector-field action. The fourth-order pole in the TT operator product carries its local coefficient.

The pole filtration makes the distinction visible. The central term belongs to the fourth normal symbol. The transformation law mixes it with lower terms through the choice of projective connection. The invariant object is the chiral operation of the Virasoro algebra together with its coordinate-change action.

15.8 Independent geometric filtrations

The theory carries several canonical filtrations: pole order measures local analytic singularity; coradical degree measures coalgebraic decomposition; collision codimension measures boundary depth; transferred-operation arity measures homotopy complexity; graph genus measures sewing topology; and cohomological degree measures deformation theory. A first-order system can combine simple poles with a large representation category, while the Virasoro fourth-order self-OPE enters a deformation complex with its own independent grading.

Every filtration has a geometric source and a corresponding spectral sequence. Relations among two filtrations are expressed by explicit comparison maps and convergence hypotheses.

16 The formal disc and vertex algebras

16.1 Translation-equivariant chiral operations

Let V be a vector superspace with an even endomorphism ∂. On the affine line, translation-invariant rational functions on configurations are generated by the differences $z_i - z_j$ and their inverses. The algebraic chiral operad has n-ary component

$$\mathcal{P}^{\mathrm{ch}}_V(n) = \mathrm{Hom}_{\mathcal{D}^T_n}\left(V^{\otimes n} \otimes \mathbb{C}[z_i - z_j, (z_i - z_j)^{-1}], V[\lambda_1, \ldots, \lambda_n]/\langle \partial + \textstyle\sum_i \lambda_i \rangle\right), \tag{115}$$

where $\mathcal{D}^T_n$ is the algebra of translation-invariant differential operators. The variables λ_i record normal derivatives. The relation in the target expresses simultaneous translation covariance.

The operad (115) is the formal-coordinate presentation of the Beilinson–Drinfeld chiral endomorphism operad. It contains the regular multiplication and all finite principal parts. Composition expands rational functions in the domains corresponding to nested collisions.

16.2 The universal Lie superalgebra of an operad

For a linear operad $\mathcal{P}$, set

$$W(\mathcal{P}) = \bigoplus_{k \geq -1} W_k(\mathcal{P}), \qquad W_k(\mathcal{P}) = \mathcal{P}(k+1)^{\Sigma_{k+1}}.$$

Operadic insertion defines a pre-Lie product and hence a Lie superbracket. For $\mathcal{P} = \mathcal{P}^{\mathrm{ch}}_{\Pi V}$, write $W^{\mathrm{ch}}(V)$. An odd binary operation is an element $X \in W^{\mathrm{ch}}_1(\Pi V)$.

The following theorem is the algebraic selection principle.

Theorem 16.1 (Bakalov–De Sole–Heluani–Kac). *An odd element $X \in W^{\mathrm{ch}}_1(\Pi V)$ satisfies*

$$[X, X] = 0 \tag{116}$$

if and only if X defines a nonunital vertex algebra structure on V with translation operator ∂. The differential $\mathrm{ad}\, X$ *on $W^{\mathrm{ch}}(V)$ is the vertex-algebra deformation differential.*

The theorem includes all coefficients of the integral lambda bracket. Skewsymmetry, the normally ordered product, the noncommutative Wick formula, and the Borcherds identity are different coefficient forms of the single equation (116) [6].

16.3 The coderivation identity

Let $\mathcal{C}$ be a conilpotent dg cooperad and let $\alpha : \mathcal{C} \to \mathcal{P}^{\mathrm{ch}}$ be an operadic twisting morphism, for example the canonical twisting morphism from the bar cooperad. An operation X then determines, through α, a coderivation d_X of the cofree $\mathcal{C}$-coalgebra $\mathrm{cofree}_{\mathcal{C}}(sV)$. The commutator of coderivations corresponds to the convolution bracket.

Proposition 16.2 (Square of the vertex coderivation). *In every such bar-coalgebra model, an odd chiral operation X satisfies*

$$d_X^2 = \frac{1}{2}[d_X, d_X] = \frac{1}{2} d_{[X,X]}. \tag{117}$$

Consequently d_X is a differential exactly when the image of X is Maurer–Cartan in the chiral convolution Lie algebra; for the binary operation of Theorem 16.1, this is exactly the vertex-algebra condition.

Proof. Coderivations of a cofree conilpotent coalgebra are determined by their corestriction to cogenerators. The corestriction of $[d_X, d_Y]$ is the signed sum of single insertions represented by the convolution pre-Lie product; antisymmetrizing gives $[X, Y]$. Setting $Y = X$ gives the formula. $\square$

Combining Theorems 15.1, 16.1 and 16.2 gives the geometric statement: the full pole-filtered collision operation produces a square-zero coderivation exactly when the vertex identities hold. The geometry represents the coefficients; the operadic bracket supplies the identity.

16.4 The Borcherds identity

For homogeneous $a, b, c \in V$, the coefficient form of the vertex Jacobi identity is

$$\sum_{j\geq 0}(-1)^j\binom{n}{j}\big(a_{(m+n-j)}b_{(k+j)}c-(-1)^n(-1)^{p(a)p(b)}b_{(n+k-j)}a_{(m+j)}c\big)=\sum_{j\geq 0}\binom{m}{j}(a_{(n+j)}b)_{(m+k-j)}c. \tag{118}$$

The binomial coefficients arise from translating one formal variable relative to another. The normal-symbol filtration supplies every power of the relative coordinate and therefore reconstructs all translated expansions in (118). Its first graded piece is the simple-residue channel, and its higher graded pieces supply the higher modes.

At the level of distributions, the same identity is the equality of three expansions of a rational distribution on the complement of the three diagonals. The chiral operad records the expansions and their common meromorphic origin.

16.5 Vacuum and conformal structure

A vacuum vector $\mathbf{1}$ and the creation axiom extend the nonunital structure to a vertex algebra. A conformal vector ω supplies operators $L_n = \omega_{(n+1)}$ with Virasoro commutation relations and $L_{-1} = \partial$. The central charge is the coefficient of the fourth-order pole in

$$Y(\omega, z)Y(\omega, w) \sim \frac{c/2}{(z-w)^4} + \frac{2Y(\omega, w)}{(z-w)^2} + \frac{\partial_w Y(\omega, w)}{z-w}.$$

The vacuum, translation, and conformal vector are distinct structures. Chiral Koszul duality applies to the underlying chiral Lie object; conformal blocks and modular anomalies use the conformal structure.

16.6 The classical limit

Let $F_\bullet V$ be a filtration compatible with all vertex products, for example the Li filtration. The associated graded is commutative with a derivation and a lambda bracket. The regular normally ordered product becomes the commutative product; the first nontrivial singular symbols become the Poisson vertex bracket.

Theorem 16.3 (Classical symbol operad)**.** *There is a canonical morphism from the pole-associated graded of the chiral operad to the classical operad $\mathcal{P}^{\mathrm{cl}}$. An odd Maurer–Cartan element in $W(\mathcal{P}^{\mathrm{cl}}_{\Pi V})$ is* *a Poisson vertex algebra structure. For a vertex algebra equipped with a compatible filtration, the image of the principal chiral symbol is the Maurer–Cartan element of the associated Poisson vertex algebra.*

The theorem makes quantization a lifting problem in the chiral deformation complex. At each order, the Maurer–Cartan equation defines a canonical degree-two extension class; its vanishing permits the next lift, and the degree-one cohomology acts transitively on the set of lifts. This is filtered Maurer–Cartan deformation theory.

16.7 Deformations

Fix a vertex operation X. The complex

$$\Big(W^{\mathrm{ch}}(\Pi V), d_X = [X, -]\Big) \tag{119}$$

controls formal deformations. If

$$X_\hbar = X + \hbar X_1 + \hbar^2 X_2 + \cdots ,$$

then the coefficient of $\hbar$ in $[X_\hbar, X_\hbar] = 0$ says $d_X X_1 = 0$. Gauge-equivalent first-order deformations differ by a coboundary. The second-order extension class is $[\frac{1}{2}[X_1, X_1]] \in H^2$.

This cohomology classifies local deformations of the chiral operation. Hochschild cohomology classifies deformations of an associative envelope, and cohomology of moduli records variation in families of curves. Explicit comparison morphisms relate these three complexes in settings where the corresponding envelopes and families have been constructed.

16.8 Homotopy chiral structures

Let $\mathcal{P}_\infty \to \mathcal{P}^{\mathrm{ch}}$ be a cofibrant resolution and let $\mathcal{C}_\infty$ be a conilpotent cooperadic model carrying its canonical twisting morphism to $\mathcal{P}_\infty$. A homotopy chiral algebra is an algebra over $\mathcal{P}_\infty$, equivalently a Maurer–Cartan element in the completed convolution L_∞-algebra

$$\mathrm{Hom}_\Sigma(\overline{\mathcal{C}_\infty}, \mathrm{End}_V).$$

The higher operations are indexed by cells in a resolution of collision composition. Their identities are the components of the Maurer–Cartan equation. This is the correct setting for transferred structures and for BRST models before cohomology concentration.

A strict vertex algebra is a homotopy chiral algebra whose chosen cofibrant model has only the strict generators. Homotopy transfer carries it to a quasi-isomorphic homotopy chiral algebra whose higher operations are explicit sums over trees determined by the contraction. The cohomology class of each first unresolved higher operation is independent of the representative and gives the corresponding extension class.

16.9 The term "chiral Koszul duality"

The adjective "chiral" refers to the chiral monoidal category and its diagonal operation. Chiral Koszul duality is the Chevalley–primitives equivalence on the Ran space. A vertex algebra is its formal-coordinate realization. A quadratic presentation can additionally define a presentation-dependent quadratic coalgebra and a Koszul comparison map to the full chiral bar object.

17 The chiral Chevalley coalgebra on the Ran space

17.1 The Ran space

The Ran space of X is the prestack of nonempty finite subsets of X:

$$\mathrm{Ran}(X) = \operatorname*{colim}_{I \in \mathrm{Fin}^{\mathrm{surj,op}}} X^I.$$

A point of X^I maps to the subset of its coordinate values; coincident coordinates are allowed in the colimit. The transition maps are partial diagonals. Thus the Ran space remembers every finite collision simultaneously.

The category $D(\mathrm{Ran}\, X)$ carries two monoidal structures. The star tensor product is induced by union of arbitrary finite subsets, including intersecting subsets. The chiral tensor product is induced by union restricted to disjoint subsets. Chiral Lie algebras are Lie algebra objects for the chiral tensor product. Factorization objects are those whose restriction to disjoint configurations is the tensor product of their one-point pieces.

The Cartesian powers X^I and their diagonal transition maps present the D-module category of the Ran prestack. Every finite-set formula above is a labelled chart of the resulting Ran object.

17.2 Chevalley chains

Let $\mathfrak{g}$ be a chiral Lie algebra. Its chiral Chevalley chains are the cofree commutative coalgebra on $\mathfrak{g}[1]$ with the coderivation induced by the chiral bracket:

$$\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) = \left(\mathrm{coSym}^{\mathrm{ch}}(\mathfrak{g}[1]), d_{\mathfrak{g}} + d_{\mu}\right). \tag{120}$$

Pullback to X^I gives the partition decomposition of Theorem 13.4. The factorization coproduct is induced by the decomposition of a finite set into disjoint subsets.

The primitive filtration is the filtration by the number of factors. Its associated graded is the cofree commutative coalgebra with only the internal differential. This filtration is complete in the conilpotent direction: every element has finite primitive length in a labelled finite component.

Proposition 17.1 (Geometric realization of Chevalley chains)**.** *The forest incidence objects $W_I^{\mathrm{FM}}(\mathfrak{g})$ assemble, under disjoint union of labels and diagonal pullback, into a simplicial resolution of* $\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})$ *on* $\mathrm{Ran}(X)$*.*

Proof. Let $\iota_I : C_I^{\mathrm{coll}}(\mathfrak{g}) \to W_I^{\mathrm{FM}}(\mathfrak{g})$ insert the terminal corolla in every block. Then $\epsilon_I \iota_I = \mathrm{id}$. For a surjection of label sets $\phi : I \twoheadrightarrow J$, let T_ϕ be the canonical Ran transition on the partition collision complexes and define the transition on the resolution by

$$\widetilde{T}_\phi = \iota_J \, T_\phi \, \epsilon_I.$$

Because $\epsilon\iota = \mathrm{id}$, the maps $\widetilde{T}_\phi$ are strictly functorial under composition, and ϵ_I is a natural transformation. Disjoint union tensors the tree complexes and preserves terminal corollas, so these transitions are compatible with the factorization coproduct. They therefore assemble the labelled resolutions over the colimit presentation of $\mathrm{Ran}(X)$. $\square$

17.3 Primitives

Let C be a coaugmented conilpotent commutative coalgebra in the chiral monoidal category. The primitive functor is defined invariantly as the right adjoint to the chiral Chevalley functor. With the Chevalley convention of (120), its value is shifted by $[-1]$ to produce the corresponding chiral Lie object.

In a strict conilpotent dg-coalgebra model, let

$$\overline{C} = \mathrm{Cofib}(\mathbf{1} \longrightarrow C)$$

be the coaugmentation coideal and let $\overline{\Delta} : \overline{C} \to \overline{C} \otimes^{\mathrm{ch}} \overline{C}$ be the reduced coproduct. When strict primitives compute the derived right adjoint—in particular for cofree conilpotent models and their fibrant replacements—one may represent the primitive complex by

$$\mathrm{Prim}(C) \simeq \mathrm{Fib}\Big(\overline{C} \xrightarrow{\overline{\Delta}} \overline{C} \otimes^{\mathrm{ch}} \overline{C}\Big). \tag{121}$$

For a cofree conilpotent coalgebra this complex is the cogenerator complex.

The Chevalley and primitive constructions form an adjunction

$$\mathrm{coChev}^{\mathrm{ch}} : \mathsf{Lie}_{\mathrm{ch}}(X) \rightleftarrows \mathrm{Coalg}^{\mathrm{ch,conil}}_{\mathsf{Com}}(\mathrm{Ran}\, X) : \mathrm{Prim}[-1]. \tag{122}$$

The shift depends on the Chevalley convention. The substantive point is that the bracket and the reduced coproduct are Koszul-dual operations.

17.4 Francis–Gaitsgory duality

The chiral monoidal category on the Ran space is pro-nilpotent with respect to cardinality. Tensor products increase the cardinality filtration, so the bar and Chevalley series converge in the pro-nilpotent completion.

Theorem 17.2 (Francis–Gaitsgory chiral Koszul duality)**.** *In the factorization and conilpotent subcategories of the pro-nilpotent chiral monoidal category of D-modules on* $\mathrm{Ran}(X)$ *considered by Francis and Gaitsgory, the chiral Chevalley functor and the primitive functor are inverse equivalences between chiral Lie algebras and conilpotent factorization coalgebras:*

$$\mathsf{Lie}_{\mathrm{ch}}(X) \simeq \mathrm{CoFact}_{\mathrm{conil}}(\mathrm{Ran}\, X). \tag{123}$$

The unit and counit of the adjunction are equivalences.

This theorem is the invariant core of chiral Koszul duality [38]. It is formulated directly in the chiral monoidal category from the Chevalley and primitive adjunction. Quadratic presentations and continuous linear duals define subsequent refinements.

17.5 Chiral homology

The chiral homology of $\mathfrak{g}$ on a complete curve X is the de Rham cohomology of its factorization coalgebra:

$$\int_X^{\mathrm{ch}} \mathfrak{g} := R\Gamma_{\mathrm{dR}}(\mathrm{Ran}\, X, \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})). \tag{124}$$

The cardinality filtration produces a spectral sequence whose first page is a sum of cohomologies of configuration spaces with coefficients in external tensor powers of $\mathfrak{g}$. The collision differential is induced by the chiral bracket.

For an ordinary Lie algebra $\mathfrak{g}_0$ placed constantly on a point, this reduces to the usual Chevalley complex. For a current chiral algebra on a curve, it becomes a global version of Lie algebra homology with local singularities. For modules inserted at marked points, the pointed collision complex computes derived coinvariants.

17.6 The factorization envelope

Beilinson and Drinfeld associate to a chiral Lie algebra its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$. Its factorization structure is generated by the same local operation μ and has associative multiplication. The Chevalley construction uses the commutative coalgebra structure generated by the same primitive bracket. The Chevalley coalgebra and the enveloping factorization algebra play different roles:

$$\begin{aligned} \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) &\quad \text{resolves primitives and homology,} \\ U^{\mathrm{ch}}(\mathfrak{g}) &\quad \text{acts on modules and defines centers.} \end{aligned}$$

The Poincare–Birkhoff–Witt theorem identifies the associated graded of $U^{\mathrm{ch}}(\mathfrak{g})$ with the symmetric chiral algebra under the standard filtration. Vertex algebras arising from other presentations enter through their own associative factorization envelopes.

17.7 Analytic realization

Over $\mathbb{C}$, regular holonomic D-modules correspond under Riemann–Hilbert to constructible sheaves. The chiral operation becomes a morphism of constructible objects with singular support along diagonals. On the open configuration space, horizontal sections form a local system. The difference between the algebraic and analytic descriptions is then the difference between a regular singular connection and its monodromy representation.

This passage produces two complementary realizations. The Ran coalgebra records symmetric factorization. Pullback to labelled configurations followed by horizontal sections records paths and braid monodromy. De Rham cohomology yields the invariant cohomology class of the connection, while monodromy yields its representation of the fundamental group.

17.8 Completion and cardinality

The Ran category is pro-nilpotent by cardinality, but additional completions may enter from conformal weight, pole order, or infinite-dimensional state spaces. These are independent filtrations. If C_N is an inverse system of finite-window complexes with surjective transition maps on cohomology, then the Milnor sequence

$$0 \longrightarrow \varprojlim{}^1 H^{m-1}(C_N) \longrightarrow H^m(\varprojlim C_N) \longrightarrow \varprojlim H^m(C_N) \longrightarrow 0 \tag{125}$$

shows that completion is exact when the derived limit vanishes. This is the precise content of a Mittag–Leffler argument.

The cardinality pro-nilpotence built into Theorem 17.2 supplies its convergence. Explicit mode and conformal-weight models may carry additional complete filtrations, whose exactness is governed by (125).

17.9 The Beilinson hierarchy

The objects generated by one chiral operation form a hierarchy:

$$\mathfrak{g} \longmapsto \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \int_X^{\mathrm{ch}} \mathfrak{g}, \qquad \mathfrak{g} \longmapsto U^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \mathrm{Mod}(U^{\mathrm{ch}}(\mathfrak{g})). \tag{126}$$

The first branch is coalgebraic and homological. The second is algebraic and representation-theoretic. Centers, traces, and conformal blocks are built from the second branch; primitives and factorization homology are built from the first. Chiral Koszul duality identifies the primitive object with the primitives of its factorization coalgebra. The enveloping branch remains the associative representation-theoretic realization generated by the same chiral operation.

18 Chiral Koszul duality, bar–cobar, and presentations

18.1 Three meanings of Koszul duality

The phrase "Koszul duality" is used for three related constructions.

The first is the Lie–commutative equivalence of Theorem 17.2; this equivalence is chiral Koszul duality.

The second is bar–cobar reconstruction for augmented associative or operadic algebras. An augmented algebra A has a conilpotent bar coalgebra $B(A)$, and a conilpotent coalgebra C has a cobar algebra $\Omega(C)$. Under the usual conilpotence and completeness conditions, the unit and counit are equivalences.

The third is quadratic Koszul duality. A quadratic presentation $A = T(V)/(R)$ determines a smaller coalgebra $A^{¡}$ generated by sV with corelations s^2R. The algebra is Koszul when the canonical map $A^{¡} \to B(A)$ is a quasi-isomorphism.

The first construction is intrinsic to the chiral monoidal category. The second supplies a universal resolution of augmented objects. The third supplies a smaller presentation-dependent model. Explicit comparison morphisms determine the loci on which two of these constructions agree.

18.2 Twisting morphisms

Let C be a conilpotent dg coalgebra and A an augmented dg algebra in a complete symmetric monoidal category. A twisting morphism is a degree-one map $\tau : C \to A$ satisfying

$$d\tau + \mu_A(\tau \otimes \tau)\Delta_C = 0. \tag{127}$$

This is a Maurer–Cartan equation in the convolution algebra $\mathrm{Hom}(C, A)$. It determines a coalgebra map $C \to B(A)$ and an algebra map $\Omega(C) \to A$.

Theorem 18.1 (Fundamental theorem of twisting morphisms)**.** *Assume that the bar and cobar constructions are defined and that the coradical filtrations converge. For a twisting morphism $\tau : C \to A$, the following are equivalent:*

(i) *the map $C \to B(A)$ is a quasi-isomorphism;*

(ii) *the map* $\Omega(C) \to A$ *is a quasi-isomorphism;*

(iii) *the left twisted tensor product* $A \otimes_\tau C$ *is acyclic;*

(iv) *the right twisted tensor product* $C \otimes_\tau A$ *is acyclic.*

The theorem is classical over a field [55]. In a filtered chiral setting it applies when the monoidal products preserve the stated limits, the bar and cobar objects exist, and the comparison spectral sequences converge. The Maurer–Cartan equation (127) is formal; these ambient hypotheses make its associated resolutions homologically effective.

18.3 Quadratic recognition

Let $A = T(V)/(R)$ be a connected weight-graded quadratic algebra. Its quadratic coalgebra $A^{¡}$ comes with a canonical twisting morphism. The usual Koszul criterion is diagonal concentration of bar homology.

Theorem 18.2 (Quadratic Koszul criterion)**.** *For a connected weight-graded quadratic algebra A, the following are equivalent:*

(i) $A^{¡} \to B(A)$ *is a quasi-isomorphism;*

(ii) $\Omega(A^{¡}) \to A$ *is a quasi-isomorphism;*

(iii) $\mathrm{Tor}^A_{p,q}(\mathbb{C}, \mathbb{C}) = 0$ *for* $p \neq q$*;*

(iv) *the quadratic twisted tensor complexes are acyclic.*

For a chiral algebra presented by generators and relations, the same criterion applies in each finite window once the free chiral algebra, the quadratic relation object, and the filtered comparison have been defined. Every family theorem therefore consists of a concrete presentation, an explicit comparison map, and a convergence proof.

18.4 PBW deformations

A PBW theorem identifies the associated graded of a filtered algebra with a prescribed graded algebra. Linear and constant terms determine additional components of the Koszul coderivation and may determine a curvature term. The filtered comparison is governed by the following criterion.

Criterion 18.3 (Filtered Koszul comparison)**.** *Let $q : C \to B(A)$ be a morphism from a filtered conilpotent coalgebra to the bar coalgebra of a filtered augmented algebra. Assume that*

(a) *the filtrations on C and $B(A)$ are exhaustive, separated, complete, and preserved by q;*

(b) *the induced map* $\mathrm{gr}\, q : \mathrm{gr}\, C \to B(\mathrm{gr}\, A)$ *is a quasi-isomorphism;*

(c) *the two comparison spectral sequences converge strongly.*

Then q is a quasi-isomorphism.

Proof. The cone of q inherits a complete exhaustive filtration. Its associated graded is the cone of $\mathrm{gr}\, q$ and is acyclic by hypothesis. Strong convergence therefore gives zero cohomology for the filtered cone, so q is a quasi-isomorphism. □

For a quadratic-linear-constant PBW deformation of a quadratic Koszul algebra, the appropriate C is the nonhomogeneous or curved Koszul coalgebra. The PBW compatibility identities make its coderivation square to the prescribed curvature, and the associated-graded comparison is the ordinary quadratic Koszul resolution. PBW, classical Koszulness, and these curved compatibility identities therefore imply the filtered comparison.

For a vertex algebra, the family theorem also controls derivatives, infinitely many modes, and completion in conformal weight or pole order. A PBW spanning set supplies the associated-graded input; the comparison cone and convergence argument establish Koszulness.

18.5 Verdier duality and a dual algebra

The chiral Chevalley coalgebra is formed by the Chevalley functor. When its labelled components are Verdier dualizable, Verdier duality subsequently exchanges coalgebra maps with algebra maps. Let $\mathbb{D}_{\mathrm{Ran}}$ denote Verdier duality on an appropriate constructible or holonomic subcategory.

Proposition 18.4 (Verdier-dual algebra)**.** *Let C be a factorization coalgebra whose finite labelled components are Verdier dualizable, and suppose Verdier duality commutes with the factorization restriction maps. Then*

$$C^{\vee} := \mathbb{D}_{\mathrm{Ran}} C$$

is a factorization algebra. If biduality holds componentwise and is compatible with factorization, then $C \to \mathbb{D}_{\mathrm{Ran}}\mathbb{D}_{\mathrm{Ran}} C$ is an equivalence of factorization coalgebras.

Applied to $C = \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})$, this produces a Verdier-dual factorization algebra. A recognition theorem can then identify the Verdier-dual factorization algebra with a familiar vertex algebra or reflected-level presentation. Quantum-group recognition proceeds from the braided module category and its fibre functor.

18.6 Completion

Let $F^1 A$ be the augmentation ideal. The completed bar and cobar constructions use inverse limits of finite tensor-length quotients, equivalently products when the ambient category admits the required products. Their differentials converge when they raise the filtration. The completed counit

$$\widehat{\Omega}\widehat{B}(A) \longrightarrow A$$

is an equivalence for complete augmented algebras in the standard pronilpotent setting.

For a curved algebra with curvature element m_0, the unary operation satisfies $m_1^2 = [m_0, -]$. Central scalar curvature acts through a projective connection, while inner algebraic curvature acts through this commutator formula. Positselski's coderived and contraderived categories provide the homological ambient for curved bar–cobar constructions [60]. An ordinary vertex algebra retains its square-zero chiral deformation differential.

18.7 Koszul dual modules

A twisting morphism $\tau : C \to A$ defines adjoint functors between A-modules and C-comodules. Under the hypotheses of Theorem 18.1, these functors give an equivalence on the appropriate derived categories. For a quadratic Koszul algebra, the resolution is linear and yields the familiar exchange of Ext and Tor.

In the chiral setting, modules are factorization modules at marked points. The pointed collision complex is the bar resolution. A module-level Koszul duality therefore requires the algebra comparison and the compatibility of marked-point factorization. Fusion products and conformal blocks are preserved only when a braided tensor equivalence of module categories has been established.

18.8 Maurer–Cartan moduli

Twisting morphisms form a Maurer–Cartan space in a convolution dg Lie or L_∞-algebra. Gauge equivalence is homotopy of twisting data. The tangent complex at τ is the twisted convolution complex

$$(\mathrm{Hom}(C, A), d + [\tau, -])\,.$$

Its first cohomology classifies infinitesimal twisting deformations modulo gauge, and its second contains the extension classes that govern the next order.

This Maurer–Cartan space is the natural moduli object associated with Koszul presentations. In a finite perfect family, the acyclic locus of the comparison cone is open. Globally, the derived zero locus of that cone records Koszulness together with its deformation theory. The bar coalgebra itself remains the complete structured invariant.

18.9 Recognition data for neighboring structures

Five neighboring conclusions require five corresponding data packages:

$$\begin{array}{ccl} \text{PBW presentation} & + & \text{Koszul comparison and convergence,} \\ C_2\text{-cofiniteness} & + & \text{a bar-homology calculation,} \\ \text{Verdier dualizability} & + & \text{a coherent biduality identification,} \\ \text{braided monodromy} & + & \text{Tannaka and representation-theoretic recognition,} \\ \text{equal central charges} & + & \text{an equivalence of complete chiral operations.} \end{array} \tag{128}$$

These packages promote a coarse or ambient property to the corresponding structured conclusion.

19 Duality, Hochschild theory, and the derived center

19.1 The associative envelope and its center

A chiral Lie algebra $\mathfrak{g}$ carries a diagonal bracket. Its chiral enveloping factorization algebra $U^{\mathrm{ch}}(\mathfrak{g})$ carries associative multiplication and a bimodule category. The derived center is formed from this associative envelope, or directly from any associative factorization algebra A.

Definition 19.1 (Derived chiral center)**.** For a chiral Lie algebra $\mathfrak{g}$ for which the chiral enveloping factorization algebra is defined, set

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathfrak{g}) := \mathrm{HH}^*_{\mathrm{Fact}}\left(U^{\mathrm{ch}}(\mathfrak{g}), U^{\mathrm{ch}}(\mathfrak{g})\right). \tag{129}$$

For an associative factorization algebra A, write $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{HH}^*_{\mathrm{Fact}}(A, A)$.

Hochschild cochains are derived endomorphisms of the diagonal bimodule. The bar resolution supplies a concrete model. In the chiral setting the resolution is supported on configurations with an incoming and an outgoing boundary component, and insertion of configurations produces brace operations.

19.2 The E_2 structure

The higher Deligne theorem gives Hochschild cochains of an E_1 algebra an E_2 structure. The same statement holds for associative factorization algebras in a suitable symmetric monoidal category. The product is cup product; the shifted Lie bracket is the Gerstenhaber bracket; higher braces encode homotopies.

Theorem 19.2 (Derived center structure)**.** *Let A be an associative factorization algebra whose bimodule category admits the diagonal bar resolution. Then $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ is an E_2 algebra. Its action on A is universal among homotopy-coherent central actions.*

The universal property is algebraic. An object B acting on A through the Swiss-cheese operad determines a morphism of E_2 algebras

$$B \longrightarrow Z^{\mathrm{der}}_{\mathrm{ch}}(A). \tag{130}$$

The map sends a closed operation to the family of open operations obtained by inserting it next to boundary inputs.

19.3 The center and a physical bulk

Let $\mathrm{Obs}_{\mathrm{bulk}}$ be the factorization algebra of local observables of a field theory with boundary algebra A. The bulk-to-boundary operator product defines a map

$$\mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z^{\mathrm{der}}_{\mathrm{ch}}(A). \tag{131}$$

A quasi-isomorphism in (131) reconstructs the bulk local observables from the boundary algebra. The homotopy fibre measures bulk observables whose boundary action is null-homotopic.

Proposition 19.3 (Kernel of the boundary action)**.** *Let $\mathcal{T}$ be a bulk theory with boundary algebra A. Let $\mathcal{D}$ carry a boundary condition whose boundary observables are the tensor unit, and suppose its bulk-to-boundary action is null-homotopic. Equip $\mathcal{T} \otimes \mathcal{D}$ with the product boundary condition. Its boundary algebra is canonically A, its derived center is $Z_{\mathrm{ord,ch}}(A)$, and the bulk-to-center morphism is the sum of the morphism for $\mathcal{T}$ and the null map from $\mathrm{Obs}_{\mathcal{D}}$. Hence the homotopy fibre contains $\mathrm{Obs}_{\mathcal{D}}$ as the direct product factor supplied by the product theory.*

The proposition identifies the constructive input for local holographic reconstruction: the bulk-to-boundary action is required to generate the bulk local-observable complex.

19.4 The annulus and Hochschild homology

Factorization homology of the circle with coefficients in an E_1 algebra is Hochschild homology:

$$\int_{S^1} A \simeq A \otimes^{L}_{A\otimes A^{\mathrm{op}}} A. \tag{132}$$

For a boundary theory, the annulus is obtained by thickening the circle. Its state space is therefore a trace of the boundary algebra. The pairing between Hochschild cochains and Hochschild chains is the algebraic form of inserting a bulk local operator into an annulus amplitude.

A cyclic trace on A thickens planar trees to ribbon graphs and contracts pairs of flags. It supplies the loop operation that raises ribbon genus. The underlying derived center retains its E_2 structure independently of this extra cyclic datum.

19.5 Morita invariance

Hochschild cochains and chains are Morita invariant. If A and B are derived Morita equivalent associative factorization algebras, then

$$\mathrm{HH}^*_{\mathrm{Fact}}(A, A) \simeq \mathrm{HH}^*_{\mathrm{Fact}}(B, B)$$

as E_2 algebras, and their Hochschild homologies agree.

A Koszul equivalence implemented by an invertible bimodule is a derived Morita equivalence and therefore identifies centers. A coalgebraic bar–cobar comparison first identifies the corresponding augmentation-generated module/comodule sectors; an additional Morita generator promotes this comparison to the full center equivalence.

19.6 A small convolution model

Assume A is augmented, complete, and Koszul with a perfect bar coalgebra $B(A)$. The diagonal bar resolution identifies Hochschild cochains with a twisted convolution complex

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A) \simeq \mathrm{Hom}^{\tau}(B(A), A)\,, \tag{133}$$

where the differential is

$$df = d_A f - (-1)^{|f|} f d_B + [\tau, f].$$

The braces insert one coalgebra input into another before applying f. This model is finite and explicit when the bar coalgebra is locally finite.

The formula (133) expresses associative deformation theory inside the center. A first-order deformation is a degree-two Hochschild cocycle, and its Gerstenhaber square is the degree-three extension class for the second order. The chiral deformation complex of Equation (119) deforms the diagonal chiral operation; the factorization Hochschild complex deforms its associative envelope. A morphism of deformation complexes compares the two problems.

19.7 Verdier duality and centers

Verdier duality exchanges !- and *-extensions across diagonals. For a perfect factorization algebra A, duality produces an opposite or dual factorization algebra. The induced map on bimodule categories reverses left and right actions. Consequently the Hochschild complex of the dual is the dual of the Hochschild chain complex of A under a Calabi–Yau pairing.

For a smooth proper associative factorization algebra, a degree-d Calabi–Yau structure gives a nondegenerate pairing

$$\mathrm{HH}^i(A, A) \otimes \mathrm{HH}_{d-i}(A) \longrightarrow \mathbb{C}.$$

The cyclic structure fixes the degree d. The corresponding Connes operator and Poincar'e duality transfer give the BV operator on Hochschild cohomology.

19.8 Topological enhancement

The derived centre is E_2 in the transverse directions. A coherent null-homotopy of one real translation in the complex curve adds one locally constant direction and gives an E_3 enhancement. Coherent null-homotopies for both real translations of the curve give an E_4 algebra by Dunn additivity, as in Proposition 9.8. The required null-homotopies are the precise topologization data.

In three-dimensional Chern–Simons theory, local bulk observables are E_3 because the bulk has three topological directions. Comparison with a boundary centre therefore uses the appropriate intermediate topologization of the boundary object. In a four-real-dimensional holomorphic–topological theory whose holomorphic direction is also topologized, the corresponding local structure is E_4. In either dimension, proving that the bulk-to-centre map is an equivalence is a field-theoretic calculation.

19.9 Centers at critical level

For the universal affine vertex algebra at the critical level, the Feigin–Frenkel theorem identifies the ordinary vertex-algebra center with functions on opers for the Langlands dual group on the formal disc [34]. This commutative chiral subalgebra acts on the affine algebra and hence maps into its factorization Hochschild cochains.

The two centers carry different gradings and universal properties. The vertex center consists of fields commuting with every field under nonnegative modes. The derived center contains the full self-extension complex of the diagonal bimodule. At noncritical level the vertex center can be scalar while higher Hochschild classes remain present.

Part III

Exchange, reduction, and quantum symmetry

20 Labelled chiral configurations and monodromy

20.1 Two geometric sources of order

An E_1 multiplication is represented by chambers of configurations on a real line. Monodromy is represented by paths in a configuration space of real dimension at least two. In the present theory those paths arise from moving labelled chiral insertion points on the complex curve or from adding a second transverse topological direction. The interval gives ordered fusion; the plane gives exchange.

20.2 Labels retain paths

The labelled configuration space of n points is

$$\mathrm{Conf}_n(X) = \{(x_1, \dots, x_n) \in X^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

The symmetric group acts freely, and the quotient is

$$\mathrm{UConf}_n(X) = \mathrm{Conf}_n(X)/\Sigma_n.$$

For $X = \mathbb{C}$,

$$\pi_1(\mathrm{Conf}_n(\mathbb{C})) = P_n, \qquad \pi_1(\mathrm{UConf}_n(\mathbb{C})) = B_n,$$

the pure braid and braid groups. A local system on the labelled space carries a representation of P_n. A Σ_n-equivariant local system descends to the quotient and carries the corresponding representation of B_n.

A symmetric factorization coalgebra consists of labelled components together with coherent descent under relabellings and diagonal maps. Horizontal sections on a labelled component retain the full fundamental-groupoid representation. This path information is the analytic datum from which braid operators are formed.

20.3 Derived descent

Let M be a chain complex with an action of a finite group G. Its homotopy coinvariants are

$$M_{hG} = M \otimes^L_{\mathbb{C}[G]} \mathbb{C}.$$

The augmentation ideal $I_G = \ker(\mathbb{C}[G] \to \mathbb{C})$ gives a distinguished triangle

$$M \otimes^L_{\mathbb{C}[G]} I_G \longrightarrow M \longrightarrow M_{hG}. \tag{134}$$

The first term is the canonical homotopy fibre of descent. In characteristic zero, semisimplicity of $\mathbb{C}[G]$ identifies derived coinvariants with ordinary coinvariants and identifies the fibre with the direct sum of the nontrivial isotypic components.

For configuration-space local systems, equivariant descent is governed by the extension

$$1 \longrightarrow P_n \longrightarrow B_n \longrightarrow \Sigma_n \longrightarrow 1.$$

The P_n representation records analytic continuation on the labelled cover, and the Σ_n equivariance identifies relabelled fibres. Together they define the B_n representation on the quotient.

20.4 The descent fibre and collision filtrations

For a coefficient system M_n carrying a Σ_n action, define

$$\mathcal{C}_n(M) = \mathrm{Fib}(M_n \to M_{n,h\Sigma_n}). \tag{135}$$

Equation (134) gives

$$\mathcal{C}_n(M) \simeq M_n \otimes^L_{\mathbb{C}[\Sigma_n]} I_{\Sigma_n}.$$

The group bar resolution supplies a universal chain model. In characteristic zero this model contracts to the nontrivial isotypic summands.

The proper partition complex supplies a second universal object: it resolves collision incidence and carries the multilinear Lie representation in top homology. A coefficient system equipped with a functorial collision filtration determines a comparison morphism from the partition-incidence resolution to the group-bar model. Its successive layers represent binary collision data, ternary coherence, and higher collision strata.

The first-order holonomy of a flat connection may define an r-matrix. This occurs when its pairwise residues satisfy the infinitesimal braid relations; the classical Yang–Baxter equation is then the coefficient form of flatness. The descent fibre retains precisely the isotypic and monodromy data supplied by the coefficient system.

20.5 Dual local systems

Let M be an oriented manifold and $\mathcal{L}$ a finite-rank local system with monodromy $\rho : \pi_1(M) \to GL(V)$. Verdier duality gives

$$\mathbb{D}\mathcal{L} \cong \mathcal{L}^\vee[\dim M]$$

up to the orientation line. The dual monodromy is

$$\rho^\vee(\gamma) = \rho(\gamma^{-1})^{\mathsf{t}}. \tag{136}$$

Parallel transport on the dual is contragredient, which supplies the inverse and transpose.

Corollary 20.1. *Verdier duality sends a braid operator R to $R^{-\mathsf{t}}$. A nondegenerate invariant pairing identifying a representation with its dual turns this operator into inverse braiding.*

For a quantum-group family parametrized by q, a recognition isomorphism may identify this inverse braiding with the parameter transformation $q \mapsto q^{-1}$.

20.6 Associators

The configuration space of three points has asymptotic zones corresponding to different parenthesizations. Regularized parallel transport between these zones gives an associator. A Drinfeld associator is a group-like formal series satisfying the pentagon and hexagon equations. After the standard normalization, the solution space is a torsor under GRT_1 [30, 77].

The pentagon is the arity-four coherence equation, and the hexagon couples the associator to the braiding. These equations define the associator torsor inside the space of formal series. In the de Rham realization, the KZ associator is the regularized transport between tangential base points at 0 and 1 on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its coefficients are multiple zeta values. Thus configuration-space transport carries arithmetic together with coherence.

20.7 Tannaka reconstruction

Let $\mathcal{C}$ be a rigid braided tensor category and let $F : \mathcal{C} \to \mathrm{Vect}$ be a faithful exact fibre functor. The coend

$$H = \int^{M\in\mathcal{C}} F(M)^\vee \otimes F(M) \tag{137}$$

reconstructs the coalgebra of matrix coefficients. A strong monoidal structure on F supplies multiplication, and rigidity supplies an antipode whenever the finite or continuous dual required by the coend exists. A quasi-monoidal structure supplies a coassociator. The braiding supplies a coquasitriangular form.

Theorem 20.2 (Ordered Tannaka principle)**.** *A rigid tensor-compatible monodromy category with a faithful exact strong monoidal fibre functor reconstructs a coquasitriangular bialgebra and, in the rigid finite or continuous-dual class, a Hopf algebra. A faithful exact quasi-monoidal fibre functor reconstructs a coquasitriangular coquasi-bialgebra. A twist trivializing the coassociator produces a strict Hopf presentation.*

A representation-theoretic recognition theorem identifies the reconstructed object with $U_q(\mathfrak{g})$, a Yangian, an elliptic quantum group, a Hall algebra, or another named quantum symmetry.

20.8 Surface braid groups

For a curve of genus $g > 0$, $\pi_1(\mathrm{Conf}_n(X))$ is a surface pure braid group. Its generators include pairwise collision loops and transport around the A- and B-cycles of the curve. Flat connections such as the KZB connection represent these generators by operators satisfying elliptic braid relations.

Elliptic transport also carries quasi-periodicity and a dynamical parameter. Mapping-class-group transport acts on conformal blocks as the complex structure varies. At higher genus the fundamental group of the moduli of punctured curves supplies the corresponding mapping-class-group representation.

20.9 The invariant and covariant outputs of descent

Symmetric descent preserves invariant traces, invariant vectors, dimensions of invariant subspaces, and every operation that factors through homotopy coinvariants. Periods in the invariant summand remain visible. The fibre (135) records the complementary non-invariant data.

For conformal blocks, dimension gives the Verlinde number, while the vector bundle with flat projective connection carries the mapping-class-group representation. For a braided category, the Grothendieck ring records fusion coefficients, while the associator, braiding, and twists retain the tensor coherence. These pairs are two realizations of invariant descent and its covariant refinement.

21 KZ geometry, quantum groups, Gaudin systems, and lines

21.1 The logarithmic connection

Let $\mathfrak{g}$ be a finite-dimensional complex simple Lie algebra with invariant tensor $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$. On $V_1 \otimes \cdots \otimes V_n$, define

$$\nabla_{\mathrm{KZ}} = d - \hbar \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j), \qquad \hbar = \frac{1}{k + h^\vee}. \tag{138}$$

The Sugawara construction fixes the shifted level $k + h^\vee$ once the invariant form has been normalized.

The invariant tensor satisfies the infinitesimal braid relations

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{for disjoint pairs}, \qquad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0 \tag{139}$$

for distinct i, j, k.

Theorem 21.1 (Flatness of the KZ connection)**.** *The connection* (138) *is flat.*

Proof. Write $\nabla = d - A$. The logarithmic one-forms are closed, so the curvature is $A \wedge A$. Terms supported on disjoint index pairs vanish by the first relation in (139). For each triple i, j, k, the remaining coefficient is a linear combination of

$$d\log z_{ij} \wedge d\log z_{jk}, \qquad d\log z_{jk} \wedge d\log z_{ki}, \qquad d\log z_{ki} \wedge d\log z_{ij}.$$

The Arnold relation makes their cyclic sum zero, and the second infinitesimal braid relation makes the corresponding operator coefficients equal in the required combination. Hence the curvature vanishes. □

The proof separates two exact inputs. The Arnold identity belongs to the configuration-space forms. The infinitesimal braid identities belong to the Casimir operators. Their tensor product is the flatness equation.

21.2 From affine currents to KZ

For the affine vertex algebra $V_k(\mathfrak{g})$, current fields satisfy

$$J^a(z)J^b(w) \sim \frac{k\,(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \tag{140}$$

Ward identities for primary insertions turn the simple-pole current action into the differential operator in (138). The double pole enters the Sugawara stress tensor, whose normal ordering contributes the shift $h^\vee$. Thus the complete affine chiral operation produces the representation-theoretic KZ connection on conformal blocks.

21.3 Drinfeld–Kohno

Set $q = \exp(\pi i \hbar)$ in the standard convention. The KZ associator is the regularized transport between tangential asymptotic zones. The Drinfeld–Kohno theorem compares this braided tensor structure with the universal R-matrix and associator of the Drinfeld–Jimbo quantum group [51, 30, 50].

Theorem 21.2 (Drinfeld–Kohno)**.** *For generic $\hbar$, the braid-group representations obtained from KZ monodromy are naturally equivalent, as braided tensor data, to those obtained from finite-dimensional $U_q(\mathfrak{g})$-modules with $q = e^{\pi i \hbar}$.*

The theorem supplies the quantum-group recognition of ordered affine monodromy. The reconstructed object includes the associator, braiding, tensor product, and representation category.

21.4 The classical r-matrix

The coefficient of the KZ one-form is

$$r_{ij}(z_i - z_j) = \hbar \frac{\Omega_{ij}}{z_i - z_j}. \tag{141}$$

The first nontrivial coefficient of the flatness equation is the classical Yang–Baxter equation. The monodromy is the path-ordered exponential of the connection, and its semiclassical expansion begins with (141).

The same rational kernel appears in the KZ connection, the rational Gaudin model, and the line-operator OPE of four-dimensional Chern–Simons theory after their coefficient spaces and couplings have been identified. The comparison maps in those identifications are part of the corresponding recognition theorems.

21.5 Gaudin Hamiltonians

Fix distinct points $z_1, \dots, z_n$. The Gaudin Hamiltonians are

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}. \tag{142}$$

The infinitesimal braid relations imply $[H_i, H_j] = 0$. These operators are the residues of the KZ connection in the variables z_i, so their commutativity is the residue form of flatness.

For highest-weight modules, the Bethe ansatz constructs simultaneous eigenvectors from critical points of a master function. In the $\mathfrak{sl}_2$ case, Bethe roots t_a satisfy

$$\sum_{i=1}^{n} \frac{\lambda_i}{t_a - z_i} - \sum_{b \neq a} \frac{2}{t_a - t_b} = 0. \tag{143}$$

Schechtman–Varchenko hypergeometric integrals construct horizontal KZ sections from the same master function. KZ transport and the Gaudin spectrum are therefore two functorial outputs of one flat connection.

21.6 Four-dimensional Chern–Simons theory

Costello–Witten–Yamazaki construct a gauge theory that is topological in two real directions and holomorphic along a complex spectral curve. Wilson lines at distinct holomorphic positions form a meromorphic tensor category. Their OPE carries a spectral-parameter R-matrix, and isotopy of three lines gives the Yang–Baxter equation [19, 20].

The difference of holomorphic line positions is the spectral parameter. The one-loop graph fixes the first quantum correction. Rational, trigonometric, and elliptic choices of spectral curve and differential produce the corresponding Yangian, quantum-loop, and elliptic line algebras on the anomaly-free perturbative sector. This field-theoretic construction supplies a second geometric realization of ordered monodromy.

21.7 Yangians and recognition

A Yangian $Y_\hbar(\mathfrak{g})$ is a filtered deformation of $U(\mathfrak{g}[t])$ equipped with rational difference R-matrices on its evaluation representations. The following criterion isolates the data required to recognize it.

Criterion 21.3 (Yangian recognition). *Let H be a filtered quasi-bialgebra reconstructed from an ordered tensor category. Suppose*

(i) $\operatorname{gr} H \cong U(\mathfrak{g}[t])$ *as a bialgebra;*

(ii) *a specified family of fundamental evaluation modules is faithful on the chosen completion;*

(iii) *their intertwiners agree with the rational Yangian R-matrices;*

(iv) *the RTT or Drinfeld relations generate the kernel of the resulting representation map.*

Then H is isomorphic, up to the associator twist of the fibre functor, to the corresponding completed Yangian.

The criterion turns a rational kernel into an algebra identification by combining PBW flatness, faithful representations, and a presentation theorem.

21.8 Elliptic transport

On an elliptic curve, the KZB connection replaces $d\log(z_i - z_j)$ by elliptic kernels with prescribed quasi-periodicity. Fay identities and dynamical correction terms give its flatness. Its monodromy represents elliptic braid transport and the modular group. Tannaka reconstruction then produces elliptic or dynamical quantum symmetry in the representation category selected by the connection.

Rational, trigonometric, and elliptic quantum groups arise from the corresponding rational, cylindrical, and elliptic base geometries. Degeneration of the curve and compatible degeneration of the coefficient category provide the comparison functors among them.

21.9 Knots and ribbon traces

A ribbon category assigns invariants to framed links by composing braidings, evaluation and coevaluation maps, and twists. The Reshetikhin–Turaev construction applied to $U_q(\mathfrak{g})$ produces quantum link invariants [63]. Chern–Simons theory supplies their path-integral realization, and KZ transport supplies the equivalent monodromy realization.

Closing a braid applies the categorical trace. The result is a scalar link invariant. Before closure, the braid representation retains the full ordered operator. This passage from operator to trace is the knot-theoretic instance of invariant descent.

21.10 Duality of monodromy

Equation (136) sends KZ monodromy to its inverse transpose. The invariant forms on representations identify this transformation with inverse braiding in the dual braided category. For the affine KZ family, the level reflection

$$k \longmapsto -k - 2h^\vee$$

sends $\hbar = (k + h^\vee)^{-1}$ to $-\hbar$ and therefore sends q to q^{-1}. The Drinfeld–Kohno equivalence carries this transformation to the corresponding dual quantum-group representation.

22 Quantum line defects and meromorphic tensor categories

22.1 Fusion and exchange

Let $A \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$. Modules placed consecutively on the transverse line fuse by relative tensor product:

$$M \star_A N := M \otimes^L_A N.$$

Excision for three consecutive intervals supplies associativity. A second transverse direction adds paths that exchange two intervals and upgrades ordered fusion to braided fusion.

Theorem 22.1 (Exchange-data theorem)**.** *The forgetful functor*

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$$

sends a two-dimensional exchange structure to its one-dimensional ordered multiplication. An E_2 lift supplies coherent exchange operators, the braid relation, and a graded-commutative product on homology. Consequently a discrete E_1 algebra admits such a lift only when its degree-zero multiplication is commutative and the higher exchange coherences have been specified.

Proof. The little two-disks operad contains a path interchanging two disks; its induced operation gives the homotopy commutativity of an E_2 multiplication. The homology operad is the Gerstenhaber operad, whose product is graded commutative. Hence the degree-zero product of a discrete E_2 algebra is commutative. Geometrically, the ordered configuration space of two intervals has fixed-order chambers, while a configuration in the plane contains an exchange path. The chosen E_2 lift records that path and all its higher coherences. □

Fusion is therefore the E_1 datum. Exchange is supplied by an E_2 enhancement, by a meromorphic flat connection, or by the categorical center of the line category.

22.2 The meromorphic line category

Let $\mathcal{L}$ be a category of line defects parametrized by the spectral curve X. For distinct spectral points, factorization defines fusion. An exchange structure is a horizontal family of isomorphisms

$$R_{L,M}(x, y) : L_x \star M_y \longrightarrow M_y \star L_x, \qquad x \neq y,$$

satisfying the braid relation on $\mathrm{Conf}_3(X)$.

Definition 22.2 (Meromorphic tensor category)**.** A meromorphic tensor category on X consists of a factorization category on labelled configurations, an associative fusion product on ordered disjoint blocks, and braid descent on the locus of distinct points. The descent maps extend meromorphically toward collision divisors and satisfy the pentagon and hexagon identities.

The associator compares asymptotic zones for four insertions. The braiding transports two insertions around one another. The polar part of the exchange operator records the chiral singularity, while its homotopy class records the transverse topology.

Theorem 22.3 (Line-category recognition)**.** *Let $\mathcal{T}$ be a renormalized quantum field theory on $X \times \mathbb{R}^2$, holomorphic on X and topological on $\mathbb{R}^2$. Suppose a class of line defects satisfies the following properties: anomaly-free fusion and exchange Ward identities, perfect morphism complexes over the meromorphic functions on X, meromorphic extension of the line OPE across collision divisors, and closure under fusion and duality. Then these defects form a rigid meromorphic tensor category.*

Proof. Factorization in one transverse direction gives fusion, and isotopy of three parallel lines gives associativity. A path in the configuration space of two points in the transverse plane gives exchange. Isotopy of three exchange paths gives the braid relation. The four-line configuration spaces give the pentagon and hexagon identities. Holomorphic dependence on spectral position and locality of counterterms give meromorphic extension along diagonals. Reversal of line orientation gives categorical duals under the perfectness hypothesis. □

22.3 Centers and categorical exchange

The internal Hochschild center $Z_{\mathrm{ord,ch}}(A)$ is an E_2 algebra in $\mathrm{FactD}(X)$. Its multiplication therefore carries little-two-disks exchange. Its module category is E_1-monoidal. A braided line category arises from the Drinfeld center of that monoidal category or from modules over an E_3 enhancement.

Corollary 22.4 (Exchange through a complete center)**.** *Suppose the bulk-to-boundary map*

$$\beta_{\mathcal{T}} : \mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z_{\mathrm{ord,ch}}(A)$$

is an equivalence of local E_2 algebras. Suppose a monoidal-affineness theorem identifies the selected bulk line category with $Z(\mathrm{Bimod}_A)$ and identifies this categorical center with modules over an E_3 topological enhancement of $Z_{\mathrm{ord,ch}}(A)$. Then the corresponding perfect line categories are equivalent as braided monoidal categories.

The local-observable equivalence and the monoidal-affineness theorem play complementary roles: the first identifies bulk local operations, and the second identifies extended line defects.

22.4 Tannaka reconstruction

Choose a faithful exact meromorphic fibre functor

$$F : \mathcal{L} \longrightarrow \mathrm{Vect}_K, \qquad K = \mathbb{C}(X).$$

The coend

$$H_{\mathcal{L}} = \int^{L \in \mathcal{L}} F(L)^{\vee} \otimes_K F(L)$$

represents matrix coefficients of tensor automorphisms of F. Fusion, associator, and braiding give multiplication, coassociator, and the universal R-form.

Theorem 22.5 (Quantum symmetry of line defects)**.** *A strong monoidal fibre functor makes $H_{\mathcal{L}}$ a coquasitriangular bialgebra; rigidity supplies a Hopf antipode in the finite or continuous-dual class. A quasi-monoidal fibre functor makes $H_{\mathcal{L}}$ a coquasitriangular coquasi-bialgebra whose coassociator is the transported tensor constraint. Whenever the relevant dual exists, the dual presentations are quasitriangular Hopf or quasi-Hopf algebras. Braided tensor equivalences compatible with the fibre functor preserve the reconstructed object.*

This theorem reconstructs quantum symmetry from the line category. PBW and presentation theorems identify the resulting object with a Yangian, quantum-loop algebra, elliptic quantum group, Hall algebra, or another named algebra.

22.5 Rational gauge theory and the Yangian

In four-dimensional Chern–Simons theory on $\Sigma \times X$, the theory is topological on Σ and holomorphic on X. Parallel Wilson lines form a meromorphic tensor category. For the rational curve $X = \mathbb{C}$, perturbation theory begins with

$$R_{V,W}(z) = 1 + \hbar \frac{\Omega_{V,W}}{z} + O(\hbar^2). \tag{144}$$

Cancellation of the line anomaly deforms the classical current algebra to the Yangian, and the Wilson-line OPE realizes its RTT or Drinfeld coproduct.

Theorem 22.6 (Rational Wilson-line algebra)**.** *For the anomaly-free perturbative Wilson-line categories constructed in rational four-dimensional Chern–Simons theory by Costello–Witten–Yamazaki, the completed algebra of line operators is the $\hbar$-adic Yangian deformation of $U(\mathfrak{g}[z])$ on the stated representation class, and exchange is governed by the rational universal R-matrix. The spectral parameter is the difference of holomorphic line positions [19, 20].*

The gauge-theory Ward identities, anomaly calculation, and PBW identification provide the field-theoretic recognition data.

22.6 Trigonometric and elliptic geometries

The holomorphic spectral curve selects the meromorphic kernel. The cylinder $\mathbb{C}^\times$ supplies trigonometric periodicity and quantum-loop symmetry. An elliptic curve supplies quasi-periodicity, modular transport, and the dynamical variables of elliptic quantum groups.

Principle 22.7 (The spectral curve selects the quantum group)**.** *For the standard rational, trigonometric, and elliptic meromorphic line categories, Tannaka reconstruction followed by presentation recognition gives rational Yangian, quantum-loop, and elliptic or dynamical quantum symmetry, respectively. A flat family of spectral curves, line categories, fibre functors, and meromorphic connections induces the corresponding degeneration functor among these symmetries.*

22.7 Shifted Yangians and boundary conditions

A line ending on a boundary or meeting an 't Hooft defect carries magnetic coweight data. The coweight changes the allowed pole orders of Drinfeld currents and produces a shifted Yangian. Geometrically, the shift is a modification of the factorization module at the marked transverse boundary condition.

For a coweight μ, the pointed module bar is

$$\mathrm{Bar}_X^{\mathrm{ord}}(M_-, A, M_+) = M_- \otimes_A^L M_+,$$

where the endpoint modules encode the boundary charges. Choosing both endpoints to be the augmentation boundary gives the unshifted sector.

Criterion 22.8 (Shifted recognition)**.** *A line algebra is the shifted Yangian of coweight μ when the coweight filtration of currents, shifted coproduct, PBW associated graded, fundamental line modules, and rational exchange kernel agree with the standard shifted presentation. These data determine the shift together with the algebra.*

22.8 KZ and gauge-theory exchange

Affine conformal blocks carry the KZ connection with coupling $(k + h^\vee)^{-1}$. Four-dimensional Chern–Simons lines carry a rational exchange connection with gauge coupling $\hbar$. A braided tensor equivalence matching generators and Casimir normalization identifies the two constructions.

Theorem 22.9 (Comparison through the line category)**.** *Let $\mathcal{L}_{\mathrm{KZ}}$ be a rigid braided tensor subcategory of the Drinfeld–KZ category at a nonresonant shifted level, and let $\mathcal{L}_{\mathrm{gauge}}$ be a rigid Wilson-line* *subcategory of a rational gauge theory. A braided tensor equivalence matching tensor generators and the first-order Casimir kernel identifies their braid monodromies. Compatible Tannaka fibre functors reconstruct the same quasitriangular Hopf or quasi-Hopf object up to the transported associator twist.*

Proof. A tensor equivalence preserves the coend reconstruction. Matching the Casimir coefficient fixes the formal deformation parameter. Agreement of the braid operator on tensor generators extends by rigidity, tensor products, and direct summands to the generated category. The transported associator records the relation between strict and KZ tensor structures. □

Thus the tensor equivalence is the comparison datum that turns the common rational kernel into an identity of quantum symmetries.

22.9 Koszul duality of line couplings

Let A be the algebra of local observables near a line and let $\varepsilon : A \to \mathbf{1}$ describe the distinguished trivial line. Deformations supported on that line are governed by the augmentation module, whose derived endomorphism algebra is

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

Theorem 22.10 (Koszul duality for the augmentation-generated line sector)**.** *Let A be a complete augmented E_1 algebra of local observables, and suppose the augmentation module $\mathbf{1}$ is compact in the selected module category. Put $A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$. Derived Morita theory identifies the thick subcategory generated by $\mathbf{1}$ with $\mathrm{Perf}(A^!)$. When $\mathbf{1}$ is a compact generator of a presentable completion, the equivalence extends to that completion. The two-sided bar $\mathbf{1} \otimes_A^L \mathbf{1}$ computes the derived matrix coefficients of the augmentation line.*

Proof. The augmentation module is the distinguished line. The Yoneda functor $\mathrm{RHom}_A(\mathbf{1}, -)$ is fully faithful on its thick closure and identifies the generator with the free rank-one $A^!$-module. Closure under cones, shifts, and retracts gives the equivalence with $\mathrm{Perf}(A^!)$. Compact generation gives the Ind-completed statement. The two-sided bar computes the derived relative tensor product of the two augmentation modules. □

A separate representation-theoretic calculation can then identify $A^!$ with a named Yangian or another line algebra.

22.10 Transfer matrices and scalar observables

Let V be an auxiliary object of a rigid line category and let $M_1, \dots, M_n$ be quantum spaces. The monodromy operator

$$T_V(z) = R_{V,M_n}(z - z_n) \cdots R_{V,M_1}(z - z_1)$$

obeys the RTT relation. Taking the categorical trace over V gives commuting transfer matrices:

$$[t_V(z), t_W(w)] = 0.$$

The braid relation and cyclicity of the trace prove the commutator identity.

The ordered operator $T_V(z)$ retains exchange data. Its categorical trace retains the commuting spectral family. Bethe equations and Baxter relations describe that family in representation categories where highest-weight or prefundamental constructions provide the required eigenvectors.

22.11 Knots and framing

Closing a braid uses duals and a ribbon twist. The Reshetikhin–Turaev invariant is the categorical trace of the closed braid. In Chern–Simons theory the twist is the framing anomaly. The three structures therefore enter in the order

$$\text{fusion} \quad\longrightarrow\quad \text{braiding} \quad\longrightarrow\quad \text{ribbon trace.}$$

They are governed by E_1, E_2, and cyclic or framed data. Geometrically they correspond to an ordered world-line, an exchange path, and a closed framed world-line.

23 BRST descent and W-algebras

23.1 A differential compatible with chirality

Let (C, Q) be a cochain complex equipped with a chiral operation X. The derivation condition is

$$[Q, X] = 0 \tag{145}$$

in the chiral deformation Lie algebra. Together with $Q^2 = 0$ and $[X, X] = 0$, it gives

$$[Q + X, Q + X] = 0.$$

Thus the BRST differential and the chiral operation are the unary and binary components of one Maurer–Cartan element.

Proposition 23.1 (BRST descent of chiral operations)**.** *Let Q be a square-zero derivation of a chiral algebra C. Then $H_Q(C)$ carries the induced chiral algebra structure, and the operation depends only on cohomology classes.*

Proof. Equation (145) sends cocycles to cocycles. Replacing one input by a Q-boundary changes the output by a Q-boundary. The chiral Jacobi identity therefore descends to cohomology. □

The same proof applies to formal-coordinate vertex operations and supplies the algebraic core of quantum Hamiltonian reduction.

23.2 The BRST–collision bicomplex

The collision complex of C carries the BRST differential induced by Q and the collision differential induced by X. Equation (145) makes them anticommute. Hence

$$\left(C_I^{\mathrm{coll}}(C), Q + d_{\mathrm{coll}}\right) \tag{146}$$

is a bicomplex.

The BRST filtration gives

$$E_1 \cong C_I^{\mathrm{coll}}(H_Q(C)).$$

The collision filtration gives the BRST cohomology of the collision resolution. These two spectral sequences compare reduction before and after the bar construction.

Theorem 23.2 (Bar–BRST comparison)**.** *Suppose the tensor products and sums defining the coefficient objects are Q-acyclic outside degree zero. Suppose the bicomplex (146) is bounded below in one grading and complete with respect to an exhaustive filtration in the other. Then*

$$C_I^{\mathrm{coll}}(H_Q^0(C)) \xrightarrow{\cong} H_Q^0\Big(C_I^{\mathrm{coll}}(C), Q\Big)\,, \tag{147}$$

all other vertical BRST cohomology groups vanish, and the total complex is quasi-isomorphic to $C_I^{\mathrm{coll}}(H_Q^0(C))$ with its induced collision differential.

Proof. The acyclicity hypothesis allows vertical cohomology to commute with the tensor products defining every coefficient object. It gives (147) and concentration in vertical degree zero. The BRST-first spectral sequence therefore has one row. Boundedness and completeness identify its abutment with the total cohomology. □

More generally, the full E_1 page $C_I^{\mathrm{coll}}(H_Q(C))$ is the derived BRST bar object. Vertical concentration is the criterion that identifies it with the ordinary bar of $H_Q^0(C)$.

23.3 Quantum Drinfeld–Sokolov reduction

Let $\mathfrak{g}$ be a simple Lie algebra and $f \in \mathfrak{g}$ nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from $V_k(\mathfrak{g})$, charged ghosts, neutral fields when required, and a differential determined by f. The affine W-algebra is

$$\mathcal{W}^k(\mathfrak{g}, f) = H^0_{\mathrm{BRST}}(C^k_{\mathrm{DS}}(\mathfrak{g}, f)). \tag{148}$$

For the universal affine W-algebra, the standard BRST complex is concentrated in degree zero, and the associated graded is described by functions on the arc space of the Slodowy slice [2, 3].

Proposition 23.1 carries the chiral operation to the cohomology. The nonlinear OPEs of the W-generators are the homological images of affine currents and ghosts. Composite fields are therefore intrinsic outputs of the reduction.

23.4 Homotopy transfer

Choose a contraction

$$(H, 0) \underset{p}{\overset{i}{\rightleftarrows}} (C, Q), \qquad Qi = 0, \quad pQ = 0, \quad Qh + hQ = \mathrm{id} - ip.$$

Homotopy transfer expresses the induced higher operations on H as sums over rooted trees: vertices carry the original chiral operation, internal edges carry h, leaves carry i, and the root carries p.

A contraction adapted to a strict cohomology model yields a transferred Maurer–Cartan element gauge equivalent to the strict vertex operation. For a general contraction, the higher tree operations represent a homotopy chiral model. Their first nontrivial cohomology classes are the strictification classes, while their chain representatives vary with i, p, h.

The independently constructed strict vertex algebra $H_Q^0(C)$ and the transferred homotopy chiral structure are related by a $\mathcal{P}_\infty$ quasi-isomorphism whenever the chosen operadic resolution and contraction satisfy the homotopy-transfer hypotheses. This quasi-isomorphism is the comparison datum that removes transfer-dependent higher operations from the strict model.

23.5 Feigin–Frenkel duality for principal W-algebras

For principal f, Feigin–Frenkel duality relates principal W-algebras for Langlands-dual Lie algebras. With lacing number $r^\vee$, the levels satisfy

$$r^\vee(k+h^\vee)(\ell+{}^L h^\vee)=1, \tag{149}$$

and there is an isomorphism

$$\mathcal{W}^k(\mathfrak{g})\cong\mathcal{W}^\ell({}^L\mathfrak{g}). \tag{150}$$

This multiplicative shifted-level relation is the standard principal W-algebra duality.

Screening operators and free-field realizations construct the duality. A comparison with chiral bar objects additionally requires a morphism from the BRST resolution to the corresponding chiral Chevalley or associative bar construction.

23.6 Screening operators

A free-field realization embeds a W-algebra into an intersection of kernels of screening operators. A screening is a contour integral of a vertex operator and commutes with the W-algebra modulo total derivatives. The screening complex realizes the reduced algebra in cohomology.

Geometrically, each screening contour is a cycle in a configuration space. Charges determine its local system, monodromy, and intersection pairing. Multiple screenings produce hypergeometric local systems and quantum-group actions. This gives an ordered configuration-space realization of BRST reduction.

23.7 Critical level

At $k=-h^\vee$, the affine algebra remains a well-defined chiral algebra and acquires the Feigin–Frenkel center. On the formal disc D,

$$Z(V_{-h^\vee}(\mathfrak{g}))\cong\mathcal{O}(\mathrm{Op}_{^L G}(D))\,. \tag{151}$$

The center of the completed critical enveloping algebra over the punctured disc is described by functions on opers on $D^\times$ with the corresponding completion and central-character conditions [34, 36]. The vertex center is a commutative chiral algebra and supplies the local algebraic input to geometric Langlands.

The Sugawara conformal vector contains the factor $(k+h^\vee)^{-1}$, so the critical specialization changes the conformal and KZ descriptions. The KZ coupling has a pole at the critical level. Critical representation theory is therefore organized by opers and the enlarged center.

23.8 Structures transported by reduction

Every chain map commuting with Q descends to BRST cohomology. This applies directly to the chiral operation, compatible module actions, and Q-invariant pairings. Additional structures descend through the corresponding compatible data:

structure	descent datum
quadratic presentation	a Q-stable generators-and-relations model
Koszul comparison	a filtered quasi-isomorphism compatible with Q
braid local system	a flat connection commuting with Q
Verdier identification	a Q-compatible perfect pairing
positive-genus sewing	clutching maps commuting with Q

Theorem 23.2 supplies the bar comparison; the rows of the table supply its structured refinements.

23.9 Field-theoretic meaning

Classical Hamiltonian reduction imposes moment-map constraints and divides by their gauge symmetry. BRST replaces that quotient by the cochain complex of constraints and ghosts. Quantum Drinfeld–Sokolov reduction is its chiral realization. The resulting Virasoro or W-algebra is the algebra of gauge-invariant boundary observables.

The ghost system contributes its own central extension. Nilpotence of the BRST charge is the cancellation equation for the total anomaly of matter, gauge, and ghost sectors. This equality lives in the BRST deformation complex and descends to the reduced chiral algebra through the same chain maps as the OPE.

Part IV

Two genera and two master equations

24 Two genera and the two-edge master equation

24.1 Two loop-forming operations

Let A be an ordered chiral algebra. A Calabi–Yau trace on the transverse E_1 multiplication closes an interval into a circle and produces Hochschild chains, annuli, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues marked points of algebraic curves by plumbing and produces conformal blocks. Both constructions contract pairs of flags. Their flags carry distinct colours because the two contractions live on different moduli spaces and carry different determinant lines.

Definition 24.1 (Transverse Calabi–Yau structure)**.** A transverse Calabi–Yau structure of degree $d_\perp$ on a smooth proper E_1 algebra A is an S^1-equivariant trace

$$\mathrm{tr}_\perp : \mathrm{HH}_\bullet^{\mathrm{FactD}(X)}(A) \longrightarrow \mathbf{1}[-d_\perp]$$

whose induced bimodule morphism $A \to A^\vee[-d_\perp]$ is an equivalence. The associated pairing is cyclic and perfect.

Definition 24.2 (Chiral modular sewing structure)**.** A chiral modular sewing structure of degree d_X consists of perfect pairings on the insertion objects and operations assigned to stable pointed complex curves, compatible with factorization D-module maps, separating and nonseparating clutching, symmetric-group actions, and the determinant twists of the sewing complex.

The transverse structure is an open topological field-theory datum internal to the chiral coefficient. The chiral structure varies the complex curve. Their simultaneous presence gives a bidegree (g, h), with chiral genus g and ribbon genus h.

24.2 Bicoloured stable graphs

A bicoloured stable graph Γ has

$$E(\Gamma) = E_X(\Gamma) \sqcup E_\perp(\Gamma).$$

An X-edge represents a chiral plumbing contraction. A $\perp$-edge represents a transverse cyclic contraction. Vertices carry operations compatible with their incident flags, and transverse flags retain cyclic order. The orientation line is

$$\mathrm{or}(\Gamma) = \det(E_X(\Gamma)) \otimes \det(E_\perp(\Gamma)). \tag{152}$$

Contracting a loop of colour X raises g by one; contracting a loop of colour $\perp$ raises h by one.

Let G^{bi} be the groupoid of bicoloured stable graphs. For a coefficient system $A(\Gamma)$, define the complete graph object

$$\mathbb{F}^{\mathrm{bi}}(A) = \prod_{[\Gamma]\in\mathsf{G}^{\mathrm{bi}}} (\mathrm{or}(\Gamma) \otimes A(\Gamma))_{\mathrm{Aut}(\Gamma)}\,. \tag{153}$$

This direct construction records the two edge colours and their common vertex operations.

24.3 The two boundary operators

Let ∂_X be the signed sum of contractions of one X-edge and $\partial_\perp$ the signed sum of contractions of one $\perp$-edge.

Proposition 24.3 (Independent graph boundaries)**.** *With the orientation convention* (152)*,*

$$\partial_X^2 = 0, \qquad \partial_\perp^2 = 0, \qquad \partial_X\partial_\perp + \partial_\perp\partial_X = 0.$$

Hence $\partial_X + \partial_\perp$ is square-zero.

Proof. For two edges of one colour, the two contraction orders induce opposite orientations in the determinant of that edge set. For edges of different colours, both orders produce the same final graph, while exchanging the two odd determinant factors in (152) changes the sign. ☐

A geometric realization supplies coefficient maps representing these combinatorial contractions on ribbon and Deligne–Mumford boundary strata.

24.4 The bicoloured Feynman transform

Let $\mathcal{P}^{\mathrm{bi}}$ be a complete dg bicoloured modular cooperad with two edge-decomposition maps, labelled X and $\perp$, whose determinant twists satisfy Proposition 24.3. Let $\mathrm{End}_A^{\mathrm{cyc}}$ be the cyclic endomorphism object defined by the two perfect pairings. The completed convolution complex is

$$\mathfrak{g}_A^{\mathrm{bi}} = \prod_{g,h,n} \mathrm{Hom}_{\Sigma_n}\left(\mathcal{P}^{\mathrm{bi}}(g,h,n), \mathrm{End}_A^{\mathrm{cyc}}(g,h,n)\right). \tag{154}$$

Joining two vertices by an edge gives brackets $[-,-]_X$ and $[-,-]_\perp$. Joining two flags of one vertex gives loop operators Δ_X and $\Delta_\perp$.

Theorem 24.4 (Two-edge master equation). *A morphism from the bicoloured Feynman transform $F(\mathcal{P}^{\mathrm{bi}})$ to $\mathrm{End}_A^{\mathrm{cyc}}$ is equivalent to an element*

$$\Theta_A^{\mathrm{bi}} \in \mathfrak{g}_A^{\mathrm{bi}}[[\hbar_X, \hbar_\perp]]$$

satisfying

$$d\Theta_A^{\mathrm{bi}} + \frac{1}{2}[\Theta_A^{\mathrm{bi}}, \Theta_A^{\mathrm{bi}}]_X + \frac{1}{2}[\Theta_A^{\mathrm{bi}}, \Theta_A^{\mathrm{bi}}]_\perp + \hbar_X \Delta_X \Theta_A^{\mathrm{bi}} + \hbar_\perp \Delta_\perp \Theta_A^{\mathrm{bi}} = 0. \tag{155}$$

Proof. The Feynman transform is freely generated by corollas, and its differential is the internal differential plus all one-edge decompositions. A morphism is determined by its values on corollas. Compatibility with the transform differential gives exactly (155). Conversely, the equation extends the corolla assignment uniquely to a chain morphism on the complete free graph object. □

Thus the two sewing structures are precisely the input from which the common master equation is formed.

24.5 The two loop operators

On the suspended convolution complex, both loop contractions are odd.

Proposition 24.5 (Supercommutation of loop contractions). *With the determinant convention* (152),

$$\Delta_X \Delta_\perp + \Delta_\perp \Delta_X = 0.$$

After passage to unsuspended scalar amplitudes, the induced even contractions commute.

Bidegree $(0, 1)$ contains the annulus and Hochschild trace. Bidegree $(1, 0)$ contains genus-one chiral sewing and its projective connection. Bidegree $(1, 1)$ contains both loop operations, and its mixed codimension-two identity is their supercommutation.

24.6 Geometric realization

Ribbon surfaces or cyclically ordered boundary configurations realize $\perp$-edges. Stable pointed curves and clutching morphisms realize X-edges. A correspondential model places both degenerations in one space and compares the two contraction orders by a Cartesian square carrying the product orientation.

Criterion 24.6 (Geometric two-edge realization). *A pair of geometric sewing theories realizes the two-edge master equation when:*

(i) *every codimension-one boundary of either colour carries a coefficient contraction;*

(ii) *same-colour codimension-two boundaries satisfy the modular or ribbon boundary identity;*

(iii) *mixed codimension-two boundaries form Cartesian comparison squares with orientation* (152)*;*

(iv) *the coefficient contractions commute over those squares;*

(v) *the graph filtration is complete.*

Under these conditions, the geometric boundary differential maps to the differential of the bicoloured Feynman transform, and Theorem 24.4 applies.

24.7 Open topological and chiral conformal genera

A smooth proper Calabi–Yau E_1 algebra defines an oriented open topological field theory. Its value on the circle is

$$\int_{S^1} A \simeq \mathrm{HH}_\bullet(A),$$

and ribbon graphs compute its amplitudes. This gives the row $g = 0$ with arbitrary ribbon genus h.

A chiral conformal theory assigns blocks to complex curves and varies with complex structure through a projectively flat connection. This gives the column $h = 0$ with arbitrary chiral genus g.

Proposition 24.7 (Bigrading of the two genera)**.** *The pair (g, h) is the universal grading of the bicoloured graph complex. The projection to g records Deligne–Mumford plumbing, and the projection to h records ribbon-loop contraction. A dynamical duality between the two expansions is represented by a morphism that mixes these gradings and intertwines the corresponding sewing operators.*

Thus a particular string or gauge duality can identify selected combinations of g and h through an explicit comparison functor.

24.8 The double expansion

For a matrix realization, a connected ribbon graph carries the power N^{2-2h-b}, where b is the number of boundaries. Chiral sewing carries the power g_s^{2g-2}. The connected potential therefore has the bivariate expansion

$$\mathcal{F}_A(N, g_s) = \sum_{g,h,b} N^{2-2h-b} g_s^{2g-2} \mathcal{F}_{g,h,b}(A). \tag{156}$$

Each coefficient is a trace of the (g, h, b) component of Θ_A^{bi}. The two parameters retain the matrix-planar and worldsheet-genus gradings separately.

24.9 Duality of the two-edge theory

Theorem 24.8 (Duality of bicoloured amplitudes)**.** *Suppose:*

(i) A and $A^!$ are joined by an invertible bimodule Morita context on the selected complete perfect sector;

(ii) their transverse Calabi–Yau traces correspond;

(iii) their chiral sewing systems are identified by a tensor equivalence preserving projective connections, pairings, and determinant twists.

Then the equivalence induces an isomorphism of two-edge convolution complexes carrying Θ_A^{bi} to $\Theta_{A^!}^{\mathrm{bi}}$.

Proof. Morita invariance identifies Hochschild chains, ribbon contractions, and $\Delta_\perp$. The sewing equivalence identifies chiral contractions and Δ_X. Preservation of determinant lines identifies the orientation signs. Hence the equivalence intertwines every term of (155). □

Applying a compatible trace to this structured equivalence produces its numerical complementarity identities.

24.10 Physical meaning

The operator $\Delta_\perp$ contracts two open or line-defect states and creates a loop in the transverse topological theory. The operator Δ_X contracts two chiral insertions through a plumbing fixture and creates a handle on the complex curve. Their supercommutation expresses locality for the two independent degenerations. The bicoloured graph operation retains the edge colour, contraction kernel, and moduli space of every loop.

25 Cyclicity, stable graphs, and the master equation

25.1 The pairing that creates a handle

A genus-zero operation has inputs and outputs. Joining two flags requires a contraction kernel. Let V carry a nondegenerate graded-symmetric pairing

$$\langle -, - \rangle : V \otimes V \longrightarrow \mathbb{C}[d] \tag{157}$$

invariant under the genus-zero operations. The inverse pairing converts an output into an input and contracts internal edges. A cyclic chiral algebra, a conformal-block pairing, or a field-theoretic propagator supplies this datum.

25.2 Modular operads

A modular operad has operations indexed by genus and labelled legs. Joining legs on two vertices adds their genera; joining two legs of one vertex creates a handle. Stable graphs encode all iterated compositions [43].

Let $\mathcal{P}$ be a dg modular cooperad and let $\mathrm{End}_V^{\mathrm{mod}}$ be the endomorphism modular operad determined by (157). The completed convolution space

$$\mathfrak{g}_{\mathcal{P},V} = \prod_{g,n} \mathrm{Hom}_{\Sigma_n}\Big(\mathcal{P}(g,n), \mathrm{End}_V^{\mathrm{mod}}(g,n)\Big) \tag{158}$$

contains the internal differential, the bracket obtained by joining two vertices, and the loop operator Δ obtained by joining two legs of one vertex.

25.3 The Feynman transform

The Feynman transform $F\mathcal{P}$ is the free modular operad on the shifted dual of $\mathcal{P}$, with determinant twists and differential given by the internal differential plus one-edge decompositions. A morphism $F\mathcal{P} \to \mathrm{End}_V^{\mathrm{mod}}$ is determined by its values on corollas.

Theorem 25.1 (Feynman transform and master equation)**.** *Giving V an algebra structure over $F\mathcal{P}$ is equivalent to giving an element $\Theta \in \mathfrak{g}_{\mathcal{P},V}[[\hbar]]$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0. \tag{159}$$

Proof. The internal differential of the free transform gives $d\Theta$. A one-edge decomposition with two vertices gives the bracket term, with factor $1/2$ for interchange of the vertices. A loop decomposition at one vertex gives $\Delta\Theta$. Compatibility of a morphism with the two differentials is precisely (159); conversely the equation extends the corolla assignment uniquely to the free transform. □

The theorem converts a constructed system of sewing operations into one Maurer–Cartan element. The geometric boundary identities then prove its master equation.

25.4 Stable curves

The boundary strata of $\overline{\mathcal{M}}_{g,n}$ are indexed by stable graphs. Separating nodes correspond to edges joining distinct vertices, and nonseparating nodes correspond to loops. Clutching morphisms realize graph contraction.

Borel–Moore chains on the clutching spaces, with determinant twists, form a modular operad under proper pushforward. Aritywise finite duality, or a specified continuous dual, gives the corresponding modular cooperad. A compatible system of multilinear maps on V then defines an algebra over the Feynman transform. Every codimension-two stratum appears with the two induced orientations of its parent boundary divisors, producing the cancellation in (159).

Principle 25.2 (Sewing principle). *The coefficient theory supplies a map for every clutching morphism. The oriented boundary geometry of $\overline{\mathcal{M}}_{g,n}$ supplies the relations among those maps.*

25.5 Graph filtration and extension classes

Filter $\mathfrak{g}_{\mathcal{P},V}$ by the number of internal edges, or by any complete stable-graph complexity filtration compatible with the bracket. Fix a genus-zero Maurer–Cartan element Θ_0. A partial lift

$$\Theta_{\le r} = \Theta_0 + \Theta_1 + \cdots + \Theta_r$$

has curvature in filtration $r+1$. Its leading part is closed for $d_{\Theta_0} = d + [\Theta_0, -]$.

Theorem 25.3 (Filtered extension theorem). *The leading curvature of $\Theta_{\le r}$ defines a canonical class*

$$\mathfrak{o}_{r+1}(\Theta_{\le r}) \in H^2\Big(\mathrm{gr}_F^{r+1}\,\mathfrak{g}_{\mathcal{P},V}, d_{\Theta_0}\Big)\,. \tag{160}$$

An extension to filtration $r+1$ exists precisely when this class vanishes. When it vanishes, gauge-equivalence classes of extensions form a torsor for the corresponding H^1.

Proof. The Bianchi identity makes the leading curvature closed. Changing a partial lift changes that curvature by a coboundary. A primitive for the curvature is exactly the next Maurer–Cartan coefficient. Two primitives differ by a closed degree-one element, and lower-filtration gauge transformations identify elements differing by a coboundary. □

The graph filtration is intrinsic to the modular convolution complex. A scalar or one-dimensional projection inherits this tower when the projection is a morphism of filtered deformation complexes.

25.6 Connected graph cumulants

Let a cyclic trace assign a scalar weight $W(\Gamma)$ to each stable graph, and form the complete formal sum

$$Z = 1 + \sum_{\Gamma} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}\Gamma|} W(\Gamma). \tag{161}$$

The exponential formula gives

$$\log Z = \sum_{\Gamma \text{ connected}} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}\Gamma|} W(\Gamma). \tag{162}$$

Thus connected amplitudes are the cumulants of the full partition function. The trace and the graph weights specify this scalar projection of the modular object.

25.7 Graphwise duality

Let V and $V^\vee$ be paired cyclic objects. Transpose each vertex operation and contract each internal edge with the inverse pairing. Determinant lines supply the corresponding signs.

Proposition 25.4 (Graphwise transpose)**.** *Suppose the coefficient systems are aritywise perfect, continuous duality commutes with the stable-graph completion, and the cyclic pairing identifies their endomorphism modular operads. Then vertexwise transposition and edgewise inverse pairing induce a filtered isomorphism of modular convolution complexes. The isomorphism intertwines the master equations and sends each connected graph weight to its transpose.*

Every numerical complementarity formula obtained from this proposition is determined by the chosen duality, cyclic trace, and characteristic-class functional. Level, central charge, and Hodge classes arise from their respective functionals on the structured graph object.

25.8 The Batalin–Vilkovisky realization

In the BV formalism, fields form a shifted symplectic space. The symplectic pairing gives the bracket, and the regularized contraction kernel gives the second-order operator Δ. The quantum action S satisfies

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \tag{163}$$

which expands to

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0.$$

A morphism from the renormalized BV deformation complex to $\mathfrak{g}_{\mathcal{P},V}$ identifies this equation with (159). In perturbative theories, compactified configuration spaces construct the morphism by extending graph weights to collision boundaries.

25.9 Anomaly classes

The renormalized quantum master expression

$$\mathcal{A} = QS + \frac{1}{2}\{S, S\} + \hbar\Delta S$$

is a degree-one local cocycle. Its cohomology class is independent of the choice of finite counterterm. Vanishing of $[\mathcal{A}]$ permits a counterterm that restores the quantum master equation.

For a family of conformal blocks, the corresponding anomaly is central curvature of the projectively flat connection. A line bundle with opposite curvature produces a flat tensor-product connection. The moduli-space realization of this anomaly line is developed in Sections 26 and 27.

26 Conformal blocks, sewing, and the anomaly line

26.1 Coinvariants and blocks

Let V be a conformal vertex algebra and let $M_1, \dots, M_n$ be modules inserted at marked points of a smooth curve C. Global chiral currents act on $M_1 \otimes \cdots \otimes M_n$. Coinvariants are the quotient by this action, and conformal blocks are their duals. Variation of the pointed curve produces sheaves on moduli.

Damiolini–Gibney–Tarasca construct sheaves of coinvariants on moduli of pointed stable curves and prove factorization and vector-bundle results under their finiteness and semisimplicity hypotheses [22, 24]. For strongly rational vertex operator algebras, Damiolini–Woike construct the associated modular functor [25]. Its genus-zero part is the vertex tensor category, and its positive-genus sewing maps give the coefficient system required by the sewing principle.

26.2 Projective flatness

Infinitesimal variation of complex structure acts through the stress tensor. The Virasoro central term makes this action projective, so the conformal-block sheaf is a module for an Atiyah algebra. Equivalently, its connection has scalar curvature.

Let $\lambda = c_1(\mathbb{E})$ be the Hodge class and let ψ_i be the cotangent-line classes. If a_i is the conformal weight of M_i, the interior projective class is

$$\mathrm{rk}(\mathbb{V}) \left(\frac{c}{2}\lambda + \sum_i a_i \psi_i \right). \tag{164}$$

On the stable compactification, factorization through intermediate simple modules gives the boundary divisor terms [22, 23].

Theorem 26.1 (Chern-class form of the anomaly)**.** *For a conformal-block bundle satisfying the factorization hypotheses, the first Chern class is*

$$c_1(\mathbb{V}) = \mathrm{rk}(\mathbb{V}) \left(\frac{c}{2}\lambda + \sum_i a_i \psi_i \right) + \sum_\delta b_\delta [\delta],$$

where each boundary coefficient b_δ is the factorization trace determined by the intermediate modules and their conformal weights, with sign fixed by the clutching convention. In particular, the Hodge coefficient is $\mathrm{rk}(\mathbb{V})c/2$.

The formula separates the local central charge from the global modular data. Rank, insertion weights, fusion rules, and boundary channels determine the remaining terms.

26.3 The anomaly line

Let E be a projectively flat bundle with

$$\nabla_E^2 = \Omega \, \mathrm{id}_E.$$

Let L be a line bundle with connection satisfying $\nabla_L^2 = -\Omega$. Then

$$\nabla^2_{E\otimes L} = 0.$$

When the required class is rational, the same construction is expressed by a twisted D-module or by a line on the corresponding gerbe.

Proposition 26.2 (Anomaly cancellation by a line)**.** *Tensoring a projectively flat conformal-block system by a line or twisted line with opposite Atiyah class produces a flat system. Sewing remains compatible when the anomaly lines satisfy the clutching multiplicativity of the Hodge and cotangent-line bundles.*

This operation changes the connection on the family of conformal blocks and leaves the internal vertex-algebra differential unchanged.

26.4 Sewing and the master equation

Choose a perfect pairing between a module and its contragredient. Sewing two marked points inserts the inverse pairing and sums over a homogeneous basis. For a local plumbing parameter q, the propagator is

$$q^{L_0 - c/24}.$$

Analytic convergence gives the sewn block for $|q|$ small, and factorization identifies its boundary value with the sum over intermediate modules.

The modular-functor sewing maps define an algebra over chains on the surface operad. After tensoring by the anomaly line of Proposition 26.2, Theorem 25.1 packages these maps into a flat modular Maurer–Cartan element.

26.5 Genus one

Zhu's theorem gives modular transformation laws for trace functions of rational vertex operator algebras [78]. For a holomorphic VOA,

$$Z_V(\tau) = \mathrm{Tr}_V\, q^{L_0 - c/24}$$

is a modular function with its multiplier. For a rational VOA, the vector of irreducible characters carries a finite-dimensional representation of $SL_2(\mathbb{Z})$.

The factor $q^{-c/24}$ and the multiplier record the projective anomaly. When the representation category is modular, the modular S-matrix and the fusion coefficients satisfy the Verlinde formula [47, 73]. The dimension of a block is the numerical trace of this modular functor, while the $SL_2(\mathbb{Z})$ representation retains its transport.

26.6 Prime forms and analytic kernels

On a compact Riemann surface, the prime form $E(z, w)$ is the canonical $(-\frac{1}{2}, -\frac{1}{2})$ form with a simple zero on the diagonal. A theta characteristic gives a standard construction of it. The differential $d_z \log E(z, w)$ has the required local simple pole and prescribed quasi-periodicity. Adding the harmonic term gives the single-valued Green kernel. Fay identities control products of prime forms and Szegő kernels [32].

These kernels represent propagators and connections analytically. Their quasi-periodicity produces monodromy and scalar curvature in the moduli direction. The fibrewise OPE differential and the moduli connection occupy separate bidegrees of the total superconnection.

26.7 Free bosons and determinant lines

For a rank-r free boson, the oscillator character is

$$\eta(\tau)^{-r}.$$

This factor is the regularized determinant of the chiral oscillator operator. The nonchiral partition function includes

$$(\operatorname{Im}\tau)^{-r/2}|\eta(\tau)|^{-2r}.$$

On higher-genus surfaces, the determinant line of the $\bar{\partial}$ operator carries the Quillen metric, and the family index theorem computes its curvature.

This determinant geometry is Polyakov's conformal anomaly in line-bundle form. Free fields are governed entirely by the Gaussian determinant and zero modes. Interacting theories add conformal blocks and graph weights built from their full OPE coefficients.

26.8 Lattice theories and Siegel theta series

For an even positive-definite lattice L of rank r, the genus-one character of the lattice VOA is

$$Z_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^r}. \tag{165}$$

At genus g, the zero modes contribute the Siegel theta series $\Theta_L^{(g)}(\Omega)$, and the oscillators contribute determinant factors. The sequence of Siegel theta series is therefore a genus-filtered arithmetic invariant of the lattice theory.

For the two even unimodular lattices of rank sixteen, the degree-g Siegel theta series agree for $g \leq 3$. In degree four their difference is proportional to the Schottky form, whose restriction to the Jacobian locus vanishes. Thus the genus-two lattice partition functions coincide, and the ambient degree-four Siegel modular form detects a distinction before restriction to Jacobians. Every separation statement consequently specifies both genus and moduli locus.

26.9 Genus two and higher genus

Genus two introduces the three-dimensional Siegel upper half-space, separating and nonseparating boundary divisors, and stable graphs with loops. The off-diagonal entries of the period matrix couple channels that are independent at genus one. Rational VOAs provide finite-dimensional blocks with mapping-class-group monodromy; lattice VOAs provide explicit Siegel theta sections.

A complete genus expansion depends on the anomaly line, fusion channels, vertex operations, pairings, and propagators. Central charge fixes the Hodge component of the anomaly. The remaining graph weights retain the interacting OPE and representation theory.

27 Anomalies, curvature, and invertible field theories

27.1 The mathematical type of an anomaly

An anomaly is the cohomology class measuring the extension required for a classical symmetry to act after quantization. Its domain fixes its mathematical type. The present theory uses the following classes:

local central extension,	$[\omega_{\mathrm{loc}}] \in H^2_{\mathrm{Lie}}(\mathfrak{s};\mathbb{C})$,
projective curvature of a family of states,	$[\Omega_{\mathrm{fam}}] \in H^2(B;\mathbb{C})$,
BV quantum extension class,	$[\mathfrak{A}_{\mathrm{BV}}] \in H^1_{\mathrm{loc}}(\mathrm{Obs}^{\mathrm{cl}})$,
BRST square class,	$[Q^2] \in H^2(\operatorname{End} C)$,
framing anomaly,	$\alpha_{\mathrm{fr}} \in \pi_0 \operatorname{InvFT}^{\mathrm{fr}}_{d+1}$,
curvature of a curved Koszul object,	$m_0 \in C^2$.

A transgression or comparison morphism can carry one of these classes into another. The domain, codomain, and gluing law are part of every such identification.

27.2 Local central extensions

The affine current algebra has OPE

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w},$$

while the Virasoro algebra has

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The parameters k and c classify central extensions in different chiral Lie algebras. For the Sugawara conformal vector of an affine theory,

$$c_{\mathrm{Sug}} = \frac{k\ \dim \mathfrak{g}}{k+h^\vee}.$$

Thus the chosen affine level and invariant form determine a Virasoro central charge on the noncritical locus.

A central extension is one coefficient of the complete chiral operation. Its cocycle invariance is one of the identities that make the collision and ordered bar differentials square to zero.

Proposition 27.1 (Central terms and the ordered bar)**.** *Let A be an augmented dg E_1 algebra in* $\mathrm{FactD}(X)$ *whose chiral coefficient carries a central extension compatible with the multiplication and differential. The normalized ordered bar coderivation satisfies $b^2 = 0$. The central coefficient enters through the multiplication of A or through its coefficient differential.*

Proof. The simplicial bar identities follow from associativity, the unit and augmentation laws, and the chain-map property of multiplication. The central extension is already a term of that multiplication and satisfies the same identities. Normalization therefore gives the standard square-zero bar coderivation. □

27.3 Projective curvature over moduli

Let $\mathbb{V} \to B$ be a bundle of conformal blocks. A projectively flat connection satisfies

$$\nabla_{\mathbb{V}}^2 = \Omega\,\mathrm{id}_{\mathbb{V}}.$$

The class $[\Omega]$ is the central extension of the tangent action represented by the Atiyah algebra of $\mathbb{V}$. For pointed stable curves, the interior contribution is

$$\mathrm{rk}(\mathbb{V})\left(\frac{c}{2}\lambda + \sum_i a_i\psi_i\right),$$

and factorization supplies the boundary divisor coefficients.

Theorem 27.2 (Type of the conformal anomaly)**.** *The conformal anomaly of a family of chiral blocks is the Atiyah-algebra extension class of the block bundle. A projective connection represents the same class by its scalar curvature. In the total superconnection, this curvature has horizontal degree two, while the internal block differential has vertical degree one.*

Proof. The internal differential is defined on each fixed fibre. Variation of the pointed curve acts through the stress-tensor Ward identity and produces a connection on the family. The Virasoro central term is the scalar commutator of two infinitesimal base variations. Hence it represents the Atiyah class and occupies the horizontal component of the superconnection square. □

27.4 The superconnection equation

The fibre and base data combine in a superconnection

$$\mathbb{D} = d_{\mathrm{fib}} + \nabla + \mathbb{A}_2 + \mathbb{A}_3 + \cdots.$$

Its square has total degree two. Flatness is $\mathbb{D}^2 = 0$, and projective flatness is $\mathbb{D}^2 = \Omega\mathrm{id}$.

Proposition 27.3 (Cancellation by an anomaly line)**.** *Let L be a line bundle with connection satisfying $\nabla_L^2 = -\Omega$. The tensor-product superconnection on $\mathbb{V} \otimes L$ is flat. A rationally integral class is treated by the corresponding gerbe of twisted bundles.*

The existence theorem for L is the integral-lift statement for the anomaly class. A chosen root of a determinant line supplies the corresponding fractional twist.

27.5 Polyakov's Weyl anomaly

Let $g = e^{2\sigma}\hat{g}$ be conformally related metrics on a closed surface. For a two-dimensional conformal field theory of central charge c, the renormalized effective action transforms by

$$W[e^{2\sigma}\hat{g}] - W[\hat{g}] = \frac{c}{24\pi}\int \left(|\nabla_{\hat{g}}\sigma|^2 + R_{\hat{g}}\sigma\right)\,d\mathrm{vol}_{\hat{g}}, \tag{166}$$

up to the stated sign convention and local topological terms. Differentiation gives

$$\langle T^{\mu}{}_{\mu}\rangle = -\frac{c}{24\pi}R.$$

The family index theorem identifies the curvature of the determinant line over moduli, producing the $c\lambda/2$ term in the chiral block connection.

The stress tensor links the Weyl response to the chiral OPE. The determinant line carries the family curvature. The transverse E_1 bar remains the interval-fusion construction.

27.6 BRST anomaly

A BRST operator is a degree-one endomorphism Q. Its square is the anomaly endomorphism. In the bosonic string, the matter and bc ghost central extensions cancel when

$$c_{\mathrm{matter}} + c_{bc} = 0, \qquad c_{bc} = -26,$$

together with the intercept condition.

Theorem 27.4 (BRST cancellation is tensorial)**.** *Let V and G be conformal complexes with commuting BRST data. The tensor-product differential is*

$$Q_{V\otimes G} = Q_V \otimes 1 + 1 \otimes Q_G.$$

Its square is the sum of the anomaly endomorphisms of V and G. Thus tensor product adds BRST anomaly classes, and their sum vanishes exactly when $Q_{V\otimes G}$ is square-zero.

27.7 BV anomaly

For a classical interaction I, define the renormalized quantum master expression

$$\mathfrak{A}(I_\hbar) = QI_\hbar + \frac{1}{2}\{I_\hbar, I_\hbar\} + \hbar\Delta I_\hbar.$$

At the first new order in $\hbar$, this expression is closed for the classical BV differential. Its cohomology class is invariant under finite counterterms. Vanishing of the class permits the next-order quantization.

Proposition 27.5 (Locality of the BV anomaly)**.** *The BV anomaly is represented by a local functional. In configuration-space renormalization, graph weights extend to Fulton–MacPherson or Axelrod–Singer compactifications. The codimension-one boundary restrictions and local counterterm faces assemble into the cocycle representing* $[\mathfrak{A}_{\mathrm{BV}}]$.

The incidence complex supplies the universal collision combinatorics. Propagators, densities, gauge fixing, and graph weights supply the field-theoretic coefficient system.

27.8 Framing anomaly

Chern–Simons theory assigns amplitudes to framed three-manifolds and framed links. Changing the framing multiplies the partition function and Wilson-line observables by phases determined by the chiral central charge of the modular tensor category. The ribbon twist records the same phase on line defects.

The Chern–Simons/WZW correspondence relates this three-dimensional framing phase to the two-dimensional WZW central charge. Constructing that relation uses the boundary theory and its determinant line.

27.9 Curved Koszul duality

A curved dg algebra has curvature m_0 and satisfies

$$d^2(x) = [m_0, x].$$

The A_∞ version distributes the same identity among all higher operations. A central scalar curvature has zero commutator and therefore leaves $d^2 = 0$ while still entering the curved Maurer–Cartan equation.

Quadratic-linear-constant Koszul duality places the constant part of the relations in m_0. A family comparison can transgress this algebraic class to a determinant or Hodge class when the comparison map has been constructed.

Proposition 27.6 (Anomaly type decomposition)**.** *The anomaly data of an ordered chiral theory form a tuple*

$$\Big([\omega_{\mathrm{loc}}], [\Omega_{\mathrm{fam}}], [\mathfrak{A}_{\mathrm{BV}}], [Q^2], \alpha_{\mathrm{fr}}, m_0\Big)$$

in the product of their native cohomology groups and Picard spectra. A scalar invariant is obtained only after choosing a character or trace on one component. Comparison theorems among components are morphisms between the corresponding deformation or invertible-theory objects.

Proof. Affine level varies with the local current cocycle. Virasoro central charge varies with the stress-tensor extension. Conformal-block curvature also depends on insertion weights and boundary factorization channels. The BRST class depends on the ghost complex, the framing class on a ribbon three-dimensional theory, and m_0 on the inhomogeneous algebraic presentation. Each component therefore defines an independent coordinate until a specified comparison morphism relates it to another. □

27.10 Anomalies as invertible theories

For local anomalies admitting the anomaly-field-theory description, an invertible theory in one higher dimension represents the anomaly. The anomalous theory is relative to its state line, and cancellation is a trivialization of the tensor product anomaly.

Chiral conformal blocks use determinant and Hodge lines. Fermions use determinant lines of Dirac operators. Chern–Simons theory uses the framing line. Tensor product tensors these invertible theories and adds their characteristic classes.

Principle 27.7 (Anomaly comparison)**.** *An identification of two anomalies is a morphism of invertible theories preserving state lines, connections, and gluing. Every resulting equality of numerical coefficients is the image of that morphism under a characteristic-class or holonomy functional.*

27.11 The anomaly square

A field theory with chiral boundary data may construct the commutative square

$$\begin{tikzcd} \text{local central extension} \arrow[r, "\text{Ward identity}"] \arrow[d, "\text{BV descent}"'] & \text{Atiyah algebra of blocks} \arrow[d, "c_1"] \\ \text{local BV anomaly} \arrow[r, "\text{family index}"'] & \text{curvature of determinant line.} \end{tikzcd} \tag{167}$$

For free chiral fields and WZW models, Ward identities and family-index calculations construct these arrows. For a general ordered chiral algebra, a stress tensor, a family of state spaces, and a BV realization supply the remaining corners.

The square is the unified language: every object retains its mathematical type, and every passage between types is an explicit morphism.

Part V

Exact mathematics and exact physics

28 Exact examples

The examples in this section separate the structures carried by an ordered chiral algebra. Each family is described through its diagonal singularity, transverse multiplication, exchange data, and sewing theory. The comparison is structural: the same geometric language accommodates distinct operator products and retains their complete operations.

28.1 Heisenberg currents

Let $\mathcal{H}_k$ be generated by a current J with

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \tag{168}$$

The singular operation is central and occupies the second normal symbol of the diagonal. Its Lie conformal bracket is

$$[J_\lambda J] = k\lambda.$$

The chiral Jacobi identity follows from centrality. The chiral Chevalley coalgebra is the cofree commutative factorisation coalgebra on the suspended current, with differential determined by the central cocycle.

Fock modules provide the natural boundary labels. Their conformal blocks on a compact curve are Fock spaces built from H^0 and H^1 of the curve, and the oscillator contribution to the partition function is a determinant. After rescaling a current with $k \neq 0$, the standard conformal vector has central charge one. The level k and the Virasoro central charge therefore occupy different geometric slots: k is the coefficient of the current two-point cocycle, while $c = 1$ is the anomaly coefficient of the stress tensor.

28.2 Affine Kac–Moody currents

For the universal affine algebra $V_k(\mathfrak{g})$,

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \tag{169}$$

The first normal symbol is the Lie bracket and the second is the invariant bilinear form. The chiral Jacobi identity has two coefficient equations:

$$[a,[b,c]] + [b,[c,a]] + [c,[a,b]] = 0, \qquad ([a,b],c) = (a,[b,c]).$$

The first equation makes the ordinary Chevalley differential square to zero. The second makes the level cocycle compatible with the bracket. Together they define the affine chiral Lie algebra.

At $k \neq -h^\vee$, the Sugawara vector has central charge

$$c_{\mathrm{aff}} = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \tag{170}$$

The KZ connection of Section 21 is the flat connection induced by current-algebra Ward identities. At positive integral level, the integrable modules of the simple affine VOA $L_k(\mathfrak{g})$ form a modular tensor category; its conformal blocks satisfy factorisation and the Verlinde formula.

For simply laced $\mathfrak{g}$ at level one, the Frenkel–Kac construction embeds the affine currents in a lattice VOA [35]. The exponential fields reproduce the root currents, and their simple-pole coefficients reproduce the original Lie bracket. The lattice realization changes the presentation while preserving the affine operator product.

28.3 Virasoro

The Virasoro vertex algebra is generated by T with

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{171}$$

The three terms occupy the fourth, second, and first normal symbols. Conformal covariance fixes their relative form. The fourth symbol is the central extension of the Lie algebra of vector fields, the second records the conformal weight of T, and the first generates translation.

At level four in the vacuum Verma module, the quasi-primary

$$\Lambda =: TT : -\frac{3}{10}\partial^2 T$$

has norm proportional to $c(5c+22)$. The factors c and $5c+22$ locate the corresponding null-vector loci. The Gram form belongs to representation theory and supplies a concrete test for degeneration of the vacuum module.

For a conformal-block bundle, the Hodge coefficient of the projective connection is $c/2$. The (b,c) ghost system of weights $(2,-1)$ has central charge -26, so a matter theory with $c = 26$ gives a tensor product with vanishing total Virasoro anomaly. The affine reflection $c \mapsto 26 - c$ has midpoint 13; a self-duality at that parameter would require an explicit equivalence of the corresponding chiral or module theories.

28.4 Free fermions

A charged free fermion has odd fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The singular operation is the first normal symbol. Wick's theorem expresses every $2n$-point function as a determinant of the two-point kernel. Spin structure enters through the choice of square root of the canonical bundle, and modular transformation acts on the resulting determinant line.

Boson–fermion correspondence identifies the charged free-fermion VOA with the rank-one lattice VOA attached to the integral lattice. The Heisenberg current appears as $:\psi\psi^*:$, while the lattice exponentials recover the fermion fields. After the lattice extension, the correspondence is an isomorphism of complete vertex-algebra structures, including the current spaces and all vertex operations.

28.5 The $\beta\gamma$ and bc systems

A bosonic first-order system has

$$\beta(z)\gamma(w) \sim \frac{1}{z-w},$$

and the fermionic bc system has the same kernel with odd parity. Their oscillator algebras are respectively Weyl and Clifford algebras. For conformal weights $(\lambda, 1-\lambda)$,

$$c_{\beta\gamma} = 2(6\lambda^2 - 6\lambda + 1), \qquad c_{bc} = -2(6\lambda^2 - 6\lambda + 1). \tag{172}$$

The signs arise from the supertrace in the determinant line. Pairing the two systems with the same weights cancels their determinant anomalies.

The elementary pole controls the basic contraction. Composite fields, modules, and BRST reductions produce further operations through Wick expansion and homotopy transfer. Pole order and deformation cohomology are therefore separate gradings of the theory.

28.6 Lattice vertex algebras

Let L be an even positive-definite lattice. The lattice VOA V_L has central charge $\operatorname{rk} L$. It contains Heisenberg currents and exponential fields e^α. Their products have the form

$$Y(e^\alpha, z)Y(e^\beta, w) = (z-w)^{(\alpha,\beta)}\epsilon(\alpha,\beta)\big(Y(e^{\alpha+\beta}, w) + \cdots\big).$$

The commutator of the cocycle ϵ is fixed by the parity determined by the lattice form. The cocycle is part of the ordered extension data and controls the exchange phase of exponentials.

For even unimodular L, the VOA is holomorphic. Its genus-g vacuum amplitude contains the Siegel theta series of L and an oscillator determinant. Replacing ϵ by its inverse produces the contragredient exchange phase. A self-contragredient lattice theory is obtained when an isometry and cocycle equivalence identify these data.

28.7 Leech and Moonshine

The Leech lattice VOA V_Λ and the Moonshine VOA $V^\natural$ both have central charge 24, while

$$(V_\Lambda)_1 \cong \mathbb{C}^{24}, \qquad (V^\natural)_1 = 0.$$

The first space supplies a rank-twenty-four Heisenberg current algebra; the second begins its positive spectrum at weight two. Their operator products, automorphism groups, and module-theoretic realizations are correspondingly different.

For $g \in \mathbb{M}$, the McKay–Thompson series

$$T_g(\tau) = \mathrm{Tr}_{V^\natural}\Big(g\, q^{L_0 - 1}\Big)$$

is an equivariant graded trace. The vertex algebra supplies the state space and multiplication; the trace records the character of the Monster action on each conformal-weight space. Moonshine therefore enriches the chiral algebra by a distinguished finite-group action and its modular trace functions.

28.8 Principal W-algebras

The principal W-algebra $\mathcal{W}^k(\mathfrak{g})$ is the degree-zero cohomology of quantum Drinfeld–Sokolov reduction in the concentration range described in Section 23. At generic level, its standard strong generators have conformal weights $d_i + 1$, where d_i are the exponents of $\mathfrak{g}$. Their OPEs contain derivatives and normally ordered composites, and the complete collection is governed by the chiral Maurer–Cartan equation (116).

For $\mathfrak{sl}_2$, the principal reduction is Virasoro. For $\mathfrak{sl}_3$, the algebra is generated by T and a weight-three field W; the WW OPE contains T, its derivatives, and the quasi-primary Λ. Jacobi and conformal covariance determine the coefficients once the normalizations of T, W, and the central charge are fixed.

Feigin–Frenkel duality uses the shifted-level relation (149). It identifies principal W-algebras for Langlands-dual Lie algebras and therefore identifies their central charges and complete vertex operations. KZ inverse braiding belongs to the affine representation category and enters through a different comparison theorem.

28.9 Critical affine algebras

At $k = -h^\vee$, the affine OPE (169) remains defined. The Sugawara denominator vanishes, and the conformal vector obtained from the quadratic current becomes unavailable. The Feigin–Frenkel center expands to the commutative algebra of functions on Langlands-dual opers. The usual WZW normalization of the KZ connection also reaches its singular parameter. Critical level is therefore the locus at which the conformal, differential-equation, and representation-theoretic structures reorganize around opers.

28.10 Structural comparison

family	diagonal coefficient	exchange realization	sewing realization
Heisenberg	central second symbol k	scalar charge monodromy	Fock determinant
affine $V_k(\mathfrak{g})$	Lie bracket and level cocycle	KZ braid transport	WZW conformal blocks
Virasoro	first, second, and fourth symbols	monodromy of Virasoro blocks	projective connection with Hodge term
first-order systems	perfect first-symbol pairing	Weyl or Clifford statistics	determinant lines
lattice V_L	lattice powers and cocycle	lattice exchange phases	Siegel theta series
principal W	nonlinear finite strong-generation data	screening and block monodromy	W-conformal blocks in finite regimes
critical affine	critical current operation	oper differential equations	Feigin–Frenkel centre

Every row realizes the same five-carrier language through a different coefficient system. The diagonal determines the local singular operation, the interval determines ordered composition, the plane determines exchange, the circle determines trace, and the stable curve determines sewing.

29 Strictly noncommutative chiral algebras

29.1 Free ordered chiral algebras

Let $M \in \mathrm{FactD}(X)$. The free augmented ordered chiral algebra is

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus M^{\otimes 3} \oplus \cdots , \tag{173}$$

with concatenation. The tensor products are taken in $\mathrm{FactD}(X)$, so each word retains the full chiral coefficient system of its letters.

Proposition 29.1 (Bar homology of the free algebra)**.** *Assume the tensor products in* (173) *are exact. The reduced bar of $T_X(M)$ contracts onto $M[1]$. Equivalently,*

$$\mathbf{1} \otimes^L_{T_X(M)} \mathbf{1} \simeq \mathbf{1} \oplus M[1]$$

as a conilpotent coalgebra, with zero reduced coproduct on $M[1]$.

Proof. The word-length decomposition gives a fibre sequence of left $T_X(M)$-modules

$$T_X(M) \otimes M \xrightarrow{\text{mult}} T_X(M) \longrightarrow \mathbf{1}.$$

On the summand of words of positive length, multiplication is the canonical identification obtained by separating the last letter; the cofiber is the empty word. Applying $\mathbf{1} \otimes^L_{T_X(M)} -$ leaves the two-term complex $M \to \mathbf{1}$ with zero reduced differential, hence the class $M[1]$. Equivalently, the standard extra degeneracy of the normalized bar contracts every bar word having an internal cut and leaves the one-letter cogenerator. □

Thus the Koszul dual algebra of a free associative object is the square-zero extension

$$T_X(M)^! \simeq \mathbf{1} \oplus M^\vee[-1]$$

when M is dualizable. Noncommutative freedom on one side becomes a first-order infinitesimal object on the other.

29.2 Square-zero chiral algebras

Conversely, let

$$A = \mathbf{1} \oplus M, \qquad M^2 = 0.$$

The bar multiplication differential vanishes, and

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = \bigoplus_{r\geq 0} M[1]^{\otimes r}$$

with deconcatenation. Its continuous dual is the completed tensor algebra

$$A^! \simeq \widehat{T}_X(M^\vee[-1]).$$

This pair is the local model of an infinitesimal boundary coupling and its algebra of iterated defect insertions.

29.3 Quiver chiral algebras

Let Q be a finite quiver, let

$$S = \bigoplus_{i\in Q_0} \mathbf{1}e_i,$$

and assign an object $M_a \in \mathrm{FactD}(X)$ to every arrow $a : i \to j$. Put

$$M = \bigoplus_{a\in Q_1} e_{t(a)} M_a e_{s(a)}.$$

The relative tensor algebra

$$A_Q = T_S(M) \tag{174}$$

is an E_1 algebra in factorization D-modules. Multiplication concatenates composable arrows and is zero otherwise. The augmentation $A_Q \to S$ forgets nontrivial paths.

The relative bar

$$B_S(A_Q) = S \otimes^L_{A_Q} S$$

retains the ordered path structure. For the relation-free quiver, it contracts to $S \oplus M[1]$. If relations $R \subset M \otimes_S M$ are imposed, the quadratic coalgebra begins

$$S \oplus M[1] \oplus R[2] \oplus \cdots .$$

Example 29.2 (Two boundary conditions)**.** For the quiver $1 \rightleftarrows^v_u 2$, the objects M_u and M_v are boundary-changing fields. The composites uv and vu occupy the distinct idempotent summands $e_2A_Qe_2$ and $e_1A_Qe_1$; their values are independent boundary endomorphisms. Their chiral insertion points may collide on X, while the order of composition is fixed by the transverse interval. This is strict noncommutative chirality in its smallest form.

A cyclic potential W determines a cyclic functional on the completed path algebra. After choosing the standard Ginzburg or cyclic A_∞ construction, its cyclic derivatives enter the differential and are the classical equations of motion of the corresponding open-string field theory. The ribbon genus of Section 24 is then the genus of the resulting open-string Feynman surface.

29.4 Matrix and Chan–Paton extensions

Let V be a commutative algebra object in $\mathrm{FactD}(X)$ and let E be a finite-dimensional vector space. The matrix extension

$$A = \mathrm{End}(E) \otimes V \tag{175}$$

is an E_1 algebra in $\mathrm{FactD}(X)$. Its multiplication is matrix multiplication times the chiral factorization product. The OPE coefficients are $\mathrm{End}(E)$-valued and preserve their order.

The algebra (175) is Morita equivalent to V. Hence

$$Z_{\mathrm{ord,ch}}(A) \simeq Z_{\mathrm{ord,ch}}(V)$$

and their perfect module categories have equivalent centres. The open amplitudes differ by Chan–Paton traces, producing powers of $\dim E$ on ribbon graphs.

Proposition 29.3 (Chan–Paton weight)**.** *For a connected ribbon graph Γ contributing to a cyclic amplitude of* (175)*, contraction of matrix indices contributes*

$$(\dim E)^{F(\Gamma)},$$

where $F(\Gamma)$ is the number of index faces. For a closed ribbon surface of genus h with b boundary components, this is $(\dim E)^{2-2h-b}$ after the standard normalization of vertices and propagators.

This is the exact algebraic origin of the ’t Hooft expansion. It belongs to the transverse ribbon genus and is independent of the genus of the chiral curve.

29.5 Universal enveloping ordered chiral algebras

Let $\mathfrak{g}$ be a Lie algebra object in $\mathrm{FactD}(X)$. Its universal enveloping algebra

$$U_X(\mathfrak{g}) \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$$

is defined by the usual coequalizer of the tensor algebra by

$$x \otimes y - (-1)^{|x||y|} y \otimes x - [x, y].$$

The Poincare–Birkhoff–Witt filtration has

$$\mathrm{gr}\, U_X(\mathfrak{g}) \simeq \mathrm{Sym}_X(\mathfrak{g})$$

under the standard flatness hypotheses.

Theorem 29.4 (Associative bar and internal Chevalley chains)**.** *For a flat Lie object* $\mathfrak{g}$ *in* $\mathrm{FactD}(X)$*, the canonical twisting morphism induces a quasi-isomorphism*

$$\mathrm{Chev}_X(\mathfrak{g}) \xrightarrow{\sim} \mathrm{Bar}_X^{\mathrm{ord}}(U_X(\mathfrak{g})).$$

Proof. Filter both sides by symmetric degree. The associated graded map is the ordinary Koszul resolution of $\mathrm{Sym}_X(\mathfrak{g})$ by the exterior coalgebra on $\mathfrak{g}[1]$. PBW identifies the associated graded of the enveloping algebra. Completeness and exactness of the filtration lift the associated-graded quasi-isomorphism. □

This theorem is the internal Lie–associative Koszul comparison in the pointwise coefficient category. It is distinct from the Beilinson–Drinfeld chiral Chevalley construction on the Ran space. The internal Lie bracket maps to the commutator in its enveloping algebra, and internal Lie chains compute the ordered bar of that enveloping algebra.

29.6 Affine currents as chiral coefficients

The affine current algebra is a Beilinson–Drinfeld chiral Lie algebra: its simple pole is the Lie bracket and its double pole is the invariant cocycle. Its chiral enveloping construction produces the universal affine chiral coefficient. This construction takes place in the chiral pseudo-tensor category and is distinct from the internal enveloping algebra of Theorem 29.4, which begins with a Lie object for the pointwise tensor product.

To obtain a strictly noncommutative ordered chiral algebra from affine data one must supply a transverse associative object: for example an endomorphism algebra of affine boundary conditions, a quiver of affine modules, or a free tensor algebra generated by the affine factorisation coefficient. The resulting internal bar resolves that transverse multiplication. The chiral Chevalley complex resolves the affine chiral Lie operation. A comparison between them is supplied by an enveloping or Morita theorem with a specified monoidal functor. The simple-pole bracket satisfies Jacobi, and invariance of the level cocycle makes the central extension a chiral Lie algebra.

29.7 Weyl and Clifford chiral algebras

Let M carry a perfect pairing $\omega : M \otimes M \to \mathbf{1}[d]$. The tensor algebra modulo

$$xy - (-1)^{|x||y|}yx = \hbar\,\omega(x, y) \tag{176}$$

produces a Weyl or Clifford object according to parity. In the chiral category, M may be a factorization D-module and ω may contain a higher normal symbol along the diagonal.

The relation is quadratic-linear-constant. Its Koszul dual is curved: the constant term becomes the curvature m_0 of the dual coalgebra. The curvature belongs to the algebraic presentation. A Hodge class on moduli belongs instead to the horizontal curvature of a family connection; a comparison requires a morphism between these two deformation complexes.

Example 29.5 (The $\beta\gamma$ system)**.** The chiral Weyl algebra generated by β and γ has singular relation

$$\beta(z)\gamma(w) \sim \frac{1}{z - w}.$$

With matrix Chan–Paton factors, the ordered product is matrix multiplication while the singular coefficient is the Weyl pairing. The two operations are visibly independent.

Example 29.6 (The *bc* system)**.** The odd analogue has Clifford relation. Its determinant line contributes the ghost conformal anomaly. The Weyl–Clifford Koszul exchange is a statement about quadratic relations; the cancellation of their conformal anomalies is a separate determinant-line calculation.

29.8 Chiral differential operators

For a smooth variety Y, a trivialization of the chiral-differential-operator obstruction gerbe produces a sheaf of chiral differential operators. It supplies a chiral factorisation coefficient whose local fields include functions and vector fields on Y. The vertex or chiral operation supplies the diagonal coefficient. A transverse E_1 multiplication is supplied by an associative endomorphism, quiver, matrix, or tensor-algebra construction in the module category.

Strict ordered examples are obtained from associative endomorphism objects in a category of chiral-differential-operator modules, from quiver endomorphisms of boundary conditions, or from matrix and tensor-algebra extensions of a dualizable coefficient. When Y carries a group action, BRST or quantum Hamiltonian reduction acts on such an ordered endomorphism algebra provided the BRST operator is a derivation of the transverse product. The resulting comparison is governed by Section 23.

29.9 Boundary endomorphism algebras

Let $\mathcal{B}$ be a factorization category of boundary conditions on X. For objects b_i, set

$$A_{ij} = \mathrm{RHom}_{\mathcal{B}}(b_i, b_j).$$

Composition gives

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

The direct sum

$$A_{\mathcal{B}} = \bigoplus_{i,j} A_{ij}$$

is an algebra over the semisimple idempotent object $S = \oplus_i \mathbf{1} e_i$. Its relative bar computes chains of boundary changes. This is the categorical form of the quiver example and the native algebra of open strings.

Theorem 29.7 (Morita reconstruction from a boundary generator)**.** *Let $b = \oplus_i b_i$ be a compact generator of a stable factorization category $\mathcal{B}$. Then*

$$A_b = \mathrm{RHom}_{\mathcal{B}}(b, b)$$

is an E_1 algebra in $\mathrm{FactD}(X)$ *and*

$$\mathcal{B} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

under the usual compact-generation hypotheses.

The theorem explains why noncommutative chiral algebras are primary in boundary physics. The category of boundary conditions comes first; its endomorphism algebra is a coordinate chart on that category.

29.10 Factorization quantum groups

A factorization quantum group combines an E_1 algebra object, a compatible coalgebra or tensor-category structure, and meromorphic exchange. In the ordered language these pieces occupy different levels:

E_1 multiplication	:	fusion,
factorization coproduct	:	splitting of lines or representations,
E_2 enhancement	:	exchange,
ribbon trace	:	closure and link invariants.

A Yangian possesses all these structures in a compatible completed sense. An ordered chiral algebra supplies fusion. Compatible coproduct, exchange, and ribbon trace are successive enrichments.

29.11 A noncommutative Koszul pair

Let V be dualizable and let $R \subset V \otimes V$ be a factorization subobject. Define

$$A = T_X(V)/(R), \qquad A^{!,\mathrm{quad}} = T_X(V^\vee[-1])/(R^\perp).$$

The canonical twisting morphism $A^{¡} \to A$ gives the Koszul complex

$$K(A) = A \otimes A^{¡}.$$

Theorem 29.8 (Quadratic recognition in the ordered chiral category)**.** *Assume positive weight, degreewise dualizability, and strong convergence of the weight filtration. The following are equivalent:*

(i) $A^{¡} \to \mathrm{Bar}_X^{\mathrm{ord}}(A)$ *is a quasi-isomorphism;*

(ii) $\mathrm{Cobar}_X(A^{¡}) \to A$ *is a quasi-isomorphism;*

(iii) the left and right quadratic Koszul complexes are acyclic off weight zero;

(iv) $\mathrm{Tor}^A(\mathbf{1}, \mathbf{1})$ *is concentrated on the quadratic diagonal.*

The proof is the fundamental theorem of twisting morphisms internal to $\mathrm{FactD}(X)$. The chiral coefficient changes the tensor products and support conditions; the fundamental theorem of twisting morphisms preserves the equivalence of the four tests in this internal setting. When these conditions hold and finite duals exist, $A^{!,\mathrm{quad}}$ identifies with the boundary endomorphism algebra $\mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$; before that comparison the two symbols denote different constructions.

29.12 What these examples establish

The examples exhibit three independent kinds of noncommutativity.

Quiver and matrix algebras have noncommutative transverse multiplication even when their chiral coefficient is free or commutative.

Internal enveloping algebras have noncommutative transverse multiplication generated by an internal Lie object. Affine current algebras supply chiral coefficients; a strictly ordered affine example requires the additional boundary, quiver, tensor, or endomorphism multiplication described above.

Yangians and line categories add meromorphic braid exchange to noncommutative fusion.

The ordered chiral language places all three in typed branches joined by explicit comparison functors. That separation is the condition under which their Koszul dualities can be stated correctly.

30 Arithmetic traces and automorphic forms

Arithmetic enters after a geometric object has been paired with a cycle, transported around a loop, or traced on a state space. The resulting numbers and functions retain the part of the chiral theory selected by that operation. Ordered holonomy produces iterated periods. Modular sewing produces characters, theta series, determinant functions, and automorphic forms. The source, cycle, and trace determine the arithmetic object.

30.1 Iterated periods and the KZ associator

On $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, the KZ equation is

$$\frac{dG}{dz} = \left(\frac{x}{z} + \frac{y}{z-1}\right) G.$$

Tangential base points at 0 and 1 define regularized fundamental solutions. Their comparison is the Drinfeld associator $\Phi_{\mathrm{KZ}}(x, y)$. Chen's expansion expresses its coefficients as regularized iterated integrals of dz/z and $dz/(1-z)$, hence as multiple zeta values.

The pentagon relation compares the five parenthesizations of four ordered factors. Geometrically, it is the flat-holonomy identity around the real associahedral chamber in $\overline{\mathcal{M}}_{0,5}$. The full complex boundary of $\overline{\mathcal{M}}_{0,5}$ has ten irreducible divisors; the real chamber selects the five divisors met by that ordering. Ordered configuration geometry therefore carries an arithmetic weight filtration given by iterated-integral length.

Passing to the Grothendieck ring retains fusion multiplicities. Retaining the braided tensor category preserves the associator and its periods. These two traces reveal different layers of the same representation theory.

30.2 Verlinde numbers

Let $\mathcal{C}$ be a modular tensor category with simple objects indexed by λ and modular matrix S. The modular functor assigns to a genus-g surface with labels $\lambda_1, \ldots, \lambda_n$ a vector space of dimension

$$\dim \mathbb{V}_g(\lambda_1, \ldots, \lambda_n) = \sum_{\mu} (S_{0\mu})^{2-2g-n} \prod_{i=1}^{n} S_{\lambda_i \mu}. \tag{177}$$

For $\widehat{\mathfrak{sl}}_2$ at level k and vacuum labels,

$$\dim \mathbb{V}_g = \left(\frac{k+2}{2}\right)^{g-1} \sum_{j=1}^{k+1} \sin\left(\frac{\pi j}{k+2}\right)^{2-2g}. \tag{178}$$

The cosecant power sum is a polynomial in $k+2$ whose coefficients contain Bernoulli numbers. Equation (178) is the numerical trace of the modular functor, obtained from the modular S-matrix and equivalent geometric quantizations of the moduli space of flat connections.

The ordered bar, the modular functor, and the Verlinde number occupy successive levels. A comparison from the bar to conformal blocks produces the state space; the categorical trace of sewing then produces its dimension.

30.3 Hodge integrals

The Hodge bundle $\mathbb{E}_g$ on $\overline{\mathcal{M}}_{g,1}$ has top Chern class λ_g. Faber and Pandharipande evaluate

$$\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2}\lambda_g \tag{179}$$

and prove the generating identity

$$1+\sum_{g\geq 1}\left(\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2}\lambda_g\right)t^{2g} = \frac{t/2}{\sin(t/2)}[33]. \tag{180}$$

A field theory reaches this series when its graph trace factors through the class $\psi_1^{2g-2}\lambda_g$. The comparison consists of a graph-weight morphism followed by integration over the stated moduli class. Different coefficient systems produce different stable-graph sums around the same tautological geometry.

30.4 Lattice theta series

For an even positive-definite lattice L, the degree-g Siegel theta series is

$$\Theta_L^{(g)}(\Omega) = \sum_{x_1,\ldots,x_g\in L} \exp\left(\pi i \sum_{a,b=1}^{g}(x_a,x_b)\Omega_{ab}\right). \tag{181}$$

For even unimodular L, this is a Siegel modular form of weight $\mathrm{rk}(L)/2$. The genus-g vacuum amplitude of the lattice VOA contains $\Theta_L^{(g)}$ multiplied by the oscillator determinant.

The theta trace sums the lattice zero modes. The cocycle governing exponential-field multiplication remains in the vertex-algebra coefficient and disappears from this particular scalar trace. For the two even unimodular lattices of rank sixteen, the theta series agree in degrees one, two, and three. Their degree-four difference is the Schottky form on Siegel space, and its restriction to the genus-four Jacobian locus vanishes [59, 66]. The ambient Siegel space and the Jacobian locus therefore supply two distinct detection problems.

30.5 Symmetric products and the DMVV lift

Let

$$\phi_M(\tau,z) = \sum_{m,\ell} c(m,\ell)q^m y^\ell$$

be the elliptic genus of a compact Calabi–Yau target M. Dijkgraaf–Moore–Verlinde–Verlinde prove

$$\sum_{N\geq 0} p^N \phi_{\mathrm{Sym}^N M}(\tau, z) = \prod_{n>0,\, m\geq 0,\, \ell\in\mathbb{Z}} \left(1 - p^n q^m y^\ell\right)^{-c(nm,\ell)} \text{[26]}. \tag{182}$$

Taking the logarithm converts the product into a sum of Hecke transforms. This is the second-quantized passage from connected single-particle data to a symmetric Fock space.

Cofree commutative coalgebras and connected-graph exponentials obey the same exponential law. A geometric identification with (182) is supplied by a functor from the sigma-model or BPS state category to the relevant factorisation coalgebra, together with equality of the corresponding traces.

30.6 Borcherds products

A weak Jacobi form with integral principal part determines, under the Borcherds lift, an automorphic product whose exponents are its Fourier coefficients. Gritsenko and Nikulin use this mechanism to construct automorphic corrections of Lorentzian Kac–Moody algebras [44]. In the standard normalization, the weight of the product equals one half of the constant Fourier coefficient of the input.

For the K3 elliptic genus, the multiplicative lift gives the Igusa cusp form Φ_{10}, and the symmetric-product generating function is expressed through $1/\Phi_{10}$. The automorphic weight, the VOA central charge, the lattice rank, and a Koszul deformation class are separate outputs of separate traces. A theorem relating any two of them consists of an explicit comparison of their source classes and normalizations.

30.7 Geometric quantum groups

Maulik and Okounkov construct Yangian actions from stable envelopes on Nakajima varieties; the ratio of stable envelopes in two chambers is an R-matrix [56]. Braverman–Finkelberg–Nakajima construct Coulomb-branch algebras, including shifted Yangians in quiver-gauge examples [10]. These constructions supply multiplication, coproduct or convolution, filtrations, and representation spaces from algebraic geometry.

A K3 surface supplies the Mukai lattice of signature $(4, 20)$, Hilbert-scheme Fock spaces, elliptic genera, and moduli spaces of sheaves. A Hall-to-chiral comparison becomes an equivalence after one constructs a factorisation functor preserving the product, completed coproduct, cyclic pairing, and exchange operator. The recognition criterion of Theorem 33.4 states these hypotheses as one recognition criterion.

30.8 Moonshine traces

The Monster action on $V^\natural$ gives McKay–Thompson series

$$T_g(\tau) = \mathrm{Tr}_{V^\natural}\left(g q^{L_0 - 1}\right).$$

Borcherds' proof of monstrous moonshine passes through a generalized Kac–Moody algebra and its denominator identity. The local chiral algebra supplies vertex operators and the graded Monster representation; the equivariant trace supplies the modular function. Keeping both structures produces an equivariant ordered chiral algebra together with a family of arithmetic trace maps.

30.9 Genus and the ambient moduli space

Genus one records characters, one-cycle monodromy, and ordinary theta series. Genus two introduces interacting periods and Siegel modular forms. Higher genus adds further cycle couplings. Every detection statement therefore has three coordinates:

function or section, ambient moduli space, restriction locus.

An L-value arises from a modular form after an integral representation, Mellin transform, or Rankin–Selberg unfolding identifies the relevant period. A graph amplitude reaches the same value through a proved equality of integrals.

30.10 Realization diagram

The arithmetic layer is organized by the sequence

$$\begin{gathered}\text{algebraic coefficient} \longrightarrow \text{de Rham connection} \longrightarrow \text{Betti local system} \\ \longrightarrow \text{period, trace, or determinant.}\end{gathered} \tag{183}$$

Chiral Koszul duality acts on the algebraic and factorisation objects. Riemann–Hilbert carries them to local systems. Parallel transport, integration, and trace produce the arithmetic functions. Compatibility of these arrows is the exact mathematical form of an arithmetic duality statement.

31 Quantum field theory in the same language

31.1 Local observables

A perturbative quantum field theory assigns a cochain complex $\mathrm{Obs}(U)$ to every open set U. Disjoint supports give multiplication maps

$$\mathrm{Obs}(U_1) \otimes \cdots \otimes \mathrm{Obs}(U_r) \longrightarrow \mathrm{Obs}(V), \qquad \bigsqcup_i U_i \subset V.$$

Descent and locality make Obs a factorization algebra [14, 15]. Its differential contains the equations of motion, gauge symmetry, and quantum corrections.

For a holomorphic theory on a complex curve, the factorization maps depend holomorphically on insertion points. Their short-distance expansion gives the chiral operator product. For a holomorphic–topological theory on $X \times M$, the holomorphic coordinate supplies the chiral coefficient and the locally constant directions in M supply little-disks operations. A boundary parallel to X inherits the corresponding ordered chiral algebra when the boundary condition preserves these structures.

31.2 Renormalized configurations

A Feynman graph Γ with vertex set I defines a differential form or distribution W_Γ on $\mathrm{Conf}_I(M)$. Collision diagonals describe ultraviolet singularities. Fulton–MacPherson or Axelrod–Singer compactification replaces each collision by a boundary screen recording relative positions. Renormalization constructs an extension

$$\widetilde{W}_\Gamma \in \mathcal{D}'(\mathrm{FM}_I(M))$$

whose boundary restrictions are local counterterm expressions.

The forest formula and the stratification of $\mathrm{FM}_I(M)$ share the same nested-forest incidence category. Their coefficient systems carry different information: $\widetilde{W}_\Gamma$ contains the propagator, gauge fixing, densities, and Lie-theoretic tensors; the chiral Chevalley complex contains diagonal-supported coefficient operations. Equality of the two chain models is supplied by a map matching every boundary asymptotic with the corresponding chiral operation.

31.3 Classical and quantum master equations

Let $(\mathcal{E}, Q, \omega)$ be a BV complex of fields, with odd Poisson bracket $\{-,-\}$. A classical interaction I satisfies

$$QI + \frac{1}{2}\{I, I\} = 0.$$

A renormalized quantum interaction satisfies

$$QI + \frac{1}{2}\{I, I\} + \hbar\Delta I = 0.$$

At order $\hbar^m$, the left side is a local cocycle in the BV deformation complex. Its cohomology class is the order-m anomaly; a counterterm is a cochain whose differential equals that cocycle.

A modular Feynman transform has the same algebraic form because its bracket joins two vertices and its loop operator joins two flags of one vertex. When renormalized Feynman weights define a morphism from the modular cooperad of compactified configurations to the BV endomorphism operad, Stokes' theorem carries the geometric boundary identity to the quantum master equation.

31.4 Three-dimensional Chern–Simons theory

Chern–Simons theory on a three-manifold with a chiral boundary polarization induces a Wess–Zumino–Witten boundary theory. Boundary currents satisfy the affine Kac–Moody OPE. In the abelian theory, the boundary Green kernel gives the current two-point coefficient directly. In the nonabelian theory, the one-loop shift $k \mapsto k + h^\vee$ appears in the effective coupling used by KZ transport and surgery formulas.

For compact simply connected gauge group and positive integral level, canonical quantization assigns conformal-block state spaces to surfaces. Wilson lines carry integrable representation labels. The Chern–Simons/WZW and Drinfeld–Kohno comparison theorems identify exchange with quantum-group braiding, while the ribbon trace gives Reshetikhin–Turaev link invariants [74, 52, 51, 63]. These identifications relate the boundary chiral algebra, surface state spaces, line category, and framing anomaly by explicit functors.

31.5 Four-dimensional Chern–Simons theory

Four-dimensional Chern–Simons theory is topological on an oriented surface Σ and holomorphic on a curve X. A Wilson line is supported on a curve in Σ and at a point of X; the point of X is the spectral parameter. Exchange of two lines produces the classical kernel $\Omega/(z_1 - z_2)$ in the rational theory. Isotopy of three lines gives the classical Yang–Baxter equation, and perturbative quantization gives the quantum equation.

Costello–Witten–Yamazaki compute rational, trigonometric, and elliptic exchange according to the meromorphic one-form on X, and they identify the rational line algebra with the completed Yangian in the perturbative anomaly-free sector [19, 20]. Transfer matrices arise by closing auxiliary lines. Their commutativity is the isotopy invariance of separated line networks.

31.6 BRST reduction and worldsheet ghosts

Gauge fixing introduces ghosts and a square-zero BRST operator Q. For a boundary algebra A, derivation compatibility gives the total bar differential

$$D = b_A + Q_{\mathrm{Bar}}, \qquad D^2 = 0.$$

The resulting spectral sequence compares the ordered bar after BRST reduction with the BRST cohomology of the ordered bar. Drinfeld–Sokolov reduction is the current-algebra example; its higher W-operations are the homotopy-transferred composites of currents, ghosts, and the contraction homotopy.

For the bosonic string, the (b, c) ghost system has central charge -26. A matter CFT with central charge 26 gives vanishing total Virasoro central charge. The state-space BRST operator also contains the normal-ordering constant, and its nilpotence fixes the standard intercept. Globally, the matter and ghost determinant lines tensor to a flat anomaly line.

31.7 Polyakov and Quillen geometry

For a free scalar on a Riemann surface,

$$Z[g] = (\det{}'\Delta_g)^{-1/2}$$

up to the zero-mode volume. Under a Weyl rescaling $g \mapsto e^{2\sigma} g$, zeta regularization gives the Polyakov action. A CFT of central charge c multiplies the universal local functional by c. The same c appears in the fourth-order pole of the stress tensor because both coefficients represent the Virasoro central extension.

Over a family of curves, the determinant is a line bundle with Quillen connection. The family index theorem computes its curvature. This constructs the bridge from the local stress-tensor cocycle to the Hodge class of the family. Affine level, ribbon-loop grading, curved Koszul presentation, and automorphic weight remain separately typed data and enter only through further comparison maps.

31.8 Brown–Henneaux and Cardy

Three-dimensional gravity with negative cosmological constant has asymptotic Virasoro symmetry

$$c = \frac{3\ell}{2G} \tag{184}$$

by the Brown–Henneaux theorem [12]. For a unitary modular-invariant CFT with vacuum dominance in the high-temperature channel, the Cardy formula gives the asymptotic density of states. Substitution of (184) reproduces the Bekenstein–Hawking entropy of the BTZ black hole in the semiclassical regime.

The protected algebraic data in this comparison are the boundary Virasoro algebra, its module spectrum, and modular covariance. A complete gravitational theory additionally supplies reality conditions, the Hilbert space, the integration cycle, and a bulk path integral. The bulk-to-centre map below isolates the local protected sector selected by the boundary condition.

31.9 Entanglement in two-dimensional CFT

For the vacuum of a unitary two-dimensional CFT on the line, the replica construction gives

$$S_A = \frac{c+\bar{c}}{6}\log\frac{L}{\epsilon} + \mathrm{const} \tag{185}$$

for an interval of length L and ultraviolet cutoff ϵ. Twist fields implement the cyclic replica boundary condition, and their conformal weights determine the logarithmic coefficient. In a parity-invariant theory $c = \bar{c}$, this coefficient is $c/3$.

Equation (185) is one trace of the stress-tensor anomaly. The full Rényi family is a collection of twist-field correlators, and the entanglement spectrum depends on the complete state space and operator product.

31.10 Bulk actions and derived centres

A boundary condition gives an action of bulk observables on boundary observables. Locality makes that action central and therefore produces

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord,ch}}(A). \tag{186}$$

The derived centre is the universal recipient of such central actions. A complete boundary condition is one for which β is an equivalence on the chosen protected local sector. Morita-equivalent boundary conditions then yield equivalent centres and hence equivalent protected bulk actions.

Costello and Gaiotto construct examples of this mechanism in twisted supersymmetric gauge theories, where boundary vertex algebras and protected bulk observables are compared directly [16]. A tensor factor whose boundary action is trivial lies in the fibre of (186); this fibre measures the protected bulk information omitted by the chosen boundary.

31.11 Celestial and twistor realizations

Celestial and twistor constructions map scattering observables to chiral fields on a complex curve. In the examples of Costello–Paquette, gauge-theory amplitudes arise from current-algebra correlators and their deformations [17]. The construction specifies the background, asymptotic states, and the transform from scattering data to chiral operators. Soft currents become particular fields, and OPE associativity constrains loop corrections in that image.

A gravitational realization follows the same pattern: a transform identifies a protected set of asymptotic observables with a chiral coefficient, and the resulting Ward identities determine the corresponding soft-current algebra. The transform and its image specify the precise relation between OPE coefficients and soft factors.

31.12 Calabi–Yau categories and topological strings

A Calabi–Yau A_∞ category has a cyclic pairing. Costello associates to suitable Calabi–Yau categories a two-dimensional topological conformal field theory whose closed states are Hochschild chains [18]. The Connes operator and cyclic pairing supply the BV structure, while ribbon graphs describe the open-string amplitudes.

A holomorphic field theory on a Calabi–Yau variety may also produce factorization algebras, Hall products, and BPS operator algebras. A chiral algebra on a curve emerges after a specified pushforward, boundary condition, compactification, or dimensional reduction. The functorial source theorem of Section 33 records the structures preserved by that passage.

31.13 Comparison diagram

The mathematical and physical models meet through explicit arrows:

$$\begin{tikzcd} \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) \arrow[r] \arrow[d, "\Phi_{\mathrm{an}}"'] & \text{algebraic factorisation coalgebra} \arrow[d, "\Phi_{\mathrm{obs}}"] \\ \text{renormalized collision complex} \arrow[r] & \mathrm{Obs}_{\partial}\,. \end{tikzcd} \tag{187}$$

The left vertical arrow is analytification or Riemann–Hilbert followed by the field-theoretic coefficient map. The right vertical arrow is the comparison of factorisation structures. When both are equivalences, the chiral and BV master equations are the same identity in two models. Equation (186) then supplies the independently typed bulk action.

32 The physical derivation of ordered chirality

32.1 The local experiment

Consider a cohomological quantum field theory on

$$X \times \mathbb{R}_t,$$

holomorphic along the complex curve X and topological along the oriented coordinate t. Choose local observables $\mathcal{O}_i(z_i, t_i)$ in the chamber

$$t_1 < t_2 < \cdots < t_n.$$

Topological descent identifies all configurations inside this chamber. Holomorphic dependence moves the z_i freely on the complement of their collision diagonals. Correlators are therefore locally constant in the ordered t-chamber and meromorphic or distributional in the z-variables.

Their short-distance expansion has the form

$$\mathcal{O}_1(z_1, t_1) \cdots \mathcal{O}_n(z_n, t_n) \sim \sum_{\alpha} C_\alpha(z_1, \ldots, z_n) \mathcal{O}_\alpha(z, t), \tag{188}$$

where C_α is a chiral distribution and the word of observables follows the orientation of the real line.

Principle 32.1 (Ordered OPE principle)**.** *Holomorphic collision determines the coefficient distribution. Topological descent determines the ordered associative word. Locality identifies the two descriptions through the interchange law.*

A formal coordinate expands the chiral distribution into modes. A chamber in $\mathrm{Conf}_n(\mathbb{R})$ expands the transverse factor into a time-ordered product. The invariant object preceding both presentations is an E_1 algebra in factorisable D-modules.

32.2 Descent and the E_1 operation

Let Q be the cohomological supercharge. Topological invariance along t is represented by a homotopy operator G_t satisfying

$$P_t = [Q, G_t].$$

For a Q-closed observable,

$$\partial_t \mathcal{O} = [Q, G_t \mathcal{O}].$$

Integration along a path inside an ordered chamber gives a chain homotopy between the endpoint insertions. Coherent descendants over compactified spaces of interval embeddings produce operations m_r. Stokes' theorem on associahedra gives the Stasheff identities among these operations.

Theorem 32.2 (Descent produces ordered multiplication)**.** *Let* Obs *be a factorization algebra on $X \times \mathbb{R}$ that is holomorphic in X and locally constant in the real direction. Restriction to product disks defines an E_1 algebra in holomorphic factorization observables on X. A coherent family of descent homotopies extending $P_t = [Q, G_t]$ gives a chain-level A_∞ model, and its product on Q-cohomology depends only on the ordered chamber.*

Proof. Locally constant factorization algebras on an oriented line are E_1 algebras. Applying this recognition theorem in the coefficient category of holomorphic factorization observables on X gives the first assertion. Integration of the descent forms over the cells of compactified interval configurations gives the higher products. Their boundary integrals are the signed sums of parenthesized composites, which are the Stasheff relations. □

32.3 The distributional coefficient

The coefficient C_α in (188) is a distribution with prescribed singular support. Ward identities give differential equations on the complement of the diagonals, and contact terms give distributions supported on the diagonals. A right D-module records both pieces in one object.

For a holomorphic current,

$$\bar{\partial} J = 0$$

on separated insertions, while distributionally

$$\bar{\partial}_z \frac{1}{z-w} = \pi \delta^{(2)}(z-w)$$

in the stated normalization. The principal part and the contact Ward identity are two images of the same distribution under the de Rham differential. The extension across the diagonal carries the complete family of higher principal parts and their derivatives.

32.4 Boundary conditions and open strings

Let the theory occupy $X \times \mathbb{R}_{\geq 0}$ and let b be a boundary condition at $t = 0$. Boundary operators form the endomorphism algebra of b. For a collection $\{b_i\}$, a boundary-changing field lies in

$$A_{ij} = \mathrm{RHom}(b_i, b_j),$$

and composition is

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

The real normal direction orders this open-string product, while the insertion point on X supplies the chiral coefficient.

A strip with boundary conditions b_i, b_j has state complex A_{ij}. Gluing strips composes states. A cyclic Calabi–Yau trace closes a strip into an annulus and identifies its state complex with Hochschild chains. Ribbon graphs then organize the perturbative open-string amplitudes.

32.5 Interval amplitudes

Place the augmentation boundary at the two ends of an interval. Factorization homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes^{L}_{A} \mathbf{1}.$$

A cellular subdivision gives the two-sided bar complex, whose differential fuses adjacent insertions. The construction depends only on the factorization product and interval excision.

When the field theory has an action functional and a gauge fixing, Feynman expansion supplies a second chain model. A comparison sends a bar word to its time-ordered amplitude. Homological perturbation and factorization excision prove that this map computes the same derived tensor product. The propagator enters the coefficient of the amplitude, while the bar differential remains the composition law of boundary operations.

32.6 Exchange from a second topological direction

Place the theory on $X \times \mathbb{R}^2$. Two parallel lines now admit an exchange path. Parallel transport gives

$$R_{12}(z_1, z_2) : L_{z_1} \otimes L_{z_2} \longrightarrow L_{z_2} \otimes L_{z_1}.$$

Isotopy of three lines gives the Yang–Baxter equation. Collision of the spectral points produces the meromorphic singularity in $z_1 - z_2$.

The little-disks index counts topological directions:

$$\begin{aligned} \mathbb{R}^1 &: \ E_1 \text{ fusion}, \\ \mathbb{R}^2 &: \ E_2 \text{ braid exchange}, \\ \mathbb{R}^3 &: \ E_3 \text{ higher linking operations}. \end{aligned}$$

The holomorphic curve remains a coefficient direction until a descent homotopy renders its translations topological.

32.7 Four-dimensional Chern–Simons theory

Let Σ be an oriented surface, X a complex curve carrying a meromorphic one-form ω, and $\mathfrak{g}$ a Lie algebra with invariant form. The action is

$$S[A] = \frac{1}{2\pi\hbar} \int_{\Sigma \times X} \omega \wedge \left(\frac{1}{2}(A, dA) + \frac{1}{6}(A, [A, A]) \right).$$

The theory is topological on Σ and holomorphic on X. A Wilson line lies on a curve in Σ and at a spectral point of X.

Tree-level exchange gives

$$r(z) = \frac{\Omega}{z}$$

in the rational case. Three-line isotopy gives the classical Yang–Baxter equation. Perturbative quantization gives an R-matrix, and the one-loop line-gauge anomaly determines the Yangian deformation of the rational current algebra [19, 20]. Trigonometric and elliptic choices of (X, ω) give the corresponding periodic exchange kernels. The spectral curve therefore selects the meromorphic quantum symmetry.

32.8 Three-dimensional gauge theory and boundary algebras

A holomorphic twist of a three-dimensional supersymmetric theory on $X \times \mathbb{R}_{\geq 0}$ may preserve a boundary supercharge. Boundary local operators form a chiral algebra on X. Boundary line junctions inherit an ordered multiplication from the normal direction, and bulk lines act by modules.

In a Lagrangian presentation, boundary fields and ghosts form a BRST complex. Quantum existence of the boundary condition is the vanishing of its gauge-anomaly class. Different complete boundary conditions can present Morita-equivalent ordered chiral algebras with the same protected bulk centre.

32.9 Drinfeld–Sokolov reduction

Affine currents coupled to ghosts implement a moment-map constraint. The BRST charge has the form

$$Q = \oint \left(\sum_a c^a (J_a - \chi_a) - \frac{1}{2} \sum_{a,b,c} f_{ab}{}^c : c^a c^b b_c : + \cdots \right).$$

Current-algebra Jacobi, invariance of the level cocycle, and the ghost terms imply $Q^2 = 0$. The reduced chiral algebra is the cohomology $H_Q(A)$.

Derivation compatibility extends Q to the ordered bar:

$$D = b_A + Q_{\mathrm{Bar}}, \qquad D^2 = 0.$$

The filtration by ghost or conformal weight yields the comparison spectral sequence. In the concentration range, its degree-zero edge map identifies the bar of the W-algebra with the BRST cohomology of the affine bar. The nonlinear W-operations are the transferred composites of current multiplication, ghost vertices, and the contraction homotopy.

32.10 Polyakov and Quillen

For a free scalar on a Riemann surface,

$$Z[g] = (\det{}'\Delta_g)^{-1/2}$$

up to the zero-mode volume. Under $g \mapsto e^{2\sigma} g$, zeta regularization gives

$$\log \frac{Z[e^{2\sigma} g]}{Z[g]} = \frac{1}{24\pi} \int \left(|\nabla\sigma|^2 + R\sigma \right) d\mathrm{vol}$$

with the sign fixed by the chosen Laplacian convention. A theory of central charge c multiplies this local functional by c. The fourth-order stress-tensor pole carries the same coefficient because it represents the same Virasoro central extension.

On a family of curves, the determinant forms a line with Quillen connection. The family index theorem computes its curvature and identifies its Hodge component. This is the precise path from the local Virasoro cocycle to a cohomology class on moduli. Affine level, ribbon-loop degree, curved Koszul presentation, and automorphic weight enter through their own coefficient systems.

32.11 Worldsheet sewing

Choose local coordinates z, w at two punctures and identify them by $zw = q$. Sewing inserts

$$q^{L_0 - c/24}$$

between dual states. The factor $q^{-c/24}$ is the cylinder vacuum energy, and contraction with the inverse two-point pairing sums the intermediate states. Analytic sewing gives convergence for $|q|$ in the plumbing domain.

A nonseparating plumbing raises the genus of the chiral curve. An annular trace raises the ribbon genus of the transverse E_1 theory. Their simultaneous occurrence gives the bidegree (g, h) of Section 24.

32.12 The Chern–Simons/WZW comparison

For compact simply connected gauge group at positive integral level, polarization and framing data connect five carriers:

$$\begin{array}{ccl} \text{current OPE} & : & \text{affine chiral } D\text{-module}, \\ \text{boundary composition} & : & E_1 \text{ algebra}, \\ \text{Wilson exchange} & : & E_2 \text{ line category}, \\ \text{surface state space} & : & \text{WZW conformal blocks}, \\ \text{framing phase} & : & \text{invertible anomaly theory}. \end{array}$$

Canonical quantization identifies the Chern–Simons surface state space with the WZW conformal-block space. Drinfeld–Kohno identifies the braid transport with quantum-group braiding. Reshetikhin–Turaev closure identifies the ribbon trace with link invariants. The framing dependence matches the ribbon twist and the projective anomaly of the conformal blocks.

32.13 Large N and two genera

For N coincident branes, matrix index contractions give a connected ribbon graph the weight N^{2-2h-b} after the standard 't Hooft normalization. A closed-string worldsheet of genus g carries the coupling g_s^{2g-2}. Before an open–closed transition, the amplitude has the natural expansion

$$\sum_{g,h,b} g_s^{2g-2} N^{2-2h-b} \mathcal{F}_{g,h,b}.$$

The pair (g, h) is the geometric bigrading of the bicoloured graph theory. An open–closed duality supplies a dynamical transformation between the two expansions and a relation among their parameters.

32.14 Holographic reconstruction

Let A be the ordered chiral algebra of a boundary condition. The central bulk action gives

$$\beta : \mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z_{\mathrm{ord,ch}}(A).$$

A complete boundary condition is defined by the equivalence of β on the selected protected local sector. A Morita context between two boundary algebras identifies their bimodule categories and transports this centre equivalence. A Koszul equivalence of augmentation-generated sectors supplies the corresponding local completion near the chosen boundary state.

The reconstructed object is the protected local bulk algebra. A complete physical theory further includes a Hilbert space, reality structure, global sectors, nonperturbative spectrum, and integration cycle.

32.15 The field-theoretic dual

The augmentation $A \to \mathbf{1}$ is a distinguished boundary or defect. Its derived endomorphism algebra

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$$

acts on local deformations of that defect. The bar

$$\mathbf{1} \otimes_A^L \mathbf{1}$$

is the interval state complex between two copies of the defect. Double centralization reconstructs the derived completion of the theory around the chosen augmentation. This is the field-theoretic meaning of augmentation Koszul duality.

32.16 Physical synthesis

A holomorphic–topological quantum field theory with product-generated protected observables, a factorisable D-module realization, coherent topological descent, and vanishing quantization anomaly produces an ordered chiral algebra. The complex curve supplies the diagonal coefficient; the oriented line supplies E_1 fusion; a second topological direction supplies exchange; a cyclic trace supplies ribbon loops; variation of the curve supplies conformal sewing; and each anomaly is the state of the invertible theory governing the corresponding gluing law.

33 Higher-dimensional sources and reduction to a curve

33.1 Factorisation pushforward

Let Y be a compact framed r-manifold, let X be a smooth complex curve, and write

$$p : Y \times X \times \mathbb{R}^q \longrightarrow X \times \mathbb{R}^q$$

for the projection. Let $\mathcal{F}$ be a factorisation algebra on the source, locally constant in the Y- and $\mathbb{R}^q$-directions and holomorphic in the X-direction. Its factorisation pushforward is defined on product opens by

$$(p_*^{\mathrm{fact}}\mathcal{F})(U \times V) := \int_{Y\times U\times V} \mathcal{F}. \tag{189}$$

The excision maps of factorisation homology supply the structure maps of the pushforward.

Theorem 33.1 (Topological pushforward to ordered chirality)**.** *Assume that factorisation homology for $\mathcal{F}$ satisfies framed excision and Fubini, that the pushforward is product-generated, and that its holomorphic coefficient has the regular-holonomic factorisable D-module realisation of Theorem 10.1. For $q = 1$, the object (189) determines an algebra*

$$p_*^{\mathrm{fact}}\mathcal{F} \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

More generally, q transverse locally constant directions determine an E_q-algebra in $\mathrm{FactD}(X)$*.*

Proof. Fubini identifies integration over a product open with iterated factorisation homology. Framed excision gives the gluing maps in the Y- and $\mathbb{R}^q$-directions. Holomorphic differential equations in X commute with the topological pushforward by the assumed D-module realisation. The little q-disks action in the locally constant directions therefore acts by morphisms of chiral factorisation objects. For $q = 1$, Theorem 10.2 identifies this action with an ordered chiral algebra. □

The theorem expresses dimensional reduction by one formula: factorisation homology integrates the compact topological directions, the residual little-disks operad records transverse composition, and the curve retains the chiral coefficient.

33.2 The dimensional rule

Let a locally constant factorisation algebra carry an E_m action in m framed topological directions. Dunn additivity and factorisation Fubini identify integration over a framed r-manifold, $r \leq m$, with an E_{m-r} object on the residual directions. In the presence of a holomorphic curve, the output has the form

$$\int_Y \mathcal{F} \in \mathrm{Alg}_{E_{m-r}}(\mathrm{FactD}(X)).$$

The pair (m, r) and the tangential structure of Y determine the surviving operad.

33.3 Calabi–Yau categories as factorisation sources

A smooth proper Calabi–Yau A_∞ category $\mathcal{C}$ carries a nondegenerate cyclic trace. Its morphism complexes form the open state spaces, Hochschild chains form the closed state complex, the Connes operator records circle rotation, and the cyclic pairing contracts boundary labels in surface sewing. A holomorphic family of such categories over X, or a holomorphic field theory with boundary category $\mathcal{C}$, upgrades these topological operations to chiral coefficients.

Theorem 33.2 (Categorical source theorem)**.** *Let $\mathcal{C}$ be a stable* $\mathrm{FactD}(X)$*-linear category with an E_1 composition direction, and let $b \in \mathcal{C}$ be a compact generator. Put*

$$A_b := \mathrm{RHom}_{\mathcal{C}}(b, b).$$

Then A_b is an E_1-algebra in $\mathrm{FactD}(X)$. *If the usual compact Morita hypotheses hold, the Yoneda functor is an equivalence*

$$\mathcal{C} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X)).$$

A Calabi–Yau trace on $\mathcal{C}$ induces the transverse cyclic structure of Theorem 24.1 on A_b.

Proof. Composition in the enriched category gives the multiplication on A_b, and the $\mathrm{FactD}(X)$-linearity makes multiplication a morphism of chiral coefficients. Compact generation identifies $\mathcal{C}$ with modules over the endomorphism algebra by enriched Morita theory. The cyclic trace evaluates cyclic strings of endomorphisms and is invariant under the E_1 composition, hence supplies the stated cyclic structure. □

The theorem separates the two pieces of input with geometric precision: the factorisation family produces chirality, while the Calabi–Yau trace produces cyclic sewing.

33.4 Naturality of deformation and obstruction classes

Let $\mathfrak{g}$ and $\mathfrak{h}$ be complete filtered L_∞ algebras and let

$$F : \mathfrak{g} \longrightarrow \mathfrak{h}$$

be a filtration-preserving L_∞ morphism. For a Maurer–Cartan element α, the twisted linear term

$$F_1^\alpha : (\mathfrak{g}, d_\alpha) \longrightarrow (\mathfrak{h}, d_{F_*(\alpha)})$$

is a chain map.

Proposition 33.3 (Transgression of obstructions)**.** *Let $\alpha_{\le r}$ solve the Maurer–Cartan equation modulo $F^{r+1}\mathfrak{g}$. Let*

$$o_{r+1}(\alpha) \in H^2\Big(\mathrm{gr}^{r+1}\,\mathfrak{g}, d_\alpha\Big)$$

be its extension obstruction. The obstruction of the transported solution $F_*(\alpha_{\le r})$ is

$$(F_1^\alpha)_* o_{r+1}(\alpha) \in H^2\Big(\mathrm{gr}^{r+1}\,\mathfrak{h}, d_{F_*(\alpha)}\Big)\,.$$

Proof. Take the filtration-$r+1$ component of the L_∞ identity applied to $\alpha_{\le r}$. Terms of lower filtration cancel because $\alpha_{\le r}$ satisfies the Maurer–Cartan equation to order r. The remaining term is the twisted linear image of the curvature representative. Passing to cohomology gives the formula. □

A geometric source therefore reaches chiral deformation theory through a morphism of controlling L_∞ algebras. The same morphism transports its Maurer–Cartan solutions, gauge equivalences, and successive obstruction classes.

33.5 Cohomological Hall algebras and their quantum doubles

Let $\mathcal{M}$ be a moduli stack of objects in an abelian or derived category. The correspondence of short exact sequences

$$\mathcal{M} \times \mathcal{M} \xleftarrow{p} \mathcal{E} \xrightarrow{q} \mathcal{M}$$

defines the Hall product $q_* p^*$ on a homology theory for which these operations exist. Ordered extension flags make this product associative and generally noncommutative.

Criterion 33.4 (Quantum-group recognition from Hall geometry)**.** *Suppose the Hall algebra is equipped with*

(i) a continuous coassociative coproduct compatible with multiplication;

(ii) a nondegenerate Hopf pairing and the associated double;

(iii) a PBW basis together with a presentation of the relation ideal;

(iv) a universal or categorical R-matrix satisfying the quasitriangular identities;

(v) a complete representation category on which all structure maps converge.

Then these data define a quasitriangular topological bialgebra; rigidity of the representation category supplies the antipode and hence a topological Hopf algebra. An identification with a named Yangian, quantum loop algebra, or Hall double follows from an isomorphism of the resulting presentation data.

The ordered chiral pushforward carries the Hall multiplication into $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$. Coproduct, pairing, double, and exchange descend through their own factorisation-compatible correspondences, so the criterion is checked map by map after pushforward.

33.6 Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the Braverman–Finkelberg–Nakajima convolution construction produces the quantized Coulomb branch. Type-A quiver examples identify these algebras with truncated shifted Yangians; the coweight shift records magnetic and boundary data. In a holomorphic–topological twist, the same convolution acts on line defects, and the residual holomorphic coordinate becomes the spectral parameter. Boundary reduction to X therefore yields a factorisation family of line modules whose ordered fusion is an E_1 algebra in $\mathrm{FactD}(X)$. The BFN comparison with a Drinfeld presentation is precisely the recognition morphism naming this algebra as a shifted Yangian.

33.7 K3 geometry and its comparison diagram

Let S be a K3 surface. Its Mukai lattice

$$\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

has signature $(4, 20)$. Nakajima correspondences act by a Heisenberg algebra on

$$\bigoplus_{n \geq 0} H^*(S^{[n]}).$$

The elliptic genera of symmetric powers satisfy the Dijkgraaf–Moore–Verlinde–Verlinde product formula, and the reciprocal Igusa cusp form is the second-quantized K3 elliptic genus.

Principle 33.5 (K3 comparison principle)**.** *K3 geometry supplies the structured chain*

$$\textit{Mukai lattice} \longrightarrow \textit{Nakajima–Hall correspondences} \longrightarrow \textit{line algebra},$$

$$\textit{K3 field theory on } S \times X \times \mathbb{R} \xrightarrow{\int_S} \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \xrightarrow{\mathrm{Tr}} \textit{elliptic genus or Siegel form.}$$

A common ordered chiral object is established by constructing comparison functors between the two lines and proving compatibility with multiplication, pairing, and trace.

The principle turns the K3 programme into a precise comparison theorem. A field theory on $S \times X \times \mathbb{R}$ produces the lower middle object by factorisation pushforward; Hall or BPS correspondences produce the upper line algebra; the equivalence of these E_1 objects carries their traces to the automorphic side.

33.8 Borcherds products and scalar traces

A Borcherds lift sends an admissible vector-valued modular or Jacobi form to an automorphic product. The constant term fixes the weight, and in the generalized Kac–Moody cases the product is a denominator identity. A categorical or chiral theory reaches this automorphic product through a trace, index, or partition function

$$\mathsf{Obj} \longmapsto K_0(\mathsf{Obj}) \longmapsto \mathrm{Tr}(q^{L_0} \cdots).$$

This composite retains the graded trace data. Multiplication, associators, extensions, and line exchange remain in the structured source and are compared through functors preceding the trace.

33.9 Omega backgrounds and BPS algebras

An Omega background makes a rotational symmetry equivariantly exact and localizes a protected sector to a noncommutative algebra. Equivariant weights become deformation parameters, while positions in a residual holomorphic curve become spectral parameters. The factorisation OPE of protected operators supplies the multiplication. Coupling the bulk sector to a defect and taking derived endomorphisms of the trivial defect supplies the augmentation-boundary Koszul dual. Geometric correspondences or representation-theoretic generators then provide the comparison that identifies the protected algebra with a named BPS algebra or Yangian.

33.10 Shifted symplectic sources

Let $\mathcal{M}$ be an n-shifted symplectic derived stack and let $L_0, L_1 \to \mathcal{M}$ be Lagrangian morphisms. Their derived intersection

$$L_0 \times^{\mathbb{R}}_{\mathcal{M}} L_1$$

is $(n-1)$-shifted symplectic and hence carries a P_n algebra of functions. An E_n quantisation of this P_n structure, varying factorisably over X, defines an E_n algebra in $\mathrm{FactD}(X)$. In the case $n = 1$, the intersection is 0-shifted symplectic and its quantisation is associative, producing the ordered chiral object of this paper.

This construction displays the two required stages: shifted symplectic geometry supplies the classical Poisson operation, and deformation quantisation supplies the associative factorisation algebra.

33.11 The source-to-chiral comparison

Suppose factorisation pushforward constructs

$$A_{\mathrm{geom}} \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)),$$

and a Hall, BPS, or localization construction produces

$$A_{\mathrm{src}} \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

An E_1-morphism compatible with chiral coefficients,

$$\phi : A_{\mathrm{src}} \longrightarrow A_{\mathrm{geom}},$$

induces the functorial bar square

$$\begin{tikzcd} A_{\mathrm{src}} \arrow[r, "\phi"] \arrow[d, "\mathrm{Bar}_X^{\mathrm{ord}}"'] & A_{\mathrm{geom}} \arrow[d, "\mathrm{Bar}_X^{\mathrm{ord}}"] \\ \mathrm{Bar}_X^{\mathrm{ord}}(A_{\mathrm{src}}) \arrow[r, "\mathrm{Bar}(\phi)"'] & \mathrm{Bar}_X^{\mathrm{ord}}(A_{\mathrm{geom}}). \end{tikzcd} \tag{190}$$

If ϕ is an equivalence on a complete sector, the lower map is an equivalence of conilpotent bar coalgebras on that sector. When the structures in Theorem 33.4 are also intertwined, ϕ identifies the corresponding line categories and their reconstructed quantum groups.

33.12 The higher-dimensional synthesis

Higher-dimensional geometry enters ordered chiral Koszul duality by three exact operations:

$$\begin{gathered} \text{factorisation pushforward,} \qquad \text{boundary endomorphisms,} \\ \text{morphisms of deformation complexes.} \end{gathered} \tag{191}$$

Cyclic traces create surface contractions. Hall correspondences create associative multiplication. Omega backgrounds create equivariant deformation parameters. Shifted symplectic forms create Poisson brackets, and quantisation creates associative operator products. The residual curve carries their holomorphic singularities and spectral dependence. Their common target is the category of E_1 algebras in factorisable D-modules.

34 The single language

34.1 The structure theorem

Theorem 34.1 (Noncommutative chiral Koszul geometry)**.** *Let X be a smooth complex algebraic curve and let*

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}).$$

The following constructions are canonical in their stated ambients.

(i) *The tensor-product operad* $E_1 \otimes \mathsf{Ch}_X$ *presents* A *on ordered sequences of finite chiral configurations. Chiral operations act inside each block, and the* E_1 *operation composes adjacent blocks.*

(ii) *The pointwise tensor and chiral convolution form the duoidal structure of Section 3. The multiplication of* A *is a morphism of chiral factorisation coalgebras. Hence the ordered bar is a coassociative coalgebra object in chiral factorisation coalgebras.*

(iii) *The normalized ordered bar is the interval object*

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1}.$$

Its simplicial differential multiplies adjacent transverse factors and applies the augmentation at the two ends. The complete chiral singularity remains in the coefficient object.

(iv) *For connected positive-weight objects, and for complete Mittag–Leffler limits of finite weight windows, internal associative bar and cobar are inverse on the augmentation-complete and conilpotent subcategories. The boundary algebra*

$$A^! := \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$$

acts on the augmentation boundary, and the double centralizer $A^{!!}$ *is the derived augmentation completion under the hypotheses of Section 7.*

(v) *A Koszul twisting equivalence identifies augmentation-complete* A*-modules with conilpotent comodules over* $C = \mathrm{Bar}_X^{\mathrm{ord}}(A)$*. Its two-sided form carries the diagonal bimodule to the diagonal bicomodule, relative tensor product to derived cotensor product, and Hochschild theories to their coHochschild models.*

(vi) *Francis–Gaitsgory cardinality pro-nilpotence applies to* $D(\mathrm{Ran}\, X)$ *with chiral convolution. In that monoidal category, enhanced operadic bar and cobar give the corresponding reduced associative equivalence, and chiral Chevalley chains give the chiral Lie–factorisation-coalgebra equivalence. A strong monoidal realization compares this Ran construction with the internal ordered bar whenever its closure and continuity hypotheses hold.*

(vii) *The Boardman–Vogt resolution of* $E_1 \otimes \mathsf{Ch}_X$ *is indexed by levelled forests with transverse and chiral edges. The two edge-contraction differentials anticommute, and their codimension-two faces encode associativity, chiral coherence, and interchange.*

(viii) *The normal filtration of each collision divisor records all singular OPE coefficients. The first normal symbol is logarithmic residue; the higher symbols are higher principal parts; the regular quotient carries the normally ordered term. On the formal disc, translation equivariance turns this filtered chiral operation into the full vertex operation, whose Maurer–Cartan equation is the integral Jacobi identity and hence the Borcherds identity.*

(ix) *An* E_2 *enhancement or a flat connection on two-dimensional configuration spaces supplies exchange. A rigid meromorphic line category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra. A quasi-monoidal fibre functor reconstructs a coquasi-bialgebra. Rigidity supplies the antipode, and a presentation comparison recognizes a named Yangian, quantum loop algebra, or elliptic quantum group.*

(x) *A square-zero BRST derivation extends factorwise to the ordered bar and anticommutes with its differential. The resulting bicomplex has the spectral sequence of Section 23; concentration of BRST cohomology yields the degree-zero Drinfeld–Sokolov comparison.*

(xi) *A transverse Calabi–Yau trace creates ribbon contractions. A chiral modular sewing action creates Deligne–Mumford clutching contractions. When both structures are supplied compatibly, the two-coloured completed graph transform carries the master equation (155), with independent ribbon and chiral genus gradings.*

(xii) *A perturbatively renormalized holomorphic–topological BV theory satisfying Theorem 10.5 produces A. Extension of graph weights to compactified configurations gives the mixed-forest action, and the BV quantum master equation gives its compatibility with boundary factorisation.*

Proof. Items (i)–(iii) are the operadic straightening, duoidal interchange, and interval-bar constructions of Part I. Items (iv)–(vi) are the internal bar–cobar, boundary double-centralizer, twisting, and Ran comparison theorems, each on its named complete subcategory. Items (vii)–(viii) follow from the mixed Boardman–Vogt resolution and the principal-part description of chiral operations. Item (ix) is braided Tannaka reconstruction followed by presentation recognition. Item (x) is the totalization of two anticommuting differentials. Item (xi) is the Feynman-transform identity for the two supplied sewing actions. Item (xii) is the factorisation form of perturbative BV quantization. Every construction is functorial in the maps defining its input data. ☐

34.2 The invariant object

The common source is

$$\boxed{A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}).} \tag{192}$$

Equivalently, A is an algebra over $E_1 \otimes \mathsf{Ch}_X$. On the regular-holonomic, product-generated analytic locus, it is the algebra of observables of a factorisation theory on $X^{\mathrm{an}} \times \mathbb{R}$ that is holomorphic along X and locally constant along $\mathbb{R}$.

Two elementary experiments generate this language:

$$\text{two observables collide on } X, \qquad \text{two intervals compose in } \mathbb{R}.$$

The collision experiment creates the chiral coefficient. The interval experiment creates noncommutative fusion. A second transverse direction creates exchange paths while preserving both primitive operations.

34.3 The functorial atlas

geometric datum	mathematical and physical construction
formal coordinate with translation	mode expansion and the full vertex operation
normal filtration of a diagonal	all polar modes, the regular term, and the Poisson-vertex associated graded
framed interval	ordered bar coalgebra and interval boundary amplitude
augmentation boundary	$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$ and derived completion
second transverse direction	exchange paths, braid monodromy, and Yang–Baxter flatness
rigid line category with fibre functor	Tannaka bialgebra or quasi-bialgebra and quantum-group recognition
Calabi–Yau trace	Hochschild chains, annular closure, and ribbon genus
family of stable curves	conformal blocks, plumbing, and chiral genus
conformal vector and determinant line	projective connection and conformal anomaly
BRST derivation	derived Hamiltonian reduction and W-algebra cohomology
BV propagator and compactified graph weights	perturbative amplitudes and the quantum master equation
higher-dimensional factorisation source	pushforward to an ordered chiral algebra
trace or index	characters, Verlinde numbers, theta series, and automorphic sections

The atlas orders the constructions by information content. Each row is a functor from the structured object in the preceding rows. Numerical invariants occur after a trace, determinant, rank, or index has been chosen.

34.4 Local geometry and global transport

The complete OPE is the filtered diagonal distribution carried by the chiral D-module. Arnold's relation governs the differential forms on three-point configuration space. The coefficient Jacobi or Borcherds identity governs the chiral operation. Pairing these two identities gives the corresponding square-zero configuration-space differential.

The Fulton–MacPherson face category resolves collision incidence. Its screen spaces carry the relative geometry of a collision and contribute their own cohomology. The interval bar uses the pointwise coefficient product, while the Francis–Gaitsgory chiral bar uses chiral convolution. A strong monoidal comparison relates them on a common complete subcategory.

One transverse direction supplies fusion. Two transverse directions supply exchange. Tannaka reconstruction turns tensor-compatible exchange into a bialgebra. Stable-curve sewing turns clutching maps into a modular master equation. An invertible anomaly theory records projective curvature. The factorisation centre receives every central bulk action through its universal map.

34.5 The mathematical and physical realizations

Mathematically, (192) combines the Ran geometry of diagonals with the operadic geometry of intervals. Two-coloured forests resolve their simultaneous homotopies. Bar and cobar resolve the augmentation boundary. Hochschild cochains encode central operations. Configuration-space transport encodes exchange, and modular operads encode sewing.

Physically, the same object is the protected local algebra of a holomorphic–topological system. Chiral collision is the short-distance OPE. The E_1 product is ordered defect fusion. The interval bar is evolution between augmentation boundary states. The centre is the universal recipient of bulk observables acting centrally on the boundary. Braiding is physical exchange in a transverse plane. Ribbon contraction closes an open line, and Deligne–Mumford contraction plumbs a complex worldsheet. The BV master equation states the compatibility of renormalized amplitudes with every compactification boundary.

A factorisation-observable comparison identifies these realizations. Its source is the algebraic construction, its target is the renormalized observable theory, and its compatibility with multiplication, collision, and gluing is the content of the physical equivalence.

34.6 The hierarchy of scalarisation

Corollary 34.2 (Functorial scalarisation)**.** *Every scalar invariant considered in this paper factors through a specified trace-type functor. Typical examples have the form*

$$A \longmapsto \int_\Sigma A \longmapsto K_0\Big(\int_\Sigma A\Big) \longmapsto \mathrm{Tr}(q^{L_0}\cdots) \in \mathbb{C},$$

or

$$A \longmapsto \mathbb{V}_{g,n}(A) \longmapsto \det \mathbb{V}_{g,n}(A) \longmapsto c_1(\det \mathbb{V}_{g,n}(A))\,.$$

An equality of scalar invariants is therefore the image of an equality or comparison at an earlier structured stage.

Proof. Characters and partition functions are traces on state spaces. Verlinde numbers are ranks of conformal-block bundles. Hodge and anomaly classes are characteristic classes of determinant lines. Theta and Borcherds products are graded traces or automorphic lifts of specified coefficient data. Each construction is functorial, so every scalar identity is obtained by applying its defining functor to the corresponding structured map. □

The resulting hierarchy retains the full object at its source and places every numerical shadow at the end of an explicit geometric path.

Coda: the geometry remembers

A quantum field carries its collision law. On a complex curve, that law is a D-module operation supported on a diagonal. Its normal filtration records the complete singular expansion, and its regular quotient records the composite field.

A noncommutative product carries an oriented interval. Composition of adjacent intervals is the E_1 multiplication. The interval with augmentation conditions at its endpoints is the bar construction, and its cuts form the deconcatenation coproduct.

Exchange carries a plane. Paths in the configuration space of the plane produce braid monodromy. A tensor-compatible family of such paths forms a braided line category, and Tannaka reconstruction gathers all of its matrix coefficients into a bialgebra.

Trace carries a circle. Closing an interval produces Hochschild chains. The cyclic pairing thickens compositional graphs into ribbon surfaces and assigns amplitudes to their gluings.

Chiral genus carries a stable complex curve. Plumbing coordinates join punctures, conformal blocks propagate intermediate states, and clutching strata compose the resulting operations. Their oriented codimension-two incidence is the modular master equation.

An anomaly carries an invertible field theory. Its state on a family is a line with connection, and its curvature is the local index class. Evaluation on a surface and choice of normalization produce the familiar numerical coefficient.

Koszul duality carries a boundary. The interval bar records resolved states between augmentation boundaries. The algebra $A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$ records endomorphisms of that boundary. On the Koszul–Morita locus, the equivalence of boundary categories transports their centres and deformation theories.

Factorisation is the common grammar. Disjoint regions multiply. Diagonals carry singular products. Cuts produce relative tensor products. Gluings produce cotensor products and traces. Motion produces monodromy. Degeneration produces master equations.

The governing equations arise from the smallest spaces that carry their geometry. The interval carries associativity. Three colliding points carry chiral Jacobi coherence. Three moving points in a plane carry Yang–Baxter flatness. Compactified graphs carry the Batalin–Vilkovisky equation. Stable curves carry modular sewing.

The theory has one invariant source and many exact realizations:

$$\boxed{A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X)).} \tag{193}$$

The diagonal supplies chirality. The interval supplies noncommutativity. The plane supplies exchange. The circle supplies trace. The stable curve supplies sewing. The bar supplies resolution. The centre supplies universal bulk action. The determinant line supplies anomaly. The BV complex supplies quantum consistency.

Together these geometries form one language for ordered quantum fields: local enough to compute, global enough to sew, and structured enough to preserve every operation carried by the theory.

A Orientation lines and the collision differential

The square-zero identities used throughout the paper are identities of oriented correspondences. This appendix fixes one convention and proves the two cancellations on which the collision complex and the corolla-normalized forest resolution rest. A coherent change of convention conjugates every differential by diagonal signs and yields an isomorphic complex.

A.1 Cohomological suspension and unordered tensor products

Let (K, d_K) be a cochain complex. We write $sK = K[1]$ in the cohomological convention

$$(sK)^n = K^{n+1}, \qquad |sx| = |x| - 1, \qquad d_{sK}(sx) = -s(d_K x).$$

For a finite set S, the unordered tensor product is

$$\bigotimes_{s\in S}^{\text{unord}} K_s := \left(\bigoplus_{\alpha:\{1,\ldots,|S|\}\xrightarrow{\sim} S} K_{\alpha(1)} \otimes \cdots \otimes K_{\alpha(|S|)} \right)_{\Sigma_{|S|}}, \tag{194}$$

where a permutation acts by the Koszul symmetry. An ordering of S identifies (194) with an ordinary tensor product; the coinvariant construction makes the resulting object independent of that temporary ordering.

Let $\mu : \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ be a degree-zero graded Lie bracket. Its suspended form

$$\bar{\mu} : s\mathfrak{g} \otimes s\mathfrak{g} \longrightarrow s\mathfrak{g}$$

has cohomological degree $+1$ and is defined by

$$\bar{\mu}(sx, sy) = (-1)^{|x|} s[x, y]. \tag{195}$$

With this convention, graded antisymmetry of $[-,-]$ becomes graded symmetry of $\bar{\mu}$ with respect to the shifted degrees:

$$\bar{\mu}(u, v) = (-1)^{|u||v|} \bar{\mu}(v, u).$$

The Jacobi identity becomes

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)} \bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)} \bar{\mu}(\bar{\mu}(w, u), v) = 0. \tag{196}$$

The same formulas hold for a chiral Lie bracket after the tensor products are replaced by chiral products and the outputs are pushed forward to the common small diagonal.

A.2 Determinant lines of collision chains

Let I be finite and let $\Pi(I)$ be its partition lattice, ordered by refinement. A cover $\pi \lessdot \rho$ merges exactly two blocks. Write $e_{\pi,\rho}$ for the one-dimensional conormal direction of that merger. For a saturated chain

$$\gamma = (\pi_0 \lessdot \pi_1 \lessdot \cdots \lessdot \pi_r)$$

define

$$\mathrm{or}(\gamma) = \bigwedge^r \left(e_{\pi_0,\pi_1} \oplus \cdots \oplus e_{\pi_{r-1},\pi_r} \right).$$

Deleting the j-th edge gives the simplicial boundary sign $(-1)^{j-1}$. Interchanging two independent collision directions reverses the determinant orientation.

For a partition π, let

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!}\left(\bigotimes_{B\in\pi}^{\text{unord}} s\mathfrak{g}\right).$$

If $\pi \lessdot \rho$ merges blocks B and C, the map $d_{\pi,\rho} : C_\pi(\mathfrak{g}) \to C_\rho(\mathfrak{g})$ is the identity on the other blocks and $\bar{\mu}$ on the B, C factors, tensored with the oriented conormal line. The total collision differential is

$$d_{\text{coll}} = d_{\mathfrak{g}} + \sum_{\pi \lessdot \rho} d_{\pi,\rho}. \tag{197}$$

Lemma A.1 (Internal–collision cancellation)**.** *If $d_{\mathfrak{g}}$ is a derivation of the chiral bracket, then*

$$d_{\mathfrak{g}} d^{(1)}_{\text{coll}} + d^{(1)}_{\text{coll}} d_{\mathfrak{g}} = 0,$$

where $d^{(1)}_{\text{coll}}$ is the sum over covers.

Proof. On the two merged factors this is the suspended form of

$$d[x, y] = [dx, y] + (-1)^{|x|}[x, dy].$$

The identity $d(sx) = -s(dx)$ gives the sign required when the degree-+1 map $\bar{\mu}$ is interchanged with the unary differential. Untouched factors contribute the usual tensor-product signs. Because the operation is defined on the unordered tensor product, the calculation is independent of the temporary ordering used to write it. ☐

A.3 Rank-two intervals in the partition lattice

Every coefficient of $(d^{(1)}_{\text{coll}})^2$ is supported on a rank-two interval $[\pi, \sigma]$ of $\Pi(I)$. Such an interval has exactly one of two forms: two disjoint pairs of blocks are merged, or three blocks are merged into one.

Proposition A.2 (Disjoint-pair cancellation)**.** *Let B, C, D, E be four distinct blocks of π. The two composites that merge B with C and D with E in the two possible orders sum to zero.*

Proof. The two coefficient operations act on disjoint factors. The transposition of the two odd collision directions can be recorded either as the interchange sign of the two degree-one maps or as the reversal

$$e_{BC} \wedge e_{DE} = -e_{DE} \wedge e_{BC}$$

of the determinant line. The interchange sign and the determinant reversal are two presentations of the same transposition. In the total oriented convention the two paths induce opposite maps to the common codimension-two component. ☐

Proposition A.3 (Triple-collision coefficient)**.** *Let B, C, D be three blocks of π, and let σ be obtained by replacing them with $B \cup C \cup D$. The component $C_\pi(\mathfrak{g}) \to C_\sigma(\mathfrak{g})$ of $(d^{(1)}_{\text{coll}})^2$ is the three-term suspended chiral Jacobiator, equivalently one half of the Nijenhuis–Richardson bracket $[\mu, \mu]$.*

Proof. The three intermediate partitions are obtained by first merging B, C, first merging C, D, or first merging D, B. For a temporary ordering u, v, w of the suspended factors, the oriented sum is

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)}\bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)}\bar{\mu}(\bar{\mu}(w, u), v),$$

which is (196). The factor 1/2 is the conventional conversion between the square of an odd coderivation and one half of its graded commutator with itself. □

Theorem A.4 (Square of the collision differential)**.** *For a graded chiral antisymmetric operation μ compatible with $d_{\mathfrak{g}}$,*

$$d_{\mathrm{coll}}^2 = \frac{1}{2} d_{[\mu,\mu]}.$$

Consequently $d_{\mathrm{coll}}^2 = 0$ if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal square is zero and the mixed terms vanish by Theorem A.1. Disjoint mergers cancel by Theorem A.2; triple mergers give the Jacobiator by Theorem A.3. Conversely, the component from the discrete three-block partition to the one-block partition is precisely the Jacobiator. □

A.4 Signs in the corolla-normalized forest resolution

For a nonempty block B, let $K(B) = N_*(N\,\mathrm{Tree}(B))$ be the normalized nerve complex from Section 14. Its vertical differential ∂_B is the alternating simplicial boundary and satisfies $\partial_B^2 = 0$. The augmentation and terminal-corolla section

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C} \xrightarrow{\iota_B} K(B)$$

are chain maps. For disjoint blocks B, C, set

$$m_{B,C} = \iota_{B\cup C}(\epsilon_B \otimes \epsilon_C) : K(B) \otimes K(C) \longrightarrow K(B \cup C). \tag{198}$$

The maps $m_{B,C}$ are symmetric and strictly associative, and commute with the vertical differentials.

The total complex

$$W_I^{\mathrm{FM}}(\mathfrak{g}) = \bigoplus_{\pi\in\Pi(I)} C_\pi(\mathfrak{g}) \otimes \bigotimes_{B\in\pi}^{\mathrm{unord}} K(B)$$

has vertical differential $\partial = \sum_B \partial_B$ and horizontal differential obtained by tensoring $d_{\pi,\rho}$ with $m_{B,C}$. The mixed commutator vanishes because (198) is a chain map. Strict associativity of the $m_{B,C}$ reduces the horizontal square to the partition-lattice calculation above. Hence the total square is zero exactly when the chiral Jacobi identity holds.

Vertical contractions resolve a rooted tree inside a fixed block. Horizontal collision maps merge blocks through the partition differential and insert the terminal corolla of the new block. These two operations define the vertical and horizontal directions of the bicomplex.

A.5 Residues and coorientations

Let Y be smooth and $D_1, \ldots, D_r$ a normal-crossing divisor. A local logarithmic form can be written

$$\alpha = \sum_{J\subseteq\{1,\ldots,r\}} \left(\bigwedge_{j\in J} \frac{dt_j}{t_j} \right) \wedge \alpha_J.$$

The iterated residue along an ordered tuple is the corresponding coefficient. Therefore

$$\mathrm{Res}_{D_i} \mathrm{Res}_{D_j} = -\,\mathrm{Res}_{D_j} \mathrm{Res}_{D_i} \tag{199}$$

when the residue maps include their conormal orientation lines. This concerns the logarithmic conormal factors. Higher OPE poles are first paired with normal jets, as in Section 15, and then subjected to the same residue orientation law.

A.6 Stable graphs

For a stable graph Γ, set $\mathrm{or}(\Gamma) = \det \mathbb{R}^{E(\Gamma)}$. A separating clutching inserts an edge joining two vertices; a nonseparating clutching inserts a loop. Ordering the newly inserted edges gives the graph differential. The determinant-line calculation proves that two successive node-forming operations anticommute. The automorphism group of Γ acts on $\mathrm{or}(\Gamma)$, and the modular convolution complex takes coinvariants with this twist.

In a BV presentation, joining different vertices gives the bracket and joining two flags of one vertex gives the loop operator Δ. With the standard modular-operad determinant convention, the codimension-two relations give

$$d^2 = 0, \qquad d\Delta + \Delta d = 0, \qquad \Delta^2 = 0,$$

together with the second-order identity expressing the bracket as the deviation of Δ from being a derivation. These equations are determinant-line orientation identities for two node-forming parameters.

B Low-arity calculations

Low arity displays every operation in its smallest geometric carrier. This appendix computes the collision complex in arities two, three, and four, identifies the Arnold and infinitesimal-braid cancellations, and records the first pole symbols, BRST descent maps, and modular sewings.

B.1 Two labels

Let $I = \{1, 2\}$. The partition lattice has two elements,

$$1|2 < 12.$$

For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is

$$\Delta_{1|2,!}(s\mathfrak{g} \boxtimes s\mathfrak{g}) \xrightarrow{\bar{\mu}} \Delta_{12,!}(s\mathfrak{g}). \tag{200}$$

Since $\Delta_{1|2}$ is the identity of X^2 and Δ_{12} is the diagonal, the map is exactly the chiral bracket

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

written after suspension. Arity two records the binary operation and its transposition law. The first quadratic coherence appears in arity three.

On a formal disc, a translation-invariant binary chiral operation is represented by

$$\{a_\lambda b\} + :ab:, \qquad \{a_\lambda b\} = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} b.$$

The summand $a_{(n)}b$ is the coefficient of the normal principal part of order $n+1$. Thus the arity-two chain already contains the full singular OPE and the regular product.

Example B.1 (Heisenberg)**.** For the rank-one Heisenberg chiral algebra with field J and

$$J(z)J(w) \sim \frac{k}{(z-w)^2},$$

the singular operation is the second normal symbol $k\,N^{\otimes 2}_{\Delta/X^2}$. Pairing this symbol with the dual second normal jet and then applying logarithmic residue extracts the coefficient k.

Example B.2 (Affine currents)**.** For a finite-dimensional Lie algebra $\mathfrak{g}$ with invariant form $(,)$,

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}.$$

The first normal symbol is the Lie bracket and the second is the invariant form multiplied by the level. They have different tensorial weights and satisfy different identities: Jacobi for the first, invariance for the second.

B.2 Three labels and the Jacobiator

For $I = \{1,2,3\}$, the partition lattice is

$$1|2|3 < \{12|3, 13|2, 23|1\} < 123.$$

Suppressing the diagonal pushforwards, the collision complex has the form

$$(s\mathfrak{g})^{\otimes 3}_{\Sigma_0} \xrightarrow{d_1} \left((s\mathfrak{g})^{\otimes 2}\right)^{\oplus 3} \xrightarrow{d_2} s\mathfrak{g}. \tag{201}$$

The labels remain fixed throughout (201); the subscript Σ_0 records this labelled convention. For homogeneous $u, v, w \in s\mathfrak{g}$,

$$d_1(u \otimes v \otimes w) = \begin{pmatrix} \bar{\mu}(u,v) \otimes w \\ (-1)^{|v||w|}\bar{\mu}(u,w) \otimes v \\ (-1)^{|u|(|v|+|w|)}\bar{\mu}(v,w) \otimes u \end{pmatrix},$$

where the three rows correspond to $12|3$, $13|2$, and $23|1$. With boundary orientations chosen cyclically,

$$d_2(x_{12|3}, x_{13|2}, x_{23|1}) = \bar{\mu}(x_{12}, x_3) - \bar{\mu}(x_{13}, x_2) + \bar{\mu}(x_{23}, x_1),$$

with the Koszul permutations implicit. Multiplication gives the suspended Jacobiator. This is the smallest calculation in which the square of the collision differential contains information.

B.3 Arnold's relation

On $\mathrm{Conf}_3(\mathbb{C})$ put

$$\omega_{ij} = d\log(z_i - z_j).$$

Set $x = z_1 - z_2$ and $y = z_2 - z_3$, so that $z_1 - z_3 = x + y$. Then

$$\omega_{12} = \frac{dx}{x}, \qquad \omega_{23} = \frac{dy}{y}, \qquad \omega_{13} = \frac{dx + dy}{x + y}.$$

A direct calculation gives

$$\begin{aligned}\omega_{12} \wedge \omega_{23} - \omega_{12} \wedge \omega_{13} - \omega_{13} \wedge \omega_{23} &= \left(\frac{1}{xy} - \frac{1}{x(x+y)} - \frac{1}{y(x+y)}\right) dx \wedge dy \\ &= 0.\end{aligned}$$

Equivalently,

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0. \tag{202}$$

Relation (202) lies in the coefficient algebra of configuration-space forms. The Jacobi identity lies in the algebra of coefficient operations. A flat connection pairs the two by tensoring ω_{ij} with endomorphisms satisfying the infinitesimal braid relations.

B.4 The KZ curvature in arity three

Let V_1, V_2, V_3 be $\mathfrak{g}$-modules and let Ω_{ij} denote the invariant Casimir acting in the i-th and j-th factors. Put

$$A = \hbar(\Omega_{12}\omega_{12} + \Omega_{13}\omega_{13} + \Omega_{23}\omega_{23}), \qquad \nabla = d - A.$$

Since $d\omega_{ij} = 0$, the curvature is $F_\nabla = A \wedge A$. The terms with identical pairs vanish. After collecting coefficients,

$$\begin{aligned}\frac{1}{\hbar^2} F_\nabla = [\Omega_{12}, \Omega_{23}]\, \omega_{12} \wedge \omega_{23} + [\Omega_{12}, \Omega_{13}]\, \omega_{12} \wedge \omega_{13} \\ + [\Omega_{13}, \Omega_{23}]\, \omega_{13} \wedge \omega_{23}.\end{aligned} \tag{203}$$

Invariance of the Casimir gives

$$[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0, \qquad [\Omega_{13}, \Omega_{12} + \Omega_{23}] = 0.$$

Hence the three commutators in (203) are the same up to signs, and the curvature is that common commutator multiplied by (202). Therefore $F_\nabla = 0$.

This calculation displays the division of labor:

$$\text{Arnold relation} \quad \times \quad \text{invariance and Jacobi of } \mathfrak{g} \quad = \quad \text{flatness of KZ.}$$

The geometric factor governs forms, the algebraic factor governs endomorphisms, and their tensor product gives flatness.

B.5 The full vertex identity in arity three

For a vertex algebra, the coefficient operation contains the complete family of modes $a_{(n)}b$ and the regular product. Write

$$Y(a, z)Y(b, w) \sim \sum_{n \geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}.$$

The equality between the expansions in the regions $|z_1 - z_2| < |z_2 - z_3|$ and $|z_2 - z_3| < |z_1 - z_3|$ is the formal-distribution Jacobi identity. Multiplying by powers of the differences and comparing coefficients gives the Borcherds identity

$$\begin{aligned}\sum_{i\geq 0}\binom{m}{i}(a_{(n+i)}b)_{(m+k-i)}c = \sum_{i\geq 0}(-1)^i\binom{n}{i}\Big(&a_{(m+n-i)}b_{(k+i)}c \\ &- (-1)^n b_{(n+k-i)}a_{(m+i)}c\Big),\end{aligned} \tag{204}$$

with the usual parity modification in the super case. The binomial coefficients come from the expansion of translated formal variables. The chiral operad encodes (204) as a single Maurer–Cartan equation. Configuration-space forms realize the domains and boundary combinatorics; the chiral operation supplies the mode coefficients and their translation laws.

B.6 Four labels

The partition lattice of four labels has ranks $0, 1, 2, 3$. The rank-one elements are the six partitions with one pair and two singletons. Rank two has two species:

$$12|34 \quad \text{and its two analogues,} \qquad 123|4 \quad \text{and its three analogues.}$$

Rank three is 1234. The two species of rank-two partitions separate the two mechanisms in the square of the differential:

- The interval from $1|2|3|4$ to $12|34$ contains two paths. Their cancellation is the disjoint-pair sign.
- The interval from $1|2|3|4$ to $123|4$ contains three paths. Their sum is the Jacobiator on the first three factors, tensored with the identity on the fourth.

The next differential, from rank two to rank three, tests compatibility of Jacobi with a fourth insertion. The same chiral Jacobi identity controls the next differential because the collision complex is the Chevalley complex of the chiral Lie algebra. In the Fulton–MacPherson resolution, however, the rank-three geometry has nontrivial screens. Their cohomology contributes additional form classes, while the algebraic Chevalley differential continues to act on the coefficient factors.

B.7 The screen of three points

Consider three points on a complex tangent line modulo translation and dilation. Fix the first two at 0 and 1; the third gives a coordinate

$$u \in \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

A compactification is $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, with boundary at $0, 1, \infty$. Its open homotopy type is a wedge of two circles, and its cohomology is

$$H^0 \cong \mathbb{C}, \qquad H^1 \cong \mathbb{C}\, d\log u \oplus \mathbb{C}\, d\log(1-u).$$

The relation among the three boundary residues is the global residue theorem. The combinatorial forest category over the triple collision does have a terminal corolla and therefore contractible nerve.

This explicit example proves why the incidence resolution and the geometric screen complex must be distinguished.

B.8 Virasoro pole symbols

The Virasoro OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{205}$$

Let D be the pair diagonal. The three singular terms of (205) lie in

$$N_{D/X^2}^{\otimes 4}, \qquad N_{D/X^2}^{\otimes 2} \otimes \langle T \rangle, \qquad N_{D/X^2} \otimes \langle \partial T \rangle.$$

The regular term lies in filtration degree zero. A change of coordinate acts triangularly on this filtered object; the Schwarzian term records the extension by the fourth-order central symbol. The full filtered operation retains the three singular symbols of (205), their coordinate-change extensions, and the regular term.

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity for this polynomial is equivalent to the Virasoro Lie conformal algebra identity. The normally ordered product is then part of the corresponding vertex algebra operation. This gives a direct formal-disc test of the chiral selection theorem.

B.9 A BRST bicomplex

Let (A, d_A) be a dg chiral algebra and let $Q : A \to A$ be a degree-one square-zero derivation anticommuting with d_A:

$$Q^2 = 0, \qquad d_A Q + Q d_A = 0, \qquad Q\mu = \mu(Q \otimes 1 + 1 \otimes Q).$$

On the collision complex, Q acts on every coefficient factor. Denote this extension by Q_{coll}. The derivation identity gives

$$Q_{\mathrm{coll}} d_{\mathrm{coll}} + d_{\mathrm{coll}} Q_{\mathrm{coll}} = 0.$$

Hence

$$D = d_A + d_{\mathrm{coll}} + Q_{\mathrm{coll}}$$

satisfies $D^2 = 0$. Filter by ghost number. If the spectral sequence is bounded below and its first page is concentrated in ghost degree zero, then

$$H^0_Q(C^{\mathrm{coll}}(A)) \cong C^{\mathrm{coll}}(H^0_Q(A)).$$

This is the low-arity mechanism behind bar–BRST interchange. In Drinfeld–Sokolov reduction the complex is infinite and filtered, so convergence must be proved from the good grading and PBW filtration; the algebraic identity above remains exact at every finite stage.

B.10 The first modular sewings

The stable pairs with smallest complexity are $(0, 3)$ and $(1, 1)$. The space $\overline{\mathcal{M}}_{0,3}$ is a point, so a cyclic three-point operation is a tensor

$$\theta_{0,3} \in (V^{\otimes 3})^{\Sigma_3}$$

with the appropriate degree and determinant twist. Contracting two legs with the inverse pairing produces a genus-one one-point tensor

$$\Delta\theta_{0,3} \in V.$$

The complex compactification $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ has three boundary divisors, and the global residue theorem relates their logarithmic residues. A real associahedral interval inside $\overline{\mathcal{M}}_{0,4}(\mathbb{R})$ has two endpoints; Stokes' theorem on that interval gives equality between the two adjacent parenthesizations. Analytic continuation, or the symmetric operadic action, relates the three complex channels. With coefficient tensors inserted, these statements give the associativity or Jacobi relation appropriate to the chosen operad.

A basic codimension-two modular relation occurs when a loop contraction and a separating contraction meet. Its algebraic image is the BV second-order identity. For an algebra over the modular operad, all contractions are defined and the relation follows from the operadic differential. A chiral algebra enters this theorem after a nondegenerate cyclic pairing and convergent sewing maps have been constructed.

B.11 The free boson determinant

Let q satisfy $|q| < 1$. On the one-particle space of a chiral boson, sewing a cylinder acts diagonally with eigenvalues q^n, $n \geq 1$. Its Fredholm determinant is

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

Second quantization gives the oscillator partition function

$$\mathrm{Tr}_{\mathrm{Fock}}\, q^{L_0 - 1/24} = q^{-1/24} \prod_{n \geq 1} (1 - q^n)^{-1} = \eta(\tau)^{-1}, \qquad q = e^{2\pi i \tau}.$$

This equality identifies three computations of the same free-field amplitude: the representation trace, the Fredholm determinant of the sewing operator, and the Gaussian functional integral.

B.12 A diagnostic table

Low-arity datum	Mathematical identity	Physical reading
pair collision	chiral operation supported on Δ	local OPE/contact term
triple collision	chiral Jacobi or Borcherds identity	independence of nested short-distance expansion
Arnold form	relation in $H^2(\mathrm{Conf}_3)$	cancellation of configuration-space propagators
Casimir coefficient	infinitesimal braid relation	flat KZ transport
three-point screen	$\mathcal{M}_{0,4}$ geometry	relative configuration in a collision
loop contraction	BV operator Δ	one-loop sewing
anomaly cocycle	curvature of the quantum master equation	obstruction class for quantization
Fredholm determinant	trace-class sewing identity	Gaussian partition function

The indicated coefficient map realizes each row. Geometry supplies the carrier, algebra supplies the local operation, and quantum field theory represents both by observables and amplitudes.

C Low-arity calculations for the ordered theory

C.1 Arity zero and one

Let $A \to \mathbf{1}$ be an augmented ordered chiral algebra and write $\bar{A}$ for the augmentation ideal. The reduced bar begins

$$B_0 = \mathbf{1}, \qquad B_1 = \bar{A}[1].$$

The internal differential on B_1 is the suspension of the differential of A. Arity one carries the suspended internal differential and the primitive coaction.

The coaugmentation $\mathbf{1} \to B$ represents the empty ordered configuration. The projection $B \to \bar{A}[1]$ is the primitive quotient. The reduced coproduct vanishes on B_1.

C.2 Arity two

In arity two,

$$B_2 = \bar{A}[1] \otimes \bar{A}[1].$$

For homogeneous $a, b \in \bar{A}$, the multiplication face is

$$b_2[a|b] = (-1)^{|a|}[ab]$$

with the suspension convention of Section A. The internal differential is

$$b_1[a|b] = [da|b] + (-1)^{|a|-1}[a|db].$$

The identity $b_1 b_2 + b_2 b_1 = 0$ is the Leibniz rule for multiplication.

Geometrically, the configuration of two ordered intervals has one collision face. The chiral coefficient of ab may have a pole along $z_1 = z_2$. The bar face composes the two letters through the E_1 multiplication; the chiral coefficient retains its pole filtration.

C.3 Arity three and associativity

There are two multiplication faces:

$$[a|b|c] \longmapsto [ab|c], \qquad [a|b|c] \longmapsto [a|bc].$$

Applying the multiplication differential twice gives

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

The two boundary points of the compactified one-dimensional moduli of three ordered intervals have opposite orientations. Associativity makes the coefficient operations equal. Hence $b_2^2 = 0$.

For an A_∞ algebra, the compactification is the interval K_3 in the convention of Section 6. Its interior operation is m_3, and the boundary identity is

$$dm_3 + m_2(m_2 \otimes 1) - m_2(1 \otimes m_2) + \text{terms containing } m_1 = 0.$$

The strict ordered chiral algebra is the case $m_{r\geq 3} = 0$.

C.4 The first mixed square

Take four product disks

$$U_i \times I_i \subset X \times \mathbb{R}, \qquad I_1 < I_2 < I_3 < I_4.$$

Suppose the first two chiral points collide while the last two transverse intervals become adjacent. There are two codimension-two paths:

chiral collision then interval fusion, interval fusion then chiral collision.

The coefficient maps agree by the interchange law (75). The conormal orientation lines occur in opposite orders, so the two boundary contributions cancel in the total differential.

This calculation totalizes the chiral and ordered resolutions while preserving their two differential directions.

C.5 Exchange begins in two transverse dimensions

The ordered chamber

$$\{t_1 < t_2\} \subset \mathrm{Conf}_2(\mathbb{R})$$

is contractible. Every loop in this chamber is null-homotopic, so the chamber carries trivial monodromy. Exchange of the two labels requires an ambient path around the collision locus.

In two transverse dimensions,

$$\mathrm{Conf}_2(\mathbb{R}^2) \simeq S^1.$$

A half-rotation exchanges the two points; its square is the generator of the pure braid group $P_2 \cong \mathbb{Z}$. Thus the R-matrix first appears after the E_2 thickening.

C.6 The free algebra

Let $A = T_X(M)$. In bar degree two the map

$$M^{\otimes p} \,|\, M^{\otimes q} \longmapsto M^{\otimes (p+q)}$$

concatenates words. The contracting homotopy splits off the first letter of every word of total tensor length at least two. Consequently only the one-letter class survives. This gives the contraction of Theorem 29.1 explicitly.

C.7 The square-zero algebra

For $A = \mathbf{1} \oplus M$ with $M^2 = 0$, all multiplication faces vanish. Hence

$$H_\bullet \operatorname{Bar}_X^{\mathrm{ord}}(A) \cong \bigoplus_{r \geq 0} H_\bullet(M[1]^{\otimes r}).$$

The deconcatenation coproduct cuts a word at every position. Continuous duality gives the completed free algebra on $M^\vee[-1]$.

C.8 A quiver calculation

For the quiver $1 \xrightarrow{u} 2 \xrightarrow{v} 3$, the relative bar over $S = \mathbf{1}e_1 \oplus \mathbf{1}e_2 \oplus \mathbf{1}e_3$ has

$$B_1 = M_u \oplus M_v, \qquad B_2 = M_v \otimes M_u.$$

The multiplication face sends $[v|u]$ to the path vu. In the path algebra, vu is the basis path from vertex 1 to vertex 3, and the two-term resolution is exact. If the relation $vu = 0$ is imposed, $[v|u]$ becomes the quadratic Koszul cogenerator. The calculation is internal to $\mathrm{FactD}(X)$, so M_u and M_v may carry singular chiral coefficients.

C.9 The enveloping algebra

Let $A = U_X(\mathfrak{g})$. Antisymmetrization sends

$$x_1 \wedge \cdots \wedge x_r \longmapsto \sum_{\sigma \in \Sigma_r} \operatorname{sgn}(\sigma)[x_{\sigma(1)}|\cdots|x_{\sigma(r)}].$$

The bar multiplication differential becomes the Chevalley differential

$$\sum_{i<j} (-1)^{\epsilon_{ij}} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_r.$$

In arity three its square is the Jacobiator. PBW proves that antisymmetrization is a quasi-isomorphism.

C.10 BRST in arity two

Let Q be an odd derivation of A with $Q^2 = 0$. On $[a|b]$,

$$Q_{\mathrm{bar}}[a|b] = [Qa|b] + (-1)^{|a|-1}[a|Qb].$$

The mixed commutator with the bar face is

$$(Q_{\mathrm{bar}}b + bQ_{\mathrm{bar}})[a|b] = \pm(Q(ab) - (Qa)b - (-1)^{|a|}a(Qb)) = 0.$$

Thus $b + Q_{\mathrm{bar}}$ is square-zero. The ghost-number filtration compares its cohomology with the bar of $H_Q(A)$ through the associated spectral sequence.

C.11 The annulus and the torus

The transverse annulus amplitude is

$$\mathrm{HH}_\bullet(A) = A \otimes^L_{A\otimes A^{\mathrm{op}}} A.$$

It closes an ordered interval and raises the ribbon genus. The genus-one chiral block is obtained by sewing two marked points of a chiral sphere and raises the curve genus. Their lowest chain models both contain a trace, but they act on different moduli spaces. In the bimodular complex they occupy bidegrees $(0,1)$ and $(1,0)$.

C.12 The first Tannaka coefficient

Let a meromorphic line category have exchange

$$R(z) = 1 + \hbar r(z) + O(\hbar^2).$$

The quantum Yang–Baxter equation at order $\hbar^2$ gives

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0.$$

For $r(z) = \Omega/z$, the coefficient of each commutator is paired with the Arnold three-term relation. This is the first nontrivial compatibility between chiral collision geometry and transverse exchange. The E_1 bar supplies ordered fusion, and the E_2 line category supplies the exchange operation entering the equation.

D The stratified Fulton–MacPherson spectral sequence

The Fulton–MacPherson space carries an incidence complex and a geometric screen complex. The incidence differential records how boundary strata meet, while the screen coefficients record their internal cohomology. A stratification spectral sequence relates the two.

D.1 A filtered space with normal-crossing boundary

Let Y be a smooth complex variety and

$$D = \bigcup_{a\in A} D_a$$

a simple normal-crossing divisor. For $J \subseteq A$, put

$$D_J = \bigcap_{a \in J} D_a, \qquad D_J^\circ = D_J \setminus \bigcup_{b \notin J} D_b.$$

Let $j : Y \setminus D \hookrightarrow Y$. The increasing filtration by depth of support on the boundary gives a Cousin-type spectral sequence for $Rj_*\mathbb{C}$ or, algebraically, for the logarithmic de Rham complex. In one standard indexing,

$$E_1^{-p,q} = \bigoplus_{|J|=p} H^{q-2p}(D_J)(-p) \Longrightarrow H^{q-p}(Y \setminus D). \tag{206}$$

The first differential is the alternating sum of Gysin or residue maps. Its signs are the determinant signs of Section A. Deligne's mixed Hodge theory identifies (206) with the weight spectral sequence when Y is proper.

For coefficients in a complex $\mathcal{L}$ constructible along the boundary stratification, the same filtration gives

$$E_1^{-p,q}(\mathcal{L}) = \bigoplus_{|J|=p} \mathbb{H}^{q-2p}(D_J, \mathcal{L}|_{D_J})(-p) \Longrightarrow \mathbb{H}^{q-p}(Y \setminus D, \mathcal{L}). \tag{207}$$

Nothing in the construction requires $\mathcal{L}$ to be constant. In chiral applications it is assembled from external tensor powers of the coefficient D-module and from principal-part lines along the normal directions.

D.2 Boundary strata of Fulton–MacPherson space

Let X be a smooth curve and $Y_I = \mathrm{FM}_I(X)$. Boundary divisors are indexed by subsets $S \subseteq I$, $|S| \geq 2$, subject to the convention determined by the chosen compactification. A collection $\mathcal{N}$ of subsets labels a nonempty stratum exactly when it is nested: any two members are disjoint or one contains the other. The nested set is equivalently a rooted collision forest F.

The open stratum associated with F has the form of an iterated bundle whose base records the root configurations on X and whose fibers are screen configuration spaces. On a curve, the screen attached to a vertex with r incoming branches is the compactification of r points on an affine line modulo translations and dilations. Its interior is

$$\mathrm{Conf}_r(\mathbb{A}^1)/(\mathbb{G}_a \rtimes \mathbb{G}_m).$$

For $r = 2$ the screen is a point. For $r = 3$ its interior is $\mathcal{M}_{0,4}$ and its compactification is $\overline{\mathcal{M}}_{0,4}$. For larger r it has nontrivial cohomology in several degrees.

A stratum combines two data: the forest gives its incidence type, and the screen factors give its internal topology.

D.3 The forest resolution as the row of constants

Let $p : Y_I \to X^I$ be the blowdown. Consider the filtration of a geometric coefficient complex $\mathcal{K}_I$ by boundary depth. The E_1 page is a sum over forests,

$$E_1^{p,q} = \bigoplus_{\substack{F \in \mathrm{For}(I) \\ |E(F)|=p}} R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) \otimes \mathrm{or}(F), \tag{208}$$

where p_F is the projection of the stratum to the corresponding partial diagonal. The differential contracts one edge and applies the induced restriction or residue map.

The row $q = 0$ contains the locally constant functions on every screen. When the screen coefficient complexes are concentrated in degree zero and their local systems are constant, barycentric replacement and terminal-corolla normalization identify this row with the combinatorial forest resolution $W_I^{\mathrm{FM}}(\mathfrak{g})$ of Section 14. The augmentation to the partition collision complex is a quasi-isomorphism because every rooted-tree category has a terminal corolla.

Criterion D.1 (Incidence sufficiency)**.** *The incidence forest complex computes the proper pushforward of $\mathcal{K}_I$ if the following two conditions hold:*

(i) $R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) = 0$ *for every* $q > 0$ *and every forest* F*;*

(ii) *the degree-zero pushforwards identify with the coefficient system used in the forest complex, compatibly with the vertical incidence differential and the corolla-normalized horizontal collision merge.*

The spectral-sequence comparison proves the criterion because the two conditions concentrate (208) in one row. Constant coefficients on complex screens usually contribute higher rows, beginning with the triple-collision screen.

D.4 The first nonconstant row

For a triple-collision vertex, the screen interior is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore the $q = 1$ row of (208) already appears in arity three. Its logarithmic generators may be taken to be

$$d\log u, \qquad d\log(1-u).$$

Their residues at $0, 1, \infty$ satisfy one linear relation. This row carries the first period data of ordered configurations. Integrating its forms along paths gives logarithms; iterating at higher arity gives multiple polylogarithms and, for KZ transport, multiple zeta values.

The chiral Chevalley differential uses the diagonal coefficient operation and partition incidence. Configuration-space integral realizations add the screen periods. The spectral sequence places these two contributions in separate rows.

D.5 Orlik–Solomon cohomology

For $X = \mathbb{A}^1$, the cohomology algebra of $\mathrm{Conf}_n(X)$ is the Orlik–Solomon algebra of the braid arrangement. It is generated in degree one by ω_{ij} with relations

$$\omega_{ij} = \omega_{ji}, \qquad \omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

A broken-circuit-free (NBC) basis gives the Poincare polynomial

$$P_{\mathrm{Conf}_n(\mathbb{C})}(t) = \prod_{j=1}^{n-1}(1 + jt). \tag{209}$$

The coefficient $(n-1)!$ in top degree is the top Betti number of the configuration space. A geometric bar model tensors or convolves this cohomology with the algebraic coefficient complex and applies the total differential. Its homology is computed from that total complex.

D.6 Nested residues

On a normal-crossing chart with local boundary parameters t_e indexed by the edges of a collision forest, a logarithmic form has an expansion

$$\alpha = \sum_{J\subseteq E(F)} \left(\bigwedge_{e\in J} \frac{dt_e}{t_e} \right) \wedge \alpha_J .$$

The depth filtration counts $|J|$. The first differential of the residue spectral sequence extracts $\alpha_{\{e\}}$ and restricts it to $t_e = 0$. A second residue extracts $\alpha_{\{e,f\}}$. Because

$$\frac{dt_e}{t_e} \wedge \frac{dt_f}{t_f} = -\frac{dt_f}{t_f} \wedge \frac{dt_e}{t_e},$$

the codimension-two contribution cancels between the two orders. When the coefficient system itself changes under contraction, the remaining coefficient is the associator or Jacobiator of that system.

For higher poles, ordinary Poincare residue still acts only on the logarithmic conormal factor. The order-m_e coefficient is first paired with the dual m_e-jet of the normal line, producing a logarithmic coefficient; ordinary residue then removes dt_e/t_e. Invariantly this is the pairing

$$N_e^{\otimes m_e} \otimes N_e^{\vee \otimes m_e} \longrightarrow \mathcal{O}$$

followed by Poincare residue. Different pole orders remain in different associated-graded normal lines.

D.7 The Rees spectral sequence

Let $F_{\mathbf{m}}$ be the multi-pole filtration along the collision divisors. The multi-Rees object

$$\mathcal{R}(\mathcal{K}_I) = \bigoplus_{\mathbf{m}\geq 0} F_{\mathbf{m}} \mathcal{K}_I \, \mathbf{t}^{\mathbf{m}}$$

interpolates between the filtered chiral coefficient system and its normal-cone associated graded. Filtering the geometric complex simultaneously by boundary depth and pole order gives a bicomplex. One spectral sequence first resolves collision incidence and then pole order; the other first passes to normal symbols and then resolves the forest geometry. Under boundedness in each finite arity and pole window, the two converge to the same total cohomology.

This interchange explains the relation between chiral and Poisson vertex structures. The pole associated graded is classical; the extension data between pole orders is quantum. Fulton–MacPherson depth is independent of pole degree; it records how many nested collisions occur. The two filtrations define an intrinsic bidegree: pole order measures singularity, while Fulton–MacPherson depth measures nested collision.

D.8 A comparison theorem for finite windows

Fix a finite set I, a maximum pole order M, and a bounded cohomological range. Let $\mathcal{K}_I^{\leq M}$ be the truncated principal-part coefficient system. Suppose there is a morphism

$$\Phi_I : W_I^{\mathrm{FM}}(\mathfrak{g}; \leq M) \longrightarrow Rp_* \mathcal{K}_I^{\leq M} \tag{210}$$

that is compatible with the vertical incidence differential and the corolla-normalized collision merge, and induces an isomorphism on every stratumwise cohomology group in the chosen range.

Theorem D.2 (Finite-window comparison)**.** *Under the preceding hypotheses,* (210) *is a quasi-isomorphism in the chosen cohomological range.*

Proof. Filter source and target by boundary depth. The map induces a morphism of the corresponding spectral sequences. By hypothesis it is an isomorphism on E_1 in the range. Since the filtration is finite for fixed I, the comparison theorem for spectral sequences gives an isomorphism on the associated graded of cohomology, hence on cohomology itself. □

The theorem is local in arity, pole order, and cohomological degree. An unbounded completion is computed by the derived inverse limit. Surjective transition maps give the Mittag–Leffler condition and annihilate $\varprojlim^1$; in general the $\varprojlim^1$ term records the additional completion cohomology.

D.9 Renormalization and compactified configurations

Let M be a real manifold and consider a perturbative quantum field theory. A Feynman amplitude associated with a graph Γ is initially a differential form or distribution on the configuration space of its vertices away from the diagonals. Pulling it to a compactification separates the asymptotic regimes of every collision forest. Renormalization is an extension across the boundary satisfying locality and compatibility with disjoint unions.

The forest formula of perturbative renormalization has the same incidence category as the Fulton–MacPherson boundary. A divergent subgraph corresponds to a cluster of vertices; nested and disjoint divergent subgraphs correspond to nested and disjoint boundary divisors. Counterterms live on the associated diagonals. Stokes' theorem on the compactification expresses the Ward-identity anomaly as a sum of boundary integrals. The quantum master equation extends across the boundary when this local cocycle is zero in anomaly cohomology, equivalently when a counterterm provides its primitive.

This physical realization uses more than the incidence complex:

(i) a propagator or parametrix;

(ii) a gauge-fixing operator;

(iii) an extension of singular distributions;

(iv) a renormalization prescription;

(v) a proof that the counterterms preserve the required symmetries.

When these data are fixed, the geometric boundary complex computes the anomaly class. Its vanishing or exactness is the additional cohomological condition that produces a quantum solution.

D.10 Degeneration of curves

For a family of curves approaching a node, the relevant compactification contains two kinds of boundary simultaneously: collisions of marked points in the fibers and degeneration of the curve. Logarithmic Fulton–MacPherson spaces resolve the first relative to the second. A local model of the node is

$$xy = q,$$

where q is the smoothing parameter. Collision parameters in the two branches and the nodal parameter q define independent conormal directions. The total residue complex therefore has a multigrading.

Separating degeneration corresponds to two components joined at the node. Nonseparating degeneration identifies two points on one component. The associated maps on conformal blocks are the two sewing operations of a modular operad. The screen cohomology of point collisions and the vanishing-cycle or nearby-cycle data of the node are distinct. A correct global comparison must include both.

D.11 Consequences

The spectral sequence proves four statements that are used repeatedly in the body of the paper.

(1) The partition collision complex is canonical and computes chiral Chevalley chains.

(2) The forest incidence complex is a canonical resolution of the partition complex.

(3) The geometric Fulton–MacPherson complex contains additional screen cohomology.

(4) A stratumwise comparison morphism and the spectral-sequence theorem identify the geometric and algebraic complexes.

From the physical side, the same four statements say that locality determines the pattern of counterterms, while the propagator and its periods determine their numerical amplitudes. The first is combinatorial and functorial. The second is analytic and theory-dependent. Chiral Koszul duality is the algebraic local-to-global theorem; configuration integrals are geometric realizations of its coefficient maps.

E Formal neighborhoods, normal jets, and degeneration coordinates

The local geometry of a chiral operation is the geometry of a divisor and its formal neighborhood. The formulas in this appendix make explicit how pole coefficients, delta derivatives, coordinate changes, and sewing parameters are different expressions of normal geometry.

E.1 The formal completion of a diagonal

Let X be a smooth curve and let $\Delta \subset X \times X$ be the diagonal. Its ideal sheaf $\mathcal{I}_\Delta$ satisfies

$$\mathcal{I}_\Delta/\mathcal{I}_\Delta^2 \cong \Omega^1_X, \qquad N_{\Delta/X^2} \cong T_X.$$

The m-th infinitesimal neighborhood is

$$\Delta^{(m)} = \mathrm{Spec}(\mathcal{O}_{X^2}/\mathcal{I}_\Delta^{m+1}).$$

The associated graded algebra of the formal completion is

$$\mathrm{gr}_{\mathcal{I}}\,\mathcal{O}_{\widehat{X^2}_\Delta} \cong \mathrm{Sym}_{\mathcal{O}_X}(\mathcal{I}_\Delta/\mathcal{I}_\Delta^2) \cong \mathrm{Sym}_{\mathcal{O}_X}(\Omega^1_X). \tag{211}$$

Dualizing gives the algebra of normal differential operators

$$\mathrm{Sym}_{\mathcal{O}_X}(T_X).$$

A coordinate difference $t = z_1 - z_2$ trivializes the conormal line and identifies the formal completion with $\mathcal{O}_X[[t]]$. A different coordinate changes t by a formal power series with invertible linear term. The formal neighborhood is invariant, and each coordinate series is one of its trivializations.

E.2 Principal parts

Let $D \subset Y$ be a smooth divisor with local equation $t = 0$. For $m \geq 0$, the sheaf $\mathcal{O}_Y(mD)$ consists locally of functions with poles of order at most m. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \xrightarrow{\sigma_{m+1}} N_{D/Y}^{\otimes(m+1)} \longrightarrow 0. \tag{212}$$

To verify the target, write a section as

$$f = t^{-(m+1)}a_{-(m+1)} + t^{-m}a_{-m} + \cdots .$$

Modulo $\mathcal{O}_Y(mD)$ only the leading coefficient remains. If $t' = ut + O(t^2)$, then

$$(t')^{-(m+1)} = u^{-(m+1)}t^{-(m+1)} + O(t^{-m}),$$

so $a_{-(m+1)}$ transforms as the $(m+1)$-st power of the normal line. The transition calculation proves (212) intrinsically.

The pole filtration

$$F_m\mathcal{O}_Y(*D) = \mathcal{O}_Y(mD)$$

has associated graded

$$\mathrm{gr}^F_m\,\mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes m}, \qquad m \geq 1. \tag{213}$$

A complete OPE is a filtered morphism with coefficients in $\mathcal{O}_Y(*D)$. Its mode of order $m-1$ is the image under σ_m.

E.3 Differential operators and delta derivatives

Let $i : D \hookrightarrow Y$ be a smooth divisor. Distributions supported on D are generated by normal derivatives of the delta distribution. Algebraically, the Kashiwara equivalence identifies right D_Y-modules supported on D with right D_D-modules. Locally, a supported distribution has the form

$$T = \sum_{m=0}^{M} T_m \, \partial_t^m \delta(t). \tag{214}$$

The coefficient pairing of principal parts with normal jets is the formal residue identity

$$\mathrm{Res}_{t=0}\big(t^n t^{-m-1} \, dt\big) = \delta_{mn}. \tag{215}$$

The dual jet functional may be represented distributionally by a normalized derivative of $\delta(t)$. In complex analysis, (215) is Cauchy's coefficient formula; in D-module theory it is the pairing between local cohomology along the diagonal and normal differential operators.

For a vertex algebra, the singular commutator is

$$[Y(a,z), Y(b,w)] = \sum_{m\geq 0} Y(a_{(m)}b, w) \frac{1}{m!} \partial_w^m \delta(z-w). \tag{216}$$

Equation (216) is the distributional form of the chiral operation. It contains every singular mode. The regular extension of the chiral operation supplies the normally ordered product, while the supported commutator supplies the singular modes.

E.4 The Rees deformation

The Rees algebra of the ideal of a divisor is

$$\mathcal{R}(\mathcal{I}_D) = \bigoplus_{m\geq 0} \mathcal{I}_D^m u^{-m}.$$

Its relative spectrum over $\mathbb{A}^1_u$ is the deformation to the normal cone. In a local chart one may write

$$t = ux. \tag{217}$$

For $u \neq 0$, the fiber is isomorphic to the original neighborhood. For $u = 0$, the coordinate x is the normal coordinate. The corresponding construction for the pole filtration uses inverse powers and interpolates between meromorphic kernels and normal symbols.

For several normal-crossing divisors D_e with local equations t_e, the multi-Rees space has equations

$$t_e = u_e x_e.$$

Its deepest special fiber is the direct sum of the normal lines. On a Fulton–MacPherson stratum indexed by a forest F, the indices e are the internal edges of F. This proves the normal-bundle splitting

$$N_{D_F/\mathrm{FM}_I(X)} \cong \bigoplus_{e\in E(F)} N_e. \tag{218}$$

Operadic substitution on the associated graded is substitution of the normal coordinates x_e along contracted subtrees.

E.5 Coordinate changes and triangularity

Let $w = f(z)$ with $f'(z) \neq 0$. Near $z_1 = z_2 = z$, put $t = z_1 - z_2$. Taylor expansion gives

$$f(z+t) - f(z) = f'(z)t + \frac{1}{2}f''(z)t^2 + \frac{1}{6}f'''(z)t^3 + O(t^4), \tag{219}$$

$$\frac{1}{f(z+t) - f(z)} = \frac{1}{f'(z)t} - \frac{f''(z)}{2f'(z)^2} + O(t), \tag{220}$$

$$\frac{1}{(f(z+t) - f(z))^2} = \frac{1}{f'(z)^2t^2} - \frac{f''(z)}{f'(z)^3t} + O(1). \tag{221}$$

A pole of order m therefore transforms into a linear combination of poles of orders at most m, with leading factor $f'(z)^{-m}$. The associated graded is tensorial, while the filtered object is triangular.

This is exactly the transformation law of vertex modes. The top pole of an OPE between primary fields transforms by the tensor power prescribed by its conformal weights. Lower poles acquire corrections from derivatives of f. The complete chiral operation is coordinate-free because these corrections are the transition functions of the filtered principal-parts sheaf.

E.6 The Schwarzian derivative

The Schwarzian derivative is

$$\{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2.$$

It obeys the cocycle identity

$$\{f \circ g, z\} = \{f, g(z)\}g'(z)^2 + \{g, z\}. \tag{222}$$

A stress tensor of central charge c transforms as

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12}\{f, z\}. \tag{223}$$

The first term is the tensorial transformation of a quadratic differential. The second is an affine cocycle. Hence stress tensors form an affine torsor for projective connections, modeled on the vector space of quadratic differentials.

Equation (223) follows from the fourth-order central term of the Virasoro OPE. The central term is a fourth normal symbol, while the Schwarzian is the extension class that mixes it into filtration degree zero under coordinate change. This is a precise example of why the associated graded and the filtered chiral operation contain different information.

E.7 The affine connection and the shifted level

For affine currents, the invariant form gives the second normal symbol. The KZ connection contains instead the Casimir divided by $k + h^\vee$:

$$\nabla = d - \frac{1}{k + h^\vee}\sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

The level k and the shifted denominator $k+h^\vee$ arise in different calculations. The first is the coefficient of the central extension in the current OPE. The second is the coefficient of the Sugawara stress tensor and the KZ transport. The Sugawara and Ward-identity calculations relate these coefficients while retaining their distinct definitions.

At $k=-h^\vee$, the Sugawara construction is singular. The current OPE and its level remain defined. The KZ connection in this normalization is singular. The Feigin–Frenkel centre appears. These facts are compatible because they concern different pieces of structure.

E.8 Prime form and normalized bidifferential

Let X be a compact Riemann surface of genus g, choose a symplectic basis of homology, and let Ω be the period matrix. The prime form $E(p,q)$ is a $(-1/2,-1/2)$-form with a simple zero on the diagonal. Locally,

$$E(p,q)=\frac{z(p)-z(q)}{\sqrt{dz(p)}\sqrt{dz(q)}}\big(1+O((z(p)-z(q))^2)\big).$$

The normalized fundamental bidifferential is

$$B(p,q)=d_p d_q \log E(p,q)+\sum_{i,j=1}^{g} c_{ij}\,\omega_i(p)\omega_j(q), \tag{224}$$

where the coefficients c_{ij} are chosen, according to convention, so that the A-periods vanish. It has a double pole with biresidue one on the diagonal. Changing the symplectic basis acts by a modular transformation and changes the decomposition into prime-form and holomorphic pieces, while the normalized bidifferential transforms covariantly.

The prime form is a section of a line bundle with a simple diagonal zero and prescribed quasi-periodicity. A propagator constructed from it requires a choice of normalization, spin structure, or Green function appropriate to the theory. Its local singular symbol is universal; its harmonic completion is global.

E.9 Plumbing coordinates

A nodal curve has local equation $xy=0$. A smoothing is

$$xy=q, \qquad |q|<1. \tag{225}$$

Removing discs around $x=0$ and $y=0$ and identifying the annuli by $y=q/x$ produces the smooth fiber. The parameter q is a normal coordinate to the Deligne–Mumford boundary divisor.

For a conformal field theory, sewing inserts a complete set of states:

$$\mathcal{A}_q=\sum_{\alpha} q^{L_0-c/24}|\alpha\rangle\langle\alpha|, \tag{226}$$

with the dual basis defined by the two-point pairing. In a rational theory the sum is organized by finitely many simple modules and convergent graded traces. In a general vertex algebra the expression may be formal. The algebraic clutching map and the analytic convergence of (226) are separate assertions.

The factor $q^{-c/24}$ is the vacuum energy on the cylinder. Under a modular change of plumbing coordinate it contributes to the projective anomaly. The same central charge appears in the Virasoro OPE because both are representations of the same central extension of vector fields. This equality identifies the local Virasoro cocycle with the cylinder vacuum-energy contribution of the same conformal theory.

E.10 Separating and nonseparating nodes

A separating node joins curves of genera g_1 and g_2. Sewing gives a bilinear composition

$$\mathbb{V}_{g_1,n_1+1} \otimes \mathbb{V}_{g_2,n_2+1} \longrightarrow \mathbb{V}_{g_1+g_2,n_1+n_2}.$$

A nonseparating node identifies two marked points of one curve and contracts them with the inverse pairing:

$$\mathbb{V}_{g-1,n+2} \longrightarrow \mathbb{V}_{g,n}.$$

These are the two modular-operad compositions. Their codimension-two compatibility is the geometric source of the bracket and loop terms in the quantum master equation.

The normal parameter q belongs to the base of the family of curves. Collision parameters t_e belong to relative configuration spaces in the fibers. In a degeneration with colliding insertions, both occur. A type-correct superconnection places the fiber differential, the logarithmic base form dq/q, and Hodge curvature in their respective bidegrees before forming its total square.

E.11 Tangential base points

The spaces $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and $\mathcal{M}_{0,n}$ carry nontrivial fundamental groups and boundary monodromy. Iterated integrals between boundary points require tangential base points. A tangent vector at 0 specifies the asymptotic direction of approach and regularizes logarithmic divergences. The KZ associator is the regularized parallel transport from the tangent vector at 0 to the tangent vector at 1.

The choice is geometric: it is a nonzero vector in the normal line to a boundary divisor. Changing it conjugates the regularized monodromy by an exponential of the residue. This is the local origin of gauge dependence of associators. The braid category is invariant up to braided equivalence; a particular power-series associator is a representative.

E.12 Reality and positivity

An algebraic chiral operation supplies complex-linear OPE data. A unitary conformal field theory enriches it with an antilinear involution and a positive Hermitian form compatible with the modes. The adjoint relation is

$$L_n^\dagger = L_{-n}$$

for the Virasoro modes in radial quantization, with analogous formulas for primary fields. Reflection positivity is an independent analytic axiom relating the involution, the Euclidean geometry, and the state-space norm.

Similarly, a Euclidean Green function depends on a real structure and metric, while the singular chiral kernel depends only on complex geometry. The chiral algebra determines protected holomorphic quantities. Reconstruction of a unitary full conformal field theory requires left–right pairing, modular invariance, and positivity. These are additional structures.

E.13 Local geometry summarized

The formal neighborhood of a diagonal contains the jets. The pole filtration contains the singular modes. The Rees deformation contains the classical limit. The Fulton–MacPherson normal bundle contains nested collision scales. The plumbing parameter contains nodal sewing. A projective connection contains the stress-tensor anomaly. Each object is a line or filtered sheaf attached to a geometric normal direction. This is the local geometry in which the coordinate formulas of chiral algebra and the short-distance formulas of quantum field theory become intrinsic.

F Field-theoretic derivations

This appendix derives the physical structures used in the paper from actions and Ward identities. Every identification is stated after the boundary observables, pairings, and comparison maps have been specified.

F.1 Classical BV theory

A classical gauge theory in the Batalin–Vilkovisky formalism consists of a graded space of fields $\mathcal{E}$, an odd symplectic pairing of cohomological degree -1, and an action $S \in \mathcal{O}(\mathcal{E})$ of degree zero. The odd symplectic form induces a Poisson bracket $\{-,-\}$ of degree one. The classical master equation is

$$QS + \frac{1}{2}\{S, S\} = 0, \tag{227}$$

where Q is the linearized differential. Equivalently, the Hamiltonian vector field

$$d_S = Q + \{S, -\}$$

satisfies $d_S^2 = 0$.

The Taylor coefficients of S define cyclic multilinear operations. Equation (227) is their L_∞ relation. If the fields are local sections of bundles on a manifold, the operations are supported on the small diagonal. Thus the local L_∞ algebra of fields is the field-theoretic analogue of a diagonal-supported chiral operation.

Gauge fixing chooses a Lagrangian complement and a parametrix for the kinetic operator. This choice is needed to define Feynman amplitudes. Different gauge-fixing choices produce canonically homotopy-equivalent local operations and factorisation observables.

F.2 Quantum master equation and anomaly

A regularized BV Laplacian Δ is a second-order operator of degree one. The quantum master equation is

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \tag{228}$$

or equivalently

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0. \tag{229}$$

In an interacting field theory the action is replaced by a scale-dependent effective interaction and Δ by a scale-dependent regularized operator. Renormalization constructs a compatible family of effective solutions precisely when the anomaly class vanishes in local cohomology [14, 15].

Suppose a solution has been constructed modulo $\hbar^N$. The coefficient of $\hbar^N$ in the quantum-master curvature is closed for the classical differential d_S. Its cohomology class is independent of changes of counterterm. Its cohomology class is the anomaly. A primitive for this class is the counterterm extending the solution to the next order. The resulting extension theory is the BV realization of the filtered Maurer–Cartan theory of Section 25.

For a cyclic local field theory, the stable-graph master equation and the BV quantum master equation are the same graph expansion: vertices are Taylor coefficients of S, internal edges are propagator contractions, and loops contract two flags of one connected graph. A boundary chiral algebra enters this identity through a factorisation comparison with the boundary observables of that field theory.

F.3 Factorization observables

For an open set U, let $\mathrm{Obs}^{\mathrm{cl}}(U)$ be the Chevalley–Eilenberg cochains of compactly supported fields in U. Extension by zero and disjoint union define structure maps

$$\mathrm{Obs}^{\mathrm{cl}}(U_1) \otimes \cdots \otimes \mathrm{Obs}^{\mathrm{cl}}(U_n) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

for disjoint $U_i \subset V$. The cosheaf and descent properties make $\mathrm{Obs}^{\mathrm{cl}}$ a factorization algebra. A perturbative quantization deforms it to Obs^q over $\mathbb{C}[[\hbar]]$.

If translations in some directions are homotopically trivial, the factorization algebra is locally constant in those directions and determines an E_n algebra. If dependence on one complex direction is holomorphic, restriction to a holomorphic curve yields a chiral factorization algebra. Boundary conditions replace the bulk field complex by a relative complex and produce a boundary factorization algebra.

A comparison with an algebraic chiral algebra is a quasi-isomorphism

$$\Phi : U^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{Obs}^q_\partial$$

compatible with factorization. Once Φ is given, the algebraic chiral bracket equals the renormalized boundary OPE in cohomology.

F.4 Chern–Simons theory and affine currents

Let M be an oriented three-manifold, G a compact Lie group with invariant form $(,)$, and $k \in \mathbb{Z}$. The Chern–Simons action is

$$S_{\mathrm{CS}}(A) = \frac{k}{4\pi} \int_M \left((A, dA) + \frac{2}{3}(A, A \wedge A) \right). \tag{230}$$

On a manifold with boundary, variation gives

$$\delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_M (\delta A, F_A) + \frac{k}{4\pi} \int_{\partial M} (A, \delta A).$$

A boundary condition chooses a Lagrangian subspace for the boundary symplectic form. In complex coordinates on the boundary, fixing one component produces a chiral boundary theory. The residual gauge symmetry has current algebra

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{f^{ab}{}_c J^c(w)}{z-w}. \tag{231}$$

The double pole is the boundary central extension determined by the coefficient of (230). The simple pole is the Lie bracket. The one-loop shift by $h^\vee$ enters the effective coupling and the Sugawara construction, while the classical boundary central extension retains the coefficient k in (231).

Wilson lines are labelled by representations of G. Their expectation values depend on framing. Canonical quantization on a surface gives the space of conformal blocks of the affine algebra. Witten's path-integral derivation of knot invariants identifies the braiding and ribbon trace with quantum-group data [74, 63]. The Drinfeld–Kohno theorem supplies the algebraic comparison of braid categories.

F.5 One-loop exactness in a holomorphic gauge

A partially holomorphic gauge can make Chern–Simons theory a deformation of a topological–holomorphic BF theory. In the gauge constructed by Gwilliam and Williams, the perturbative quantization is one-loop exact and the bulk–boundary relation with chiral WZW theory is explicit [45]. One-loop exactness means that the relevant local cohomology eliminates higher-loop counterterms in that gauge. Boundary correlators continue to contain the tree-level nonlinear vertices and representation-theoretic state sums.

The exactness result supplies a rigorous model in which the boundary current OPE, the anomaly shift, and the factorization algebra of observables can be compared at chain level. It is one of the clearest physical realizations of chiral Koszul duality: the local Lie algebra of gauge transformations passes to Chevalley cochains, while boundary factorization organizes their states and operators.

F.6 Four-dimensional Chern–Simons theory

Let Σ be an oriented real surface and C a complex curve with meromorphic one-form ω. Four-dimensional Chern–Simons theory has action

$$S_{4d}(A) = \frac{1}{2\pi\hbar}\int_{\Sigma\times C} \omega \wedge \left((A, dA) + \frac{2}{3}(A, A\wedge A)\right). \tag{232}$$

The theory is topological along Σ and holomorphic along C. Wilson lines supported on curves in Σ and located at points of C possess a spectral parameter. Crossing two lines gives an R-matrix. The quantum Yang–Baxter equation follows from isotopy invariance of three line crossings. Rational, trigonometric, and elliptic kernels arise from the geometry of (C,ω) [19, 20].

The line-operator OPE gives a tensor category with spectral parameters. A Yangian or quantum loop algebra is recognized when the category is generated by evaluation representations and the R-matrix and relations agree with the corresponding presentation. The field theory provides the monodromy and operator products; the algebraic recognition identifies the named Hopf algebra.

F.7 The Polyakov anomaly

For a two-dimensional conformal field theory on a Riemann surface with metric g, a Weyl rescaling $g \mapsto e^{2\sigma} g$ changes the partition function by the infinitesimal anomaly

$$\delta_\sigma \log Z[g] = \frac{c}{24\pi} \int_X \sigma R_g \, d\mathrm{vol}_g + \text{boundary terms}. \tag{233}$$

Integrating gives the Polyakov action, up to local counterterms [65]. The coefficient c is the Virasoro central charge. On the moduli space of complex structures, the same anomaly is the curvature of a determinant or Hodge line.

For a free scalar, Gaussian integration gives

$$Z[g] \propto (\det{}' \Delta_g)^{-1/2}$$

with zero-mode factors. The variation of the zeta-regularized determinant reproduces (233) with $c = 1$. For free fermions, ghosts, and systems of several fields, determinants and Pfaffians give the corresponding central charges. The equality between OPE central charge and determinant-line curvature is a local index theorem for that field theory.

F.8 BRST quantization of the bosonic string

In conformal gauge, the bosonic string contains a matter CFT of central charge c_m and a (b, c) ghost system of central charge -26. The BRST current has the form

$$j_{\mathrm{BRST}} = c \left(T_m + \frac{1}{2} T_{gh} \right) + \text{derivative term}.$$

The BRST charge $Q = \oint j_{\mathrm{BRST}}$ satisfies

$$Q^2 \propto (c_m - 26) \sum_{n \in \mathbb{Z}} (n^3 - n) c_{-n} c_n. \tag{234}$$

Therefore $Q^2 = 0$ precisely when the total Virasoro central charge vanishes, apart from the intercept condition in the state-space formulation. This is anomaly cancellation. It is an exact equality of central extensions in the coupled matter–ghost chiral algebra.

The number 26 is the opposite of the central charge of the reparametrization ghost system of weights $(2, -1)$. Each BRST reduction carries the anomaly polynomial determined by its own ghost representation.

F.9 Conformal blocks as states of quantization

Canonical quantization of Chern–Simons theory on a surface Σ produces a Hilbert space depending on the complex structure used in geometric quantization. Projective flatness identifies these spaces as the complex structure varies. The resulting projective bundle is identified with affine conformal blocks. Sewing surfaces corresponds to gluing three-manifolds and summing over intermediate representations.

For a modular tensor category with simple objects i, the dimension of the genus-g state space is

$$\dim \mathcal{H}_g = \sum_i S_{0i}^{2-2g}. \tag{235}$$

This is the Verlinde formula. The modular functor enriches these dimensions by mapping-class-group representations and gluing maps [73, 47].

F.10 Cardy asymptotics

Let

$$Z(\tau,\bar{\tau}) = \mathrm{Tr}_{\mathcal{H}}\, q^{L_0-c/24}\bar{q}^{\bar{L}_0-\bar{c}/24}$$

be a modular-invariant torus partition function with a discrete spectrum, nonnegative degeneracies, and a unique vacuum that dominates the low-temperature limit. Applying the modular transformation $\tau \mapsto -1/\tau$ and a saddle-point estimate gives the asymptotic density

$$\log\rho(h,\bar{h}) \sim 2\pi\sqrt{\frac{c_{\mathrm{eff}}h}{6}} + 2\pi\sqrt{\frac{\bar{c}_{\mathrm{eff}}\bar{h}}{6}}. \tag{236}$$

The assumptions are essential. Logarithmic theories, continuous spectra, noncompact targets, and additional light states can change the asymptotics.

For asymptotically AdS_3 gravity with Brown–Henneaux boundary conditions,

$$c = \frac{3\ell}{2G}.$$

When the gravitational spectrum satisfies the hypotheses leading to (236), the Cardy entropy agrees with the Bekenstein–Hawking entropy of the BTZ black hole. The equality follows from the Brown–Henneaux identification together with modular invariance, the spectral hypotheses, and the Cardy saddle [12].

F.11 Entanglement entropy

For the vacuum of a unitary two-dimensional CFT on the line, the replica method computes the entropy of an interval of length L with ultraviolet cutoff ϵ:

$$S_A = \frac{c+\bar{c}}{6}\log\frac{L}{\epsilon} + s_0. \tag{237}$$

The proof uses twist fields of dimensions

$$h_n = \frac{c}{24}\left(n - \frac{1}{n}\right), \qquad \bar{h}_n = \frac{\bar{c}}{24}\left(n - \frac{1}{n}\right),$$

and analytic continuation in the replica number. Equation (237) uses the state, interval geometry, and full left–right CFT. The holomorphic chiral algebra supplies the central charge and holomorphic twist-field constraints, while the full state supplies the density matrix.

Relative entropy, modular Hamiltonians, and topological entanglement are computed from their respective states and replica or operator-algebraic constructions. A chiral obstruction tower reaches these quantities through an explicit comparison with the corresponding replica correlators.

F.12 Boundary algebras in supersymmetric gauge theory

Topological or holomorphic twists of supersymmetric gauge theories often produce protected boundary operator algebras. Costello and Gaiotto construct vertex algebras associated with boundary conditions in three-dimensional $\mathcal{N} = 4$ gauge theories and relate their modules to bulk line defects [16]. Costello, Dimofte, and Gaiotto construct boundary chiral algebras for holomorphic twists of three-dimensional $\mathcal{N} = 2$ theories.

These constructions supply examples of the comparison map Φ. Mirror symmetry or bulk duality can induce equivalences of boundary chiral algebras or their module categories. The chosen twist and boundary condition specify the protected sector, while the untwisted theory carries the complementary observables and dynamics.

F.13 Celestial and twistor chiral algebras

In suitable four-dimensional gauge theories, scattering amplitudes can be organized by a chiral algebra on the celestial sphere or on twistor space. Costello and Paquette derive form-factor integrands from correlators of a chiral algebra obtained from a six-dimensional holomorphic theory [17]. The OPE associativity constrains loop corrections; anomalies of the chiral algebra correspond to obstructions in the bulk theory.

The exact object is a protected holomorphic observable algebra. Mellin transformation converts momentum variables to conformal weights in the celestial basis, and soft theorems provide current insertions. A reconstruction theorem for the complete gravitational S-matrix additionally controls infrared divergences, reality conditions, and the inverse transform.

F.14 Calabi–Yau categories and closed strings

A smooth and proper Calabi–Yau dg category has a nondegenerate cyclic trace. Its Hochschild chains carry the Connes operator, and its Hochschild cochains carry a Gerstenhaber structure. Costello constructs a topological conformal field theory from Calabi–Yau categorical data [18]. The cyclic trace permits the contraction of graph edges, exactly as in the modular envelope.

A field theory or factorisation algebra on the Calabi–Yau geometry, together with a compactification or defect and a restriction map to the curve, produces the chiral algebra. The induced morphism of deformation complexes carries cyclic classes to chiral operations. Euler characteristics and Borcherds-product weights are traces taken after this geometric morphism.

F.15 The physical synthesis

The equations derived above form one coherent structure:

$$\begin{aligned}\text{local action} \longrightarrow \text{BV Maurer–Cartan element} &\longrightarrow \text{factorization observables,}\\ &\longrightarrow \text{boundary chiral operation.}\end{aligned}$$

A propagator resolves internal edges. Fulton–MacPherson geometry resolves collisions. The quantum master equation governs boundary strata of Feynman graphs. KZ transport governs exchange of current-algebra insertions. Conformal blocks govern sewing of states. Determinant lines govern projective anomalies. Each passage is exact when its comparison map has been constructed.

The resulting unity is physical in Feynman's sense and mathematical in Beilinson's sense: local interactions are followed through their singular support, collision maps, and gluing correspondences until the comparison functors identify their realizations.

G Exact models seen from every side

A general language is convincing only when its principal objects agree in examples by independent calculations. This appendix treats five models. In each case the chiral operation, collision coalgebra, ordered transport, modular amplitude, and field-theoretic realization are kept distinct and then compared by explicit formulas.

G.1 The rank-one Heisenberg algebra

Let H_k be generated by a field J with OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \tag{238}$$

Equivalently,

$$[J_\lambda J] = k\lambda.$$

The chiral Jacobi identity is immediate because the lambda bracket is central. The full vertex algebra is the symmetric algebra on the negative modes J_{-n}, $n \geq 1$, applied to the vacuum.

Normal geometry. The singular operation in (238) is the second normal symbol of the diagonal. Under $z \mapsto f(z)$, a weight-one current transforms as a one-form, and the coefficient k pairs the two normal lines. Let $\mathfrak{h}_{\mathrm{ch}}$ denote the abelian current chiral Lie algebra and let

$$0 \longrightarrow \omega_X \mathbf{K} \longrightarrow \widehat{\mathfrak{h}}_{\mathrm{ch}} \longrightarrow \mathfrak{h}_{\mathrm{ch}} \longrightarrow 0$$

be its central extension. The bracket of two current sections lands in the central line through the second normal symbol. The Chevalley differential of $\widehat{\mathfrak{h}}_{\mathrm{ch}}$ contracts two current cogenerators to the central cogenerator; the quotient $\mathfrak{h}_{\mathrm{ch}}$ has the zero bracket differential.

Fock modules. For a momentum α, the Fock module $\mathcal{F}_\alpha$ is generated by a vector $|\alpha\rangle$ such that

$$J_n|\alpha\rangle = 0 \quad (n > 0), \qquad J_0|\alpha\rangle = \alpha|\alpha\rangle.$$

The oscillator modes satisfy

$$[J_m, J_n] = km\,\delta_{m+n,0}.$$

After choosing a conformal vector

$$T = \frac{1}{2k} : JJ :$$

for $k \neq 0$, the central charge is one. The character is

$$\operatorname{ch} \mathcal{F}_\alpha(\tau) = \frac{q^{\alpha^2/(2k)-1/24}}{\eta(\tau)}. \tag{239}$$

The level k controls the momentum quadratic form, while the oscillator determinant has central charge one. This is an explicit demonstration that level and Virasoro anomaly are different invariants.

Sewing. Let T_q be the one-particle sewing operator with eigenvalues q^n. Then

$$\det(1 - T_q) = \prod_{n\geq 1}(1 - q^n).$$

The bosonic Fock functor turns a one-particle determinant into its inverse, giving (239). On a compact Riemann surface of genus g, the nonzero-mode Gaussian integral gives a zeta-regularized determinant of the scalar Laplacian, while zero modes give a volume factor. A compactified boson with momentum lattice Λ multiplies the oscillator determinant by a Siegel theta series.

Field theory. Abelian Chern–Simons theory with action

$$\frac{k}{4\pi}\int_M A \wedge dA$$

and a chiral boundary condition has boundary current algebra (238). A Wilson line of charge q acquires the braiding phase determined by the bilinear form $q_1 q_2 / k$ after the standard normalization of charges and level. The boundary algebra, line category, and Gaussian determinant therefore arise from different operations in one free field theory.

Koszul reading. The quotient $\mathfrak{h}_{\mathrm{ch}}$ has a cofree commutative Chevalley coalgebra with zero bracket differential. For the central extension $\widehat{\mathfrak{h}}_{\mathrm{ch}}$, the Chevalley coderivation contracts two current cogenerators to the central cogenerator. In the level-k vacuum representation the central element acts as $k\mathbf{1}$; after reducing by the vacuum line, this contraction is represented as curvature in the reduced bar model. The completed cobar reconstructs the corresponding chiral enveloping algebra. Linear duality exchanges symmetric algebras and symmetric coalgebras under finite-weight completion.

G.2 Affine Kac–Moody currents

Let $\mathfrak{g}$ be simple and let $(,)$ be an invariant form. The universal affine vertex algebra at level k has currents with OPE

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \tag{240}$$

The lambda bracket is

$$[J^a_\lambda J^b] = J^{[a,b]} + k\lambda(a,b).$$

Its Jacobi identity has two coefficients. The coefficient independent of λ is Jacobi in $\mathfrak{g}$. The coefficient linear in the formal variables is invariance of $(,)$. The simple-pole bracket satisfies Jacobi, and the central term is an invariant two-cocycle compatible with that bracket.

Collision complex. For the centrally extended chiral Lie algebra, the partition collision differential has two components: the simple-pole bracket merges currents to $J^{[a,b]}$, and the central cocycle merges them to the central line. The simple-pole component squares to zero by Jacobi; compatibility with the central component is exactly invariance of $(,)$. A geometric coefficient model that includes all principal parts carries both the first and second normal symbols.

Sugawara tensor. For $k \neq -h^\vee$, the stress tensor is

$$T(z) = \frac{1}{2(k+h^\vee)} \sum_a : J^a J_a :(z), \tag{241}$$

with central charge

$$c = \frac{k \dim \mathfrak{g}}{k+h^\vee}. \tag{242}$$

The shift by $h^\vee$ comes from normal ordering. Equation (242) is the affine conformal anomaly and is rational in k. Linear functions of $k+h^\vee$ describe different affine invariants.

KZ transport. For modules $V_1, \ldots, V_n$, conformal blocks satisfy the KZ equation

$$\left(\partial_{z_i} - \frac{1}{k+h^\vee} \sum_{j\neq i} \frac{\Omega_{ij}}{z_i - z_j}\right) F = 0.$$

Flatness is the calculation of Section B. The monodromy is a braid-group representation. Under the Drinfeld–Kohno comparison, it agrees with the braiding of the corresponding quantum-group category at a parameter fixed by the normalization of the invariant form and the exponentiation of $(k+h^\vee)^{-1}$.

Chern–Simons theory. Three-dimensional Chern–Simons theory at integral level produces the same conformal blocks after quantization on a surface. Wilson-line braiding gives the same ribbon category. The bulk–boundary comparison identifies the current algebra, conformal blocks, and Wilson-line ribbon category at the common quantized level. Framing dependence corresponds to the ribbon twist and the gravitational Chern–Simons counterterm.

Critical level. At $k = -h^\vee$, (241) is undefined. The affine algebra itself remains defined and develops the Feigin–Frenkel centre. The KZ equation in the displayed normalization is singular. The centre is identified with functions on opers for the Langlands dual group. These statements locate criticality in the Sugawara denominator, KZ normalization, and enlarged centre while retaining the affine level $k = -h^\vee$.

Koszul and representation-theoretic dualities. Chiral bar–cobar reconstruction applies in the pro-nilpotent Ran category. Quadratic PBW recognition can be tested after choosing a presentation. Kazhdan–Lusztig equivalence, Drinfeld–Kohno monodromy, Feigin–Frenkel duality, and quadratic Koszul duality are different theorems. They can be composed only when their hypotheses and categories match.

G.3 The Virasoro chiral algebra

The Virasoro algebra is the central extension of vector fields on the circle. Its OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{243}$$

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity is a polynomial identity in two formal variables. It expresses the cocycle identity of the Gelfand–Fuchs central extension and the Lie bracket of vector fields.

Normal symbols. The three singular terms of (243) occupy normal degrees four, two, and one. The normally ordered product occupies degree zero. A logarithmic residue detects only the degree-one component after a logarithmic form has been supplied. The chiral operation remembers all four pieces and their extensions.

Projective connections. The transformation law

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12}\{f, z\}$$

shows that the stress tensor is an affine projective connection. The Schwarzian cocycle is the coordinate form of the Virasoro central extension. On moduli, it gives a projective connection on conformal blocks. The Hodge coefficient is proportional to c, while insertions and boundary factorization contribute further classes.

Verma modules and null vectors. A highest-weight module $M(c, h)$ is generated by v with

$$L_0 v = hv, \qquad L_n v = 0 \quad (n > 0).$$

The Kac determinant detects singular vectors. Passing from the universal Virasoro algebra to a minimal-model quotient changes the homological algebra because relations generated by null vectors are imposed. Each simple quotient has its own Koszul comparison complex, whose null-vector relations determine the result.

The bosonic string. The (b, c) reparametrization ghosts have central charge -26. Coupling them to a matter theory gives a nilpotent BRST charge when the total central charge and intercept anomaly vanish. The number 26 is therefore the cancellation value of a specific ghost system. The value 13 is the midpoint of the reflection $c \mapsto 26 - c$. A Virasoro self-duality at this point is an additional isomorphism of chiral algebras and modules.

Polyakov and determinant geometry. The Weyl anomaly has coefficient c. For free fields it is computed from determinants. The curvature of the determinant line agrees with the central extension represented by (243). This is the rigorous bridge between local OPE and global moduli curvature. A full nonchiral CFT further supplies the antiholomorphic sector, state space, modular invariant, and positivity; a gravitational path integral further supplies bulk fields and integration cycles.

Three-dimensional gravity. Brown–Henneaux boundary conditions produce two Virasoro algebras with

$$c = \bar{c} = \frac{3\ell}{2G}.$$

Under modular invariance, a sparse light spectrum, and vacuum dominance, Cardy asymptotics reproduce the BTZ entropy. The chiral algebra supplies the asymptotic symmetry and central extension. The spectrum and modular invariant are additional physical data.

G.4 Free fermions and ghost systems

A free complex fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The associated Clifford algebra is quadratic. Wick's theorem expresses every correlator as a determinant. On a Riemann surface, a choice of spin structure enters the Szego kernel. The partition function is a determinant or Pfaffian of the Dirac operator.

A (b, c) system with conformal weights $(\lambda, 1-\lambda)$ has stress tensor

$$T = (1-\lambda){:}(\partial b)c{:} - \lambda{:}b\partial c{:}$$

and central charge

$$c_{bc} = 1 - 3(2\lambda - 1)^2. \tag{244}$$

A bosonic (β, γ) system with the same weights has the opposite determinant statistics and central charge

$$c_{\beta\gamma} = -c_{bc}.$$

The cancellation in a tensor product is a determinant identity and a Virasoro central-charge identity. The associated graded polynomial and exterior algebras are ordinary Koszul duals. The Weyl and Clifford algebras are PBW deformations of these two algebras and are related by the corresponding curved or inhomogeneous Koszul formalism when their constant quadratic terms are retained. Their Koszul relation concerns the filtered quadratic algebras; their conformal module categories remain separate representation-theoretic objects.

Ghost number provides a second grading. BRST reduction uses the ghost differential and the vertex differential simultaneously. Spectral-sequence concentration can make the reduction exact. A nonconcentrated spectral sequence retains higher BRST cohomology as part of the reduced derived object.

G.5 Lattice vertex algebras

Let Λ be an even lattice with bilinear form $(,)$ and let ε be a normalized two-cocycle on Λ. The lattice vertex algebra is

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

For $\alpha, \beta \in \Lambda$, the vertex operators have leading product

$$Y(e^\alpha, z)Y(e^\beta, w) = (z-w)^{(\alpha,\beta)}\varepsilon(\alpha, \beta)Y(e^{\alpha+\beta}, w) + \cdots. \tag{245}$$

Evenness gives locality. The cocycle controls the ordered phase. Replacing ε by a cohomologous cocycle gives an isomorphic algebra; replacing it by its inverse reverses the commutator phase.

The genus-one character is

$$\operatorname{ch} V_\Lambda(\tau) = \frac{\Theta_\Lambda(\tau)}{\eta(\tau)^{\operatorname{rk}\Lambda}}. \tag{246}$$

For an even unimodular positive-definite lattice, the chiral central charge is $\operatorname{rk}\Lambda$ and the theory is holomorphic. At genus g, the zero-mode sum is the degree-g Siegel theta series

$$\Theta_\Lambda^{(g)}(\Omega) = \sum_{x_1,\ldots,x_g\in\Lambda} \exp\left(\pi i \sum_{i,j}(x_i,x_j)\Omega_{ij}\right).$$

The oscillator and lattice pieces are independent factors.

For the two even unimodular rank-sixteen lattices $E_8 \oplus E_8$ and D_{16}^+, the Siegel theta series agree in degrees $g = 1, 2, 3$. In degree four their difference is a nonzero scalar multiple of the Schottky form on $\mathfrak{H}_4$; its restriction to the genus-four Jacobian locus vanishes [59, 66]. Their genus-two theta series coincide. The degree-four difference is the Schottky form on $\mathfrak{H}_4$, and its restriction to the smooth genus-four Jacobian locus is zero.

The Leech lattice VOA and the Moonshine VOA both have central charge 24. Their weight-one spaces distinguish them: the Leech theory has a rank-twenty-four Heisenberg sector, while the Moonshine weight-one space is zero. Thus equal central charge can accompany different OPEs and automorphism groups.

G.6 Principal W-algebras

Let $\mathfrak{g}$ be simple and f principal nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine algebra, charged ghosts, and neutral fields when required. Its degree-zero cohomology is the principal W-algebra $\mathcal{W}^k(\mathfrak{g}, f)$ under the standard hypotheses.

For the universal principal W-algebra at generic level, the standard strong generators have conformal weights $d_i + 1$, where d_i are the exponents of $\mathfrak{g}$; special levels may impose decoupling relations or collapse generators. Their OPEs contain nonlinear normally ordered polynomials. These terms are higher transferred operations of the BRST contraction, extending the simple-pole Lie bracket by normally ordered composites.

Feigin–Frenkel duality relates principal W-algebras for Langlands dual Lie algebras at levels satisfying

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1, \tag{247}$$

with the lacing convention included. The relation is multiplicative in the two shifted levels and exchanges $\mathfrak{g}$ with its Langlands dual.

If a filtered BRST contraction is compatible with the chiral bar construction and its spectral sequence converges, then bar and BRST cohomology commute. This transports collision operations and Koszul comparisons through the reduction. The specified contraction, filtration, and convergence theorem determine the nilpotent orbit and completion on which the comparison holds.

G.7 Five comparisons in one table

Model	highest local pole	ordered datum	modular datum	physical realization
Heisenberg	double pole k	abelian phase	oscillator determinant and zero modes	abelian Chern–Simons boundary
affine	double pole plus Lie pole	KZ braid monodromy	affine conformal blocks	nonabelian Chern–Simons/WZW
Virasoro	fourth-order central pole	monodromy of Virasoro blocks	projective connection and determinant line	conformal anomaly; AdS_3 asymptotic symmetry
free ghosts	simple contraction	fermionic or bosonic exchange	determinant/Pfaffian	gauge fixing and BRST
lattice	integral power and cocycle	cocycle phase	Siegel theta series	compactified free boson
W-algebra	nonlinear higher-spin OPE	module-category braiding when constructed	W-conformal blocks	Hamiltonian reduction

Each column is supplied by its own construction. The local pole controls the singular operation; modules and a flat connection control ordered transport; sewing controls modular data; an action and boundary condition control the physical realization. The comparison maps carry one chiral operation into all four outputs while preserving their object types.

G.8 What the examples establish

The examples test the common language by independent constructions whose comparison maps are known.

For Heisenberg theory, the boundary quantization map identifies the current OPE with the abelian Chern–Simons boundary bracket, while Gaussian sewing identifies the oscillator trace with the determinant formula.

For affine theory, the Ward identities produce the KZ connection; Drinfeld–Kohno identifies its braid category with the corresponding quantum-group category, and the Chern–Simons/WZW comparison identifies the same braid data with Wilson-line exchange.

For Virasoro theory, the fourth normal symbol gives the central cocycle, the Ward identity gives the Schwarzian transformation, and the family index theorem identifies its moduli-space image with determinant-line curvature and the Weyl anomaly.

For ghost systems, the Clifford or Weyl quadratic relations determine the parity signs, the corresponding determinant ratio gives the central charge, and the BRST square measures the same total anomaly after the BRST comparison is made.

For lattice theories, the group-algebra cocycle determines ordered phases, Poisson summation gives the modular transformation of the theta series, and the compactified-boson path integral yields the same theta factor after zero-mode and oscillator separation.

Each agreement is mediated by an explicit construction. This is the standard of evidence appropriate to chiral Koszul duality: one operation, calculated in several faithful realizations, with every passage preserving the category of the object.

H The typed reconstruction atlas

A single mathematical language becomes powerful when every object retains its ambient category and every transition is carried by a named functor. This appendix collects the constructions of the paper in that form and places their historical and physical realizations on the same diagram.

H.1 The primitive object

Fix a smooth complex curve X. The primitive datum is

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}). \tag{248}$$

It consists of a factorisable D-module, an associative multiplication

$$\star : A \otimes^{\mathrm{pt}} A \longrightarrow A,$$

and an augmentation $A \to \mathbf{1}$. The factorisation structure carries holomorphic collision. The multiplication carries transverse ordered composition. The requirement that $\star$ be a morphism in $\mathrm{FactD}(X)$ is their interchange law.

A chiral Lie coefficient

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

produces the Chevalley branch

$$\mathfrak{g} \xrightarrow{\mathrm{coChev}^{\mathrm{ch}}} \mathrm{CoFact}_{\mathrm{conil}}(\mathrm{Ran}\, X).$$

A strong monoidal realization compares this branch with the ordered interval construction when both arise from the same field or categorical source.

H.2 The construction table

input geometry or algebra	output	structure governing the passage
oriented interval with augmentation endpoints	$\mathrm{Bar}_X^{\mathrm{ord}}(A)$, a conilpotent coassociative coalgebra	simplicial two-sided bar and interval excision
augmentation boundary	$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$	derived endomorphisms and completion
quadratic presentation $T(V)/(R)$	$A^{¡} \to \mathrm{Bar}_X^{\mathrm{ord}}(A)$	Koszul twisting morphism and filtration comparison
chiral Lie coefficient	commutative factorisation coalgebra on $\mathrm{Ran}\, X$	chiral Chevalley chains and Ran pronilpotence
formal coordinate with translation	vertex or nonlocal-vertex operation	expansion of the filtered diagonal distribution
holonomic or constructible duality datum	Verdier-dual factorisation object	dualisability, base change, and continuity
second transverse direction	exchange local system	E_2 geometry or Yang–Baxter-flat transport
rigid tensor category with fibre functor	bialgebra, Hopf algebra, or quasi-Hopf form	Tannaka coend and rigidity
cyclic trace	Hochschild chains and ribbon sewing	annular closure and pair-of-pants composition
stable-curve sewing	modular graph operations	separating and nonseparating clutching
conformal vector	projective connection and determinant anomaly	Virasoro uniformization and family index theory
renormalized BV theory	factorisation observables and graph weights	propagator, compactification, counterterms, and QME

Each row defines a functorial construction. Comparison theorems arise by supplying a morphism between two outputs in the same target category and proving that it preserves the displayed structure.

H.3 Bar, boundary dual, Verdier dual, and centre

The interval bar is the coalgebraic state object

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1}.$$

The augmentation-boundary algebra is

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

The quadratic dual is obtained from a chosen quadratic presentation and maps canonically to the full bar. Verdier duality is a contravariant sheaf-theoretic functor that converts a coalgebraic factorisation object into an algebraic one when tensor, diagonal, and limit compatibilities are present. The derived centre is the internal Hochschild cochain object

$$Z_{\mathrm{ord,ch}}(A) = \mathrm{HH}^{\bullet}_{\mathrm{FactD}(X)}(A, A),$$

carrying its E_2 structure.

The comparison maps among these objects are:

$$A^{\text{¡}} \longrightarrow \mathrm{Bar}^{\mathrm{ord}}_X(A), \qquad \mathrm{Cobar}^{\mathrm{ord}}_X \mathrm{Bar}^{\mathrm{ord}}_X(A) \longrightarrow A, \qquad A \longrightarrow A^{!!},$$

together with Verdier and Hochschild functors on the subcategories where their hypotheses hold. These maps express quadratic recognition, bar–cobar reconstruction, derived completion, geometric duality, and central deformation theory, respectively.

H.4 Four differential geometries

Four operators carry degree two information.

(i) A cochain differential satisfies $d^2 = 0$.

(ii) A curved A_∞ structure satisfies the Stasheff identities with a nullary operation m_0; in particular m_1^2 is the inner operation prescribed by m_0 and m_2.

(iii) A connection satisfies $\nabla^2 = F_\nabla$, where F_∇ is a two-form on the base with values in endomorphisms.

(iv) A regularized BV operator satisfies the quantum master equation precisely when its anomaly cocycle is cancelled by a quantum interaction or counterterm.

The corresponding remedies live in their native geometries. A curved algebra is treated in a curved or second-kind derived category. Projective curvature is cancelled by an inverse anomaly line with opposite Atiyah class. A BV anomaly is removed by a counterterm whose differential is the anomaly representative. Writing these operators in separate complexes preserves their degrees and makes every cancellation type-correct.

H.5 Residue, principal part, and propagator

Poincaré residue extracts the coefficient of a logarithmic conormal form. The principal-part symbol of order m is a section of $N_{D/Y}^{\otimes m}$. The full pole filtration is the direct sum of these normal symbols together with their extension data. A propagator is a Green kernel selected by a kinetic operator and gauge-fixing condition; its singular symbol lies in the principal-part filtration, while its regular and harmonic pieces encode global analytic data.

On an affine two-point screen, $d\log(z - w)$ generates the logarithmic cohomology class. Pairing the order-m principal part with a dual normal jet and then applying residue extracts the corresponding coefficient. This two-stage operation gives every OPE mode its intrinsic tensor weight.

H.6 Ordering, exchange, and reconstruction

Labelled configurations carry pure-braid transport. Descent to unlabelled configurations extends this transport to the braid group when the permutation endpoints are identified. Symmetric descent factors the braid action through the symmetric group. These stages are governed by the exact sequence

$$1 \longrightarrow P_n \longrightarrow B_n \longrightarrow \Sigma_n \longrightarrow 1.$$

A Drinfeld associator is a group-like series satisfying pentagon and hexagon identities and comparing parenthesized transports. Verdier duality sends monodromy ρ to ρ^{-T}. A bilinear form may identify inverse transpose with inverse. A rigid braided category and fibre functor then reconstruct the corresponding bialgebra by its Tannaka coend. A presentation theorem names the reconstructed object as a Yangian, quantum loop algebra, or elliptic quantum group.

H.7 Modular sewing

The boundary divisors of $\overline{\mathcal{M}}_{g,n}$ are indexed by one-edge stable graphs. Codimension-two strata are indexed by two-edge graphs. A cyclic coefficient system assigns contractions to the flags of every graph. Separating contractions give the convolution bracket; self-contractions give the loop operator Δ. The oriented incidence identity then reads

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0.$$

For conformal blocks, representation theory supplies the spaces of states, the two-point pairing supplies contractions, and analytic sewing supplies the plumbing expansion. The projectively flat connection has a first Chern class built from the Hodge class, cotangent-line classes weighted by conformal dimensions, and boundary classes weighted by factorisation channels. This formula records the complete projective anomaly of the modular functor.

H.8 The historical line

Historical note H.1 (Distributions and operator products). Dirac made support and differentiation part of operator calculus. Schwartz supplied the intrinsic theory of distributions. Wilson organized short-distance singularities by local operator coefficients. Borcherds gathered all mode identities into one formal-distribution law. Beilinson and Drinfeld placed that law on a curve as a diagonal D-module correspondence.

Historical note H.2 (Compactification and factorisation). Fulton and MacPherson resolved simultaneous collisions by adding relative screens. Kontsevich used the same geometry to organize configuration integrals. The Ran space assembled all finite subsets into one factorisation variable. Francis and Gaitsgory proved chiral Koszul duality in the resulting pro-nilpotent chiral monoidal category.

Historical note H.3 (Braids, gauge theory, and quantum groups). Knizhnik and Zamolodchikov constructed flat current-algebra transport. Kohno and Drinfeld identified its braid category with quantum-group representation theory. Witten realized the same braid and ribbon operations through Wilson lines in Chern–Simons theory. Costello, Witten, and Yamazaki produced rational, trigonometric, and elliptic line algebras from four-dimensional gauge theory.

Historical note H.4 (Sewing, anomaly, and BV). Polyakov represented the conformal anomaly by geometric curvature. Segal formulated conformal field theory through sewing. Tsuchiya, Ueno, and Yamada constructed conformal blocks in families. Getzler and Kapranov encoded stable sewing by modular operads. Batalin and Vilkovisky encoded gauge symmetry by master equations, and Costello's renormalization theory realized those equations on compactified graph configurations.

H.9 The physical comparison

Let Obs_∂ be the boundary factorisation algebra of a holomorphic–topological quantum field theory on $X^{\mathrm{an}} \times \mathbb{R}$. A physical realization of A is a quasi-isomorphism

$$\Phi : A^{\mathrm{an}} \xrightarrow{\simeq} \mathrm{Obs}_\partial \tag{249}$$

compatible with E_1 fusion and chiral factorisation. Under Φ, the following pairs are two realizations of one operation:

algebraic geometry	quantum-field-theoretic realization
factorisable D-module	cochain complex of protected boundary observables
diagonal chiral operation	renormalized boundary OPE
principal-part filtration	hierarchy of contact singularities
partition collision complex	local short-distance factorisation complex
Fulton–MacPherson screen	relative ultraviolet configuration
ordered interval product	fusion of ordered defects
exchange local system	transport of line defects
cyclic pairing	two-point function used in sewing
modular graph contraction	fixed-topology amplitude
BV anomaly class	obstruction cocycle for quantum gauge symmetry
Hochschild centre map	central action of a bulk observable on the boundary

The map (249) identifies the two columns and carries every algebraic construction to its observable-theoretic counterpart.

H.10 Minimal constructive data

Proposition H.5 (Quantum-group data)**.** *A quantum-group reconstruction is determined by a braided rigid category, a strong or quasi-monoidal fibre functor, and the resulting Tannaka coend. The multiplication, coproduct, braiding, associator, and antipode are the corresponding tensor-structural maps.*

Proposition H.6 (Modular coefficient data)**.** *A modular master element is determined by a cyclic coefficient object together with separating and nonseparating clutching maps compatible on codimension-two strata. The Feynman-transform differential then proves the quantum master equation.*

Proposition H.7 (Bulk comparison data)**.** *A physical bulk acting on the boundary defines a central map*

$$\mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z_{\mathrm{ord,ch}}(A).$$

An open–closed equivalence is the assertion that this map is a quasi-isomorphism on the selected protected sector.

Proposition H.8 (Scalar projection data)**.** *Central charge, level, determinant curvature, ghost anomaly, and automorphic weight are obtained from their native structured objects by specified traces or characteristic-class maps. A relation among two such numbers is the image of a comparison morphism between those native objects.*

H.11 The common diagram

$$
\begin{tikzcd}
 & A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X)) \arrow[dl, "\mathrm{Bar}^{\mathrm{ord}}_X"'] \arrow[d, "\mathrm{RHom}_A(\mathbf{1},-)"] \arrow[dr, "\mathrm{HH}^\bullet_{\mathrm{FactD}}"] \arrow[dd, bend right] & & \\
\mathrm{Coalg}^{\mathrm{conil}}_{\mathrm{coAss}}(\mathrm{FactD}(X)) \arrow[r, bend left, "\mathrm{Cobar}^{\mathrm{ord}}_X"] & A^! = \mathrm{RHom}_A(\mathbf{1},\mathbf{1}) & Z_{\mathrm{ord,ch}}(A) \arrow[r, "E_2/E_3 \text{ extension}"] & \mathcal{L}_A \arrow[d, "\mathrm{Tannaka}"] \\
 & \mathrm{Obs}_\partial(X \times \mathbb{R}) \arrow[ur, "\text{bulk action}"'] & & \mathcal{H}_A.
\end{tikzcd}
\tag{250}
$$

The chiral Lie branch, cyclic sewing branch, and higher-dimensional source branch attach to this diagram through the comparison maps constructed in Parts II, IV, and V. Every arrow has a source category, a target category, and a geometric operation that defines it.

H.12 The final principle

Holomorphic collision and transverse associative composition form the primitive pair. Their mixed operad governs ordered chiral fields. Chiral Koszul duality resolves collision data on the Ran space. The interval bar resolves augmentation-boundary states. Exchange geometry creates braid transport. Cyclicity creates loops. Stable curves create sewing. The derived centre creates universal central action. Traces and determinants produce the numerical invariants.

The unity of the subject is the commutativity of these constructions. The same ordered chiral object enters every branch, and each branch retains the additional geometry that gives its operations physical and mathematical meaning.

I Historical lineage of the language

I.1 Distributions and local fields

Dirac's delta distribution made singular support a precise mathematical datum [27]. Quantum field theory inherited this lesson: products of fields are defined away from coincidence and acquire contact terms on diagonals. The chiral D-module operation is the algebraic form of that distributional fact. It records the singular product and its support before a coordinate expansion is chosen.

Wilson's operator product expansion separated short-distance coefficients from local operators [75]. Borcherds and the vertex-algebra tradition encoded the consistency of those coefficients in formal distributions. Beilinson and Drinfeld replaced the formal coordinate by a curve and the formal delta function by a morphism of D-modules supported on the diagonal. The present paper uses the Beilinson–Drinfeld object as the chiral coefficient of a genuinely associative ordered product.

I.2 Time ordering and the associative direction

Feynman's time-ordered product records the temporal order of operator insertions [39]. It depends on an ordering of insertion times, and its perturbative expansion is indexed by graphs. Stasheff's associahedra later supplied the homotopy geometry of all parenthesizations [68, 69]. Factorization homology identifies the same geometry with the bar construction on an interval.

The ordered chiral algebra combines these two historical lines. Dirac and Beilinson govern support on the complex diagonal. Feynman and Stasheff govern composition along the ordered real direction.

I.3 Configuration spaces and renormalization

Arnold computed the cohomology of configuration spaces and isolated the three-term relation that underlies the flatness of logarithmic connections [4]. Fulton and MacPherson compactified configurations by replacing collisions with screens [40]. Axelrod–Singer and Kontsevich used related compactifications to make perturbative gauge-theory and deformation-quantization integrals finite and compatible with boundary strata [9, 53].

The compactification has two roles that must remain distinct. Its face category resolves collision incidence. Its screen spaces carry nontrivial cohomology and propagator integrals. The first is combinatorial; the second contains the analytic weight of Feynman graphs.

I.4 Chiral algebras and Koszul duality

Koszul, Priddy, Quillen, and the operadic tradition developed the passage between algebraic operations and coalgebraic resolutions. Beilinson–Drinfeld placed local Lie operations on the Ran space. Francis–Gaitsgory proved the corresponding chiral Koszul duality in a pro-nilpotent monoidal category.

The associative extension considered here uses the same Ran geometry with one additional ordered operadic direction. Its bar is the interval bar internal to factorization D-modules. Its dual is the derived endomorphism algebra of the augmentation boundary. This is the form of Koszul duality that appears in defect physics.

I.5 Anomalies and geometry

Polyakov identified the Weyl anomaly with a local action for the conformal factor [65]. Quillen and the family index theorem identified the same anomaly with curvature of a determinant line [62]. Belavin and Knizhnik related determinant geometry to string measures on moduli [11]. These are different realizations of a central extension of conformal transformations.

The language of the present paper preserves the chain of maps: local stress-tensor cocycle, Ward identity, Atiyah algebra of conformal blocks, determinant-line curvature. The structured chain remains primary, and scalar quantities arise from its traces and characteristic classes.

I.6 Gauge theory, knots, and quantum groups

Witten's Chern–Simons theory connected three-dimensional gauge fields, two-dimensional current algebras, conformal blocks, braid groups, and knot invariants [74]. Drinfeld and Jimbo supplied the quantum groups [29, 48]; Kohno identified their braid representations with KZ monodromy [51]. The category of lines and its exchange determine the reconstructed quantum group; dimensions of invariant subspaces are numerical traces of this category.

Costello, Witten, and Yamazaki added a holomorphic spectral curve to topological gauge theory [19, 20]. Wilson-line OPEs then produced rational, trigonometric, and elliptic quantum groups. Gauge-theoretic constructions of Gaiotto, Moore, and Neitzke and of Nekrasov placed line defects, framed BPS states, Omega backgrounds, and BPS/CFT Ward identities in the same protected sector

[41, 57]; Coulomb-branch convolution gives a complementary algebraic realization [10]. The spectral curve carries chiral dependence; transverse topology carries fusion and exchange.

I.7 Factorization algebras and BV theory

Costello and Gwilliam formulated perturbative quantum field theory as a factorization algebra of observables [14, 15]. The BV classical and quantum master equations became local cohomological identities. Boundary and defect observables became factorization modules. This made the passage from an action functional to a chiral or topological operator algebra a functor with a defined source and target.

The present construction is compatible with that formalism. An ordered chiral algebra is the algebraic local observable structure on a holomorphic–topological product. A BV theory is one realization of it, together with fields, pairing, integration, and renormalization.

I.8 Open–closed theories and centres

Hochschild cochains have long governed deformations of associative algebras. Deligne's conjecture and the Swiss-cheese operad identified their E_2 structure and their universal action on an E_1 algebra. In open–closed field theory, the centre is the universal closed algebra acting on the open sector.

This universal property is exact but limited. It reconstructs the protected bulk from a boundary only when the bulk-to-centre map is an equivalence. Witten's open-string field theory and Costello's Calabi–Yau categories supply settings in which Hochschild chains and cyclic operations are physical closed states [76, 18]. Twisted holography supplies settings in which a boundary algebra determines a protected bulk sector.

I.9 The resulting synthesis

The historical constructions meet in one typed sentence. A noncommutative chiral algebra is an E_1 algebra in factorization D-modules; its chiral coefficient is a diagonal-supported distribution, its multiplication is ordered fusion, its bar is interval factorization homology, its centre is the universal exchange algebra, its quantum group is Tannaka reconstruction of a braided line category, its ribbon loops come from a cyclic trace, its worldsheet sewing comes from conformal blocks, and its anomalies are invertible theories identified by transgression.

Every clause has an independent historical origin. Their unity is the geometry of support, order, exchange, and sewing.

References

[1] D. Ayala and J. Francis, *Factorization homology of topological manifolds*, J. Topol. **8** (2015), 1045–1084.

[2] T. Arakawa, *Representation theory of W-algebras*, Invent. Math. **169** (2007), 219–320.

[3] T. Arakawa, *Introduction to W-algebras and their representation theory*, in *Perspectives in Lie Theory*, Springer INdAM Ser. 19, Springer, 2017, 179–250.

[4] V. I. Arnold, *The cohomology ring of the colored braid group*, Mat. Zametki **5** (1969), 227–231.

[5] A. Beilinson and V. Drinfeld, *Chiral Algebras*, Amer. Math. Soc. Colloq. Publ. 51, 2004.

[6] B. Bakalov, A. De Sole, R. Heluani, and V. G. Kac, *An operadic approach to vertex algebra and Poisson vertex algebra cohomology*, Jpn. J. Math. **14** (2019), 249–342.

[7] B. Bakalov, A. De Sole, R. Heluani, and V. G. Kac, *Chiral versus classical operad*, Int. Math. Res. Not. IMRN (2020), no. 19, 6463–6488.

[8] B. Bakalov, A. De Sole, and V. G. Kac, *Computation of cohomology of vertex algebras*, Jpn. J. Math. **16** (2021), 81–154.

[9] S. Axelrod and I. M. Singer, *Chern–Simons perturbation theory II*, J. Differential Geom. **39** (1994), 173–213.

[10] A. Braverman, M. Finkelberg, and H. Nakajima, *Coulomb branches of 3d $\mathcal{N} = 4$ quiver gauge theories and slices in the affine Grassmannian*, Adv. Theor. Math. Phys. **23** (2019), 75–166.

[11] A. A. Belavin and V. G. Knizhnik, *Algebraic geometry and the geometry of quantum strings*, Phys. Lett. B **168** (1986), 201–206.

[12] J. D. Brown and M. Henneaux, *Central charges in the canonical realization of asymptotic symmetries*, Comm. Math. Phys. **104** (1986), 207–226.

[13] D. Butson and S. Nair, *On the deformation theory of chiral quantizations*, arXiv:2606.27341, 2026.

[14] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory, Vol. 1*, New Mathematical Monographs 31, Cambridge Univ. Press, 2017.

[15] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory, Vol. 2*, New Mathematical Monographs 41, Cambridge Univ. Press, 2021.

[16] K. Costello and D. Gaiotto, *Vertex operator algebras and 3d $\mathcal{N} = 4$ gauge theories*, J. High Energy Phys. 05 (2019), 018.

[17] K. Costello and N. M. Paquette, *Celestial holography meets twisted holography: 4d amplitudes from chiral correlators*, J. High Energy Phys. 10 (2022), 193.

[18] K. Costello, *Topological conformal field theories and Calabi–Yau categories*, Adv. Math. **210** (2007), 165–214.

[19] K. Costello, E. Witten, and M. Yamazaki, *Gauge theory and integrability, I*, ICCM Not. **6** (2018), 46–119.

[20] K. Costello, E. Witten, and M. Yamazaki, *Gauge theory and integrability, II*, ICCM Not. **6** (2018), 120–146.

[21] P. Deligne, *Équations différentielles à points singuliers réguliers*, Lecture Notes in Math. 163, Springer, 1970.

[22] C. Damiolini, A. Gibney, and N. Tarasca, *Conformal blocks from vertex algebras and their connections on moduli spaces of curves*, Geom. Topol. **25** (2021), 2235–2286.

[23] C. Damiolini, A. Gibney, and N. Tarasca, *Vertex algebras of CohFT-type*, in *Facets of Algebraic Geometry, Vol. I*, LMS Lecture Note Ser. 472, Cambridge Univ. Press, 2022, 164–189.

[24] C. Damiolini, A. Gibney, and N. Tarasca, *On factorization and vector bundles of conformal blocks from vertex algebras*, Ann. Sci. Éc. Norm. Supér. **57** (2024), 241–292.

[25] C. Damiolini and L. Woike, *Modular functors from conformal blocks of rational vertex operator algebras*, arXiv:2507.05845, 2025.

[26] R. Dijkgraaf, G. Moore, E. Verlinde, and H. Verlinde, *Elliptic genera of symmetric products and second quantized strings*, Comm. Math. Phys. **185** (1997), 197–209.

[27] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford Univ. Press, 1958.

[28] G. Dunn, *Tensor product of operads and iterated loop spaces*, J. Pure Appl. Algebra **50** (1988), 237–258.

[29] V. G. Drinfeld, *Quantum groups*, in *Proceedings of the ICM, Berkeley 1986*, 798–820.

[30] V. G. Drinfeld, *Quasi-Hopf algebras*, Leningrad Math. J. **1** (1990), 1419–1457.

[31] P. Etingof and D. Kazhdan, *Quantization of Lie bialgebras, I*, Selecta Math. **2** (1996), 1–41.

[32] J. Fay, *Theta Functions on Riemann Surfaces*, Lecture Notes in Math. 352, Springer, 1973.

[33] C. Faber and R. Pandharipande, *Hodge integrals and Gromov–Witten theory*, Invent. Math. **139** (2000), 173–199.

[34] B. Feigin and E. Frenkel, *Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras*, Int. J. Mod. Phys. A **7** (1992), Suppl. 1A, 197–215.

[35] I. B. Frenkel and V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980), 23–66.

[36] E. Frenkel and D. Ben-Zvi, *Vertex Algebras and Algebraic Curves*, 2nd ed., Math. Surveys Monogr. 88, Amer. Math. Soc., 2004.

[37] E. Frenkel and D. Hernandez, *Baxter's relations and spectra of quantum integrable models*, Duke Math. J. **164** (2015), 2407–2460.

[38] J. Francis and D. Gaitsgory, *Chiral Koszul duality*, Selecta Math. **18** (2012), 27–87.

[39] R. P. Feynman, *Space-time approach to non-relativistic quantum mechanics*, Rev. Mod. Phys. **20** (1948), 367–387.

[40] W. Fulton and R. MacPherson, *A compactification of configuration spaces*, Ann. of Math. **139** (1994), 183–225.

[41] D. Gaiotto, G. W. Moore, and A. Neitzke, *Framed BPS states*, Adv. Theor. Math. Phys. **17** (2013), 241–397.

[42] S. Gautam and V. Toledano Laredo, *Yangians and quantum loop algebras*, Selecta Math. **19** (2013), 271–336.

[43] E. Getzler and M. M. Kapranov, *Modular operads*, Compos. Math. **110** (1998), 65–126.

[44] V. Gritsenko and V. Nikulin, *Siegel automorphic form corrections of some Lorentzian Kac–Moody Lie algebras*, Amer. J. Math. **119** (1997), 181–224.

[45] O. Gwilliam and B. R. Williams, *A one-loop exact quantization of Chern–Simons theory*, arXiv:1910.05230, 2020.

[46] D. Hernandez and M. Jimbo, *Asymptotic representations and Drinfeld rational fractions*, Compos. Math. **148** (2012), 1593–1623.

[47] Y.-Z. Huang, *Vertex operator algebras and the Verlinde conjecture*, Comm. Contemp. Math. **10** (2008), 103–154.

[48] M. Jimbo, *A q-difference analogue of $U(\mathfrak{g})$ and the Yang–Baxter equation*, Lett. Math. Phys. **10** (1985), 63–69.

[49] V. G. Kac, *Vertex Algebras for Beginners*, 2nd ed., Univ. Lecture Ser. 10, Amer. Math. Soc., 1998.

[50] D. Kazhdan and G. Lusztig, *Tensor structures arising from affine Lie algebras, I–IV*, J. Amer. Math. Soc. **6**–**7** (1993–1994).

[51] T. Kohno, *Monodromy representations of braid groups and Yang–Baxter equations*, Ann. Inst. Fourier **37** (1987), 139–160.

[52] V. Knizhnik and A. Zamolodchikov, *Current algebra and Wess–Zumino model in two dimensions*, Nuclear Phys. B **247** (1984), 83–103.

[53] M. Kontsevich, *Deformation quantization of Poisson manifolds*, Lett. Math. Phys. **66** (2003), 157–216.

[54] J. Lurie, *Higher Algebra*, available from the author's webpage, 2017 version.

[55] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren Math. Wiss. 346, Springer, 2012.

[56] D. Maulik and A. Okounkov, *Quantum groups and quantum cohomology*, Astérisque 408, 2019.

[57] N. Nekrasov, *BPS/CFT correspondence: non-perturbative Dyson–Schwinger equations and qq-characters*, J. High Energy Phys. 03 (2016), 181.

[58] D. Mumford, *Towards an enumerative geometry of the moduli space of curves*, in *Arithmetic and Geometry, Vol. II*, Birkhäuser, 1983, 271–328.

[59] M. Oura, C. Poor, R. Salvati Manni, and D. Yuen, *Modular forms of weight 8 for* $\Gamma_g(1,2)$, Math. Ann. **346** (2010), 477–498.

[60] L. Positselski, *Two kinds of derived categories, Koszul duality, and comodule–contramodule correspondence*, Mem. Amer. Math. Soc. **212** (2011), no. 996.

[61] S. Priddy, *Koszul resolutions*, Trans. Amer. Math. Soc. **152** (1970), 39–60.

[62] D. Quillen, *Determinants of Cauchy–Riemann operators over a Riemann surface*, Funct. Anal. Appl. **19** (1985), 31–34.

[63] N. Reshetikhin and V. Turaev, *Ribbon graphs and their invariants derived from quantum groups*, Comm. Math. Phys. **127** (1990), 1–26.

[64] A. N. Schellekens, *Meromorphic* $c = 24$ *conformal field theories*, Comm. Math. Phys. **153** (1993), 159–185.

[65] A. M. Polyakov, *Quantum geometry of bosonic strings*, Phys. Lett. B **103** (1981), 207–210.

[66] S. Grushevsky and R. Salvati Manni, *The superstring cosmological constant and the Schottky form in genus 5*, Amer. J. Math. **133** (2011), 1007–1027; erratum incorporated in arXiv:0809.1391v5.

[67] G. Segal, *The definition of conformal field theory*, in *Topology, Geometry and Quantum Field Theory*, London Math. Soc. Lecture Note Ser. 308, Cambridge Univ. Press, 2004, 421–577.

[68] J. D. Stasheff, *Homotopy associativity of H-spaces. I*, Trans. Amer. Math. Soc. **108** (1963), 275–292.

[69] J. D. Stasheff, *Homotopy associativity of H-spaces. II*, Trans. Amer. Math. Soc. **108** (1963), 293–312.

[70] V. V. Schechtman and A. N. Varchenko, *Arrangements of hyperplanes and Lie algebra homology*, Invent. Math. **106** (1991), 139–194.

[71] D. Tamarkin, *Deformations of chiral algebras*, in *Proceedings of the ICM, Vol. II*, Beijing, 2002, 105–116.

[72] A. Tsuchiya, K. Ueno, and Y. Yamada, *Conformal field theory on universal family of stable curves with gauge symmetries*, Adv. Stud. Pure Math. **19** (1989), 459–566.

[73] E. Verlinde, *Fusion rules and modular transformations in 2D conformal field theory*, Nuclear Phys. B **300** (1988), 360–376.

[74] E. Witten, *Quantum field theory and the Jones polynomial*, Comm. Math. Phys. **121** (1989), 351–399.

[75] K. G. Wilson, *Non-Lagrangian models of current algebra*, Phys. Rev. **179** (1969), 1499–1512.

[76] E. Witten, *Noncommutative geometry and string field theory*, Nuclear Phys. B **268** (1986), 253–294.

[77] T. Willwacher, *M. Kontsevich's graph complex and the Grothendieck–Teichmüller Lie algebra*, Invent. Math. **200** (2015), 671–760.

[78] Y. Zhu, *Modular invariance of characters of vertex operator algebras*, J. Amer. Math. Soc. **9** (1996), 237–302.

[79] T. Torii, *On duoidal ∞-categories*, J. Homotopy Relat. Struct. **20** (2025), no. 1, 125–162.